

Lie Algebras

Joe Orii

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1 Introduction

§1.1 Definition of Lie Algebras

Definition 1.1.1. Let F be a field. A **Lie algebra** over F is an F -vector space L , together with a bilinear map called the **Lie bracket** $[\cdot, \cdot]: L \times L \rightarrow L$ that satisfies the following:

L1: $[x, x] = 0$, for all $x \in L$

L2: $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$, for all $x, y, z \in L$ (**Jacobi Identity**).

We may also call this the **commutator** of x and y .

Remark 1.1.2 — Usually, $F = \mathbb{C}$ (but it may also be $\mathbb{R}$ or $\mathbb{F}_2$, and so on), for the purpose of being algebraically closed and having characteristic 0.

Remark 1.1.3 — By bilinearity and L1 of the Lie bracket, we have that

$$0 = [x + y, x + y] = [x, y] + [y, x],$$

so

$$[x, y] = -[y, x], \quad \forall x, y \in L \quad (\text{L1}')$$

In the case that $\text{char}(F) \neq 2$, setting $x = y$ actually implies L1.

Remark 1.1.4 — Unless specifically stated otherwise, all Lie algebras here should be taken to be finite-dimensional.

Exercise 1.1.5.

1. Show that $[v, 0] = 0 = [0, v]$ for all $v \in L$.
2. Suppose that $x, y \in L$ satisfy $[x, y] \neq 0$. Show that x and y are linearly independent over F .

Solution: 1. We have that $[v, 0] = [v, 0 + 0] = [v, 0] + [v, 0]$, so we get the desired result.

2. Suppose that there exists nonzero $a, b \in L$ such that $ax + by = 0$. Then,

$$0 = [0, x] = [ax + by, x] = [by, x] = b[y, x] \neq 0,$$

which is a contradiction. □

§1.2 Examples

Example 1.2.1

Let $F = \mathbb{R}$. The vector cross product $(x, y) \mapsto x \wedge y$ defines the structure of a Lie algebra on $\mathbb{R}^3$. We denote this Lie algebra by $\mathbb{R}_\wedge^3$. Explicitly, if $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$, then

$$x \wedge y = (x_2y_3 - x_3y_2, x_3y_1 - x_1y_3, x_1y_2 - x_2y_1).$$

Exercise 1.2.2. Show that $\wedge$ is bilinear. Then, check that the Jacobi identity holds. *Hint: If $x \cdot y$ denotes the vector dot product, then recall that*

$$x \wedge (y \wedge z) = (x \cdot z)y - (x \cdot y)z, \quad \forall x, y, z \in \mathbb{R}^3.$$

Solution: Bilinearity is clear, so we check the Jacobi identity:

$$\begin{aligned} x \wedge (y \wedge z) + y \wedge (z \wedge x) + z \wedge (x \wedge y) \\ &= (x \cdot z)y - (x \cdot y)z + (y \cdot x)z - (y \cdot z)x + (z \cdot y)x - (z \cdot x)y \\ &= 0 \end{aligned}$$

□

Example 1.2.3 (Abelian Lie Algebra)

Any vector space V has a Lie bracket defined by $[x, y] = 0$ for all $x, y \in V$. This is the **abelian Lie algebra** structure on V . Particularly, the field F may be regarded as a 1-dimensional abelian Lie algebra.

Example 1.2.4 (General Linear Algebra)

Suppose that V is a f.d. vector space over F . Write $\mathfrak{gl}(V)$ for the set of all linear maps from V to V . This is again a vector space over F , and it becomes a Lie algebra, known as a **general linear algebra**. If we define the Lie bracket $[\cdot, \cdot]$ by

$$[x, y] = x \circ y - y \circ x, \quad x, y \in \mathfrak{gl}(V),$$

where $\circ$ denotes the composition of maps.

Exercise 1.2.5. Check that the Jacobi identity holds for the above.

Solution: I will write everything, since it is an exercise I should fully write out at least once in my life.

$$\begin{aligned} [x, [y, z]] &= [x, y \circ z - z \circ y] \\ &= x \circ (y \circ z - z \circ y) - (y \circ z - z \circ y) \circ x \\ &= x \circ y \circ z - x \circ z \circ y - y \circ z \circ x + z \circ y \circ x \end{aligned}$$

Next,

$$\begin{aligned} [y, [z, x]] &= [y, z \circ x - x \circ z] \\ &= y \circ (z \circ x - x \circ z) - (z \circ x - x \circ z) \circ y \\ &= y \circ z \circ x - y \circ x \circ z - z \circ x \circ y + x \circ z \circ y \end{aligned}$$

Finally,

$$\begin{aligned} [z, [x, y]] &= [z, x \circ y - y \circ x] \\ &= z \circ (x \circ y - y \circ x) - (x \circ y - y \circ x) \circ z \\ &= z \circ x \circ y - z \circ y \circ x - x \circ y \circ z + y \circ x \circ z \end{aligned}$$

We then see that this sums to 0. $\square$

Example 1.2.6

Here is a matrix version of the above. Write $\mathfrak{gl}(n, F)$ for the vector space of all $n \times n$ matrices over F with Lie bracket defined by

$$[x, y] := xy - yx,$$

where xy is the usual matrix product of x and y .

As a vector space, $\mathfrak{gl}(n, F)$ has a basis consisting of the matrix units e_{ij} for $1 \leq i, j \leq n$, where e_{ij} denotes the $n \times n$ matrix with 1 in the (i, j) -th entry and 0 elsewhere.

Exercise 1.2.7. Check that

$$[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{il}e_{kj}.$$

Solution: Clear. $\square$

Abuse of Notation 1.2.8. We may often just write $\mathfrak{gl}(n)$ instead of $\mathfrak{gl}(n, F)$.

§1.3 Subalgebras and Ideals

Definition 1.3.1. Given a Lie algebra L , we define a **Lie subalgebra** of L to be a vector subspace $K \subseteq L$ such that $[x, y] \in K$, for all $x, y \in K$.

Example 1.3.2 (Special Linear Algebra)

Let $\mathfrak{sl}(n, F) \subset \mathfrak{gl}(n, F)$ be the subspace of matrices which have trace 0. For arbitrary square matrices x and y , we can see that $xy - yx$ has trace 0, so $[x, y] = xy - yx$ defines a Lie algebra structure on $\mathfrak{sl}(n, F)$. Explicitly, [L1](#) and [L2](#) are inherited from $\mathfrak{gl}(n, F)$. This Lie algebra is known as the **special linear algebra**. As a vector space, $\mathfrak{sl}(n, F)$ has a basis consisting of the e_{ij} for $i \neq j$ with $e_{ii} - e_{i+1, i+1}$ for $1 \leq i < n$. Thus, the dimension of this Lie algebra is $n^2 - 1$.

Example 1.3.3

Let $\mathfrak{b}(n, F) \subset \mathfrak{gl}(n, F)$ be subalgebra of upper triangular matrices. This is a Lie algebra with the same Lie bracket as $\mathfrak{gl}(n, F)$. We note that the $\mathfrak{b}$ comes from Borel, but this is not important to the content.

Similarly, let $\mathfrak{n}(n, F)$ be the strictly upper triangular matrices in $\mathfrak{gl}(n, F)$ (upper triangular matrices which have diagonals 0 as well). This is again a Lie algebra with the same Lie bracket as $\mathfrak{gl}(n, F)$. In fact, for $\mathfrak{n}(2, F)$, this is an abelian Lie algebra.

Exercise 1.3.4. Show that a one-dimensional Lie algebra is an abelian Lie algebra.

Solution: If L is one-dimensional, then $L = \text{Span}\{x\}$. However, we know that $[\alpha x, \beta x] = \alpha\beta[x, x] = 0$, for $\alpha, \beta \in F$. Thus, true. $\square$

Definition 1.3.5. An **ideal** of a Lie algebra L is a vector subspace I of L such that $[x, y] \in I$, for all $x \in L, y \in I$.

Remark 1.3.6 — We note that by L1', $[x, y] = -[y, x]$, implying that there is no need to distinguish between *left* and *right* ideals.

Example 1.3.7

$\mathfrak{sl}(n, F)$ is an ideal of $\mathfrak{gl}(n, F)$, and $\mathfrak{n}(n, F)$ is an ideal of $\mathfrak{b}(n, F)$.

Remark 1.3.8 — We note that the definition tells us that every ideal is always a subalgebra, but every subalgebra need not to be an ideal. An example highlighting this would be that $\mathfrak{b}(n, F)$ is a subalgebra of $\mathfrak{gl}(n, F)$, but for $n \geq 2$, it is not an ideal. This can be explicitly seen by taking $e_{11} \in \mathfrak{b}(n, F)$ and $e_{21} \in \mathfrak{gl}(n, F)$, and computing that

$$[e_{21}, e_{11}] = e_{21} \notin \mathfrak{b}(n, F).$$

Example 1.3.9 (Trivial Ideals)

Given a Lie algebra L , L and 0 are both ideals of L . These are called the **trivial ideals** of L .

Example 1.3.10 (The Centre)

Given a Lie algebra L , the **centre** of L (which is generally non-trivial) is defined by

$$Z(L) := \{x \in L : [x, y] = 0, \forall y \in L\}.$$

It is clear that $L = Z(L)$ if and only if L is abelian. On the other hand, it takes some work to see when $Z(L) = 0$. It is clear that for any L , $Z(L)$ is an ideal of L (and so a subalgebra as well). It is also clear that for any L , $Z(L)$ is abelian.

Exercise 1.3.11. Find $Z(L)$ when $L = \mathfrak{sl}(2, F)$. *Hint: this should depend on the characteristic of F*

Solution: Recall that $\mathfrak{sl}(2, F)$ is the set of 2×2 matrices over F which have trace 0, along with the usual bracket induced by $\mathfrak{gl}(2, F)$. We note that $\mathfrak{sl}(2, F)$ is spanned by

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

so we shall consider the following bracket:

$$\begin{aligned} \left[\begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \begin{pmatrix} d & e \\ f & -d \end{pmatrix} \right] &= \begin{pmatrix} ad + bf & ae - bd \\ cd - af & ce + ad \end{pmatrix} - \begin{pmatrix} da + ce & db - ea \\ fa - dc & fb + da \end{pmatrix} \\ &= \begin{pmatrix} bf - ce & 2(ae - bd) \\ 2(cd - af) & ce - bf \end{pmatrix} \end{aligned}$$

This forces $b = c = 0$, so we get

$$Z(\mathfrak{sl}(2, F)) = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \right\}, \quad a \in F.$$

However, such matrices are not contained in $\mathfrak{sl}(2, F)$ unless $\text{char}(F) = 2$. If $\text{char}(F) \neq 2$, then $Z(\mathfrak{sl}(2, F)) = 0$. $\square$

Exercise 1.3.12. For $\mathfrak{gl}(2, \mathbb{C})$ check that

$$Z(\mathfrak{gl}(2, \mathbb{C})) = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \right\}, \quad a \in \mathbb{C}.$$

Solution: We are looking for the matrices which commute with all other matrices. This is clear $\text{Span}\{I_2\}$. $\square$

Example 1.3.13

For $\mathfrak{sl}(2, \mathbb{C})$, $Z(\mathfrak{sl}(2, \mathbb{C})) = 0$. In fact, the only ideals of $\mathfrak{sl}(2, \mathbb{C})$ are 0 and $\mathfrak{sl}(2, \mathbb{C})$ itself.

§1.4 Homomorphisms

Definition 1.4.1. If L_1 and L_2 are Lie algebras over a field F , a map $\varphi: L_1 \rightarrow L_2$ is a **homomorphism** if φ is a linear map and

$$\varphi([x, y]) = [\varphi(x), \varphi(y)], \quad \forall x, y \in L_1.$$

We say that φ is an **isomorphism** if φ is also bijective.

Remark 1.4.2 — Note that in the definition above, the first Lie bracket is taken in L_1 , and the second is taken in L_2 .

Example 1.4.3

If K is a subalgebra of L , then the natural inclusion map $K \hookrightarrow L$ is a homomorphism of Lie algebras.

Exercise 1.4.4. Show that if $\varphi: L_1 \rightarrow L_2$ is a homomorphism, then $\ker \varphi$ is an ideal of L_1 , and $\text{im } \varphi$ is a Lie subalgebra of L_2 .

Solution: It is clear that $\ker \varphi$ is a vector subspace of L . Let $x \in \ker \varphi$ and $y \in L_1$. Then, we want to show that $[x, y] \in \ker \varphi$. We have

$$\varphi([x, y]) = [\varphi(x), \varphi(y)] = [0, y] = 0,$$

so indeed, $[x, y] \in \ker \varphi$.

Next, we check that $\operatorname{im} \varphi$ is a Lie subalgebra. Let $x, y \in \operatorname{im} \varphi$. Then, we need to check that $[x, y] \in \operatorname{im} \varphi$. However, this is clear because

$$[x, y] = \varphi([\varphi^{-1}(x), \varphi^{-1}(y)]).$$

□

Exercise 1.4.5. Let L_1, L_2 be abelian Lie algebras. Show that L_1 and L_2 are isomorphic if and only if they have the same dimension.

Solution: The Lie bracket is trivial for L_1 and L_2 , so a Lie algebra isomorphism $\varphi: L_1 \rightarrow L_2$ simply reduces to a linear map, so $\dim L_1 = \dim L_2$. Conversely, the structure constants are trivial so we get an isomorphism. □

§1.4.i The Adjoint Homomorphism

Definition 1.4.6. If L is a Lie algebra, we define the **adjoint homomorphism** $\operatorname{ad}: L \rightarrow \mathfrak{gl}(L)$ by

$$(\operatorname{ad} x)(y) := [x, y],$$

for $x, y \in L$. That is, $\operatorname{ad}: x \mapsto [x, \cdot]$.

We now check that this is indeed a homomorphism. Recall that we check that ad is linear and that it is a homomorphism of Lie algebras.

First, we shall note that from the bilinearity of Lie bracket, the map $\operatorname{ad} x$ is linear for $x \in L$. For the same reason, the map $x \mapsto \operatorname{ad} x$ is itself linear.

We will now verify that ad is a homomorphism of Lie algebras; that is, that it's compatible. We have to check that

$$\operatorname{ad}([x, y]) = [\operatorname{ad} x, \operatorname{ad} y]$$

for all $x, y \in L$. In other words, for all $x, y, z \in L$, we have to check that

$$(\operatorname{ad}([x, y]))(z) = ([\operatorname{ad} x, \operatorname{ad} y])(z).$$

The LHS says that

$$(\operatorname{ad}([x, y]))(z) = [[x, y], z].$$

The RHS says that

$$[\operatorname{ad} x, \operatorname{ad} y] = \operatorname{ad} x \circ \operatorname{ad} y - \operatorname{ad} y \circ \operatorname{ad} x,$$

since the bracket is in $\mathfrak{gl}(L)$, so we have

$$([\operatorname{ad} x, \operatorname{ad} y])(z) = [x, [y, z]] - [y, [x, z]] = [x, [y, z]] + [y, [z, x]] = -[z, [x, y]] = [[x, y], z].$$

Therefore, we have equality.

Remark 1.4.7 — Note that the Jacobi identity is equivalent to ad being a homomorphism of Lie algebras. Some may even take this to be the motivation behind the Jacobi identity.

Exercise 1.4.8. Determine $\ker(\text{ad})$.

Solution: This is $Z(L)$. □

§1.5 Algebras

Definition 1.5.1. An **algebra** over a field F is a F -vector space A together with a bilinear map $A \times A \rightarrow A$, $(x, y) \mapsto xy$. Here, we say that xy is the **product** of x and y .

Remark 1.5.2 — The idea is that the bilinear map is the ‘multiplication’. However, do not be confused by the notation, for xy doesn’t mean literally multiplication, but rather the output of the bilinear map.

Remark 1.5.3 — As is clear from the name, Lie algebras are algebras which satisfy the Jacobi identity and the degeneracy condition.

Definition 1.5.4. An algebra A is said to be:

- **associative** if $(xy)z = x(yz)$, for all $x, y, z \in A$.
- **unital** or **unitary** if there is an element $1_A \in A$ such that $1_A x = x = x 1_A$, for all nonzero elements of A

Remark 1.5.5 — Usually, when we talk about algebras, we only consider Lie algebras or associative algebras (that are usually unitary as well).

Example 1.5.6

The $n \times n$ matrices over F , with the usual matrix multiplication, is an associative unitary algebra. Note that we also gave this set the structure of a Lie algebra, with Lie bracket $[x, y] = xy - yx$.

Exercise 1.5.7. Let L be a Lie algebra. Show that the Lie bracket is associative ($[x, [y, z]] = [[x, y], z]$, for all $x, y, z \in L$) if and only if for all $a, b \in L$, the commutator $[a, b]$ lies in $Z(L)$.

Solution: The converse is clear, so we prove the forward implication. In particular, associativity and the Jacobi identity tells us that $[y, [z, x]] = 0$, for all $x, y, z \in L$, so we get the forward direction as well. □

Example 1.5.8

We can generalize the previous example. Given any associative algebra A over F , then we define a new bilinear operation $[\cdot, \cdot]$ on A by

$$[a, b] := ab - ba,$$

for all $a, b \in A$. Then, A equipped with $[\cdot, \cdot]$ is a Lie algebra. Note that the Lie algebras $\mathfrak{gl}(V)$ (and $\mathfrak{gl}(n, F)$) are special cases of this. We have previously shown that this satisfies the Jacobi identity already.

§1.6 Derivations

Definition 1.6.1. Let A be an algebra over a field F . A **derivation** of A is an F -linear map $d: A \rightarrow A$ such that

$$d(ab) = ad(b) + d(a)b, \quad \forall a, b \in A.$$

We let $\text{Der}A$ denote the set of derivations of A .

Remark 1.6.2 — We note that when we write $d(ab)$, this multiplication ab is not the usual multiplication, but is the multiplication (a, b) defined by the bilinear map on the algebra A .

Remark 1.6.3 — $\text{Der}A$ is closed under addition and scalar multiplication, and also contains the zero map. Thus, $\text{Der}A$ is a F -vector subspace of $\mathfrak{gl}(A)$ (linear maps from A to A). Moreover, $\text{Der}(A)$ is a Lie subalgebra of $\mathfrak{gl}(A)$, since the first part of the following exercise implies that if D and E are derivations, then so is $[D, E]$.

Exercise 1.6.4. Let D and E be derivations of an algebra A .

1. Show that $[D, E] = D \circ E - E \circ D$ is also a derivation.
2. Show that $D \circ E$ need not be a derivation.

Solution: 1. Let $a, b \in A$. Then, we have the following:

$$\begin{aligned} [D, E](ab) &= (D \circ E)(ab) - (E \circ D)(ab) \\ &= D(aE(b) + E(a)b) - E(aD(b) + D(a)b) \\ &= (aD(E(b)) + D(a)E(b)) + (E(a)D(b) + D(E(a))b) \\ &\quad - (aE(D(b)) + E(a)D(b)) - (D(a)E(b) + E(D(a))b) \\ &= a[D, E](b) + [D, E](a)b \end{aligned}$$

2. We can consider the example below. Let $D = E$ be the usual differentiation. Then,

$$D^2(fg) = D(f'g + fg') = f''g + 2f'g' + fg'' = D^2(f)g + 2D(f)D(g) + fD^2(g).$$

□

Example 1.6.5

Let $A = C^\infty\mathbb{R}$ be the vector space of all infinitely differentiable functions $\mathbb{R} \rightarrow \mathbb{R}$. For $f, g \in A$, we define the product fg by pointwise multiplication; that is, $(fg)(x) := f(x)g(x)$. With this definition, A is an associative algebra. The usual derivative, $Df = f'$, is a derivation of A since by the product rule

$$D(fg) = (fg)' = f'g + fg' = (Df)g + f(Dg).$$

Example 1.6.6

Let L be a Lie algebra and let $x \in L$. The map $\text{ad } x: L \rightarrow L$ is a derivation of L , since by the Jacobi identity, we get

$$(\text{ad } x)[y, z] = [x, [y, z]] = [[x, y], z] + [y, [x, z]] = [(\text{ad } x)y, z] + [y, (\text{ad } x)z],$$

for all $y, z \in L$.

§1.7 Structure Constants

Let L be a finite dimensional Lie algebra over F , and choose a basis $(x_1, \dots, x_n)$. Then $[\cdot, \cdot]$ is completely determined by the products $[x_i, x_j]$. We define scalars $a_{ij}^k \in F$ such that

$$[x_i, x_j] = \sum_{k=1}^n a_{ij}^k x_k.$$

We then call the a_{ij}^k to be the **structure constants** of L with respect to the basis.

Remark 1.7.1 — We emphasize that the structure constants a_{ij}^k depend on the choice of basis of L ; that is, different bases result in different structure constants.

By L1 and L1', $[x_i, x_i] = 0$ for all i and $[x_i, x_j] = -[x_j, x_i]$ for all $i \neq j$. Therefore, we just need to know the structure constants a_{ij}^k for $1 \leq i < j \leq n$ to fully determine the structure constants.

Exercise 1.7.2. Let L_1 and L_2 be Lie algebras. Then, show that L_1 is isomorphic to L_2 if and only if there is a basis B_1 of L_1 and a basis B_2 of L_2 such that the structure constants of L_1 with respect to B_1 are equal to the structure constants of L_2 with respect to B_2 .

Solution: Let $\{x_1, \dots, x_n\} = B_1$ and $\{y_1, \dots, y_n\} = B_2$. First suppose $\varphi: L_1 \xrightarrow{\sim} L_2$, so that $\varphi(x_i) = \varphi(y_i)$. Let $[x_i, x_j] = a_{ij}^k x_k$. Then,

$$\sum_{k=1}^n a_{ij}^k y_k = \varphi\left(\sum_{k=1}^n a_{ij}^k x_k\right) = \varphi([x_i, x_j]) = [\varphi(x_i), \varphi(x_j)] = [y_i, y_j].$$

Thus, true. The converse is clear. If $[x_i, x_j] = [y_i, y_j]$, then we can construct the isomorphism by $x_i \mapsto y_i$. $\square$

Example 1.7.3

L is abelian if and only if all the structure constants are 0.

Example 1.7.4

Consider $\mathfrak{sl}(2, F) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a + d = 0 \right\}$. We can take the basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Indeed, this is a basis, because

$$be + cf + ah = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \in \mathfrak{sl}(2, F).$$

We can take the brackets to find the structure constants:

$$\begin{aligned} [e, f] &= ef - fe = h \\ [h, e] &= he - eh = 2e \\ [h, f] &= hf - fh = -2f \end{aligned}$$

Exercise 1.7.5. Let L be a Lie algebra with basis $(x_1, \dots, x_n)$. What condition does the Jacobi identity impose on the structure constants a_{ij}^k ?

Solution: We get $a_{ij}^k = a_{ji}^k$. □

Exercise 1.7.6. Prove that $\mathfrak{sl}(2, \mathbb{C})$ has no nontrivial ideals.

Solution: Suppose that $0 \neq I \subseteq \mathfrak{sl}(2, \mathbb{C})$ is an ideal. If $e \in I$, then it follows that $h \in I$ as well, since $[e, h] = -2e$ and $[e, f] = h$. Then, $[h, f] = -2f$, so $f \in I$ as well. In a similar manner, we get that if there is a basis element in I , then we actually get all. Thus, $I = \mathfrak{sl}(2, \mathbb{C})$. □

Exercise 1.7.7. Let L be the 3-dimensional **complex** Lie algebra with basis $\{x, y, z\}$ and Lie bracket defined by

$$[x, y] = z, \quad [y, z] = x, \quad [z, x] = y.$$

1. Show that L is isomorphic to the Lie subalgebra of $\mathfrak{gl}(3, \mathbb{C})$ consisting of all 3×3 antisymmetric matrices with entries in $\mathbb{C}$.
2. Find an explicit isomorphism $\mathfrak{sl}(2, \mathbb{C}) \cong L$.

Solution: 1. Note that we can take the following three matrices to be a basis of all 3×3 antisymmetric matrices with entries in $\mathbb{C}$ is

$$f = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad g = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}.$$

Then,

$$[f, g] = h, \quad [g, h] = f, \quad [h, f] = g.$$

Thus, sending $x \mapsto f$, $y \mapsto g$, $z \mapsto h$, we have the desired isomorphism.

2. Let the following three matrices be a basis for $\mathfrak{sl}(2, \mathbb{C})$.

$$a = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad b = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad c = \frac{1}{2} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

Then, $[a, b] = c$, $[b, c] = a$, $[c, a] = b$, so we're done. $\square$

Exercise 1.7.8. Let $S \in M_n(F)$. Define

$$\mathfrak{gl}_S(n, F) = \{x \in \mathfrak{gl}(n, F) : x^T S = -Sx\}.$$

1. Show that $\mathfrak{gl}_S(n, F)$ is a Lie subalgebra of $\mathfrak{gl}(n, F)$.
2. Find $\mathfrak{gl}_S(2, \mathbb{R})$ if $S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.
3. Does there exist a matrix S such that $\mathfrak{gl}_S(2, \mathbb{R})$ is equal to the set of all diagonal matrices in $\mathfrak{gl}(2, \mathbb{R})$?
4. Find a matrix S such that $\mathfrak{gl}_S(3, \mathbb{R})$ is isomorphic to the Lie algebra $\mathbb{R}_\lambda^3$.

Solution: 1. It is clear that this is a subspace, so we check that for $x, y \in \mathfrak{gl}_S(n, F)$, the bracket $[x, y] \in \mathfrak{gl}_S(n, F)$.

$$\begin{aligned} [x, y]^T S + S[x, y] &= (xy - yx)^T S + S(xy - yx) \\ &= (y^T x^T - x^T y^T) S - S(xy - yx) \\ &= (xy)^T S + S(xy) - ((yx)^T S + S(yx)) \\ &= 0. \end{aligned}$$

Thus, true.

2. Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathfrak{gl}(2, \mathbb{R})$. We check when we have

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Let us shift the RHS to the LHS and see what we get:

$$\begin{pmatrix} 0 & a \\ 0 & b \end{pmatrix} + \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix} = 0$$

This means $b = c = 0$ and $a = -d$. Therefore,

$$\mathfrak{gl}_S(2, \mathbb{R}) = \left\{ \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix} : a \in \mathbb{R} \right\}.$$

3. Suppose that $\mathfrak{gl}_S(2, \mathbb{R}) = \text{Span}\{I_2\}$. Then, we find matrices $S \in M_2(\mathbb{R})$ such that

$$(\lambda I_2)S + S(\lambda I_2) = 0, \quad \lambda \in \mathbb{R}.$$

In other words, $2\lambda S = 0$, so $S = 0$, the zero matrix. However, $\mathfrak{gl}_0(2, \mathbb{R}) = \mathfrak{gl}(2, \mathbb{R})$, which is not the set of diagonal matrices.

4. We will use the structure constants. We find the structure constants of $\mathbb{R}_\wedge^3$ using the basis $\{e_1, e_2, e_3\}$. We have:

$$e_1 \wedge e_2 = e_3, \quad e_2 \wedge e_3 = e_1, \quad e_3 \wedge e_1 = e_2$$

Let $\{x, y, z\}$ be a basis for $\mathfrak{gl}_S(3, \mathbb{R})$, so that this is subject to the conditions

$$[x, y] = z, \quad [y, z] = x, \quad [z, x] = y.$$

Then, using the previous exercise as motivation, we claim that $\mathfrak{gl}_S(3, \mathbb{R})$ is isomorphic to the Lie subalgebra of $\mathfrak{gl}(3, \mathbb{R})$ consisting of 3×3 anti symmetric matrices with entries in $\mathbb{R}$ which we shall call L . We note that a basis for L can be given by the following matrices:

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

We now verify the structure constants:

$$[A, B] = C, \quad [B, C] = A, \quad [C, A] = B.$$

Thus our claim is correct. Then, $X^T S = -XS$ using anti-symmetry of X , and we want $-XS = -SX$. It is then clear that S must be of the form λI_3 . $\square$

Exercise 1.7.9. Let V be an n -dimensional complex vector space and let $L = \mathfrak{gl}(V)$. Suppose that $x \in L$ is diagonalizable, with eigenvalues $\lambda_1, \dots, \lambda_n$. Show that $\text{ad } x \in \mathfrak{gl}(L)$ is also diagonalizable and that its eigenvalues are $\lambda_i - \lambda_j$, for $1 \leq i, j \leq n$.

Solution: It suffices to check how $\text{ad } x$ acts on the basis vectors of $\mathfrak{gl}(L)$. That is, how it acts on matrices e_{ij} . First note that we can write $x = \lambda_1 E_{11} + \dots + \lambda_n E_{nn}$. Then, we can check that $\text{ad } x(E_{ij}) = (\lambda_i - \lambda_j)(E_{ij})$. $\square$

Exercise 1.7.10. Let L be a Lie algebra. We showed that $\text{ad } x: L \rightarrow L$ are derivations of L ; these are known as **inner derivations**. Show that if $\text{IDer } L$ is the set of inner derivations of L , then $\text{IDer } L$ is an ideal of $\text{Der } L$.

Solution: First, we show additive closure.

$$(\text{ad } x + \text{ad } y)(z) = [x, z] + [y, z] = [x + y, z] = \text{ad } (x + y)(z),$$

so true. Next, we show closure under scalar multiplication.

$$\text{ad } \lambda x(y) = [\lambda x, y] = \lambda[x, y] = \lambda(\text{ad } x(y)),$$

so true. Finally, let $D \in \text{Der } L$ and $\text{ad } x \in \text{IDer } L$, and let $y \in L$. Then,

$$\begin{aligned} [\text{ad } x, D](y) &= \text{ad } x \circ D(y) - D \circ \text{ad } x(y) \\ &= [x, D(y)] - D[x, y] \\ &= [x, D(y)] - [x, D(y)] - [D(x), y] \\ &= -[D(x), y] \\ &= -\text{ad } D(x)(y). \end{aligned}$$

Thus, $[\text{ad } x, D] = -\text{ad } D(x)$, so we have containment. $\square$

2 Ideals and Homomorphisms

In this chapter, we'll see that the theory of Lie algebras play a similar role to that of normal subgroups of groups.

§2.1 Constructions with Ideals

Given I, J to be ideals of a Lie algebra L , we shall show several examples of how to construct new ideals using I and J here.

Example 2.1.1 ($I \cap J$)

We know that $I \cap J$ is a subspace of L , so we just need to check that if $x \in L$ and $y \in I \cap J$, then $[x, y] \in I \cap J$. This follows from the fact that I and J are ideals.

Example 2.1.2 ($I + J$)

We also have that $I + J$ is an ideal of L , where

$$I + J := \{x + y : x \in I, y \in J\}$$

It is easy to see that this is a vector subspace of L , and we can also see that the bracket is contained in $I + J$.

Example 2.1.3 (Commutator of Ideals)

We can also define a product of ideals. Let

$$[I, J] := \text{Span} \{[x, y] : x \in I, y \in J\}.$$

We claim that $[I, J]$ is an ideal of L .

Firstly, it is, by definition, a subspace. Secondly, if $x \in I$, $y \in J$, and $u \in L$, then the Jacobi identity gives

$$[u, [x, y]] = [x, [u, y]] + [[u, x], y].$$

Here, $[u, y] \in J$ as J is an ideal, so $[x, [u, y]] \in [I, J]$. Similarly, $[[u, x], y] \in [I, J]$. Thus, their sum belongs to $[I, J]$. A general element $t \in [I, J]$ is a linear combination of brackets $[x, y]$ with $x \in I$, $y \in J$, say $t = \sum c_i [x_i, y_i]$, where the c_i are scalars and $x_i \in I$ and $y_i \in J$. Then, for any $u \in L$, we have

$$[u, t] = \left[u, \sum c_i [x_i, y_i] \right] = \sum c_i [u, [x_i, y_i]],$$

where $[u, [x_i, y_i]] \in [I, J]$ as shown above. Thus, $[u, t] \in [I, J]$ and so $[I, J]$ is an ideal of L .

Remark 2.1.4 — It's necessary to define $[I, J]$ to be the span of the commutators of elements of I and J rather than just the set of such commutators. Below is an exercise for where the set of commutators is not itself an ideal.

Exercise 2.1.5. Recall that if L is a Lie algebra, we defined L' to be the subspace spanned by the commutators $[x, y]$ for $x, y \in L$. The purpose of this exercise is to show that the set of commutators may not even be a vector space.

Consider the ring $\mathbb{R}[X, Y]$, and let L be the set of all matrices of the form

$$A(f(x), g(y), h(x, y)) = \begin{pmatrix} 0 & f(x) & h(x, y) \\ 0 & 0 & g(y) \\ 0 & 0 & 0 \end{pmatrix}$$

1. Prove that L is a Lie algebra with the usual commutator bracket. (Note that L is infinite dimensional here)
2. Prove that

$$\begin{aligned} [A(f_1(x), g_1(y), h_1(x, y)), A(f_2(x), g_2(y), h_2(x, y))] \\ = A(0, 0, f_1(x)g_2(y) - f_2(x)g_1(y)). \end{aligned}$$

Hence, describe L' .

3. Show that if $h(x, y) = x^2 + xy + y^2$, then $A(0, 0, h(x, y))$ is not a commutator.

Solution: 1. L is still a Lie algebra then, since the calculations are analogous to that in the case of the general linear group.

2. Let $A_1 := A(f_1(x), g_1(y), h_1(x, y))$ and $A_2 := A(f_2(x), g_2(y), h_2(x, y))$. Then, $[A_1, A_2]$ is equal to:

$$\begin{pmatrix} 0 & 0 & f_1(x)g_2(y) \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & f_2(x)g_1(y) \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

so true. L' is therefore spanned by matrices of the form $A(0, 0, f_1(x)g_2(y) - f_2(x)g_1(y))$.

3. We show that we can't find polynomials $f_1(x), f_2(x), g_1(y), g_2(y) \in \mathbb{R}[X, Y]$ such that $f_1(x)g_2(y) - f_2(x)g_1(y) = x^2 + xy + y^2$. None of these polynomials can be constant, but then we can't get the terms x^2 and y^2 without leaving out xy . $\square$

§2.1.i Derived Algebra

Example 2.1.6 (Derived Algebra)

An important example of the construction above is when we take $I = J = L$.

Notation 2.1.7. We write L' for $[L, L]$.

Definition 2.1.8. Despite being an ideal of L , L' is usually known as the **derived algebra** of L . The term **commutator algebra** is also sometimes used.

Exercise 2.1.9. Show that if $L = \mathfrak{gl}(n, F)$, then $L' = \mathfrak{sl}(n, F)$.

Solution: If we take $x, y \in \mathfrak{gl}(n, F)$, then

$$[x, y] = xy - yx,$$

which has trace zero, so $L' \subseteq \mathfrak{sl}(n, F)$. Conversely, take $x \in \mathfrak{sl}(n, F)$. Then, noting that, taking any $y, z \in \mathfrak{gl}(n, F)$, the bracket $[y, z]$ has trace zero, we can let $x = [y, z]$, so indeed $\mathfrak{sl}(n, F) \subset L'$. $\square$

Exercise 2.1.10. If $F = \mathbb{C}$ (or characteristic not 2), then $\mathfrak{sl}(2, \mathbb{C})' = \mathfrak{sl}(2, \mathbb{C})$.

Solution: It is clear that $\mathfrak{sl}(2, \mathbb{C})' \subseteq \mathfrak{sl}(2, \mathbb{C})$, so we prove the reverse inclusion. The easiest way to do this is to just look at all commutator relations. Let

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then, $[e, f] = h$, $[h, e] = e$, $[h, f] = -f$. $\mathfrak{sl}(2, \mathbb{C})'$ is the span of h, e, f , which is just $\mathfrak{sl}(2, \mathbb{C})$ again. $\square$

§2.2 Quotient Algebras

Note that if I is an ideal of the Lie algebra L , then I is in particular a subspace of L . Therefore, we may consider the cosets

$$z + I = \{z + x : x \in I\},$$

for $z \in L$ and the quotient vector space

$$L/I = \{z + I : z \in L\}.$$

We now claim that the Lie bracket on L/I can be defined by

$$[w + I, z + I] = [w, z] + I,$$

for $w, z \in L$. Here, the bracket on the RHS is the Lie bracket in L . To check that this is well-defined, we must check that $[w, z] + I$ depends only on the cosets containing w and z and not on the particular coset representatives w and z . Suppose that $w + I = w' + I$ and $z + I = z' + I$. Then, $w - w' \in I$ and $z - z' \in I$. By bilinearity of the Lie bracket in L ,

$$\begin{aligned} [w', z'] &= [w' + (w - w'), z' + (z - z')] \\ &= [w, z] + [w - w', z'] + [w', z - z'] + [w - w', z - z'], \end{aligned}$$

where the final three summands all belong to I . Thus, $[w' + I, z' + I] = [w, z] + I$ as we needed. Therefore, from the first part of the below exercise, L/I is a Lie algebra. We call this the **quotient** or **factor algebra** of L by I .

Exercise 2.2.1.

1. Show that the Lie bracket defined on L/I is bilinear and satisfies the axioms of the Lie bracket.
2. Show that the linear transformation $\pi: L \rightarrow L/I$ which takes an element $z \in L$ to its

coset $z + I$ is a homomorphism of Lie algebras.

Solution: 1. Bilinearity follows trivially from the bilinearity of the Lie bracket. Next, $[x + I, x + I] = I$, so okay. Finally, we just need to verify the Jacobi identity. This also follows from the usual Jacobi identity.

2. We have

$$\pi([x, y]) = [x, y] + I = [x + I, y + I] = [\pi(x), \pi(y)].$$

□

Example 2.2.2

Consider the quotient of Lie algebras $\mathfrak{gl}(2, \mathbb{C})/\mathfrak{sl}(2, \mathbb{C})$. $\mathfrak{gl}(2, \mathbb{C})$ is four-dimensional and $\mathfrak{sl}(2, \mathbb{C})$ is three dimensional, as $\mathbb{C}$ -vector spaces. Therefore, the quotient is one-dimensional, so we get that

$$\mathfrak{gl}(2, \mathbb{C})/\mathfrak{sl}(2, \mathbb{C}) \cong \mathbb{C},$$

the abelian Lie algebra. Of course, we can think of this quotient more directly to think about if this is true. Particularly, the trace map $\mathfrak{gl}(2, \mathbb{C}) \rightarrow \mathbb{C}$ provides this explicit isomorphism.

Theorem 2.2.3 (Isomorphism Theorems)

(i) Let $\varphi: L_1 \rightarrow L_2$ be a homomorphism of Lie algebras. Then, $\ker \varphi$ is an ideal of L_1 and $\operatorname{im} \varphi$ is a subalgebra of L_2 , and

$$L_1/\ker \varphi \cong \operatorname{im} \varphi.$$

(ii) If I and J are ideals of a Lie algebra, then $(I + J)/J \cong I/(I \cap J)$.

(iii) Suppose that I and J are ideals of a Lie algebra L such that $I \subseteq J$. Then, J/I is an ideal of L/I and $(L/I)/(J/I) \cong L/J$.

Proof. That $\ker \varphi$ is an ideal of L_1 and $\operatorname{im} \varphi$ is a subalgebra of L_2 were proven previously. All the isomorphisms we need are already known for vector spaces and their subspaces. We have also proven in the previous exercise that they are also homomorphisms of Lie algebras. □

Example 2.2.4

We have that $\operatorname{tr}: \mathfrak{gl}(n, F) \rightarrow F$ is a homomorphism of Lie algebras. We have that

$$\operatorname{tr}([x, y]) = \operatorname{tr}(xy - yx) = \operatorname{tr}(xy) - \operatorname{tr}(yx) = 0,$$

and

$$[\operatorname{tr}(x), \operatorname{tr}(y)] = 0,$$

because F is abelian (it's the one-dimensional vector space). $\ker(\operatorname{tr}) = \mathfrak{sl}(n, F)$, so we have that

$$\mathfrak{gl}(n, F)/\mathfrak{sl}(n, F) \cong F.$$

Exercise 2.2.5. Show that if L is a Lie algebra, then $L/Z(L)$ is isomorphic to a subalgebra of $\mathfrak{gl}(L)$.

Solution: Consider $\text{ad}: L \rightarrow \mathfrak{gl}(L)$, which takes $x \mapsto \text{ad } x(\cdot)$. Indeed, the kernel is $Z(L)$, so by the first isomorphism theorem, we get

$$L/Z(L) \xrightarrow{\sim} \text{im}(\text{ad})$$

□

Exercise 2.2.6. Show that if $z \in L'$, then $\text{tr}(\text{ad } z) = 0$.

Solution: $z \in L'$, so $z = [u, v]$, for some $u, v \in L$. Therefore,

$$\text{tr}(\text{ad}([u, v])) = \text{tr}(\text{ad } u \circ \text{ad } v - \text{ad } v \circ \text{ad } u) = 0.$$

□

§2.3 Correspondence between Ideals

Analogous to the case of ideals in a ring, given an ideal I of a Lie algebra L , there is a bijective correspondence between ideals of the factor algebra L/I and ideals of L which contain I . If J is an ideal of L containing I , then J/I is an ideal of L/I . Conversely, if K is an ideal of L/I , then $J := \{z \in L: z + I \in K\}$. One can readily check that J is an ideal of L and that J contains I . These two maps are inverses of one another. Essentially, we're just taking images and preimages.

Example 2.3.1

Suppose that L is a Lie algebra and I is an ideal in L such that L/I is abelian. In this case, the ideals of L/I are just the subspaces of L/I . By the correspondence above, the ideals of L which contain I are exactly the subspaces of L that contain I .

§2.4 Direct Sum Algebra

Definition 2.4.1. Let L_1 and L_2 be Lie algebras, and define

$$L := \{(x_1, x_2): x_i \in L_i\}$$

be the direct sum of their underlying vector spaces. If we define

$$[(x_1, x_2), (y_1, y_2)] := ([x_1, y_1], [x_2, y_2]),$$

then L becomes a Lie algebra, the **direct sum** of L_1 and L_2 . As for vector spaces, we denote the direct sum of Lie algebra L_1 and L_2 by $L = L_1 \oplus L_2$.

Remark 2.4.2 — We may also write

$$[x_1 \oplus x_2, y_1 \oplus y_2] = [x_1, y_1] + [x_2, y_2]$$

for the condition on the Lie bracket.

Exercise 2.4.3.

1. Show that the above actually defines a Lie algebra.
2. Prove that $\mathfrak{gl}(2, \mathbb{C})$ is isomorphic to the direct sum of $\mathfrak{sl}(2, \mathbb{C})$ with $\mathbb{C}$, the 1-dimensional complex abelian Lie algebra.
3. Show that if $L = L_1 \oplus L_2$, then $Z(L) = Z(L_1) \oplus Z(L_2)$ and $L' = L'_1 \oplus L'_2$. Formulate a general version for a direct sum $L_1 \oplus \cdots \oplus L_k$.
4. Are the summands in the direct sum decomposition of a Lie algebra uniquely determined? *Hint: If you think the answer is yes, read Section 16.4 in [1]*

Solution: 1. First,

$$[(x, y), (x, y)] = ([x, x], [y, y]) = (0, 0),$$

so the degeneracy condition is clear. Next, the Jacobi identity:

$$\begin{aligned} [(x_1, x_2), [(y_1, y_2), (z_1, z_2)]] &= [(x_1, x_2), ([y_1, z_1], [y_2, z_2])] \\ &= ([x_1, [y_1, z_1]], [x_2, [y_2, z_2]]) \end{aligned}$$

$$\begin{aligned} [(y_1, y_2), [(z_1, z_2), (x_1, x_2)]] &= [(y_1, y_2), ([z_1, x_1], [z_2, x_2])] \\ &= ([y_1, [z_1, x_1]], [y_2, [z_2, x_2]]) \end{aligned}$$

$$\begin{aligned} [(z_1, z_2), [(x_1, x_2), (y_1, y_2)]] &= [(z_1, z_2), ([x_1, y_1], [x_2, y_2])] \\ &= ([z_1, [x_1, y_1]], [z_2, [x_2, y_2]]) \end{aligned}$$

Summing these, we see that from the fact that L_1 and L_2 are Lie algebras, the Jacobi identity shows component-wise. Thus, this is indeed a Lie algebra.

2. Note that $\mathfrak{gl}(2, \mathbb{C})$ has the basis $e_1 = E_{12}, e_2 = E_{21}, e_3 = E_{11} - E_{22}, e_4 = E_{11} + E_{22}$, where E_{ij} has 1 in the (i, j) -th position and 0 elsewhere. We claim that $\text{Span}\{e_1, e_2, e_3\}$ is an ideal isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. Indeed, $[e_1, e_2] = e_3, [e_2, e_3] = 2e_2, [e_1, e_3] = -2e_1$, so this is true. The center of $\mathfrak{gl}(2, \mathbb{C})$ is spanned by e_4 , so indeed it follows that

$$\mathfrak{gl}(2, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathbb{C}.$$

3. Consider $(x_1, x_2), (y_1, y_2) \in L_1 \oplus L_2$. Then,

$$[(x_1, x_2), (y_1, y_2)] = ([x_1, y_1], [x_2, y_2]).$$

Fixing (x_1, x_2) , if we want this to be equal to 0, it follows that $y_1 \in Z(L_1), y_2 \in Z(L_2)$, and the reverse inclusion is clear. It follows that $Z(L) = Z(L_1) \oplus Z(L_2)$. For the second statement, it is clear that L' is spanned by the brackets in each component, so $L' = L'_1 \oplus L'_2$. The general version for this would be $L = L_1 \oplus \cdots \oplus L_k$ implies

$$\begin{aligned} Z(L) &= Z(L_1) \oplus \cdots \oplus Z(L_k) \\ L' &= L'_1 \oplus \cdots \oplus L'_k \end{aligned}$$

4. If $L_1 \oplus L_2 = L_3 \oplus L_4$, it need not be that L_1 is equal to L_3 or L_4 (or similar statements). Note that this already fails for vector spaces. If L_1, L_2, L_3, L_4 are already abelian and 1-dimensional, this can fail: any vector space is a direct sum of subspaces in many ways (take different bases for instance). $\square$

Exercise 2.4.4. Suppose that $L = L_1 \oplus L_2$ is the direct sum of two Lie algebras.

1. Show that $\{(x_1, 0) : x_1 \in L_1\}$ is an ideal of L isomorphic to L_1 and that $\{(0, x_2) : x_2 \in L_2\}$ is an ideal of L isomorphic to L_2 . Show that the projections $p_1(x_1, x_2) = x_1$ and $p_2(x_1, x_2) = x_2$ are Lie algebra homomorphisms.

Suppose now that L_1 and L_2 have no non-trivial proper ideals.

2. Let J be a proper ideal of L . Show that if $J \cap L_1 = 0$, and $J \cap L_2 = 0$, then the projections $p_1 : J \rightarrow L_1$ and $p_2 : J \rightarrow L_2$ are isomorphisms.
3. Deduce that if L_1 and L_2 are not isomorphic as Lie algebras, then $L_1 \oplus L_2$ has only two nontrivial proper ideals.
4. Assume that the ground field F is infinite. Show that if $L_1 \cong L_2$ and $\dim L_1 = 1$, then $L_1 \oplus L_2$ has infinitely many different ideals.

Solution: 1. This is immediate.

2. Let $x = (x_1, x_2) \in J$. If $p_1(x) = 0$, then $J \cap L_1 = 0$ implies that $x_1 = 0$. Therefore, $p_2(x) = x_2$, but since the intersection is 0, it follows that x_2 is zero as well. For the same reason, p_2 is injective. Now, since p_1, p_2 are Lie algebra morphisms, it follows that $p_1(J), p_2(J)$ are subalgebras of L_1, L_2 , respectively. In fact, we claim that they are ideals (we won't check this, but is standard). Since L_1 and L_2 have no nontrivial ideals, we get surjectivity of p_1, p_2 .

3. $L_1 \oplus L_2$ has the ideals L_1, L_2 . In fact, there are no more. If J is some other proper ideal with zero intersections with L_1 and L_2 , then $L_1 \cong L_2$, a contradiction.

4. L_1 and L_2 are both abelian. Say $L_1 = \text{Span}\{v\}$, $L_2 = \text{Span}\{w\}$. Then, combinations of the form (av, bw) are ideals of L (where $a, b \in F$). The isomorphism classes can be distinguished via $\mathbb{P}^1(F)$, so there are infinitely many. $\square$

Exercise 2.4.5. Let L_1 and L_2 be Lie algebras, and let $\varphi : L_1 \rightarrow L_2$ be a surjective Lie algebra homomorphism. Determine whether the following statements are true or false.

1. $\varphi(L'_1) = L'_2$.
2. $\varphi(Z(L_1)) = Z(L_2)$.
3. if $h \in L_1$ and $\text{ad } h$ is diagonalizable, then $\text{ad } \varphi(h)$ is diagonalizable.

What about if φ is an isomorphism?

Solution: 1. Let $[x, y] \in L'_2$, $\varphi(x') = x$, $\varphi(y') = y$, so that $x', y' \in L'_1$. Then,

$$[x, y] = [\varphi(x'), \varphi(y')] = \varphi([x', y']),$$

so indeed L'_1 surjects onto L'_2 , so true.

2. We claim this is true if φ is an isomorphism. Let $x \in Z(L_1)$ and $y \in L_2$, and $y' \in L_1$ such that $\varphi(y') = y$. Then,

$$[\varphi(x), y] = \varphi([x, y']) = \varphi(0) = 0,$$

so $\varphi(Z(L_1)) \subseteq Z(L_2)$. Conversely, let $z \in Z(L_2)$ with $\varphi(z') = z$ and $x \in L_1$. Then,

$$0 = [z, \varphi(x)] = \varphi([z', x]).$$

This is where we require injectivity of φ , for then we can say that $[z', x] = 0$, so $z' \in Z(L_1)$. Indeed, if we only had surjectivity, then $[z', x]$ is not necessarily 0.

3. Let $v_1, \dots, v_n$ be a basis for L_1 such that $\text{ad } h$ is diagonalizable with respect to this basis. That is, $[h, v_i] = \lambda v_i$, for some $\lambda \in F$. Then,

$$\lambda \varphi(v_i) = \varphi([h, v_i]) = [\varphi(h), \varphi(v_i)] = \text{ad } \varphi(h)(v_i).$$

In particular, we note that by the surjectivity of φ , $\varphi(v_1), \dots, \varphi(v_n)$ must span L_2 ; taking any subset of this basis which act as a basis for L_2 , then we see that $\text{ad } \varphi(h)$ is diagonalizable with respect to it, by the calculation above. $\square$

Exercise 2.4.6. For each pair of the following Lie algebras over $\mathbb{R}$, decide whether or not they are isomorphic:

1. the Lie algebra $\mathbb{R}_\wedge^3$ where the Lie bracket is given by the vector product.
2. the upper triangular 2×2 matrices over $\mathbb{R}$.
3. the strictly upper triangular 3×3 matrices over $\mathbb{R}$.
4. $L = \{x \in \mathfrak{gl}(3, \mathbb{R}) : x^T = -x\}$

Solution: We note that $\mathbb{R}_\wedge^3 \cong L$ by computing the structure constants. Indeed taking the basis

$$(1, 0, 0), \quad (0, 1, 0), \quad (0, 0, 1)$$

for $\mathbb{R}_\wedge^3$ and the basis

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

for L shows that the structure constants are the same; thus, isomorphic. The derived algebra for $\mathfrak{b}(2, \mathbb{R})$ and $\mathfrak{n}(3, \mathbb{R})$ have dimension 1, so we see that $\mathbb{R}_\wedge^3 \cong L$ are not isomorphic to these. Now suppose that we have an isomorphism $\varphi: \mathfrak{b}(2, \mathbb{R}) \xrightarrow{\sim} \mathfrak{n}(3, \mathbb{R})$. Then, we would have that $\varphi(Z(\mathfrak{b}(2, \mathbb{R}))) = Z(\mathfrak{n}(3, \mathbb{R}))$. We can compute that

$$Z(\mathfrak{b}(2, \mathbb{R})) = 0.$$

We also compute that

$$Z(\mathfrak{n}(3, \mathbb{R})) = \text{Span} \left\{ \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\}.$$

It follows that these two are not isomorphic, since the dimension of the kernel (the nullity) differ. $\square$

Exercise 2.4.7. In Exercise 1.7.8, we defined the Lie algebra $\mathfrak{gl}_S(n, F)$ over a field F , where $S \in M_n(F)$.

Suppose that $T \in \mathfrak{gl}(n, F)$ is another $n \times n$ matrix such that $T = P^T S P$ for some invertible $n \times n$ matrix $P \in \mathfrak{gl}(n, F)$. Show that the Lie algebras $\mathfrak{gl}_S(n, F)$ and $\mathfrak{gl}_T(n, F)$ are isomorphic.

Solution: We have that $\mathfrak{gl}_S(n, F) = \{x \in \mathfrak{gl}(n, F) : x^T S = -Sx\}$. Consider the map $\varphi : \mathfrak{gl}_S(n, F) \rightarrow \mathfrak{gl}_T(n, F)$ such that $x \mapsto P^{-1}xP$. Then, we have that

$$\begin{aligned} x^T S = -Sx &\implies x^T ((P^{-1})^T T P^{-1}) = -((P^{-1})^T T P^{-1})x \\ &\implies (P^{-1}xP)^T T = -T(P^{-1}xP). \end{aligned}$$

Thus, this map actually lands in $\mathfrak{gl}_T(n, F)$. Indeed, the map $P^{-1}xP$ is bijective because P is invertible; thus, has full rank. We just check that φ respects the Jacobi identity as needed.

$$\begin{aligned} \varphi([x, y]) &= P^{-1}[x, y]P = P^{-1}(xy - yx)P = P^{-1}xyP - P^{-1}yxP \\ &= [P^{-1}xP, P^{-1}yP] = [\varphi(x), \varphi(y)]. \end{aligned}$$

□

Exercise 2.4.8. Let S be an $n \times n$ invertible matrix with entries in $\mathbb{C}$. Show that if $x \in \mathfrak{gl}_S(n, \mathbb{C})$, then $\text{tr} x = 0$.

Solution: We have the $x^T S = -Sx$. Then, $S^{-1}x^T S = -x$, so $\text{tr}(x) = \text{tr}(x^T) = \text{tr}(-x) = -\text{tr}(x)$. Thus, $\text{tr}(x) = 0$. □

Exercise 2.4.9. Let I be an ideal of a Lie algebra L . Let B be the **centralizer** of I in L ; that is,

$$B := C_L(I) := \{x \in L : [x, a] = 0, \forall a \in I\}.$$

Now suppose that

1. $Z(I) = 0$
2. if $D : I \rightarrow I$ is a derivation, then $D = \text{ad } x$ for some $x \in I$.

Show that $L = I \oplus B$.

Solution: We want to show that given $x \in B$ and $y \in L$, $[x, y] \in B$. That is, we must verify $[[x, y], a] = 0$, for all $a \in I$. For this, we use the Jacobi identity:

$$[[x, y], a] + \underbrace{[[y, a], x]}_{=0} + [[a, x], y] = 0,$$

and $[a, x] \in I$ because I is an ideal, so $[[a, x], y] = 0$. Thus, $[[x, y], a] = 0$, and we've proven true.

Note that $Z(I) = \{x \in I : [x, y] = 0, \forall y \in I\}$. Therefore, $Z(I) = 0$ implies that $I \cap B = 0$. Now take $x \in L$ and consider $\text{ad } x$. This is a derivation from L to L , but restricting to I , this is a derivation from I to I . Therefore,

$$\text{ad } x|_I = \text{ad } y|_I,$$

for some $y \in I$. Then, for all $z \in I$, $[x - y, z] = [x, z] - [y, z] = 0$. Thus, $x - y \in C_L(I)$, so we can write every $x \in L$ as $x = y + (x - y)$, and so we're done. □

3 Low Dimensional Algebras

We would like to know how many essentially different Lie algebras there are (up to isomorphism) and what approaches we can use to classify them. To get a feel of this, we first look at Lie algebras of dimensions one, two, and three. Another reason for looking at such low-dimensional Lie algebras is because they will arise as subalgebras of Lie algebras of larger dimension later.

Remark 3.0.1 — Abelian Lie algebras are easily understood; for any positive integer n , there exists an abelian Lie algebra of dimension n . We've also seen that any two abelian Lie algebras of the same dimension over the same field are isomorphic, so we understand them completely. Thus, our focus will be on non-abelian Lie algebras from now.

§3.1 Dimension 1

This is easy; any 1-dimensional Lie algebra is abelian and is unique up to isomorphism. This was a previous exercise.

§3.2 Dimension 2

Let L be a 2-dimensional Lie algebra now. One possibility is that it is abelian.

Next, we suppose that L is a non-abelian Lie algebra of dimension 2 over a field F . It is clear that we can find $x \in L'$ such that $x = [y, z]$ for some $y, z \in L$, since L is nonabelian. Then, let w be any other element of L , linearly independent of x . Then, x, w is a basis of L .

We now claim that the derived algebra of L' , must be the 1-dimensional space spanned by x . We note that $x \in L'$, and that L' is spanned by $[x, x], [x, w], [w, x], [w, w]$. The first and last are zero and the middle two are negatives of each other. Thus, L' is spanned by $[x, w]$, so it's 1-dimensional, and $x \in L'$, so our claim is true. It follows then that $[x, w] = \lambda x$, for some nonzero λ . Then, $[x, \lambda^{-1}w] = x$; write $v = \lambda^{-1}w$. Then, we've shown that if L is 2-dimensional and nonabelian, then there exists a basis x, v of L such that $[x, v] = x$. Therefore, in particular L is unique up to isomorphism.

We've shown that if a 2-dimensional non-abelian Lie algebra exists, then it must have a basis $\{x, y\}$ with the Lie bracket given by the equation above. We shall also check that defining the Lie bracket in this way really does give a Lie algebra. In other words, that the Jacobi identity holds.

Exercise 3.2.1. Check that in this case, we really did write down a Lie algebra; that is, that the Jacobi identity holds.

Therefore, we've proven the following theorem.

Theorem 3.2.2

Let F be any field. Up to isomorphism, there is a unique 2-dimensional nonabelian Lie algebra over F . This Lie algebra has basis $\{x, y\}$ such that its Lie bracket is described by $[x, y] = x$. The centre of this Lie algebra is 0.

Remark 3.2.3 — When we say ‘the Lie bracket is described by ...’, this includes the information that $[x, x] = 0$ and $[x, y] = -[y, x]$.

§3.3 Dimension 3

Now, if L is a nonabelian 3-dimensional Lie algebra over a field F , then we know only that the derived algebra L' is non-zero. It may have dimension 1 or 2, or maybe even 3. We also know that $Z(L) \subset L$ is a proper ideal of L . We shall therefore first relate L' to $Z(L)$.

§3.3.i The Heisenberg Algebra

Definition 3.3.1. Let L be any 3-dimensional Lie algebra with L' being 1-dimensional and also $L' \subset Z(L)$. There is a unique such Lie algebra, with basis f, g, z , where $[f, g] = z$ and $z \in Z(L)$. This is known as the **Heisenberg algebra**.

We shall now go on to prove that the definition above actually holds—that such an algebra exists and is unique (up to isomorphism).

We can choose $f, g \in L$ such that $[f, g] \neq 0$, since L' is assumed to be 1-dimensional; further, $[f, g]$ spans L' . We’ve also assumed that L' is contained in the centre of L , so we know that $[f, g]$ commutes with all elements of L . Now set $z := [f, g]$.

Claim 3.3.2. f, g, z are a basis for L .

Proof. Suppose that $\alpha, \beta, \gamma \in F$, and $\alpha f + \beta g + \gamma z = 0$. We get that

$$0 = [\alpha f + \beta g + \gamma z, f] = -\beta[f, g] + \gamma[z, f] = \beta z \implies \beta = 0.$$

Bracketing with g , giving $\alpha = 0$. Bracketing with z , we get $\gamma = 0$, so this is a set of 3 linearly independent elements, so we have a basis. $\square$

Thus, f, g, z is a basis with

$$[f, g] = z, \quad [f, z] = 0, \quad [g, z] = 0.$$

Thus, the Lie algebra is uniquely determined up to isomorphism. We simply have to observe that the Lie algebra of strictly upper triangular 3×3 matrices over F has this form if one takes the basis $\{e_{12}, e_{23}, e_{13}\}$ satisfies these conditions. Further, we see that L' is equal to the centre $Z(L)$ with these. We can therefore conclude that there is a unique example.

Remark 3.3.3 — We can go from the other way as well. We can define the

Heisenberg algebra is the set of matrices

$$\left\{ \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix} \right\} \subset \mathfrak{gl}(3, F)$$

with the usual Lie bracket for $\mathfrak{gl}(3, F)$. We shall compute the Lie bracket for two such matrices below:

$$\begin{aligned} \left[\begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & e & f \\ 0 & 0 & g \\ 0 & 0 & 0 \end{pmatrix} \right] &= \begin{pmatrix} 0 & 0 & ag \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & ce \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & ag - ce \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

Thus, we see that L' is 1-dimensional. It is also clear from the above computation that $L' = Z(L)$.

§3.3.ii Another Lie Algebra where $\dim L' = 1$

Naturally, we turn to the case when $\dim L' = 1$ and $L' \not\subset Z(L)$ now. We shall give an explicit construction again. We can use the direct sum construction to give on such a Lie algebra. Namely, take $L = L_1 \oplus L_2$, where L_1 is 1-dimensional and L_2 is the 2-dimensional non-abelian Lie algebra in Theorem 3.2.2. Then, we can apply Exercise 2.4.3 to say that

$$L' = L'_1 \oplus L'_2 = L'_2.$$

Therefore, L' is 1-dimensional. Moreover, $Z(L) = Z(L_1) \oplus Z(L_2) = L_1$ and thus L' is not contained in L_1 .

Remark 3.3.4 — Perhaps surprisingly, there are no other Lie algebras with this property, which is what the following theorem contains.

Theorem 3.3.5

Let F be any field. There is a unique 3-dimensional Lie algebra over F such that L' is 1-dimensional and L' is not contained in $Z(L)$. This Lie algebra is the direct sum of the 2-dimensional nonabelian Lie algebra with the 1-dimensional Lie algebra.

Proof. Choose a $0 \neq x \in L'$. By definition, $x \notin Z(L)$, so there is some $y \in L$ with $[x, y] \neq 0$. Then x, y are linearly independent (by Exercise 1.1.5). Since x spans L' , we know that $[x, y]$ is a multiple of x . We can rescale y to get $[x, y] = x$ then. (Alternatively, we can say that the subalgebra of L generated by x, y is a 2-dimensional nonabelian Lie algebra, so by Theorem 3.3.5, we can say that $[x, y] = x$.)

We now extend $\{x, y\}$ to a basis of L , say by w . Since x spans L' , there exist scalars α, β such that

$$[x, w] = \alpha x, \quad [y, w] = \beta x,$$

for some α, β . We now claim that L contains a nonzero central element z which is not in $\text{Span}\{x, y\}$.

For $z = \lambda x + \mu y + \nu w$, we can calculate the following:

$$\begin{aligned}[x, z] &= [x, \lambda x + \mu y + \nu w] = \mu x + \nu \alpha x \\ [y, z] &= [y, \lambda x + \mu y + \nu w] = -\lambda x + \nu \beta x\end{aligned}$$

Hence, if we take $\lambda = \beta$, $\mu = -\alpha$ and $\nu = 1$, then $[x, z] = [y, z] = 0$ and $z \notin \text{Span}\{x, y\}$, as $z = \beta x - \alpha y + w$ then. It's also clear that $[z, z] = 0$. The Lie algebra is determined by

$$[x, y] = x, \quad [x, z] = 0, \quad [y, z] = 0.$$

Therefore, $L = \text{Span}\{x, y\} \oplus \text{Span}\{z\}$ is a direct sum of Lie algebras of the required form. $\square$

§3.3.iii Semidirect Products

Definition 3.3.6. Let $\mathfrak{g}$ be a Lie Algebra, and $\mathfrak{a} \subset \mathfrak{g}$ an ideal. Suppose that there exists a subalgebra $\mathfrak{b}$ of $\mathfrak{g}$ such that $\mathfrak{b} \hookrightarrow \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$ is an isomorphism. Then, $\mathfrak{g}$ is the **semidirect product** of $\mathfrak{a}$ by $\mathfrak{b}$.

Remark 3.3.7 — Note that the first map in the sequence $\mathfrak{b} \hookrightarrow \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$ is the canonical inclusion and the second is the quotient.

Example 3.3.8

Consider the unique non-abelian Lie algebra, which we shall denote L here, of dimension 2 with basis x, y with $[x, y] = x$. Taking $\mathfrak{a} = \mathbb{C}x$ and $\mathfrak{b} = \mathbb{C}y$, we can write $L = \mathbb{C}x \oplus \mathbb{C}y$.

Claim 3.3.9. If $\mathfrak{g}$ is the semidirect product of $\mathfrak{a}$ by $\mathfrak{b}$, then as vector spaces, we can write

$$\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{b}.$$

Proof. First, if $x \in \mathfrak{a} \cap \mathfrak{b}$, then the image of x under the map $\mathfrak{b} \hookrightarrow \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$ is 0; in particular, injectivity tells us that $x = 0$ then. The decomposition of $\mathfrak{g}$ into the vector subspaces $\mathfrak{a}$ and $\mathfrak{b}$ then comes from the surjectivity of the map from $\mathfrak{b}$ to $\mathfrak{g}/\mathfrak{a}$. $\square$

Let us continue with our situation that we have a semidirect product $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{b}$.

Claim 3.3.10. The semidirect product $\mathfrak{g}$ is determined by the data of the Lie algebras $\mathfrak{a}, \mathfrak{b}$, and a Lie algebra homomorphism $\theta: \mathfrak{b} \rightarrow \text{Der}(\mathfrak{a})$.

Proof. For any $x \in \mathfrak{b}$, we can restrict the adjoint map $\text{ad } x: \mathfrak{g} \rightarrow \mathfrak{g}, y \mapsto [x, y]$ to $\mathfrak{a}$, since $\mathfrak{a}$ is an ideal: $\text{ad } x|_{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{a}$. Conversely, if we know the ideals $\mathfrak{a}$ and $\mathfrak{b}$, and the map $\text{ad } x|_{\mathfrak{a}}$ for $x \in \mathfrak{b}$, then we know the bracket $[\cdot, \cdot]$ on $\mathfrak{g}$ since $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{b}$ is a direct sum. This is because of the following: if we denote $a \oplus b = (a, b) \in \mathfrak{a} \oplus \mathfrak{b}$, then

$$[(a_1, b_1), (a_2, b_2)] = [a_1, a_2] + [b_1, b_2] + \text{ad } b_1(a_2) - \text{ad } b_2(a_1).$$

Since we need to understand the last two terms, the claim is true. Indeed, the map $\mathfrak{b} \rightarrow \mathfrak{gl}(\mathfrak{a})$ defined by $x \mapsto \text{ad } x|_{\mathfrak{a}}$ is a linear map.

We shall highlight that more is true; this map is a Lie algebra homomorphism and the image is contained in the derivations $\text{Der}(\mathfrak{a}) = \{D \in \mathfrak{gl}(\mathfrak{a}) : D([x, y]) = [x, Dy] + [Dx, y], \forall x, y \in \mathfrak{a}\}$. $\square$

In fact, we can go the opposite direction of the claim above:

Claim 3.3.11. Given algebras $\mathfrak{a}, \mathfrak{b}$ and a Lie algebra homomorphism $\theta : \mathfrak{b} \rightarrow \text{Der}(\mathfrak{a})$, we can construct a corresponding Lie algebra $\mathfrak{g}$, which is unique up to isomorphism.

Proof. We set $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{b}$ as vector spaces, so that $\mathfrak{b} \hookrightarrow \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$ is indeed an isomorphism. We now determine the brackets $[\cdot, \cdot]$ and that $\mathfrak{a}$ is an ideal.

- For $a_1, a_2 \in \mathfrak{a}$, $[a_1, a_2]$ is determined by the Lie algebra $\mathfrak{a}$.
- For $b_1, b_2 \in \mathfrak{b}$, $[b_1, b_2]$ is determined by the Lie algebra $\mathfrak{b}$.
- For $a \in \mathfrak{a}$ and $b \in \mathfrak{b}$, we define

$$[a, b] := \theta(b)(a),$$

since we want $\mathfrak{a}$ to be an ideal. Indeed, $\theta(b)(a) \in \mathfrak{a}$. Of course, $[a, b] = -[b, a]$.

Indeed, this is bilinear because everything inside is a linear map; further, we defined the alternating condition. What needs to be checked is the Jacobi identity: $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$.

- If $x, y, z \in \mathfrak{a}$ or $x, y, z \in \mathfrak{b}$, then this is trivial, because $\mathfrak{a}$ and $\mathfrak{b}$ are Lie algebras themselves.
- Now let $x \in \mathfrak{a}$ and $y, z \in \mathfrak{b}$. Then,

$$\begin{aligned} [x, [y, z]] + [y, [z, x]] + [z, [x, y]] &= -\theta([y, z])(x) - [y, \theta(z)(x)] + [z, \theta(y)(x)] \\ &= -\theta([y, z])(x) + \theta(y)(\theta(z)(x)) - \theta(z)(\theta(y)(x)) \\ &= -\theta([y, z])(x) + (\theta(y)\theta(z) - \theta(z)\theta(y))(x) \\ &= 0. \end{aligned}$$

Note that we can justify the second to last line is actually zero because this would require

$$\theta([y, z]) = [\theta(y), \theta(z)] = \theta(y)\theta(z) - \theta(z)\theta(y).$$

In other words, we require that θ was a Lie algebra homomorphism.

- Finally, let $x \in \mathfrak{b}$ and $y, z \in \mathfrak{a}$. Then,

$$\begin{aligned} [x, [y, z]] + [y, [z, x]] + [z, [x, y]] &= -\theta(x)([y, z]) + [y, \theta(x)(z)] - [z, \theta(x)(y)] \\ &= \theta(x)([y, z]) + [y, \theta(x)(z)] + [\theta(x)(y), z] \\ &= -\theta(x)([y, z]) + \theta(x)([y, z]) \\ &= 0. \end{aligned}$$

In particular, $\theta(x)$ is a derivation of $\mathfrak{a}$, so we were able to go from the second to third line. In other words, we require that $\theta(x)$ was actually contained in $\text{Der}(\mathfrak{a})$. $\square$

Remark 3.3.12 — We note that if θ is not a Lie algebra homomorphism or contained in $\text{Der}(\mathfrak{a})$, then the last two conclusions do not hold.

Remark 3.3.13 — The whole point of this discussion is to say that determining a Lie algebra $\mathfrak{g}$ is equivalent to finding a semidirect product structure given by the following data: an ideal $\mathfrak{a} \subset \mathfrak{g}$ and a subalgebra $\mathfrak{b}$, and a Lie algebra homomorphism $\theta: \mathfrak{b} \rightarrow \text{Der}(\mathfrak{a})$.

Exercise 3.3.14. Let V be a vector space and let φ be an endomorphism of V . Let L have underlying vector space $V \oplus \text{Span}\{x\}$. Show that if we define the Lie bracket on L by $[y, z] = 0$ for all $y, z \in V$ and $[x, y] = \varphi(y)$, for all $y \in V$, then L is a Lie algebra and $\dim L' = \text{rank } \varphi$.

Solution: Implicitly done above. □

Exercise 3.3.15. Suppose that I is an ideal of a Lie algebra L and that there is a subalgebra S of L such that $L = S \oplus I$ (as vector spaces).

1. Show that the map $\theta: S \rightarrow \mathfrak{gl}(I)$ defined by $\theta(s)x = [s, x]$ is a Lie algebra homomorphism from S into $\text{Der } I$.

In this case, we say that L is a **semidirect product** of I by S .

2. Show conversely that given Lie algebras S and I and a Lie algebra homomorphism $\theta: S \rightarrow \text{Der } I$, the vector space $S \oplus I$ may be made into a Lie algebra by defining

$$[(s_1, x_1), (s_2, x_2)] = ([s_1, s_2], [x_1, x_2] + \theta(s_1)x_2 - \theta(s_2)x_1)$$

for $s_1, s_2 \in S$, and $x_1, x_2 \in I$, and that this Lie algebra is a semidirect product of I by S . (The direct sum construction introduced in Exercise 2.4.3 is the special case where $\theta(s) = 0$ for all $s \in S$.)

3. Show that the Lie algebras in Exercise 3.3.14 may be constructed as semidirect products.
4. Investigate necessary and sufficient conditions for two semidirect products to be isomorphic.

Solution: 1. and 2. were done above (3.3.iii).

3. For this, we need to find an S, I, θ . We have that V is an ideal since $[x, y] = \varphi(y) \in V$ for all $y \in V$ and $[y, z] = 0$, for all $y, z \in V$. Thus, we let $S = \text{Span}\{x\} = \mathbb{C}x$. Then, we let $\theta: \mathbb{C}x \rightarrow \text{Der}(V)$. However, V is abelian, so $\text{Der}(V) = \mathfrak{gl}(V)$. This map sends $x \mapsto \varphi$. Thus, $\theta(\lambda x)(y) = [\lambda x, y] = \lambda[x, y] = \lambda\varphi(y)$, for all $y \in V$.

4. We first see when $\theta = 0$; that is, in the case we have a direct product. Suppose we have $A \oplus B \cong C \oplus D$. It is not true that $A \cong C$ or D , or to make analogous assertions. A more reasonable question would be as follows: let $L_1 = S_1 \oplus I$, $L_2 = S_2 \oplus I$ as vector spaces, with an associated Lie algebra homomorphism θ_1 and θ_2 . Fixing S, I , we vary θ . The classification of $\dim L = 3$ and $\dim L' = 2$. We showed that L' is abelian and we can show that L is a semidirect product. We simply just take the ideal to be $I = L'$, and we look for $L = S \oplus I$. Any S that is 1-dimensional and not contained in L' works then. We could also have $M = S' \oplus M'$. Then, we ask when is $L \cong M$? If $\varphi: L \xrightarrow{\sim} M$, then $\varphi(L') = M'$. Thus, φ induces an isomorphism $L' \xrightarrow{\sim} M'$. Therefore,

$\theta: S \rightarrow \text{Der}(L') = \mathfrak{gl}(L')$ and $\theta': S' \rightarrow \text{Der}(M') = \mathfrak{gl}(M')$. To complete this, it requires to show that L and M is classified by θ up to scalar multiples and isomorphisms $L' \xrightarrow{\sim} M'$. $\square$

§3.3.iv Lie Algebras with a 2-Dimensional Derived Algebra

Suppose $\dim L = 3$ and $\dim L' = 2$ and let $F = \mathbb{C}$ now. Then, we know that L/L' is 1-dimensional and abelian. If $x \notin L'$, then $\mathbb{C}x$ is a 1-dimensional subalgebra of L , and $L = L' \oplus \mathbb{C}x$, so L is a semidirect product, determined by $\theta: \mathbb{C}x \rightarrow \text{Der}(L')$. Here, L' plays the role of the ideal $\mathfrak{a}$ and L/L' will play the role of the subalgebra $\mathfrak{b}$.

Lemma 3.3.16

In the situation above, we have that:

- (i) The derived algebra L' is abelian.
- (ii) The linear map $\text{ad } x: L' \rightarrow L'$ is an isomorphism.

Proof. (i). Let $L' = \mathbb{C}y \oplus \mathbb{C}z$ as vector spaces, and $x \notin L'$. It suffices to show that $[y, z] = 0$. We know that $[y, z] \in L'$, so there are scalars α and β such that $[y, z] = \alpha y + \beta z$. Then, write the matrix of $\text{ad } y$ with respect to the basis $\{x, y, z\}$. It takes the form

$$\begin{pmatrix} 0 & 0 & 0 \\ * & 0 & \alpha \\ * & 0 & \beta \end{pmatrix},$$

where $*$ denotes a coefficient we don't need to write out explicitly. The top row is 0 because L' is spanned by y, z . The middle column is zero because $\text{ad } y(y)$ is zero. We see that $\text{tr}(\text{ad } y) = \beta$. As $y \in L'$, Exercise 2.2.6 tells us that $\beta = 0$. Similarly, by considering a matrix for $\text{ad } z$, we get $\alpha = 0$, so $[y, z] = 0$.

(ii). The derived algebra L' is spanned by $[x, y], [x, z], [y, z]$. However, $[y, z] = 0$, so as L' is 2-dimensional, $\{[x, y], [x, z]\}$ is a basis of L' . Therefore, the image of $\text{ad } x$ is 2-dimensional, and $\text{ad } x: L' \rightarrow L'$ is an isomorphism. $\square$

Remark 3.3.17 — Also note that the result holds for a Lie algebra over any field F . The fact that $F = \mathbb{C}$ will be required in the next step where we classify these Lie algebras.

Remark 3.3.18 — It follows that L is a semidirect product, determined by a linear map $\theta: \mathbb{C}x \rightarrow \text{Der}(L') = \mathfrak{gl}(L')$, where the last equality follows from L' being abelian.

Remark 3.3.19 — Since $\text{ad } x: L' \rightarrow L'$ is an isomorphism, i.e. $\theta(x)$ is an isomorphism; we see that L is determined by the data of the 1-dimensional vector space $\mathbb{C}x$, the 2-dimensional vector space L' , and the linear map θ , with $\theta(x)$ an isomorphism $L' \xrightarrow{\sim} L'$. Conversely, any such data determines a Lie algebra. with $\dim L' = 2$.

What remains now is to classify the map θ .

Case 1.

Claim 3.3.20. There exists $x \notin L'$ such that $\text{ad } x: L' \rightarrow L'$ is diagonalizable.

In this case, we may assume that y, z are eigenvectors of $\text{ad } x$. The associated eigenvalues must be nonzero by part (ii) of the lemma.

Suppose $[x, y] = \lambda y$. We can assume that $\lambda = 1$ by rescaling x : $[\lambda^{-1}x, y] = y$. With respect to the basis $\{y, z\}$ of L' , we get that the linear map $\text{ad } x: L' \rightarrow L'$ has the form

$$\begin{pmatrix} 1 & 0 \\ 0 & \mu \end{pmatrix}$$

for some nonzero $\mu \in \mathbb{C}$. Here, we note that this is equivalent to

$$\begin{pmatrix} 1 & 0 \\ 0 & \mu^{-1} \end{pmatrix}.$$

Exercise 3.3.21. Let us call the algebra of L_μ . Show that the Lie algebra L_μ is isomorphic to L_ν if and only if either $\mu = \nu$ or $\mu = \nu^{-1}$.

Remark 3.3.22 — The two exercises above tell us that there is an infinite family of non-isomorphic such Lie algebras.

Case 2.

Claim 3.3.23. For all $x \notin L'$, the linear map $\text{ad } x: L' \rightarrow L'$ is not diagonalizable.

Take any $x \notin L'$. As we work over $\mathbb{C}$, $\text{ad } x: L' \rightarrow L'$ must have an eigenvector, say $y \in L'$. As before, by scaling x , we can assume that $[x, y] = y$. Extending y to a basis $\{y, z\}$ of L' , we have that $[x, z] = \lambda y + \mu z$, where $\lambda \neq 0$ (otherwise, $\text{ad } x$ would be diagonalizable). By scaling z , we can arrange that $\lambda = 1$. The matrix of $\text{ad } x$ acting on L' therefore has the form

$$A = \begin{pmatrix} 1 & 1 \\ 0 & \mu \end{pmatrix}.$$

Since we’ve assumed that A is not diagonalizable, this can’t have two distinct eigenvalues. Thus, $\mu = 1$. Again, this completely determines a Lie algebra having the properties with which we started. Up to isomorphism, we get just one such algebra.

§3.3.v Lie Algebras where $L' = L$

Let us suppose that L is a complex Lie algebra of dimension 3 such that $L = L'$. We already know one example— $L = \mathfrak{sl}(2, \mathbb{C})$. We now show that up to isomorphism, this is the only one.

Claim 3.3.24. Let $x \in L$ be nonzero. Then, $\text{ad } x$ has rank 2.

Proof. Extend x to a basis of L , say $\{x, y, z\}$. Then, L' is spanned by $\{[x, y], [x, z], [y, z]\}$. However, $L' = L$, so this set must be linearly independent, and thus the image of $\text{ad } x$ has a basis $\{[x, y], [x, z]\}$ of size 2, as claimed. $\square$

Claim 3.3.25. There is some $h \in L$ such that $\text{ad } h: L \rightarrow L$ has an eigenvector with a nonzero eigenvalue.

Proof. Choose any nonzero $x \in L$. If $\text{ad } x$ has a nonzero eigenvalue, then we can take $h = x$. Otherwise, $\text{ad } x$ has rank 2 and all eigenvalues 0. Thus, the Jordan normal form of $\text{ad } x$ is

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix indicates that there for some basis of L given by $\{e_1, e_2, e_3\}$, then

$$\text{ad } x(e_1) = 0, \quad \text{ad } x(e_2) = e_1, \quad \text{ad } x(e_3) = e_2.$$

In particular, $\ker(\text{ad } x) = \mathbb{C}e_1$. Furthermore, $\text{ad } x(x) = 0$, so $x \in \ker(\text{ad } x)$ as well. Therefore, $x = \lambda e_1$ for some $\lambda \neq 0$. Then, $\text{ad } x(e_2) = e_1 = \lambda^{-1}x$. Then,

$$[e_2, x] = -[x, e_2] = -\lambda^{-1}x.$$

Thus, we can take $h = e_2$, so that $\text{ad } h$ has $-\lambda^{-1}$ as an eigenvalue. $\square$

By the previous step, we can find $h, x \in L$ such that $[h, x] = \alpha x \neq 0$. Since $h \in L$ and $L = L'$, we know from Exercise 2.2.6 that $\text{ad } h$ has trace zero. It follows that $\text{ad } h$ must have three distinct eigenvalues: $\alpha, 0, -\alpha$, where $\alpha \neq 0$. Replacing h by $\left(\frac{2}{\alpha}\right)h$, then we can get that $\alpha = 2$. In particular, there exists a nonzero e with $[h, e] = 2e$ and a nonzero f with $[h, f] = -2f$. Therefore, we have a basis e, f, h for L with

$$[h, e] = 2e, \quad [h, f] = -2f.$$

Note that this gives a basis (in particular linear independence) by Exercise 1.1.5.

To fully describe the structure of L , we need to find $[e, f]$. We get this from the Jacobi identity and bracketing focusing on the bracket $[h, [e, f]]$. The Jacobi identity gives

$$[h, [e, f]] + [e, [f, h]] + [f, [h, e]] = 0.$$

Now, we have:

$$\begin{aligned} [e, [f, h]] &= [e, 2f] = 2[e, f] \\ [f, [h, e]] &= [f, 2e] = -2[e, f] \end{aligned}$$

These implies that $[h, [e, f]] = 0$. Since $\ker(\text{ad } h) = \mathbb{C} \cdot h$, this means that

$$[e, f] = \gamma h,$$

for some $\gamma \in \mathbb{C}$. Furthermore, this γ cannot be zero because if it were, then $f \in \ker(\text{ad } e)$ and so $f \in \mathbb{C} \cdot e$, but f and e are linearly independent. Note at this point, we've only rescaled h to get 2 and -2 , so we can rescale f and e however we like. Therefore, by replacing e with $\gamma^{-1}e$, we can assume that $[e, f] = h$. Thus, $L \xrightarrow{\sim} \mathfrak{sl}(2, \mathbb{C})$ via the mapping

$$\begin{aligned} e &\mapsto \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ f &\mapsto \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \\ h &\mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned}$$

This shows that there is one and only 3-dimensional complex Lie algebra with $L' = L$ (up to isomorphism).

Remark 3.3.26 — Note that we always have the map $\text{ad}: L \rightarrow \mathfrak{gl}(L)$, and by definition, $\ker(\text{ad}) = Z(L)$. Under the assumption $L = L'$ and $\dim L = 3$, it is clear that $Z(L) = 0$, since $\text{ad}: L \hookrightarrow \mathfrak{gl}(L) \cong \mathfrak{gl}(3, \mathbb{C})$ is an injection.

§3.3.vi To Conclude...

To conclude, here is a complete classification of all 3-dimensional Lie algebras:

Abelian?	$\dim L'$	Other...
Yes	0	$\cong$
No	1	$L' \subset Z(L)$, Heisenberg Algebra
No	1	$L' \not\subset Z(L)$
No	2	$\text{ad } x: L' \rightarrow L'$ is diagonalizable, infinitely many
No	2	$\text{ad } x: L' \rightarrow L'$ is not diagonalizable
No	3	$\cong \mathfrak{sl}(2, \mathbb{C})$

§3.4 Exercises

Exercise 3.4.1. Find out where each of the following 3-dimensional complex Lie algebra appears in our classification:

- $\mathfrak{gl}_S(3, \mathbb{C})$, where $S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
- the Lie subalgebra of $\mathfrak{gl}(3, \mathbb{C})$ spanned by the matrices

$$u = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \mu & 0 \\ 0 & 0 & \nu \end{pmatrix}, \quad v = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad w = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

where λ, μ, ν are fixed complex numbers (and are not all zero).

- $\left\{ \begin{pmatrix} 0 & a & b & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} : a, b, c \in \mathbb{C} \right\}$
- $\left\{ \begin{pmatrix} 0 & 0 & a & b \\ 0 & 0 & 0 & c \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} : a, b, c \in \mathbb{C} \right\}$

Solution: 1. Take $x \in \mathfrak{gl}(3, \mathbb{C})$. Writing $x = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ the condition $x^T S = -Sx$ provides

$$\mathfrak{gl}_S(2, \mathbb{C}) = \text{Span} \left\{ \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{=e_1}, \underbrace{\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_{=e_2}, \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}}_{=e_3} \right\}$$

We compute the brackets now for L' .

$$[e_1, e_2] = -e_3, \quad [e_1, e_3] = e_2, \quad [e_2, e_3] = e_1$$

We see that $L' = L$ then.

2. We compute the brackets now for L' :

$$[u, v] = (\lambda - \nu)v, \quad [u, w] = (\mu - \nu)w, \quad [v, w] = 0$$

Thus, $L' = \mathbb{C}v + \mathbb{C}w$. Thus, we fall in the case where $\dim L' = 2$. $u \notin L'$, so consider the map $\text{ad } u: L' \rightarrow L'$. This takes the form

$$\begin{pmatrix} (\lambda - \nu) & 0 \\ 0 & (\mu - \nu) \end{pmatrix},$$

so we fall into the diagonalizable case.

3. We compute the bracket again: Let

$$A = \begin{pmatrix} 0 & a & b & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & d & e & 0 \\ 0 & 0 & f & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Then,

$$[A, B] = \begin{pmatrix} 0 & 0 & af - dc & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus, $\dim L' = 1$ and is spanned by

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

and so we verify the containment (or discontainment) in the centre. We see that the centre is spanned by the matrix

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

so indeed, $L' \subset Z(L)$ (if we denote the Lie algebra by L). This is the Heisenberg Algebra.

4. The bracket of two matrices in this Lie algebra is zero, so this is the unique 3-dimensional abelian Lie algebra. $\square$

Exercise 3.4.2. Show that over $\mathbb{R}$ the Lie algebras $\mathfrak{sl}(2, \mathbb{R})$ and $\mathbb{R}_\wedge^3$ are not isomorphic.
Hint: Prove that there is no non-zero $x \in \mathbb{R}_\wedge^3$ such that the map $\text{ad } x$ is diagonalizable.

Solution: Note that $\{f, g, h\}$ act as a basis for $\mathfrak{sl}(2, \mathbb{R})$, where $[g, f] = h$, $[h, f] = 2f$, $[h, g] = -2g$, so h is diagonalizable. Suppose that $\varphi: \mathfrak{sl}(2, \mathbb{R}) \xrightarrow{\sim} \mathbb{R}_\wedge^3$. However, $\text{ad } \varphi(h)$ is not diagonalizable, since there exists no element in $\mathbb{R}_\wedge^3$ that is diagonalizable. $\square$

Exercise 3.4.3. Show that over $\mathbb{R}$ there are exactly two non-isomorphic 3-dimensional Lie algebras with $L = L'$.

Solution: We claim that the two in Exercise 3.4.2 are the two isomorphism classes. Let L be a 3-dimensional Lie algebra with $L' = L$. Then, $L' = \text{Span}\{[x, y], [y, z], [z, x]\}$, where $\{x, y, z\}$ are some basis of L . Then, we must have that $\text{ad } x: L \rightarrow L$ has rank 2. We now focus on the map $\text{ad } x$.

We now focus on if $\text{ad } x$ is diagonalizable or not. The rational canonical form of $\text{ad } x$ takes the form

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & \alpha \\ 0 & 1 & \beta \end{pmatrix}.$$

The characteristic polynomial of this is $x(x^2 - \beta x - \alpha)$. It then depends on if the quadratic inside has real roots or not. We note that $\alpha = \beta = 0$ is not possible, for then $\dim L' \neq 3$. If it does, then this $\text{ad } x$ is diagonalizable, and it can be brought into the form

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \mu \end{pmatrix},$$

for some real μ . If not, then $\text{ad } x$ is not diagonalizable. Thus, the matrix above has no real roots since we're assuming that $\text{ad } x$ is not diagonalizable. This tells us that under some basis $\{e_1, e_2, e_3\}$, we get that $[e_1, e_2] = e_3$, $[e_1, e_3] = \alpha e_2 + \beta e_3$. We now want to compute $[e_2, e_3]$. We get that

$$\begin{aligned} [e_1, [e_2, e_3]] + [e_2, [e_3, e_1]] + [e_3, [e_1, e_2]] &= [e_1, [e_2, e_3]] + [e_2, -\alpha e_2 - \beta e_3] \\ &= [e_1, [e_2, e_3]] - \beta [e_2, e_3] = 0 \end{aligned}$$

Let $[e_2, e_3] = ae_1 + be_2 + ce_3$. Then, the condition above transforms to

$$be_3 + c(\alpha e_1 + \beta e_3) - \beta(ae_1 + be_2 + ce_3) = 0.$$

In other words,

$$e_1(c\alpha - a\beta) + e_2(b) + e_3(b) = 0.$$

Thus, $b = 0$ and $c\alpha - a\beta = 0$. We may then take $[e_2, e_3] = \alpha e_1 + \beta e_3$.

Return now to the RCF of $\text{ad } x$. This has trace 0, so $\beta = 0$. Therefore,

$$[e_1, e_2] = e_3, \quad [e_1, e_3] = \alpha e_2, \quad [e_2, e_3] = \alpha e_1.$$

We can then rescale to get the result. □

Exercise 3.4.4. Let L be a nonabelian Lie algebra. Show that $\dim Z(L) \leq \dim L - 2$.

Solution: Recall that $Z(L) = \{x \in L: [x, y] = 0, \forall y \in L\}$. L is nonabelian, so it follows that there exists two elements $x, y \in L$ such that $[x, y] \neq 0$. In particular, we have found two elements that are linearly independent and not in $Z(L)$, so $\dim Z(L) \leq \dim L - 2$. □

Exercise 3.4.5. Let L be the 3-dimensional Heisenberg Lie algebra defined over a field F . Show that $\text{Der}L$ is 6-dimensional. Identify the inner derivations as defined in Exercise 1.7.10 and show that the quotient $\text{Der}L/\text{IDer}L$ is isomorphic to $\mathfrak{gl}(2, F)$.

Solution: Recall that the 3-dimensional Heisenberg Lie algebra has the properties that $\dim L' = 1$ and $L' \subset Z(L)$. Note that $\text{Der}L = \{D \in \mathfrak{gl}(L) : D([a, b]) = [a, D(b)] + [D(a), b]\}$. Let x, y, z be our basis such that

$$[x, y] = z, \quad [x, z] = 0, \quad [y, z] = 0.$$

Now let $D \in \text{Der}L$. Since D is a linear map from A to A , we can consider how D can act on the basis elements.

$$D: x \mapsto a_1x + b_1y + c_1z$$

$$D: y \mapsto a_2x + b_2y + c_2z$$

$$D: z \mapsto a_3x + b_3y + c_3z$$

Then, we check with the brackets.

$$\begin{aligned} a_3x + b_3y + c_3z &= D(z) = D([x, y]) \\ &= [x, D(y)] + [D(x), y] \\ &= [x, a_2x + b_2y + c_2z] + [a_1x + b_1y + c_1z, y] \\ &= b_2z + a_1z. \end{aligned}$$

Similarly,

$$\begin{aligned} 0 &= D(0) = D([x, z]) \\ &= [x, D(z)] + [D(x), z] \\ &= [x, a_3x + b_3y + c_3z] + [a_1x + b_1y + c_1z, z] \\ &= b_3z. \end{aligned}$$

Finally,

$$\begin{aligned} 0 &= D(0) = D([y, z]) \\ &= [y, D(z)] + [D(y), z] \\ &= [y, a_3x + b_3y + c_3z] + [a_2x + b_2y + c_2z, z] \\ &= -a_3z \end{aligned}$$

Therefore, $a_3 = b_3 = 0$, and $c_3 = a_1 + b_2$. Therefore, the only constants which define D are $a_1, b_1, c_1, a_2, b_2, c_2$, so $\text{Der}L$ is a 6-dimensional F -vector space.

$\text{IDer} = \{\text{ad } l : l \in L\}$, so we just need to check the brackets. Here, $\text{ad } z$ is the only degenerate map, so $\text{IDer} = \{\text{ad } x, \text{ad } y\}$. We shall now show that $\text{Der}L/\text{IDer}L \cong \mathfrak{gl}(2, F)$. $\dim(\text{Der}L/\text{IDer}L) = 4 = \dim(\mathfrak{gl}(2, F))$. Let us write $D \in \text{Der}L$ as a matrix in $\mathfrak{gl}(3, F)$:

$$D = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ 0 & 0 & a_1 + b_2 \end{pmatrix}$$

Then, it's clear that every map in $\text{Der}L/\text{IDer}L$ is of the form

$$\begin{pmatrix} a_1 & b_1 & 0 \\ a_2 & b_2 & 0 \\ 0 & 0 & a_1 + b_2 \end{pmatrix},$$

since $\text{ad } x$ and $\text{ad } y$ represent the matrices

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

respectively. Therefore, the matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

span $\text{Der}L/\text{IDer}L$, and we can send each of these to a basis element E_{ij} of $\mathfrak{gl}(2, F)$, where E_{ij} has a 1 in the ij -th position and 0 elsewhere. In other words, we have the mapping

$$\begin{pmatrix} a_1 & b_1 & 0 \\ a_2 & b_2 & 0 \\ 0 & 0 & a_1 + b_2 \end{pmatrix} \mapsto \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix},$$

which has kernel $\text{IDer}L$. This is a surjection between vector spaces of the same dimension, so provides an explicit isomorphism. $\square$

Exercise 3.4.6. Find, up to isomorphism, all Lie algebras with a 1-dimensional derived algebra.

Solution: Let L be a Lie algebra with $\dim L' = 1$. Let $L' = \text{Span}\{x\}$. Then, extend this to a basis $\{x, y_1, \dots, y_n\}$, where L is $n + 1$ -dimensional. We consider when $L' \subset Z(L)$ and $L' \not\subset Z(L)$ again. $\square$

4 Solvable Lie Algebras and a Rough Classification

§4.1 Solvable Lie Algebras

Lemma 4.1.1

Let I be an ideal of a Lie algebra L . Then, L/I is abelian if and only if I contains the derived algebra L' ($L' \subseteq I$).

Proof. L/I is abelian if and only if, for all $x, y \in L$, we have

$$[x + I, y + I] = [x, y] + I = I.$$

Equivalently, for all $x, y \in L$, $[x, y] \in I$. I is a subspace of L , so this holds if and only if the space spanned by the brackets $[x, y]$ is contained in I ; that is, $L' \subseteq I$. $\square$

Remark 4.1.2 — This lemma tells us that L' is the smallest ideal of L with an abelian quotient. By the same argument, L' itself has a smallest ideal whose quotient is abelian; namely the derived algebra of L' , which we denote $L^{(2)}$, and so on.

Definition 4.1.3. We define the **derived series** of L to be the series with terms $L^{(1)} = L'$, and

$$L^{(k)} := [L^{(k-1)}, L^{(k-1)}], \quad k \geq 2.$$

Remark 4.1.4 — It is clear that $L \supseteq L^{(1)} \supseteq L^{(2)} \supseteq \dots$

Remark 4.1.5 — It is also clear the the product (or bracket) of ideals is an ideal, so $L^{(k)}$ is an ideal of L and not just an ideal of $L^{(k-1)}$.

Definition 4.1.6. The Lie algebra L is **solvable** if for some $m \geq 1$, $L^{(m)} = 0$.

Example 4.1.7

Given any Lie algebra L , the centre $Z(L)$, is a solvable ideal of L .

Example 4.1.8

The nonabelian Lie algebra of dimension 2 is solvable. Of course, the abelian Lie algebra is solvable as well. Thus, all 2-dimensional Lie algebras are solvable.

Example 4.1.9

The Heisenberg algebra, or more generally, upper triangular matrices with the subalgebra from $\mathfrak{gl}(3, F)$, are solvable.

Remark 4.1.10 — The first lemma then tells us that $L^{(0)}/L^{(1)} = L/[L, L]$ is abelian. Going one more step, $L^{(1)}/L^{(2)} = L^{(1)}/[L^{(1)}, L^{(1)}]$ is abelian. We can go on and on...

Example 4.1.11

If $L = \mathfrak{sl}(2, \mathbb{C})$, then we’ve seen in Exercise 2.1.10 that $L = L'$, and so $L^{(m)} = L$ for all $m \geq 1$, so $\mathfrak{sl}(2, \mathbb{C})$ is not solvable.

Lemma 4.1.12

L is solvable if and only if there exists ideals

$$L = I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n = 0$$

such that I_i/I_{i+1} is abelian.

Proof. The forward direction is clear by definition, so we prove the converse (just take the chain of ideals to be the derived series). We shall show that $L^{(k)}$ is contained in I_k for k between 1 and n , then since $L^{(n)} \subseteq I_n = 0$, so L is solvable.

We proceed by induction. For $k = 0$, we get that since L/I_1 is abelian, we have from Lemma 4.1.1 that $L' \subseteq I_1$. For the inductive step, we suppose that $L^{(k-1)} \subseteq I_{k-1}$, where $k \geq 2$. The Lie algebra I_{k-1}/I_k is abelian. Thus, by Lemma 4.1.1, this time applied to the Lie algebra I_{k-1} , we have $[I_{k-1}, I_{k-1}] \subseteq I_k$. However, $L^{(k-1)}$ is contained in I_{k-1} by the inductive hypothesis, so we deduce that

$$L^{(k)} = [L^{(k-1)}, L^{(k-1)}] \subseteq [I_{k-1}, I_{k-1}],$$

so $L^{(k)} \subseteq I_k$. □

Remark 4.1.13 — This proof shows that if $L^{(k)}$ is nonzero, then I_k is also nonzero. Therefore, the derived series may be thought of as the fastest descending series whose successive quotients are abelian.

Remark 4.1.14 — Lie algebra homeomorphisms are linear maps that preserve Lie brackets, so one would expect that they preserve the derived series.

Exercise 4.1.15. Suppose that $\varphi: L \rightarrow M$ is a homomorphism of Lie algebras. Show that

$$\varphi(L^{(k)}) = (\varphi(L))^{(k)}.$$

Solution: We can do this by induction.

$$\varphi(L^{(1)}) = \varphi([L, L]) = [\varphi(L), \varphi(L)] = \varphi(L)^{(1)}.$$

Now suppose this is true for some $k \geq 1$. Then,

$$\varphi(L^{(k+1)}) = \varphi([L^{(k)}, L^{(k)}]) = [\varphi(L^{(k)}), \varphi(L^{(k)})] = [(\varphi(L))^{(k)}, (\varphi(L))^{(k)}] = (\varphi(L))^{(k+1)}.$$

□

Remark 4.1.16 — This exercise suggests that the property of being solvable should be inherited by various constructions.

Lemma 4.1.17

Let L be a Lie algebra.

- (i) If L is solvable, then every subalgebra and every homomorphic image of L are solvable; that is, if $\varphi: L \rightarrow M$ is a Lie algebra homomorphism, then $\varphi(L)$ is solvable.
- (ii) Suppose that L has an ideal I . Then, L is solvable if and only if I and L/I are solvable.
- (iii) If I and J are solvable ideals of L , then $I + J$ is a solvable ideal of L .

Proof. (i) If L_1 is a subalgebra of L , then for each k it is clear that $L_1^{(k)} \subseteq L^{(k)}$, so if $L^{(m)} = 0$, then also $L_1^{(m)} = 0$. For the second part, Exercise 4.1.15 directly applies.

(ii) Only if: L/I is solvable by (i). Now assume that $I, L/I$ are solvable now. We have $(L/I)^{(k)} = (L^{(k)} + I)/I$, by applying Exercise 4.1.1 to the canonical homomorphism $L \rightarrow L/I$ or directly via induction on k . If L/I is solvable, then for some $m \geq 1$, we have $(L/I)^{(m)} = 0$. That is, $L^{(m)} + I = I$, so $L^{(m)} \subseteq I$. Since I is also solvable, then $I^{(k)} = 0$ for some $k \geq 1$. Hence, $(L^{(m)})^{(k)} \subseteq I^{(k)} = 0$. Then, an easy induction tells us that

$$(L^{(m)})^{(k)} = L^{(m+k)},$$

and so we are done.

(iii) By the second isomorphism, we get $(I + J)/J \cong I/I \cap J$, so it is solvable by Lemma 4.1.17. Since I is also solvable, (ii) of this lemma (applied to $J \subseteq I + J$) implies that $I + J$ is solvable. □

Corollary 4.1.18

Let L be a finite dimensional Lie algebra. There is a unique solvable ideal of L containing every solvable ideal of L i.e. there is a unique maximal solvable ideal I of L .

Proof. Let I be a solvable ideal of largest possible dimension; this certainly exists since L is finite dimensional, but the uniqueness is now the issue. Suppose that J is any solvable ideal. By Lemma 4.1.17, we know that $I + J$ is a solvable ideal. Now, $I \subseteq I + J$ and hence $\dim I \leq \dim(I + J)$. We chose I of maximal possible dimension and therefore we must have $\dim I = \dim(I + J)$ and hence $I = I + J$; in other words, J is contained in I . □

Definition 4.1.19. This largest solvable ideal is said to be the **radical** of L , denoted $\text{rad } L$.

Example 4.1.20

Let $L = \mathfrak{gl}(2, \mathbb{C})$; then, $L' = \mathfrak{sl}(2, \mathbb{C})$ and $L^{(2)} = \mathfrak{sl}(2, \mathbb{C})$, so L is not solvable. However, consider $\mathfrak{b}(2, \mathbb{C})$, the set of upper triangular matrices. However, this is a solvable subalgebra of dimension 3. Thus, if there was a maximal solvable subalgebra it would be this, but, for instance, the set of lower triangular matrices is also a solvable subalgebra of dimension 3. The point is that we can’t take solvable subalgebras to *build up* a new one (at least in an obvious way). Note that $\mathfrak{b}(2, \mathbb{C})$ is not an ideal of L , and only a subalgebra, this doesn’t contradict the corollary above.

The definition of $\text{rad } L$ then suggests the following definition.

Definition 4.1.21. A nonzero finite dimensional Lie algebra L is **semisimple** if it has no nonzero solvable ideals; equivalently, if $\text{rad } L = 0$.

Remark 4.1.22 — L is solvable if $\text{rad } L = L$.

Example 4.1.23

By Exercise 1.7.6, $\mathfrak{sl}(2, \mathbb{C})$ is semisimple.

Lemma 4.1.24

If L is a Lie algebra, then the factor algebra $L/\text{rad } L$ is semisimple.

Proof. Let $\bar{J}$ be a solvable ideal of $L/\text{rad } L$. By the ideal correspondence, there is an ideal J of L containing $\text{rad } L$ such that $\bar{J} = J/\text{rad } L$. By definition, $\text{rad } L$ is solvable, and $J/\text{rad } L = \bar{J}$ is solvable by hypothesis. Thus, Lemma 4.1.17 implies that J is solvable. But then J is contained in $\text{rad } L$, since $\text{rad } L$ is maximal; that is, $\bar{J} = 0$. $\square$

Remark 4.1.25 — This lemma tells us that to analyze L , it amounts to looking at solvable ideals and semisimple ideals.

Exercise 4.1.26. Prove that if L is solvable, then there exists a solvable subalgebra K of L of dimension k , for all $k \leq \dim L$.

Solution: Note that if L is solvable, then there exists a chain of ideals

$$L = I_0 \supseteq I_1 \supseteq \cdots \supseteq I_n = 0$$

such that I_i/I_{i+1} is abelian. Each I_i is an ideal, so a subalgebra of L , so is solvable in its own right. Further, quotients of solvable algebras by an ideal are solvable, so each quotient I_i/I_{i+1} is solvable in its own right. Therefore, we may apply the same lemma to I_i/I_{i+1} and repeat; however, since the quotient is abelian, it is simply a vector space, so we can just decrease the dimension by 1 at every step. Returning to the original sequence, we now have a new series, say indexed by V_i now such that each V_i/V_{i+1} is 1-dimensional. It then follows that $\dim V_i = \dim L - i$, so we are done. $\square$

§4.2 Nilpotent Lie Algebra

Definition 4.2.1. We define the **lower central series** of a Lie algebra L to be the series with terms $L^1 = L'$ and

$$L^k = [L, L^{k-1}],$$

for $k \geq 2$.

Remark 4.2.2 — It follows that $L \supseteq L^1 \supseteq L^2 \supseteq \dots$

Remark 4.2.3 — As the product of ideals is an ideal, L^k is also an ideal of L , not just an ideal of L^{k-1} .

Remark 4.2.4 — The name ‘central series’ comes from the fact that L^k/L^{k+1} is contained in the centre of L/L^{k+1} .

Definition 4.2.5. The Lie algebra L is **nilpotent** if for some $m \geq 1$, we have $L^m = 0$.

Example 4.2.6

The Lie algebra $\mathfrak{n}(n, F)$ of strict upper triangular matrices over a field F is nilpotent.

Exercise 4.2.7. Let $L := \mathfrak{n}(n, F)$ be the Lie algebra of strictly upper triangular $n \times n$ matrices over a field $\mathbb{F}$. Show that L^k has a basis consisting of all matrix units E_{ij} with $j - i > k$. Hence, show that L is nilpotent. What is the smallest m such that $L^m = 0$?

Solution: Consider the matrices E_{ij} , which has 1 at (i, j) and zero else where. We note that E_{ij} form a basis for $\mathfrak{n}(n, F)$, where $i < j$. We prove this via induction. For $k = 1$, L^1 , this is true:

$$[E_{i,i+1}, E_{i+1,j}] = E_{i,j},$$

where $j > i + 1$, so $j - i > 1$. Now assume the result is true up to some k . Then, L^k has basis E_{ij} , where $j - i > k$. Then, we consider commutators

$$[E_{i,i+k+1}, E_{i+k+1,j}] = E_{i,j},$$

where $j > i + k + 1$. Thus, $j - i > k + 1$, so the induction is complete. The fact that L is nilpotent then follows from the fact that if $k = n$, then there are no pairs of integers $1 \leq i < j \leq n$ such that $j - i > k = n$, since the greatest difference of $j - i$ is $n - 1$. It follows that the smallest m such that $L^m = 0$ is $m = n$. $\square$

Exercise 4.2.8. Show that any nilpotent Lie algebra is solvable. For this, show by induction on k that $L^{(k)} \subseteq L^k$.

Solution: We will proceed by induction to show that $L^{(k)} \subseteq L^k$, for all $k \geq 1$. For $k = 1$, we have that $L^1 = [L, L] = L^{(1)}$, so $L^{(1)} \subseteq L^1$. Now assume this holds up to some k . Then, we have that

$$L^{(k+1)} = [L^{(k)}, L^{(k)}] \subseteq [L^k, L^k] \subseteq [L, L^k] = L^{k+1}.$$

Thus, $L^{(k)} \subseteq L^k$. Given this, if L is nilpotent, the $L^k = 0$ for some k , but then $L^{(k)} = 0$. $\square$

Remark 4.2.9 — There are solvable Lie algebras which are not nilpotent. For instance, we can consider the Lie algebra $\mathfrak{b}(n, F)$ of upper triangular matrices over a field F for $n \geq 2$, as we see in the exercise below. Another example is the two-dimensional nonabelian Lie algebra, as we see below.

Example 4.2.10 (Non-Example)

Let $L = \mathbb{C}x \oplus \mathbb{C}y$ with $[x, y] = x$. Then, $L^0 = L$, $L^k = L' = \mathbb{C}x$, for all $k \geq 1$.

Exercise 4.2.11. Let $L := \mathfrak{b}(n, F)$ be the Lie algebra of upper triangular $n \times n$ matrices over a field F .

1. Show that $L' = \mathfrak{n}(n, F)$.
2. More generally, show that $L^{(k)}$ has a basis consisting of all matrix units e_{ij} with $j - i \geq 2^{k-1}$. (The commutator formula for the e_{ij} will be helpful.)
3. Hence show that L is solvable. What is the smallest m such that $L^{(m)} = 0$?
4. Show that if $n \geq 2$ then L is not nilpotent.

Solution: 1. E_{ij} , where $i \leq j$ form a basis for $\mathfrak{b}(n, F)$. Then, we consider the commutator $[E_{ij}, E_{jk}]$. If $i = j$ and $j < k$ or $i < j$ and $j = k$, then

$$[E_{ij}, E_{jk}] = E_{ik},$$

where $i < k$. If $i = j = k$,

$$[E_{ij}, E_{jk}] = 0.$$

Note that commutators of matrices of the form $[E_{ii}, E_{jj}] = 0$. Therefore, we can't attain the basis elements E_{ii} from the commutators. However, as the computation above shows, we can generate matrices of the form E_{ij} , where $i < j$, which is the basis of $\mathfrak{n}(n, F)$.

2. We prove this via induction. $L^{(1)}$ is clear from Exercise 4.2.7. Suppose this is true up to some k . Then, we want to find the basis of $L^{(k+1)}$. Consider the commutator

$$[E_{i,j}, E_{k,l}] = \delta_{j,k} E_{i,l} - \delta_{i,l} E_{k,j}.$$

Here, $j - i \geq 2^{k-1}$ and $l - k \geq 2^{k-1}$. If $j = k$ and $i \neq l$, then this produces the matrix $E_{i,l}$, where

$$l - i \geq 2 \cdot 2^{k-1} = 2^k,$$

so the induction is complete.

3. L is solvable because again we exhaust pairs of $1 \leq i \leq j \leq n$ such that $j - i \geq 2^{k-1}$, for k large. The biggest difference for $j - i$ is $n - 1$. Thus, the minimal k is when $2^{k-1} \geq n - 1$. In other words, the minimal k is $k = \lceil \log_2(n - 1) + 1 \rceil$.

4. We note that $[E_{ii}, E_{ij}] = E_{ij}$, where $j > i$, so this never terminates. Therefore, L is not nilpotent. $\square$

Lemma 4.2.12

Let L be a Lie algebra. Then,

- (i) If L is nilpotent, then any Lie subalgebra of L is nilpotent.
- (ii) If $L/Z(L)$ is nilpotent, then L is nilpotent.

Proof. (i) is clear from definition. By induction, or by a variant of Exercise 4.1.15, we can show that $(L/Z(L))^k$ is equal to $(L^k + Z(L))/Z(L)$. Thus, if $(L/Z(L))^m$ is zero, then L^m is contained in $Z(L)$ and therefore $L^{m+1} = 0$. $\square$

Remark 4.2.13 — Note that the following analogue of Lemma 4.1.17 does not hold: if I is an ideal of a Lie algebra L , then it's possible that both L/I and I are nilpotent but L is not. The 2-dimensional nonabelian Lie algebra illustrates this. Further, this remark suggests that solvability is more fundamental to the structure of Lie algebras than nilpotency.

§4.3 A Look Ahead

The previous section suggests that we may have a chance to understand all finite dimensional Lie algebras. The radical $\text{rad } L$ of any Lie algebra L is solvable, and $L/\text{rad } L$ is semisimple, so to understand L it is necessary to understand the following:

1. an arbitrary solvable Lie algebra
2. an arbitrary semisimple Lie algebra

Over $\mathbb{C}$, the answer to the first question was found by Sophus Lie, who proved that every solvable Lie algebra appears as a subalgebra of a Lie algebras of upper triangular matrices. A precise statement is given later. For (ii), we will show that every semisimple Lie algebra is a direct sum of *simple* Lie algebras.

Definition 4.3.1. The Lie algebra L is **simple** if it has no ideals other than 0 and L and it is not abelian.

Remark 4.3.2 — Note that if L is simple, then L is semisimple, so there is no obstruction to what we expect as a definition. Indeed, a simple ideals has no nonzero proper ideals, so there is definitely no solvable ideal in it.

Remark 4.3.3 — The restriction that a simple Lie algebra is not abelian removes only the 1-dimensional abelian Lie algebra. Without this restriction, this Lie algebra would be simple but not semisimple, which is not desirable.

Exercise 4.3.4. Prove directly that $\mathfrak{sl}(n, \mathbb{C})$ is a simple Lie algebra for $n \geq 2$.

Solution: Firstly, note that $\mathfrak{sl}(n, \mathbb{C})$ is clearly not abelian. What we need to show is similar to Exercise 1.7.6. The basis elements of $\mathfrak{sl}(n, \mathbb{C})$ are clear, and we should (check) get that $\mathfrak{sl}(n, \mathbb{C})' = \mathfrak{sl}(n, \mathbb{C})$. In other words, this tells us that if there is some linear

combination of the basis elements in an ideal $I \subseteq \mathfrak{sl}(n, \mathbb{C})$, and if we bracket it with all the basis elements, we would either recover the entire Lie algebra or get that that the linear combination actually had zero coefficients (the zero ideal). $\square$

We then need to find all simple Lie algebras over $\mathbb{C}$. This is a major theorem, and the proof is based on work by Killing, Engel, and Cartan. We shall eventually prove most of the following theorem.

Theorem 4.3.5 (Simple Lie Algebras)

With five exceptions, every finite dimensional simple Lie algebra over $\mathbb{C}$ is isomorphic to one of the **classical Lie algebras**:

$$\mathfrak{sl}(n, \mathbb{C}), \quad \mathfrak{so}(n, \mathbb{C}), \quad \mathfrak{sp}(2n, \mathbb{C}).$$

The five exceptional Lie algebras are known as $\mathfrak{e}_6$, $\mathfrak{e}_7$, $\mathfrak{e}_8$, $\mathfrak{f}_4$, and $\mathfrak{g}_2$

We've already introduced the family of special linear Lie algebras, $\mathfrak{sl}(n, \mathbb{C})$. The remaining families can be defined as certain subalgebras of $\mathfrak{gl}(n, \mathbb{C})$ using the construction introduced in Exercise 1.7.8.

§4.3.i $\mathfrak{so}(n, \mathbb{C})$

Recall that if $S \in \mathfrak{gl}(n, \mathbb{C})$, then we define a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{C})$ by

$$\mathfrak{gl}_S(n, \mathbb{C}) := \{x \in \mathfrak{gl}(n, \mathbb{C}) : x^T S = -Sx\}.$$

Assume first of all that $n = 2l$. Take S to be the matrix with $l \times l$ blocks:

$$S = \begin{pmatrix} 0 & I_l \\ I_l & 0 \end{pmatrix}.$$

We define $\mathfrak{so}(2l, \mathbb{C}) := \mathfrak{gl}_S(2l, \mathbb{C})$. When $n = 2l + 1$, we take

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_l \\ 0 & I_l & 0 \end{pmatrix}$$

and define $\mathfrak{so}(2l + 1, \mathbb{C}) := \mathfrak{gl}_S(2l + 1, \mathbb{C})$. These Lie algebras are known as the **special orthogonal Lie algebras**.

§4.3.ii $\mathfrak{sp}(2n, \mathbb{C})$

The Lie algebras $\mathfrak{sp}(n, \mathbb{C})$ are only defined for even n . If $n = 2l$, we take

$$S = \begin{pmatrix} 0 & I_l \\ -I_l & 0 \end{pmatrix}$$

and define $\mathfrak{sp}(2l, \mathbb{C}) := \mathfrak{gl}_S(2l, \mathbb{C})$. These Lie algebras are known as the **symplectic Lie algebras**.

It follows from Exercise 2.4.8 that $\mathfrak{so}(n, \mathbb{C})$ and $\mathfrak{sp}(n, \mathbb{C})$ are subalgebras of $\mathfrak{sl}(n, \mathbb{C})$. (This also follows from the explicit bases given in a later chapter).

§4.4 Exercises

Exercise 4.4.1. Any nilpotent Lie algebra contains an ideal of codimension 1.

Solution: Let L be a nilpotent Lie algebra. Then, $L' \neq 0$. L/L' is an abelian sub-algebra then, and consider the map $\pi: L \rightarrow L/L'$. We take some 1-dimensional subspace M/L' of L/L' such that it has codimension 1. Then, M/L' is an ideal of L/L' , so by the ideal correspondence, $M := \pi^{-1}(M/L')$ contains L' , so is an ideal of L . Then, M has codimension 1 in L . $\square$

Exercise 4.4.2. Let $x \in \mathfrak{gl}(2l, \mathbb{C})$. Show that x belongs to $\mathfrak{sp}(2l, \mathbb{C})$ if and only if it is of the form

$$x := \begin{pmatrix} m & p \\ q & -m^T \end{pmatrix},$$

where p and q are symmetric. Hence find the dimension of $\mathfrak{sp}(2l, \mathbb{C})$.

Solution: Take the block form of $x \in \mathfrak{gl}(2l, \mathbb{C})$ as

$$x = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

where $a, b, c, d \in \mathfrak{gl}(l, \mathbb{C})$. We now consider what $x^T S = -Sx$ is telling us, where $S = \begin{pmatrix} 0 & I_l \\ -I_l & 0 \end{pmatrix}$. We have as follows:

$$x^T S = \begin{pmatrix} -c^T & a^T \\ -d^T & b^T \end{pmatrix} = \begin{pmatrix} -c & -d \\ a & b \end{pmatrix} = -Sx$$

These must be equal, so this tells us that $c = c^T$, $b = b^T$, and $a = -d^T$, giving the desired form. As for the converse, we just check that x of the form provided is in $\mathfrak{sp}(2l, \mathbb{C})$, but this is a simple check. $\square$

Exercise 4.4.3 (L solvable/nilpotent $\iff \text{ad } L \subseteq \mathfrak{gl}(L)$ is solvable/nilpotent). Use Lemma 4.1.17 to show that if L is a Lie algebra, then L is solvable if and only if $\text{ad } L$ is a solvable subalgebra of $\mathfrak{gl}(L)$. Show that this result also holds if we replace solvable with nilpotent.

Solution: Note that $\text{ad } L = \{\text{ad } x: x \in L\}$ here. If L is solvable, then since $\text{ad}: L \rightarrow \mathfrak{gl}(L)$ is a homomorphism of Lie algebras, we can deduce by (i) of Lemma 4.1.17 that $\text{ad } L$ is solvable. For the converse, suppose that $\text{ad } L$ is a solvable subalgebra of $\mathfrak{gl}(L)$. Note that the centre $Z(L)$ is an ideal of L , and $L/Z(L) \cong \text{ad } L$. In particular, $Z(L)$ is solvable because $Z(L)' = [Z(L), Z(L)] = 0$. Therefore, it follows that L is solvable.

Now let us prove the same for nilpotent. The converse is easy. $L/Z(L) \cong \text{ad } L$, so if $\text{ad } L$ is nilpotent, we can directly apply Lemma 4.2.12 to say that L is nilpotent. For the forward direction, assume L is nilpotent. Then, there is some $m \geq 1$ such that $L^m = [L, [L, [\dots [L, L]]]] = 0$. We claim that $\text{ad } x^{m-1} = 0$, for any $x \in L$. Let $y \in L$. Then, we have

$$\text{ad } x^m(y) = [x, [x, [\dots [x, y]]]],$$

where there are m number of x s. Well, this is clearly equal to 0 since $L^m = 0$, so we've proven true. $\square$

Exercise 4.4.4. Let $\varphi: L \rightarrow M$ be a homomorphism of Lie algebras. Then, $\varphi(L^k) = \varphi(L)^k$.

Solution: We prove this via induction. When $k = 1$, this is clear:

$$\varphi(L^1) = \varphi([L, L]) = [\varphi(L), \varphi(L)] = \varphi(L)^1.$$

Now suppose that the result holds up to $k - 1$. Then, we have the following:

$$\begin{aligned} \varphi(L^k) &= \varphi([L, L^{k-1}]) \\ &= [\varphi(L), \varphi(L^{k-1})] \\ &= [\varphi(L), \varphi(L)^{k-1}] \\ &= \varphi(L)^k \end{aligned}$$

□

Remark 4.4.5 — We can apply the two exercises above to Exercise 4.4.3 to conclude that $\text{ad}(L)$ is a nilpotent subalgebra of $\mathfrak{gl}(M)$.

Exercise 4.4.6. Show that a Lie algebra is semisimple if and only if it has no nonzero abelian ideals (note that abelian ideals are ideals that are abelian in their own right).

Solution: Let L be a semisimple Lie algebra; that is, $\text{rad } L = 0$. Consider an abelian ideal $I \subseteq L$. Then, $[I, I] = 0$, so I is solvable, but $\text{rad } L = 0$, so $I = 0$.

Now for the converse, so assume that L has no nonzero abelian ideals. Let I be a solvable ideal of L . Then, there’s some m such that $I^{(m)} = 0$, but $I^{(m-1)} \neq 0$. However, $I^{(m-1)}$ is an abelian ideal, which is a contradiction. □

Exercise 4.4.7 (Lie Groups). We begin with the Lie algebra $\mathfrak{sl}(n, \mathbb{C})$. Recall that the $n \times n$ matrices with determinant 1 form a group under matrix multiplication, denoted $SL(n, \mathbb{C})$. Let I denote the $n \times n$ identity matrix. We ask when $I + tX$ is in $SL(n, \mathbb{C})$ for an $n \times n$ matrix X .

1. Show that $\det(I + tX)$ is a polynomial in t of degree n with the first two terms

$$\det(I + tX) = 1 + t \text{tr}(X) + \dots$$

If we neglect all powers of t except 1 and let $t \rightarrow 0$, then we obtain the statement

$$I + tX \in SL(n, \mathbb{C}) \iff X \in \mathfrak{sl}(n, \mathbb{C}).$$

Remark 4.4.8 — This could have been taken as the definition of $\mathfrak{sl}(n, \mathbb{C})$, even though, interpreted literally, it is false.

2. (a) Let S be an $n \times n$ matrix. Let $\langle \cdot, \cdot \rangle$ denote the complex bilinear form with matrix S . Show that if we let $G_S(n, \mathbb{C})$ be the set of invertible matrices A such that

$$\langle Av, Aw \rangle = \langle v, w \rangle \quad \text{for all } v, w \in \mathbb{C}^n,$$

then $G_S(n, \mathbb{C})$ is a group.

- (b) Show that if we perform the construction in part (1) with $G_S(n, \mathbb{C})$ in place of $SL(n, \mathbb{C})$, we obtain $\mathfrak{gl}_S(n, \mathbb{C})$.

3. (a) An invertible matrix A is customarily said to be **orthogonal** if $A^t A = A A^t = I$; that is, if $A^{-1} = A^t$. Show that the set of $n \times n$ orthogonal matrices with coefficients in $\mathbb{C}$ is the group $G_I(n, \mathbb{C})$ and that the associated Lie algebra $\mathfrak{g}_I(n, \mathbb{C}) = \mathfrak{gl}_I(n, \mathbb{C})$ is the space of all antisymmetric matrices.
- (b) Prove that $\mathfrak{g}_I(n, \mathbb{C}) \cong \mathfrak{so}(n, \mathbb{C})$. *Hint: Use Exercise 2.4.7*

Remark 4.4.9 — The reason for not taking this as the definition of $\mathfrak{so}(n, \mathbb{C})$ will emerge later.

4. A bilinear form on a vector space is said to be **symplectic** if

$$\langle v, v \rangle = 0 \quad \text{for all } v \in V.$$

Show that

$$S := \begin{pmatrix} 0 & I_l \\ -I_l & 0 \end{pmatrix}$$

is the matrix of a non-degenerate symplectic bilinear form on a $2l$ -dimensional space. The associated Lie algebra is $\mathfrak{gl}_S(2l, \mathbb{C}) = \mathfrak{sp}(2l, \mathbb{C})$.

Solution: 1. $\det(I + tX)$ is the characteristic polynomial of the matrix, so it is clear that this is equal to $1 + \text{ttr}(X)$.

2(a). We have that $\langle v, w \rangle = v^T S w$. First, it's clear that $I_n \in G_S(n, \mathbb{C})$, since $(I_n v)^T S(I_n w) = v^T S w$. Next, if $A \in G_S(n, \mathbb{C})$, we have that $A^{-1} \in G_S(n, \mathbb{C})$. Note that

$$(Av)^T S(Aw) = v^T A^T S A w = v^T S w,$$

so $A^T S A = S$, so $(A^{-1})^T S A^{-1} = S$, so this is clear. The fact that if $A, B \in G_S(n, \mathbb{C})$ is also clear.

2(b). We consider $(I + tX + \dots)$. Then, we have $(I + tX + \dots) \in G_S(n, \mathbb{C})$ if and only if

$$(I + tX + O(t^2))^T S (I + tX + O(t^2)) = S.$$

Therefore, if we send $t \rightarrow 0$ for all terms that are of degree two and above, then we can find the desired result. We have the following:

$$S + tSX + tX^T S + O(t^2) = S$$

Therefore, $I + tX \in G_S(n, \mathbb{C})$ if and only if $X^T S = -SX$; that is, if and only if $X \in \mathfrak{gl}_S(n, \mathbb{C})$.

3(a). It is clear that $I_n \in G_I(n, \mathbb{C})$, and that inverses are in $G_I(n, \mathbb{C})$. Next, let $A, B \in G_I(n, \mathbb{C})$. $(AB)^T AB = I = AB(AB)^T$, so indeed $AB \in G_I(n, \mathbb{C})$ as well. Let $I + tX \in G_I(n, \mathbb{C})$. This is true if and only if

$$(I + tX)^T (I + tX) = I \iff I + t(X^T + X) + O(t^2) = I.$$

This is true if and only if $X^T = -X$, so the associated Lie algebra is the space of all anti-symmetric matrices, which is $\mathfrak{gl}_I(n, \mathbb{C})$.

3(b). We rely on Exercise 2.4.7 as intended. Let $n = 2l$ first. We want to find some invertible matrix P such that $I = P^T S P$, where $S = \begin{pmatrix} 0 & I_l \\ I_l & 0 \end{pmatrix}$. With some work, this is actually possible (XXX). Now when $n = 2l + 1$. Again, we want to find some invertible

matrix P such that $I = P^T S P$, where $S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_l \\ 0 & I_l & 0 \end{pmatrix}$. Again, with some work, this would come (XXX).

4.

□

Exercise 4.4.10. Let F be a field. Exercise 2.4.7 shows that if $S, T \in \mathfrak{gl}(n, F)$ are congruent matrices (that is, there exists an invertible matrix P such that $T = P^T S P$), then $\mathfrak{gl}_S(n, F) \cong \mathfrak{gl}_T(n, F)$. Does the converse hold when $F = \mathbb{C}$? For a challenge, think about other fields.

Solution: We want to show that $\mathfrak{gl}_S(n, \mathbb{C}) \cong \mathfrak{gl}_T(n, \mathbb{C})$ implies that there exists invertible P such that $S = P^T T P$. Let $\{x_1, \dots, x_m\}$ be a basis for $\mathfrak{gl}_S(n, \mathbb{C})$ and $\{y_1, \dots, y_m\}$ be a basis for $\mathfrak{gl}_T(n, \mathbb{C})$. Let $\varphi: \mathfrak{gl}_S(n, \mathbb{C}) \xrightarrow{\sim} \mathfrak{gl}_T(n, \mathbb{C})$ be the isomorphism which maps $x_i \mapsto y_i$. Then,

$$x_i^T S = -S x_i \implies y_i^T S = -S y_i,$$

and so the basis y_i acts as a basis for $\mathfrak{gl}_S(n, \mathbb{C})$ as well. Then, in the basis y_i , we get that S and T are diagonal matrices with entries $s = (s_1, \dots, s_m)$, $t = (t_1, \dots, t_m)$, where

$$S = \sum_{i=1}^m s_i y_i, \quad T = \sum_{i=1}^m t_i y_i.$$

We then need to see when S and T are equivalent. Since S and T are diagonal, these represent bilinear forms on $\mathbb{C}^n$. Over $\mathbb{C}$, we have algebraic closure, so we get that the converse does hold. Over $\mathbb{R}$, this depends on how many negative and positives appear in s and t . □

5 Subalgebras of $\mathfrak{gl}(V)$

Let F be a field, and V be an n -dimensional F -vector space.

§5.1 Nilpotent Maps

Let L be a subalgebra of $\mathfrak{gl}(V)$. We may regard elements of L as linear transformations of V , so in addition to the Lie bracket we can also exploit compositions xy of linear maps for $x, y \in L$. Care must be taken, however, as in general this composition will not belong to L .

Lemma 5.1.1

Let $x \in L$ (where L is a subalgebra of $\mathfrak{gl}(V)$). If the linear map $x: V \rightarrow V$ is nilpotent (i.e. $x^r = 0$ for some $r \geq 1$), then $\text{ad } x: L \rightarrow L$ is also nilpotent.

Proof. Take $y \in L$ and expand $(\text{ad } x)^m(y) = [x, [x, \dots [x, y] \dots]]$. Every term in the resulting sum is of the form $x^j y x^{m-j}$ for some j between 0 and m . Suppose that $x^r = 0$ and let $m \geq 2r$. Either $j \geq r$, in which case $x^j = 0$, or $m - j \geq r$, in which case $x^{m-j} = 0$. It follows that $(\text{ad } x)^{2r} = 0$. $\square$

Remark 5.1.2 — This lemma can be a companion to the result in Exercise 1.7.9, which showed that if $L = \mathfrak{gl}(V)$ and $x =: V \rightarrow V$ is diagonalizable, then $\text{ad } x: L \rightarrow L$ is also diagonalizable.

§5.2 Weights

We now generalize the notions of eigenvalues and eigenvectors of a fixed linear map.

Definition 5.2.1. Let L be a subalgebra of $\mathfrak{gl}(V)$. $v \in V$ is an **eigenvector** for L if v is an eigenvector for every element in L . That is, $x(v) = \lambda v$ for $\lambda \in F$, for every $x \in L$ (λ depends on x).

Example 5.2.2

Let $L = \mathfrak{d}(n, F)$, the Lie subalgebra of $\mathfrak{gl}(n, F)$ consisting of diagonal matrices. Let $\{e_1, \dots, e_n\}$ be the standard basis of F^n . Then each e_i is an eigenvector for L .

Given an eigenvector, we have a function $\lambda: L \rightarrow F$, $x \mapsto \lambda(x)$ such that $xv = \lambda(x)v$. This function is linear:

$$(ax + by)\lambda(v) = a\lambda(x)v + b\lambda(y)v = (a\lambda(x) + b\lambda(y))v,$$

for any $x, y \in L$ and $a, b \in F$. By definition, $\lambda \in L^*$, the dual vector space.

Remark 5.2.3 — When L is a 1-dimensional Lie algebra $L \subset \mathfrak{gl}(V)$, i.e. $L = Fx$, for some nonzero $x: V \rightarrow V$. Then, v is an eigenvector for L if and only if v is an eigenvector of x . If $xv = \lambda v$, then λ corresponds to the function $L \rightarrow F$, $ax \mapsto a\lambda$.

Definition 5.2.4. A **weight** for a Lie subalgebra L of $\mathfrak{gl}(V)$ is a linear map $\lambda: L \rightarrow F$ such that

$$V_\lambda := \{v \in V : x(v) = \lambda(x)v, \text{ for all } x \in L\}$$

is a nonzero subspace of V . The vector space V_λ is the **weight space** associated to the weight λ .

Remark 5.2.5 — Thus V_λ is nonzero if and only if V contains a common eigenvector for the elements of L , with the eigenvalues of elements of L given by the function λ .

Example 5.2.6

If L is 1-dimensional, V_λ is an eigenspace for x ; $L = Fx$.

Exercise 5.2.7. Continue with the notation above.

1. Check that the eigenspaces V_λ defined above are vector subspaces of V .
2. Let $L = \mathfrak{d}(n, F)$, and define $\epsilon_i: L \rightarrow F$ by $\epsilon_i(a) = \alpha_i$, where a is a diagonal matrix with entries $(\alpha_1, \dots, \alpha_n)$. Show that $V_{\epsilon_i} = \text{Span}\{e_i\}$ and that V decomposes as a direct sum of the V_{ϵ_i} for $1 \leq i \leq n$.

Solution: 1. This is clear; if $v, w \in V_\lambda$ and $a, b \in F$, then $av + bw \in V_\lambda$, since

$$x(av + bw) = ax(v) + bx(w) = a\lambda(x)v + b\lambda(x)w = \lambda(x)(av + bw).$$

2. We have that

$$V_{\epsilon_i} = \{v \in V : a(v) = \alpha_i v, \forall a \in L\}.$$

Writing $v = \sum_{k=1}^n c_k e_k$, then we get that

$$\sum_{k=1}^n \alpha_k c_k e_k = \alpha_i \sum_{k=1}^n c_k e_k.$$

Thus, we must have $v = c_i e_i$. In other words, $V_{\epsilon_i} = \text{Span}\{e_i\}$. Then, $V = \text{Span}\{e_1, \dots, e_n\} = \bigoplus_{i=1}^n V_{\epsilon_i}$. $\square$

Exercise 5.2.8. Let $A = \mathfrak{b}(n, F)$ be the Lie subalgebra of $\mathfrak{gl}(n, F)$ consisting of upper triangular matrices. Show that e_1 is an eigenvector for A . Find the corresponding weight and determine its weight space.

Solution: Take a matrix in A :

$$M = \begin{pmatrix} a & * & \dots & * \\ 0 & * & & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & * \end{pmatrix}, \quad a \in F$$

Then, it's clear that $Me_1 = ae_1$, so e_1 is a common vector for any matrix in A . The corresponding weight is $\lambda: A \rightarrow F$ defined by $M \mapsto a$, as defined above. The weight space A_λ is the span of e_1 . $\square$

§5.3 The Invariance Lemma

We now prove a generalization of the following linear-algebraic result in the context of Lie subalgebras: if $a, b: V \rightarrow V$ are commuting linear transformations and W is the kernel of a , then W is b -invariant; that is, b maps W into W . Note that we can prove this easily: if $w \in W$, then $a(bw) = b(aw) = 0$ and so $bw \in W$.

Lemma 5.3.1

Suppose that A is an ideal of a Lie subalgebra L of $\mathfrak{gl}(V)$. Let

$$W = \{v \in V : a(v) = 0, \text{ for all } a \in A\}.$$

Then W is an L -invariant subspace of V , i.e. if $x \in L$, $x(W) \subseteq W$. We may call W the **weight 0 space** of A .

Proof. Take $x \in L$ and $w \in W$. Then, we must show that $a(xw) = 0$ for all $a \in A$; then, $xw \in W$. But

$$ax = ax - xa + xa = [a, x] + xa,$$

where $[a, x] \in A$ as A is an ideal. Then,

$$a(xw) = (ax)w = [a, x]w + xaw = [a, x]w,$$

since $aw = 0$. Finally, $[a, x] \in A$ as noted, so $[a, x]w = 0$. $\square$

Remark 5.3.2 — The technique used here of replacing ax with $xa + [a, x]$ is frequently used.

Exercise 5.3.3. Show that Lemma 5.3.1 really does generalize the result mentioned in the first paragraph of this section.

Solution: Take L to be $\mathfrak{gl}(V)$ and $A = L$. We are exactly in the situation described. $\square$

This has dealt with zero eigenvalues. More generally, one can prove that if $a, b: V \rightarrow V$ are commuting linear maps, $\lambda \in F$, and V_λ is the λ -eigenspace of a (that is, $V_\lambda = \{v \in V : av = \lambda v\}$), then V_λ is invariant under b .

This fact too has a generalization to Lie algebras. As before, we shall replace the linear map a by an ideal $A \subseteq \mathfrak{gl}(V)$. The subspace considered in Lemma 5.3.1 may be viewed as the 0-weight space for A . In our generalization, we allow an arbitrary weight.

Lemma 5.3.4 (Invariance Lemma)

Assume that F has characteristic zero. Let L be a Lie subalgebra of $\mathfrak{gl}(V)$ and let A be an ideal of L . Then, for any $\lambda \in A^*$, the associated weight space

$$V_\lambda = \{v \in V : av = \lambda(a)v, \text{ for all } a \in A\}$$

is an L -invariant (or L -stable) subspace of V , i.e. $\forall x \in L, v \in V_\lambda, xv \in V_\lambda$.

Proof. We need to show that if $x \in L$ and $v \in V_\lambda$, then $x(v)$ is an eigenvector for every element of A , with the eigenvalue of $a \in A$ given by $\lambda(a)$.

For $a \in A$, we have

$$a(xv) = x(av) + [a, x](v) = x\lambda(a)v + \lambda([a, x])v.$$

Note that $[a, x] \in A$ as A is an ideal. Therefore all we need show is that the eigenvalue of the commutator $[a, x]$ on V_λ is zero.

Consider the subspace $U = \text{Span}\{v, xv, x^2v, \dots\}$. This is a finite-dimensional subspace of V , since V is finite dimensional. Let n be the least number such that the vectors $x^n v \in \text{Span}\{v, xv, \dots, x^{n-1}v\}$. Then, an induction tells us that $x^m v \in \text{Span}\{v, xv, \dots, x^{n-1}v\}$, for all $m \geq n$. Then, the elements $v, xv, \dots, x^{n-1}v$ are linearly independent. Because if not, then this contradicts our choice of n .

Claim 5.3.5. For all $z \in A$, z preserves $\text{Span}\{v, xv, \dots, x^{n-1}v\}$. In fact, with respect to the basis $v, x(v), \dots, x^{n-1}(v)$, z is upper triangular with all diagonal entries equal to $\lambda(z)$:

$$\begin{pmatrix} \lambda(z) & * & \dots & * \\ 0 & \lambda(z) & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda(z) \end{pmatrix}.$$

Proof. What we're trying to show can be rephrased: for all $m \geq 0$, we need to show that $zx^m v - \lambda(z)x^m v \in \text{Span}\{v, xv, \dots, x^{m-1}v\}$; this is the key statement. We do this by induction on the number of the column.

First when $m = 0$. $zv - \lambda(z)v = 0$, i.e. $zv = \lambda(z)v$. This is true by definition since $v \in V_\lambda$. Next is the inductive step. We have

$$\begin{aligned} zx^{m+1}v &= (zx)x^m v \\ &= ([z, x] + xz)x^m v \\ &= [z, x]x^m v + x(zx^m v). \end{aligned}$$

We now use the inductive hypothesis in the third line, starting with the term on the right. We have

$$zx^m v = \lambda(z)x^m v + \text{Span}\{v, xv, \dots, x^{m-1}v\},$$

so

$$xxz^m v = \lambda(z)x^{m+1}v + \text{Span}\{xv, \dots, x^m v\}$$

by induction. Next is the term on the left. $[z, x] \in A$ (because $z \in A$), so we may apply the inductive hypothesis to this element as well. This gives the following:

$$[z, x]x^m v = \lambda([z, x])x^m v + \text{Span}\{v, \dots, x^{m-1}v\}$$

This completes the induction, since if we add the terms, we are in $\text{Span}\{v, xv, \dots, x^m v\}$. $\square$

We now have the claim in hand. Take $z = [a, x]$ then, with $a \in A$, $x \in L$. We see that the trace of the matrix of z acting on U is $m\lambda(z)$. U is invariant under the action of

$a \in A$, and U is x -invariant by construction. Thus, the trace of z on U is the trace of $ax - xa$, also viewed as a linear transformation of U , and this is obviously 0. Therefore

$$m\lambda([a, x]) = 0.$$

As F has characteristic zero, it follows that $\lambda([a, x]) = 0$. Note that this final step is where we use that $\text{char}(F) = 0$. $\square$

Remark 5.3.6 — This proof would be much easier if it were true that the linear maps y^r belonged to the Lie algebra L . Unfortunately, this is not true in general.

§5.4 Application of the Invariance Lemma

Proposition 5.4.1

Let $x, y: V \rightarrow V$ be linear maps from a complex vector space V to itself. Suppose that x and y both commute with $[x, y]$. Then, $[x, y]$ is a nilpotent map.

Proof. Since we're working over $\mathbb{C}$, it's enough to show that if λ is an eigenvalue of the linear map $[x, y]$, then $\lambda = 0$.

Let λ be an eigenvalue. Let $W = \{v \in V: [x, y]v = \lambda v\}$ be the corresponding eigenspace of $[x, y]$; this is a non-zero subspace of V . Let L be the Lie subalgebra of $\mathfrak{gl}(V)$ spanned by x , y , and $[x, y]$. As $\text{Span}\{[x, y]\}$ is an ideal of L , the Invariance Lemma implies that W is invariant under x and y .

Pick a basis of W and let X and Y be the matrices of x and y with respect to this basis. The commutator $[x, y]$ has matrix $XY - YX$ in its action on W . But every element of W is an eigenvector of $[x, y]$ with eigenvalue λ , so

$$XY - YX = \begin{pmatrix} \lambda & 0 & \cdots & 0 \\ 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda \end{pmatrix}.$$

Now taking traces gives

$$0 = \text{tr}(XY - YX) = \lambda \dim W$$

and so $\lambda = 0$. $\square$

§5.5 Exercises

Exercise 5.5.1.

1. Let L be a Lie subalgebra of $\mathfrak{gl}(V)$. Suppose that there is a basis of V such that, with respect to this basis, every $x \in L$ is represented by a strictly upper triangular matrix. Show that L is isomorphic to a Lie subalgebra of $\mathfrak{n}(n, F)$ and hence that L is nilpotent.
2. Let L be a subalgebra of $\mathfrak{gl}(V)$. Suppose there is a basis of V such that, with respect

to this basis, every $x \in L$ is represented by upper triangular matrices. Show that L is isomorphic to a subalgebra of $\mathfrak{b}(n, F)$ and hence L is solvable.

Solution: 1. We consider the map $\varphi: L \rightarrow \mathfrak{n}(n, F)$, which takes an elements $x \in L$ to its matrix in the basis V . Then, φ is clearly linear, and this implies that the Lie bracket is preserved:

$$\varphi([x, y]) = \varphi(xy - yx) = \varphi(x)\varphi(y) - \varphi(y)\varphi(x) = [\varphi(x), \varphi(y)].$$

The kernel of φ is also zero, so we indeed have that L is isomorphic to a Lie subalgebra of $\mathfrak{n}(n, F)$. Furthermore, since $\mathfrak{n}(n, F)$ is nilpotent itself, we get that L is nilpotent as well, since it is a subalgebra of a nilpotent Lie algebra.

2. We consider the map $\varphi: L \rightarrow \mathfrak{b}(n, F)$, which takes elements in L to its matrix in the basis V . The linearity and preservation of the Lie bracket is clear from the first part, so this is a Lie algebra homomorphism. The kernel is yet again zero, so we get that L is isomorphic to a Lie subalgebra of $\mathfrak{b}(n, F)$. Now, $\mathfrak{b}(n, F)$ is solvable, so it follows that L is solvable. $\square$

Exercise 5.5.2. This exercise gives an alternative approach to Proposition (5.4.1). Recall that V is a complex vector space and $x, y \in \mathfrak{gl}(V)$ are linear maps such that $[x, y]$ commutes with both x and y .

1. Let $z = [x, y]$. Show that $\text{tr} z^m = 0$ for all $m \geq 1$.
2. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $[x, y]$. Show that $\lambda_i = 0$ for all i , and deduce that $[x, y]$ is nilpotent. *Hint: Use the Vandermonde determinant*

Solution: 1. For $m = 1$, we have that $\text{tr}(z) = 0$ because $\text{tr}(xy - yx) = 0$. Suppose this is true for up to some m . Then,

$$z^{m+1} = [x, y]z^m = (xy - yx)z^m = xyz^m - yxz^m = xyz^m - yz^mx.$$

Then,

$$\text{tr}(z^{m+1}) = \text{tr}(xyz^m - yz^mx) = 0,$$

since we have $xyz^m - yz^mx = x(yz^m) - (yz^m)x$.

2. Note that we have the identity

$$\text{tr}(z^m) = \sum_{i=1}^n \lambda_i^m,$$

where λ_i are the eigenvalues of z . These exist because we're working over $\mathbb{C}$. We then have the matrix relation

$$\begin{pmatrix} \lambda_1 & \lambda_2 & \dots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \dots & \lambda_n^2 \\ \vdots & \vdots & & \vdots \\ \lambda_1^n & \lambda_2^n & \dots & \lambda_n^n \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \mathbf{0}.$$

Considering the determinant of the square matrix on the left, we get that

$$\prod_i \lambda_i \prod_{i < j} (\lambda_i - \lambda_j) = 0,$$

by the Vandermonde determinant. This can then be completed by induction. In the case of $n = 1$, we're done. Suppose that the result holds up to $n - 1$. Then, $\lambda_i = \lambda_j$, for some pair. Then, we can truncate the matrix relation then to an $(n - 1) \times (n - 1)$ matrix, so the inductive hypothesis applies, giving $\lambda_1 = \dots = \lambda_n$. Thus, all the eigenvalues of $[x, y]$ are zero, so it's JCF is strictly upper triangular. Thus, nilpotent. $\square$

Exercise 5.5.3. Let L be a Lie algebra and let A be a subalgebra of L . The **normalizer** of A , denoted $N_L(A)$, is defined by

$$N_L(A) := \{x \in L : [x, a] \in A \text{ for all } a \in A\}.$$

1. Prove that $N_L(A)$ is a subalgebra of L and that $N_L(A)$ contains A . Show, moreover, that $N_L(A)$ is the largest subalgebra of L in which A is an ideal.
2. Let $L = \mathfrak{gl}(n, \mathbb{C})$ and let A be the subalgebra of L consisting of all diagonal matrices. Show that $N_L(A) = A$. *Hint: While this can be proved directly, it follows very quickly from the ideas in this chapter. By the Invariance Lemma, any weight space for A is $N_L(A)$ -invariant. How does this restrict the possible matrices of elements of $N_L(A)$?*

Solution: 1. Let $x, y \in N_L(A)$, and $a \in A$. Then, we have the following:

$$[[x, y], a] = [x, [y, a]] + [y, [a, x]],$$

so $[y, a], [a, x] \in A$, since $x, y \in N_L(A)$. Thus, the terms on the RHS are in A . It follows that $[[x, y], a] \in A$ as well, so $N_L(A)$ is indeed a subalgebra. Further, $A \subset N_L(A)$, since if $a \in A$, then for any $b \in A$, $[a, b] \in A$. Now, this is the largest because if there is some larger subalgebra $K \subseteq L$ such that A is an ideal, then there is some element $x \in K \setminus N_L(A)$ such that $[x, a] \in A$. However, this is not possible, since $N_L(A)$ already contains all those elements.

2. We have that $A = \mathfrak{d}(n, \mathbb{C})$. First note that $A \subseteq N_L(A)$, so we just need to show the reverse inclusion. Further note that A is an ideal of $N_L(A)$. Thus, any weight space of A is $N_L(A)$ invariant, so if $x \in N_L(A)$, then $x(A_\lambda) \subseteq A_\lambda$. In particular, we know that the weights $\lambda \in A^*$ are given by $\lambda_i: A \rightarrow \mathbb{C}$, which takes a diagonal matrix $a = (\alpha_1, \dots, \alpha_n) \mapsto \alpha_i$. Then, take $e_i \in A_{\lambda_i} = \text{Span}\{e_i\}$. Then, xe_i must be of the form λe_i by $N_L(A)$ -invariance. This implies that $x_{ki} = 0$, for all $k \neq i$. In other words, x is diagonal, so is in A . Thus, $N_L(A) \subseteq A$. $\square$

Exercise 5.5.4. Show that if $a, y \in \mathfrak{gl}(V)$ are linear maps, then for any $m \geq 1$

$$ay^m = y^m a + \sum_{k=1}^m \binom{m}{k} y^{m-k} a_k$$

where $a_1 = [a, y]$, $a_2 = [a_1, y] = [[a, y], y]$, and generally $a_k = [a_{k-1}, y]$. Deduce that

$$\text{ad } y^m = \sum_{k=1}^m (-1)^{k+1} \binom{m}{k} y^{m-k} (\text{ad } y)^k.$$

In this formula, the power y^{m-k} acts on $\mathfrak{gl}(V)$ as the linear map which sends $x \in \mathfrak{gl}(V)$ to $y^{m-k}x$.

Solution: We prove the first claim via induction. For $m = 1$, we have the following:

$$ay = ya + (ay - ya) = ya + [a, y].$$

Next for the inductive step. Suppose this result holds up to $m - 1$. Then,

$$\begin{aligned}
ay^m &= y^m a + (ay^m - y^m a) \\
&= y^m a + \left(\left(y^{m-1} a + \sum_{k=1}^{m-1} \binom{m-1}{k} y^{(m-1)-k} a_k \right) y - y^m a \right) \\
&= y^m a + \left(\sum_{k=1}^{m-1} \binom{m-1}{k} y^{(m-1)-k} a_k y + y^{m-1} (ya + [a, y]) - y^m a \right) \\
&= y^m a + \left(\sum_{k=1}^{m-1} \binom{m-1}{k} y^{(m-1)-k} (ya_k + [a_k, y]) + y^{m-1} [a, y] \right) \\
&= y^m a + \left(\left(\sum_{k=1}^{m-1} \binom{m-1}{k} y^{m-k} a_k + \binom{m-1}{k} y^{(m-1)-k} a_{k+1} \right) + y^{m-1} a_1 \right) \\
&= y^m a + \left(my^{m-1} a_1 + \sum_{k=2}^{m-1} \binom{m-1}{k} y^{m-k} a_k + \binom{m-1}{k-1} y^{m-k} a_k \right) \\
&= y^m a + \left(my^{m-1} a_1 + a_m + \sum_{k=2}^{m-1} \binom{m}{k} y^{m-k} a_k \right) \\
&= y^m a + \left(\sum_{k=1}^m \binom{m}{k} y^{m-k} a_k \right)
\end{aligned}$$

Whew. Now for the next part. We have that $\text{ad } y^m(a) = y^m a - y^m a$, so it is equal to the negative of the sum on the RHS. Thus, we just need to clean the terms inside the sum. Note that we only really need to prove that $a_k = (-1)^k \cdot (\text{ad } y^k(a))$. Again, we do this by induction. For $k = 1$, this is clear. Assume true for some n (unrelated to the power n). Then,

$$a_n = [a_{n-1}, y] = [(-1)^{n-1} (\text{ad } y^{n-1}(a)), y] = (-1)^n \text{ad } y^n(a),$$

so we're done. $\square$

6 Engel's Theorem and Lie's Theorem

§6.1 Engel's Theorem

Theorem 6.1.1 (Engel's Theorem)

Let F be a field, and V be a finite dimensional F -vector space. Suppose that L is a Lie subalgebra of $\mathfrak{gl}(V)$ such that every element of L is a nilpotent linear transformation of V . Then there is a basis of V in which every element of L is represented by a strictly upper triangular matrix.

To prove Engel's Theorem, we adapt the strategy used to prove the analogous result for a single nilpotent linear transformation. The proof of this result is outlined in the following exercise.

Exercise 6.1.2. Let V be an n -dimensional vector space where $n \geq 1$, and let $x: V \rightarrow V$ be a nilpotent linear map.

1. Show that there is a non-zero vector $v \in V$ such that $xv = 0$.
2. Let $U = \text{Span}\{v\}$. Show that x induces a nilpotent linear transformation $\bar{x}: V/U \rightarrow V/U$. By induction, we know that there is a basis $\{v_1 + U, \dots, v_{n-1} + U\}$ of V/U in which $\bar{x}$ has a strictly upper triangular matrix. Prove that $\{v, v_1, \dots, v_{n-1}\}$ is a basis of V and that the matrix of x in this basis is strictly upper triangular.

Solution: 1. Consider $v, xv, x^2v, \dots$ and choose n to be minimal such that $x^n v = 0$. Then, $x(x^{n-1}v) = x^n v$, and $x^{n-1}v \neq 0$.

2. We now prove by induction on $\dim V$ that x is strictly upper triangular with respect to some basis. Let $U = Fv$ be a 1-dimensional subspace of V . Then, $xv = 0$, so x acts by zero on U . Therefore, x induces a linear map $V/U \rightarrow V/U$ which is also nilpotent. By induction, you can choose a basis of V/U such that this linear map is strictly upper triangular. Then v together with any lifting to V of this basis for V/U gives us the required basis. $\square$

The crucial step in the proof of Engel's Theorem is the analogue of the first question above. We must find a non-zero vector $v \in V$ that lies in the kernel of every $x \in L$.

Proposition 6.1.3

Suppose that L is a Lie subalgebra of $\mathfrak{gl}(V)$, where V is non-zero, such that every element of L is a nilpotent linear transformation. Then there is some non-zero $v \in V$ such that $xv = 0$ for all $x \in L$.

Proof. We proceed by induction on $\dim L$. If $\dim L = 1$, then L is spanned by a single nilpotent linear transformation, say x . By part 1 of Exercise 6.1.2 of the previous exercise there is some non-zero $v \in V$ such that $xv = 0$. An arbitrary element of L is a scalar multiple of x , so v lies in the kernel of every element of L . Now suppose that $\dim(L) > 1$.

Claim 6.1.4. Take a maximal Lie subalgebra A of L ; that is, A is a proper subalgebra of L and if $A \subset B \subset L$ is a subalgebra, then $B = A$ or $B = L$. We claim that A is an ideal of L and that $\dim A = \dim L - 1$.

Proof. Consider the quotient vector space L/A . We define a linear map

$$\varphi: A \rightarrow \mathfrak{gl}(L/A)$$

by letting $\varphi(a)$ act on $x + A$ as

$$\varphi(a)(x + A) = [a, x] + A.$$

This is well-defined, for if $x \in A$ then $[a, x] \in A$. This is also clearly a linear map. Moreover, φ is a homomorphism of Lie algebras. If $a, b \in A$ then

$$\begin{aligned} [\varphi(a), \varphi(b)](x + A) &= \varphi(a)([b, x] + A) - \varphi(b)([a, x] + A) \\ &= ([a, [b, x]] + A) - ([b, [a, x]] + A) \\ &= [a, [b, x]] - [b, [a, x]] + A \\ &= [[a, b], x] + A \end{aligned}$$

by the Jacobi identity. The last term is equal to $\varphi([a, b])(x + A)$, as required. Therefore, $\varphi(A)$ is a Lie subalgebra of $\mathfrak{gl}(L/A)$ and $\dim \varphi(A) \leq \dim A < \dim L$.

To apply the inductive hypothesis, we need to know that $\varphi(A)$ is a nilpotent linear transformation of L/A . However, $\varphi(a)$ is induced by $\text{ad } a \in \mathfrak{gl}(L)$, for $a \in A$. By Lemma 5.1.1 we know that if $a \in \mathfrak{gl}(V)$ is nilpotent, so is $\text{ad}(a) \in \mathfrak{gl}(L)$; therefore $\varphi(a)$ is as well. By the inductive hypothesis, $\varphi(A) \subset \mathfrak{gl}(L/A)$ has an eigenvector with eigenvalue 0. In other words, there is some non-zero element $y \in L \setminus A$ such that for all $a \in A$, $\varphi(a)(y + A) = [a, y] + A = 0$. That is, $[a, y] \in A$ for all $a \in A$. This means that $A \oplus \text{Span}\{y\}$ is a subalgebra of L containing A . By maximality, $A \oplus \text{Span}\{y\}$ must be equal to L . Therefore $L = A \oplus \text{Span}\{y\}$, and it follows that A is an ideal of L . $\square$

We now apply the inductive hypothesis to $A \subseteq \mathfrak{gl}(V)$. This gives us a non-zero $v \in V$ such that $av = 0$ for all $a \in A$. We now use the invariance lemma for weight 0 subspaces. L acts on the space of weight 0 eigenvectors for A , V_0 . $V_0 \neq 0$, and

$$V_0 = \{v \in V : av = 0 \text{ for all } a \in A\}.$$

By Lemma 5.3.1, V_0 is invariant under L , so in particular $y(V_0) \subseteq V_0$. Since y is nilpotent, the restriction of y to V_0 is also nilpotent. Hence there is some non-zero vector $v \in V_0$ such that $yv = 0$. Thus, $av = 0$ for all $a \in A$ and $yv = 0$. Therefore, taking $x \in L$ we can write it as $x = a + \beta y$ for some $a \in A$ and some $\beta \in F$. Doing this, we obtain

$$xv = av + \beta yv = 0.$$

This shows that v is a non-zero vector in the kernel of every element of L . $\square$

§6.1.i Proof of Engels' Theorem

Recall the main theorem we're trying to prove.

Theorem 6.1.5 (Engel's Theorem)

Let V be a vector space. Suppose that L is a Lie subalgebra of $\mathfrak{gl}(V)$ such that every element of L is a nilpotent linear transformation of V . Then there is a basis of V in which every element of L is represented by a strictly upper triangular matrix.

Proof. The remainder of the proof is closely analogous to part 2 of Exercise 6.1.2. We use induction on $\dim V$. If $V = \{0\}$, then there is nothing to do, so we assume that $\dim V \geq 1$.

By Proposition 6.1.3, there is a non-zero vector $v \in V$ such that $x(v) = 0$ for all $x \in L$. Let $U = \text{Span}\{v\}$ and let $\bar{V}$ be the quotient vector space V/U . Any $x \in L$ induces a linear transformation $\bar{x}$ of $\bar{V}$. The map $L \rightarrow \mathfrak{gl}(\bar{V})$ given by $x \mapsto \bar{x}$ is easily checked to be a Lie algebra homomorphism.

The image of L under this homomorphism is a subalgebra of $\mathfrak{gl}(\bar{V})$ which satisfies the hypothesis of Engel's Theorem. Moreover, $\dim(\bar{V}) = n - 1$, so by the inductive hypothesis there is a basis of $\bar{V}$ such that with respect to this basis all $\bar{x}$ are strictly upper triangular. If this basis is $\{v_i + U : 1 \leq i \leq n - 1\}$, then $\{v, v_1, \dots, v_{n-1}\}$ is a basis for V . As $x(v) = 0$ for each $x \in L$, the matrices of elements of L with respect to this basis are strictly upper triangular. $\square$

§6.1.ii Another POV

We now give another way to think about Engel's Theorem that does not rely on L being given to us as a subalgebra of some $\mathfrak{gl}(V)$. Recall that a Lie algebra is said to be nilpotent if for some $m \geq 1$ we have $L^m = 0$, or equivalently, if for all $x_0, x_1, \dots, x_m \in V$ we have

$$[x_0, [x_1, \dots, [x_{m-1}, x_m] \dots]] = 0.$$

Theorem 6.1.6 (Engel's Theorem Version 2)

A Lie algebra L is nilpotent if and only if for all $x \in L$ the linear map $\text{ad}x: L \rightarrow L$ is nilpotent.

Proof. We begin by proving the easier only if direction. Note that L is nilpotent if and only if there is some $n \geq 1$ such that $L^n = 0$. In other words, there is an $n \geq 1$ such that for all $x_1, \dots, x_n \in L$,

$$\text{ad}x_1 \circ \text{ad}x_2 \circ \dots \circ \text{ad}x_{n-1} = 0.$$

Therefore, it is clear that $(\text{ad}x)^n = 0$ for all $x \in L$, and so every $\text{ad}x$ is nilpotent.

We now prove the if direction. Let $\bar{L} = \text{ad}L$ denote the image of L under the adjoint homomorphism $\text{ad}: L \rightarrow \mathfrak{gl}(L)$. Then, $L/Z(L) \xrightarrow{\sim} \bar{L}$, so L is nilpotent if and only if $\bar{L}$ is nilpotent. If $\text{ad}x$ is nilpotent for every element x of L , then Engel's theorem for $\bar{L}$ tells us that there is a basis of L in which $\bar{L}$ is strictly upper-triangular. It now follows from Exercise 5.5.1 that $\bar{L}$ is nilpotent. Finally, Lemma 4.2.12 implies that L is nilpotent. $\square$

Remark 6.1.7 — It is very tempting to assume that a Lie subalgebra L of $\mathfrak{gl}(V)$ is nilpotent if and only if there is a basis of V such that the elements of L are represented by strictly upper triangular matrices. However, the 'only if' direction is

false, i.e. $L \subset \mathfrak{gl}(V)$ being nilpotent does not imply that there exists a basis of V in which all $x \in L$ are strictly upper triangular.

For example, any 1-dimensional Lie algebra is (trivially) nilpotent. Let I denote the identity map in $\mathfrak{gl}(V)$. The Lie subalgebra $\text{Span}\{I\}$ of $\mathfrak{gl}(V)$ is therefore nilpotent. In any basis of V , the map I is represented by the identity matrix, which is certainly not strictly upper triangular.

Engel's Theorem has many applications, both in the proofs to come and in more general linear algebra.

Exercise 6.1.8. Let L be a complex Lie algebra. Show that L is nilpotent if and only if every 2-dimensional subalgebra of L is abelian.

Solution: By Engel's theorem (V2), L is nilpotent if and only if $\text{ad } x$ is nilpotent for all $x \in L$. This is true if and only if all eigenvalues of $\text{ad } x$ are 0. Then, use the fact that 2-dimensional Lie algebras are of the form $\mathbb{C}x \oplus \mathbb{C}y$, with $[x, y] = \lambda y$, for $\lambda = 0$ or $\lambda = 1$. We can view this condition as y being an eigenvector for $\text{ad } x$. Then, all the eigenvalues of $\text{ad } x$ are 0 if and only if $[x, y] = 0$; that is, an abelian Lie algebra. $\square$

§6.2 Lie's Theorem

Let L be a Lie subalgebra of $\mathfrak{gl}(V)$. We would now like to understand when there is a basis of V such that the elements of L are all represented by upper triangular matrices. The answer is given below in Lie's Theorem.

Theorem 6.2.1 (Lie's Theorem)

Let V be a finite dimensional complex vector space and let L be a solvable Lie subalgebra of $\mathfrak{gl}(V)$. Then there is a basis of V in which every element of L is represented by an upper triangular matrix.

The following exercise outlines a proof of the corresponding result for a single linear transformation.

Exercise 6.2.2. Let V be an n -dimensional complex vector space where $n \geq 1$ and let $x: V \rightarrow V$ be a linear map.

1. Show that x has an eigenvector $v \in V$.
2. Let $U = \text{Span}\{v\}$. Show that x induces a linear transformation $\bar{x}: V/U \rightarrow V/U$. By induction, we know that there is a basis $\{v_1 + U, \dots, v_{n-1} + U\}$ of V/U in which $\bar{x}$ has an upper triangular matrix. Prove that $\{v, v_1, \dots, v_{n-1}\}$ is a basis of V and that the matrix of x in this basis is upper triangular.

Solution: 1. The JCF of x is diagonal, since we're working over $\mathbb{C}$. Thus, x has a nonzero eigenvector v .

2. $\square$

As in the proof of Engel's Theorem, the main step is the generalization of part 1.

Proposition 6.2.3

Let V be a non-zero complex vector space. Suppose that L is a solvable Lie subalgebra of $\mathfrak{gl}(V)$. Then there is some non-zero $v \in V$ which is a simultaneous eigenvector for all $x \in L$.

Proof. This looks similar to Proposition 6.1.3, but this time we need the full power of the Invariance Lemma, as non-zero eigenvalues are not quite as convenient as zero eigenvalues for calculations.

As before, we use induction on $\dim(L)$. If $\dim(L) = 1$, the result for a single linear transformation gives us all we need, so we may assume that $\dim(L) > 1$. Since L is solvable, we know that L' is properly contained in L . Choose a vector subspace $A \subset L$ with $L' \subseteq A \subseteq L$ and $\dim A = \dim L - 1$, so $L = A \oplus \text{Span}\{z\}$ for some $z \in L \setminus A$.

By Lemma 4.1.1 and Example 2.3.1, A is an ideal of L , so A is solvable. We may now apply the inductive hypothesis to obtain a vector $v \in V$ which is a simultaneous eigenvector for all $a \in A$, i.e. for some $\lambda \in \mathbb{C}$,

$$V_\lambda = \{v \in V \mid a(v) = \lambda(a)v \text{ for all } a \in A\}.$$

This space is non-zero, as it contains v . Now choose some $x \in L$ and $x \notin A$ (so L is spanned by elements of A and x). By the Invariance Lemma (Lemma 5.3.4, requiring characteristic 0), the space V_λ is preserved by x . Choose a non-zero $v \in V_\lambda$. Then, v is an eigenvector for every element of L .

We just present here the proof of the claim that v is a simultaneous eigenvector for all $x \in L$. Any $x \in L$ may be written in the form $x = a + \beta z$ for some $a \in A$ and $\beta \in \mathbb{C}$. We have

$$x(v) = a(v) + \beta z(v) = \lambda(a)v + \beta \mu v$$

where μ is the eigenvalue of z corresponding to v . This completes the proof. $\square$

The remainder of the proof of Lie's Theorem is analogous to the proof of Engel's Theorem; it follows by the induction on $\dim V$.

§6.2.i A Generalization of Lie's Theorem

It is natural to ask whether Lie's theorem holds for more general fields. As the eigenvalues of a matrix in upper triangular form are its diagonal entries, we will certainly need the field to contain the eigenvalues of all the elements of $\mathfrak{gl}(L)$. The simplest way to achieve this is to require that our field be algebraically closed.

The example in Exercise 6.2.4 below shows that we also need our field to have characteristic zero. This is perhaps more surprising. We used this assumption in the last step of the proof of the Invariance Lemma when we deduced from

$$m\text{tr}([x, y]) = 0$$

that $\text{tr}([x, y]) = 0$. (This would of course be inadmissible if the characteristic of the field divided m .)

Exercise 6.2.4. Let p be prime and let F be a field of characteristic p . Consider the following $p \times p$ matrices:

$$x = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 0 \end{pmatrix}, \quad y = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & p-2 & 0 \\ 0 & 0 & \cdots & 0 & p-1 \end{pmatrix}.$$

Check that $[x, y] = x$. Deduce that x and y span a 2-dimensional solvable subalgebra L of $\mathfrak{gl}(p, F)$. Show that x and y have no common eigenvector, and so the conclusions of Proposition 6.2.3 and Lie's Theorem fail. Show that the conclusion of part 1 of the following exercise also fails.

Solution: We have that

$$\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & p-2 & 0 \\ 0 & 0 & \cdots & 0 & p-1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & p-1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$$

and

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & p-2 & 0 \\ 0 & 0 & \cdots & 0 & p-1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & p-2 \\ p-1 & 0 & \cdots & 0 & 0 \end{pmatrix}$$

Thus, $[x, y] = x$, since we're in characteristic p . Then, x and y span a 2-dimensional solvable Lie algebra, since $L = \text{Span}\{x, y\}$, and $L' = \text{Span}\{x\}$, and so $L^{(2)} = 0$. The eigenvectors for y are $e_1, e_2, \dots, e_p$, but these are not eigenvectors for x , so they do not have simultaneous eigenvectors. Thus, Proposition 6.2.3 clearly fails. As for Lie's theorem, we have a solvable Lie algebra L , but x is not represented by an upper triangular matrix. $\square$

Exercise 6.2.5.

1. Let L be a solvable Lie subalgebra of $\mathfrak{gl}(V)$, where V is a complex vector space. Show that every element of L' is a nilpotent endomorphism of V .
2. Let L be a complex Lie algebra. Show that L is solvable if and only if L' is nilpotent.

Solution: 1. By Lie's theorem, we can assume that every element of L is given by an upper triangular matrix (since L is solvable). Then, if $x, y \in L$, then

$$[x, y] = xy - yx$$

is going to be strictly upper triangle. Therefore, it is nilpotent.

2. If L' is nilpotent, it is true that L' is solvable. Thus, L/L' is abelian, so solvable. Since L' and L/L' are both solvable, then L is solvable. For the converse, assume that L is solvable. What we look at is $\text{ad}: L \rightarrow \mathfrak{gl}(L)$. Then, $\text{ad } L \subset \mathfrak{gl}(L)$ is solvable. By the first part of the exercise, every element of $(\text{ad } L)'$ is nilpotent. Now, note that $(\text{ad } L)' = \text{ad}(L')$, since this is just taking the brackets. Therefore, $\text{ad}(L')$ is nilpotent. However, $\text{ad}(L') \cong L'/Z(L')$, so L' is nilpotent. $\square$

Remark 6.2.6 — Note that both results above are false if the characteristic is positive.

Exercise 6.2.7. Use Lie's theorem to prove Proposition 5.4.1: Let $x, y: V \rightarrow V$ be linear maps from a complex vector space V to itself. Then, if x and y both commute with $[x, y]$, then $[x, y]$ is a nilpotent map.

Solution: Let L be the span of $x, y, [x, y]$. We will show that L is solvable. We have that $L' = \text{Span}\{[x, y]\}$, so it is abelian. Thus, L is indeed solvable. We can apply Lie's theorem to say that there exists a basis such that each matrix $x, y, [x, y]$ is upper triangular. Then, if X and Y are the matrices of x and y in this basis, then the matrix of $[x, y]$ is $XY - YX$. This is the commutator of $[X, Y]$, which is strictly upper triangular, so we get that $XY - YX$ is a nilpotent map. Therefore, $[x, y]$ is nilpotent. $\square$

7 Some Representation Theory

§7.1 Definitions

Definition 7.1.1. Let L be a Lie algebra over a field F . A **representation** of L is a Lie algebra homomorphism $\varphi: L \rightarrow \mathfrak{gl}(V)$, where V is a finite dimensional vector space over F .

Abuse of Notation 7.1.2. For brevity, we will sometimes omit mention of the homomorphism and just say that V is a representation of L .

If V is a representation of a Lie algebra L over a field F , then we can fix a basis of V and write the linear transformations of V afforded by elements of L as matrices. Alternatively, we can specify a representation by giving a homomorphism $L \rightarrow \mathfrak{gl}(n, F)$. In this setup, a representation is sometimes called a **matrix representation**.

Suppose that $\varphi: L \rightarrow \mathfrak{gl}(V)$ is a representation. By Exercise 1.4.4, the image of φ is a Lie subalgebra of $\mathfrak{gl}(V)$ and the kernel of φ is an ideal of L . By working with $\varphi(L)$, we will in general lose some information about L . If, however the map φ is injective, then no information is lost, leading to the following definition.

Definition 7.1.3. We say that a representation is **faithful** if $\ker(\varphi) = 0$ (or, equivalently, injective), in which case we can think of L as a subalgebra of $\mathfrak{gl}(V)$.

Remark 7.1.4 — Note that if L is a simple Lie algebra, then it has no nonzero proper ideals. Thus, if φ is a faithful representation of L , then $\ker(\varphi)$ is an ideal of L (as we will see later), so $\ker(\varphi) = 0$ or L . Particularly, if φ is not the trivial representation, then $\ker(\varphi) = 0$.

§7.2 Examples

Example 7.2.1

We have already encountered the adjoint map

$$\mathrm{ad}: L \rightarrow \mathfrak{gl}(L), \quad (\mathrm{ad} x)y = [x, y].$$

This is a Lie homomorphism, so ad provides a representation of L , with $V = L$. This representation is known as the **adjoint representation**. It occurs very often and encapsulates much of the structure of L . We saw that the kernel of the adjoint representation is $Z(L)$. Hence the adjoint representation is faithful precisely when the centre of L is zero. For example, this happens when $L = \mathfrak{sl}(2, \mathbb{C})$.

Exercise 7.2.2. Consider the adjoint representation of $\mathfrak{sl}(2, \mathbb{C})$. Show that with respect to

the basis (h, e, f) given in Exercise 1.7.4, $\text{ad } h$ has matrix

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Find the matrices representing $\text{ad } e$ and $\text{ad } f$.

Solution: Note that this basis is determined by

$$[e, f] = h, \quad [h, e] = 2e, \quad [h, f] = -2f.$$

Thus, the matrix for $\text{ad } h$ is pretty clear. Next, for $\text{ad } e$, we have

$$\begin{pmatrix} 0 & 0 & 1 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

For $\text{ad } f$, we have the following:

$$\begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}$$

□

Definition 7.2.3. Suppose that L is a Lie subalgebra of $\mathfrak{gl}(V)$. The inclusion map $L \rightarrow \mathfrak{gl}(V)$ is trivially a Lie algebra homomorphism (for example, because it is the restriction of the identity map on $\mathfrak{gl}(V)$ to L). The corresponding representation is known as the **natural representation** of L .

We have seen several such representations, mostly in the form of matrices, for example, $\mathfrak{sl}(n, \mathbb{C})$, or the upper triangular matrices $\mathfrak{b}(n, \mathbb{C})$. The Lie algebras considered in Exercise 3.4.1 all have natural representations. The natural representation is always faithful.

Example 7.2.4

There is also the trivial representation. To define this representation, take $V = F$ and define $\varphi(x) = 0$ for all $x \in L$. This representation is never faithful for non-zero L .

Example 7.2.5

The Lie algebra $\mathbb{R}_\lambda^3$ introduced in chapter 1 has a trivial centre, and so its adjoint representation is a 3-dimensional faithful representation. In Exercise 1.7.8 we showed that $\mathbb{R}_\lambda^3$ is isomorphic to a subalgebra of $\mathfrak{gl}(\mathbb{R}^3)$; this gives another 3-dimensional representation. In fact, these representations are in a sense the same; we make this sense more precise below.

§7.3 Modules

Definition 7.3.1. Suppose that L is a Lie algebra over a field F . A **Lie module** for L , or alternatively an **L -module**, is a finite-dimensional F -vector space V together with a map $L \times V \rightarrow V$, $(x, v) \mapsto x \cdot v$ which satisfies the following conditions: For $\lambda, \mu \in F$, $x, y \in L$, $v, w \in V$,

- $(\lambda x + \mu y) \cdot v = \lambda(x \cdot v) + \mu(y \cdot v),$
- $x \cdot (\lambda v + \mu w) = \lambda(x \cdot v) + \mu(x \cdot w)$
- $[x, y] \cdot v = x \cdot (y \cdot v) - y \cdot (x \cdot v)$

Remark 7.3.2 — Note that the first two conditions are equivalent to saying that the map $(x, v) \mapsto x \cdot v$ is bilinear. The second condition implies that for each $x \in L$ the map $v \mapsto x \cdot v$ is a linear endomorphism of V , so elements of L act on V by linear maps. The significance of the third condition will be revealed shortly.

§7.3.i Equivalence of Modules and Representations

Given an L -module V , we can define a homomorphism of Lie algebras $\varphi: L \rightarrow \mathfrak{gl}(V)$ by $\varphi(x)(v) = x \cdot v$. Conversely, given φ , V is an L -module by $x \cdot v = \varphi(x)v$.

Just to check, we will show that we can define $\varphi(x)(v) = x \cdot v$. To show that this works, we must check that the axioms for an L -module are satisfied. First, we have

$$(\lambda x + \mu y) \cdot v = \varphi(\lambda x + \mu y)(v) = (\lambda \varphi(x) + \mu \varphi(y))(v)$$

as φ is linear. By the definition of addition and scalar multiplication of linear maps, this is $\lambda \varphi(x)(v) + \mu \varphi(y)(v) = \lambda(x \cdot v) + \mu(y \cdot v)$. The second condition is verified similarly.

$$x \cdot (\lambda v + \mu w) = \varphi(x)(\lambda v + \mu w) = \lambda \varphi(x)(v) + \mu \varphi(x)(w) = \lambda(x \cdot v) + \mu(x \cdot w)$$

By our definition and because φ is a Lie homomorphism, we have

$$[x, y] \cdot v = \varphi([x, y])(v) = [\varphi(x), \varphi(y)](v).$$

As the Lie bracket in $\mathfrak{gl}(V)$ is the commutator of linear maps, this equals

$$\varphi(x)(\varphi(y)(v)) - \varphi(y)(\varphi(x)(v)) = x \cdot (y \cdot v) - y \cdot (x \cdot v).$$

Remark 7.3.3 — This tells us that representations of L are equivalent to L -modules.

Exercise 7.3.4. Check that φ is a Lie algebra homomorphism.

Solution: Let $x, y \in L$. φ is certainly linear, so we just need to check that it respects the Lie bracket. Then,

$$\begin{aligned} \varphi([x, y])(v) &= [x, y] \cdot (v) \\ &= x \cdot y \cdot v - y \cdot x \cdot v \\ &= x \cdot \varphi(y)(v) - y \cdot \varphi(x)(v) \\ &= \varphi(x)\varphi(y)(v) - \varphi(y)\varphi(x)(v) \\ &= [\varphi(x), \varphi(y)](v). \end{aligned}$$

□

Remark 7.3.5 — It would be reasonable to ask at this point why we have introduced both representations and L -modules. The reason is that both approaches have their advantages, and sometimes one approach seems more natural than the other. For modules, the notation is easier, and some of the concepts can appear more natural. On the other hand, having an explicit homomorphism to work with can be helpful when we are more interested in the Lie algebra than in the vector space on which it acts.

A similar situation arises when we have a group acting on a set. Here one can choose between the equivalent languages of G -actions and permutation representations. Again, both have their advantages.

Exercise 7.3.6. Let L be a finite-dimensional Lie algebra. Let V be L with the L -module structure on V given by the adjoint representation of L . Show that the submodules of V are precisely the ideals of L .

Solution: $\text{ad}: L \rightarrow \mathfrak{gl}(L)$, so $V = L$ is indeed an L -module. Submodules of V are then subspaces W of V which are invariant under the action of L . That is, for each $x \in L$ and $w \in W$, we have that $x \cdot w := [x, w] \in W$. This is precisely the definition of an ideal. $\square$

§7.4 Submodules and Factor Modules

Definition 7.4.1. Suppose that V is a Lie module for the Lie algebra L . A **submodule** of V is a subspace W of V which is invariant under the action of L . That is, for each $x \in L$ and for each $w \in W$, we have $x \cdot w \in W$.

Remark 7.4.2 — In the language of representations, submodules are known as **subrepresentations**.

Example 7.4.3

Let L be a Lie algebra. We may make L into an L -module via the adjoint representation. The submodules of L are exactly the ideals of L .

Example 7.4.4

Let $L = \mathfrak{b}(n, F)$ be the Lie algebra of $n \times n$ upper triangular matrices and let V be the natural L -module, so by definition $V = F^n$ and the action of L is given by applying matrices to column vectors. Let $e_1, \dots, e_n$ be the standard basis of F^n . For $1 \leq r \leq n$, let $W_r = \text{Span}\{e_1, \dots, e_r\}$. Exercise 7.4.5 below asks you to show that W_r is a submodule of V .

Exercise 7.4.5. Let F be a field and let $L = \mathfrak{b}(n, F)$ and $V = F^n$.

1. Check that V is an L -module, where the structure is given by the natural representation; that is, by applying matrices to column vectors.
2. Let $e_1, \dots, e_n$ be the standard basis of F^n . For $1 \leq r \leq n$, let $W_r = \text{Span}\{e_1, \dots, e_r\}$. Prove that W_r is a submodule of V .

3. Show that every non-zero submodule of V is equal to one of the W_r . Deduce that each W_r is indecomposable and that if $n \geq 2$ then V is not completely reducible as an L -module.

Solution: 1. We just need to verify that for any $\lambda, \mu \in F$, $x, y \in L$, and $v, w \in V$, the Lie module axioms hold. These are all clear, since we're just applying matrices to column vectors: $(\lambda x + \mu y)v = \lambda(x \cdot v) + \mu(y \cdot v)$, $x \cdot (\lambda v + \mu w) = \lambda(x \cdot v) + \mu(x \cdot w)$, and $[x, y] \cdot v = x \cdot y \cdot v - y \cdot x \cdot v$.

2. We just need to show that W_r is invariant under the L -action. Let $x \in L$. Then, it is easy to see that $x \cdot W_r \subseteq W_r$.

3. Let W be a submodule of V , and let it contain some element $v = \sum_{i=1}^r a_i e_i$, for some $1 \leq r \leq n$. Then, since W is invariant under the L -action, we actually get that $e_r \in W$, $e_{r-1} \in W$, and so on. In particular, we get that $e_1, \dots, e_r$ are all in W and in fact these span W , so $W = W_r$. Since the submodules of V are of the form W_r , we cannot decompose W_r into a direct sum of two submodules U and W , since one of these submodules would have to contain e_r , making it equal to W_r , and so the other would be 0. Therefore, V is not completely reducible because if it were, then it would be a direct sum of irreducible submodules, but the only irreducible submodules of V that is not 0 is W_1 . $\square$

Example 7.4.6

Let L be a complex solvable Lie algebra. Suppose that $\varphi: L \rightarrow \mathfrak{gl}(V)$ is a representation of L . As φ is a homomorphism, $\text{im } \varphi$ is a solvable subalgebra of $\mathfrak{gl}(V)$. Proposition 6.2.3 (the main step in the proof of Lie's Theorem) implies that V has a 1-dimensional subrepresentation.

Definition 7.4.7. Suppose that W is a submodule of the L -module V . We can give the quotient vector space V/W the structure of an L -module by setting

$$x \cdot (v + W) := (x \cdot v) + W \quad \text{for } x \in L \text{ and } v \in V.$$

We call this module the **quotient** or **factor module** V/W .

We must first check that the action of L is well-defined. Suppose that $v + W = v' + W$. Then

$$(x \cdot v) + W - (x \cdot v') + W = x \cdot (v - v') + W = 0 + W$$

as $v - v' \in W$ and W is L -invariant. We should also check that the action satisfies the three conditions; this will be an exercise. She will see that each property follows easily from the corresponding property of the L -module V .

Example 7.4.8

If A is a subalgebra of L , we can regard L as an A -module. Then, A is a submodule of L . Thus, L/A is also an A -module. In other words, $[\cdot, \cdot]: A \times L/A \rightarrow L/A$ is well-defined.

Example 7.4.9

Suppose that I is an ideal of the Lie algebra L . We have seen that I is a submodule of L when L is considered as an L -module via the adjoint representation. The factor module L/I becomes an L -module via

$$x \cdot (y + I) := (\text{ad } x)(y) + I = [x, y] + I.$$

We can interpret this in a different way. We know that L/I is also a Lie algebra, with the Lie bracket given by

$$[x + I, y + I] = [x, y] + I.$$

So, regarded as an L/I -module, the factor module L/I is the adjoint representation of L/I on itself.

Example 7.4.10

Let $L = \mathfrak{b}(n, F)$ and $V = F^n$. Fix r between 1 and n and let $W = V_r$ be the r -dimensional submodule defined in the example.

Let $x \in L$ have matrix X with respect to the standard basis. The matrix for the action of x on W with respect to the basis $e_1, \dots, e_r$ is obtained by taking the upper left $r \times r$ block of X . Moreover, the matrix for the action of x on the quotient space V/W with respect to the basis $e_{r+1} + W, \dots, e_n + W$ is obtained by taking the lower right $(n-r) \times (n-r)$ block of X :

$$X = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1r} & & \\ 0 & a_{22} & \cdots & a_{2r} & & * \\ \vdots & \vdots & \ddots & \vdots & & \\ 0 & 0 & \cdots & a_{rr} & & \\ & & & & a_{r+1,r+1} & \cdots & a_{r+1,n} \\ & & & & \vdots & \ddots & \vdots \\ & & & & 0 & \cdots & a_{nn} \end{pmatrix}.$$

Exercise 7.4.11. This exercise generalizes the remarks made in the previous example. Let L be a Lie algebra, and let V be a finite-dimensional L -module with a submodule W of dimension m . By taking a basis of V which contains a basis of W , show that V has a basis in which the action of every $x \in L$ is represented by a ‘block matrix’

$$\begin{pmatrix} X_1 & X_2 \\ 0 & X_3 \end{pmatrix},$$

where X_1 is a square matrix of size $m \times m$. Show that X_1 is the matrix of x restricted to W and that X_3 represents the action of x on the factor module V/W .

Solution: Let $W = \text{Span}\{w_1, \dots, w_m\}$ and $V = \text{Span}\{w_1, \dots, w_m, v_1, \dots, v_n\}$. Take $x \in L$. Then, x acts on a generic $v \in V$ by this block-matrix form, since $x(W) \subseteq W$. It is easy to then see that X_1 is x restricted to W . Then, X_3 is the action of x on V/W , since if we act on the vectors $v_1 + W, \dots, v_n + W$, then we will get a vector in $\text{Span}\{v_1, \dots, v_n\} + W$. $\square$

§7.5 Irreducible and Indecomposable Modules

Definition 7.5.1. The Lie module V is said to be **irreducible**, or **simple**, if it is non-zero and it has no submodules other than $\{0\}$ and V .

Suppose that V is a non-zero L -module. We may find an irreducible submodule S of V by taking any non-zero submodule of V of minimal dimension. Note that if V is irreducible, then we just take V itself. The quotient module V/S will itself have an irreducible submodule, S' , and so on. In a sense V is made up of the simple modules $S, S', \dots$. One says that irreducible modules are the *building blocks* for all finite-dimensional modules.

Example 7.5.2

If V is 1-dimensional, then V is irreducible. For example, the trivial representation is always irreducible.

Example 7.5.3

If L is a simple Lie algebra, then L viewed as an L -module via the adjoint representation is irreducible. For example $\mathfrak{sl}(2, \mathbb{C})$ is irreducible as an $\mathfrak{sl}(2, \mathbb{C})$ -module.

Example 7.5.4

If L is a complex solvable Lie algebra then it follows from Example 7.4.6 that all the irreducible representations of L are 1-dimensional.

Given a module V , how can one determine whether or not it is irreducible? One useful criterion is given in the following exercise.

Exercise 7.5.5. Show that V is irreducible if and only if for any non-zero $v \in V$ the submodule generated by v contains all elements of V . The submodule generated by v is defined to be the subspace of V spanned by all elements of the form

$$x_1 \cdot (x_2 \cdots (x_m \cdot v) \cdots),$$

where $x_1, \dots, x_m \in L$.

Solution: Suppose that V is irreducible and let $v \in V$. Then, the submodule generated by v must be equal to V since there are no proper nonzero submodules. Now for the converse. Suppose that the submodule generated by some $v \in V$, which I denote as (v) contains all elements of V . Suppose that W is a submodule of V . Then take $w \in W$, then $x_1 \cdot (x_2 \cdots (x_m \cdot v) \cdots) = w$ for some m , so view the submodule (w) of V then. $(w) = V$ by assumption, so it follows that $W = V$, so there are no proper nonzero submodules of V . $\square$

Another criterion that is sometimes useful is given in Exercise 7.5.6:

Exercise 7.5.6. Let L be a Lie algebra and let A be an abelian subalgebra of L . Suppose that the L -module V decomposes as a direct sum of weight spaces for A . Show that any L -submodule of V has a basis of common eigenvectors for A .

Solution: We have that $V = \oplus_i V_{\lambda_i}$. Note that if we have a basis for every weight space, then these bases are common eigenvectors for A . Then, if $W \subseteq V$ is a L -submodule, then it is a union of some of these basis vectors, which is still a set of common eigenvectors for A . Therefore, we get that

$$W = \oplus_i (W \cap V_{\lambda_i}).$$

□

Definition 7.5.7. If V is an L -module such that $V = U \oplus W$, where both U and W are L -submodules of V , we say that V is the **direct sum** of the L -modules U and W .

Definition 7.5.8. The module V is said to be **indecomposable** if there are no non-zero submodules U and W such that $V = U \oplus W$.

Remark 7.5.9 — Every irreducible module is clearly indecomposable, but the converse is not usually true.

Definition 7.5.10. The L -module V is **completely reducible** if it can be written as a direct sum of irreducible L -modules; that is,

$$V = S_1 \oplus S_2 \oplus \cdots \oplus S_k,$$

where each S_i is an irreducible L -module.

Example 7.5.11

Let F be a field and let $L = \mathfrak{d}(n, F)$ be the subalgebra of $\mathfrak{gl}(n, F)$ consisting of diagonal matrices. The natural module $V = F^n$ is completely reducible. If $S_i = \text{Span}\{e_i\}$, then S_i is a 1-dimensional simple submodule of V and $V = S_1 \oplus \cdots \oplus S_n$. As the S_i are the weight spaces for L , we can view this as a reformulation of Example 5.2.2.

Example 7.5.12

If $L = \mathfrak{b}(n, F)$ where F is a field, then the natural module $V = F^n$ is indecomposable: See Exercise 7.4.5 below. Note, however, that provided $n \geq 2$, V is not irreducible since $\text{Span}\{e_1\}$ is a non-trivial submodule. So V is not completely reducible.

Exercise 7.5.13. Direct sums are not the only way in which we can construct new modules from old.

1. Let V be a module for the Lie algebra L . Show that we may make the dual space V^* into an L -module by defining

$$(x \cdot \theta)(v) = -\theta(x \cdot v) \quad \text{for } x \in L, \theta \in V^*, v \in V.$$

Show that the adjoint representation of $\mathbb{R}_\lambda^3$ is self-dual. Prove more generally that V is isomorphic to V^* if and only if there is a basis of V in which the matrices representing the action of L are all skew-symmetric.

2. Let V and W be L -modules. Show that $\text{Hom}(V, W)$, the vector space of linear maps from V to W , may be made into an L -module by defining

$$(x \cdot \theta)(v) = x \cdot (\theta(v)) - \theta(x \cdot v)$$

for $x \in L$, $\theta \in \text{Hom}(V, W)$, and $v \in V$. Show that the linear map $\theta \in \text{Hom}(V, W)$ is an L -module homomorphism if and only if $x \cdot \theta = 0$ for all $x \in L$.

Solution: 1. We first need to show that this satisfies the L -module axioms. Let $\lambda, \mu \in F$, $x, y \in L$, $\theta, \rho \in V^*$, and $v \in V$. First,

$$\begin{aligned} ((\lambda x + \mu y) \cdot \theta)(v) &= -\theta((\lambda x + \mu y) \cdot v) \\ &= -\theta(\lambda x \cdot v + \mu y \cdot v) \\ &= -\lambda\theta(x \cdot v) - \mu\theta(y \cdot v) \\ &= \lambda(x \cdot \theta)(v) + \mu(y \cdot \theta)(v) \\ &= (\lambda(x \cdot \theta) + \mu(y \cdot \theta))(v). \end{aligned}$$

Next,

$$\begin{aligned} (x \cdot (\lambda\theta + \mu\rho))(v) &= -\lambda\theta(x \cdot v) - \mu\rho(x \cdot v) \\ &= \lambda(x \cdot \theta)(v) + \mu(x \cdot \rho)(v) \\ &= (\lambda(x \cdot \theta) + \mu(x \cdot \rho))(v). \end{aligned}$$

Finally,

$$\begin{aligned} ([x, y] \cdot \theta)(v) &= -\theta([x, y] \cdot v) \\ &= -\theta(x \cdot y \cdot v - y \cdot x \cdot v) \\ &= (x \cdot \theta)(y \cdot v) - (y \cdot \theta)(x \cdot v) \\ &= -(y \cdot (x \cdot \theta))(v) + (x \cdot (y \cdot \theta))(v) \\ &= ((x \cdot (y \cdot \theta)) - (y \cdot (x \cdot \theta)))(v). \end{aligned}$$

Thus, this actually does make the dual space V^* into an L -module.

Next, we need to show that the adjoint representation of $\mathbb{R}_\wedge^3$ is self-dual. Translating this into more approachable terminology, I want to show that the module V induced by the adjoint representation of $\mathbb{R}_\wedge^3$ and its dual V^* are isomorphic as $\mathbb{R}_\wedge^3$ -modules. First, we consider the $\mathbb{R}_\wedge^3$ -module induced by the adjoint representation. That is, if $x \in \mathbb{R}_\wedge^3 = L$, and $y \in V = \mathbb{R}^3$, then

$$x \cdot y = \text{ad } x(y) = [x, y].$$

Now, recall that $\mathbb{R}_\wedge^3$ has the basis $\{e_1, e_2, e_3\}$ with

$$[e_1, e_2] = e_3, \quad [e_2, e_3] = e_1, \quad [e_3, e_1] = e_2.$$

We now want to define a homomorphism of representations $\phi: V \rightarrow V^*$. Let $x, y, v \in \mathbb{R}^3$. Then, ϕ better satisfy

$$\phi(x \cdot y)(v) = x \cdot \phi(y)(v).$$

We can expand the LHS as

$$\phi(\text{ad } x(y))(v) = \phi(x \times y)(v),$$

and the RHS as

$$(x \cdot \phi(y))(v) = \text{ad } x(\phi(y))(v) = -\phi(y)(x \cdot v) = -\phi(y)(x \times v).$$

Here, we used the fact that the dual space V^* is made into an L -module by $(x \cdot \phi(y))(v) = -\phi(y)(x \cdot v)$. Thus, we want ϕ to satisfy

$$\phi(x \times y)(v) = -\phi(y)(x \times v).$$

Then, we may consider the map $\phi: V \rightarrow V^*$ which maps

$$a \mapsto \phi(a)(*) = a \cdot (*),$$

the 'dot product by a '-map. Then, it is clear that $\phi(x \times y)(v) = -\phi(y)(x \times v)$ is true (by properties of cross products and dot products). We must now show that this is bijective. First, injectivity is clear, since if a maps to the zero map, then $a = 0$. Next, we check for surjectivity; however, this follows from the fact that the dual V^* is the same dimension as V , so we automatically get surjectivity.

Let $B = \{v_1, \dots, v_n\}$ be a basis for V and $B' = \{f_1, \dots, f_n\}$ be a basis for V^* . Then, we let $\phi: V \rightarrow V^*$ be the usual isomorphism $\phi(v_i) = f_i$, so that $f_i(v_j) = \delta_{ij}$. In particular, we have that $\phi(v_i) = [v_i]^T$, the transpose. Let $x \in L$, $v, w \in V$ then. We have that ϕ is a homomorphism of L -modules between V and W , so

$$\phi(x \cdot v)(w) = (x \cdot \phi(v))(w).$$

The LHS then gives

$$([x][v])^T[w] = [v]^T[x]^T[w].$$

The RHS is first equal to $-\phi(v)(x \cdot w)$, so we get the RHS is

$$-[v]^T[x][w] = [v]^T(-[x])[w].$$

In particular, this is true for all v and w , so $[x] = -[x]^T$.

Now for the converse. Suppose that in some basis B for V and B' for V^* , we get that $[x] = -[x]^T$. We must show that our candidate $\phi: V \rightarrow V^*$, $v \mapsto v^T(\cdot)$ actually is an isomorphism. First, we check that this is an L -module homomorphism. Let $x \in L$ and $v, w \in V$. Then:

$$\begin{aligned} \phi(x \cdot v)(w) &= [v]^T[x]^T[w] \\ &= [v]^T(-[x])[w] \\ &= -\phi(v)(x \cdot w) \\ &= (x \cdot \phi(v))(w), \end{aligned}$$

so we get the desired result. Next, this is clearly injective, because if $[v]^T(\cdot)$ is the zero map, then $v = 0$. Then, $\dim(V) = \dim(V^*)$, so we get surjectivity, and so we have an isomorphism.

2. Let $\lambda, \mu \in F$, $x, y \in L$, $\theta, \rho \in \text{Hom}(V, W)$, and $v \in V$. We first have that

$$\begin{aligned} ((\lambda x + \mu y) \cdot \theta)(v) &= (\lambda x + \mu y) \cdot (\theta(v)) - \theta((\lambda x + \mu y) \cdot v) \\ &= (\lambda x \cdot \theta(v) - \lambda \theta(x \cdot v)) + (\mu y \cdot \theta(v) - \mu \theta(y \cdot v)) \\ &= (\lambda(x \cdot \theta) + \mu(y \cdot \theta))(v). \end{aligned}$$

Next, we have

$$\begin{aligned} (x \cdot (\lambda \theta + \mu \rho))(v) &= x \cdot ((\lambda \theta + \mu \rho)(v)) - (\lambda \theta + \mu \rho)(x \cdot v) \\ &= \lambda(x \cdot (\theta(v)) - \theta(x \cdot v)) + \mu(x \cdot (\rho(v)) - \rho(x \cdot v)) \\ &= \lambda(x \cdot \theta)(v) + \mu(x \cdot \rho)(v) \\ &= (\lambda(x \cdot \theta) + \mu(x \cdot \rho))(v) \end{aligned}$$

Finally, we have

$$\begin{aligned}
 ([x, y] \cdot \theta)(v) &= [x, y] \cdot (\theta(v)) - \theta([x, y] \cdot v) \\
 &= x \cdot y \cdot (\theta(v)) - y \cdot x \cdot (\theta(v)) - \theta(x \cdot y \cdot v - y \cdot x \cdot v) \\
 &= x \cdot y \cdot (\theta(v)) \\
 &= (x \cdot y \cdot \theta)(v) - (y \cdot x \cdot \theta)(v) \\
 &= (x \cdot y \cdot \theta - y \cdot x \cdot \theta)(v)
 \end{aligned}$$

Thus, $\text{Hom}(V, W)$ is indeed a L -module.

We now show the second part. Suppose that $\theta \in \text{Hom}(V, W)$ is a L -homomorphism. Then, for $x \in L$, $v \in V$, we have

$$(x \cdot \theta)(v) = x \cdot (\theta(v)) - \theta(x \cdot v) = x \cdot (\theta(v)) - x \cdot (\theta(v)) = 0.$$

Now for the converse. Suppose that $x \cdot \theta = 0$, for all $x \in L$, then

$$\theta(x \cdot v) = x \cdot (\theta(v)),$$

so θ is an L -module homomorphism. $\square$

§7.6 Homomorphisms

Definition 7.6.1. Let L be a Lie algebra and let V and W be L -modules. An **L -module homomorphism** or **Lie homomorphism** from V to W is a linear map $\theta: V \rightarrow W$ such that

$$\theta(x \cdot v) = x \cdot \theta(v) \quad \text{for all } v \in V \text{ and } x \in L.$$

An **isomorphism** is a bijective L -module homomorphism.

Remark 7.6.2 — Let $\varphi_V: L \rightarrow \mathfrak{gl}(V)$ and $\varphi_W: L \rightarrow \mathfrak{gl}(W)$ be the representations corresponding to V and W . In the language of representations, the condition becomes

$$\theta \circ \varphi_V = \varphi_W \circ \theta.$$

Homomorphisms are in particular linear maps, so we can talk about the kernel and image of an L -module homomorphism. And as expected there are the following isomorphism theorems for L -modules.

Theorem 7.6.3 (Isomorphism Theorems)

- (i) Let $\theta: V \rightarrow W$ be a homomorphism of L -modules. Then $\ker \theta$ is an L -submodule of V and $\text{im } \theta$ is an L -submodule of W , and there is an isomorphism of L -modules,

$$V / \ker \theta \cong \text{im } \theta.$$

- (ii) If U and W are submodules of V , then $U + W$ and $U \cap W$ are submodules of V and

$$(U + W) / W \cong U / (U \cap W).$$

- (iii) If U and W are submodules of V such that $U \subseteq W$, then W/U is a submodule of V/U and the factor module $(V/U)/(W/U)$ is isomorphic to V/W .

Exercise 7.6.4. Prove the isomorphism theorems for modules by adapting the argument used to prove the corresponding theorems for Lie algebras (see Theorem 2.2.3).

§7.6.i Bijective Correspondence for Submodules

Analogous to before, we have the bijective correspondence of submodules again. If U is a submodule of V , then there is a bijective correspondence between the submodules of V/U and the submodules of V containing U . Again this is precisely analogous to the corresponding result for Lie algebras, given in the second chapter.

Example 7.6.5

Let L be the 1-dimensional abelian Lie algebra spanned, say, by x . We may define a representation of L on a vector space V by mapping x to any element of $\mathfrak{gl}(V)$. Let W be another vector space. The representations of L corresponding to linear maps $f: V \rightarrow V$ and $g: W \rightarrow W$ are isomorphic if and only if there is a vector space isomorphism $\theta: V \rightarrow W$ such that $\theta f = g\theta$, or equivalently, such that $\theta f \theta^{-1} = g$. Thus the representations are isomorphic if and only if there are bases of V and W in which f and g are represented by the same matrix.

For a concrete example, the 2-dimensional matrix representations defined by

$$x \mapsto \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad x \mapsto \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

are isomorphic because the two matrices are conjugate (as may be checked by diagonalizing the first). For a more substantial example, see Exercise 7.6.6.

Exercise 7.6.6. Let L be the 2-dimensional complex non-abelian Lie algebra. We showed that L has a basis x, y such that $[x, y] = x$. Check that we may define a representation of L on $\mathbb{C}^2$ by setting

$$\varphi(x) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \varphi(y) = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Show that this representation is isomorphic to the adjoint representation of L on itself.

Solution: It suffices to check this $\varphi: L \rightarrow \mathfrak{gl}(2, \mathbb{C})$ is a Lie algebra homomorphism. First,

$$\varphi([x, y]) = \varphi(x) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Next,

$$[\varphi(x), \varphi(y)] = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Thus, this respects the Lie bracket. Furthermore, it is clear that $\varphi(x)$ and $\varphi(y)$ are both linear maps. The adjoint representation $\text{ad}: L \rightarrow \mathfrak{gl}(L)$ is defined as follows:

$$\text{ad } x = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \text{ad } y = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Now we need to define a L -module homomorphism $\theta: \mathbb{C}^2 \rightarrow L$. We define this by

$$\theta(e_1) = x, \quad \theta(e_2) = -x + y.$$

We now check that this is actually an L -module homomorphism. We need to show that

$$\theta(\varphi(ax + by))(v) = \text{ad}(ax + by)(\theta)(v),$$

where $ax + by$ is an arbitrary element in L . We show this holds for the basis vectors e_1 and e_2 .

$$\theta(\varphi(ax + by)(e_1)) = \theta(-be_1) = -bx = \text{ad}(ax + by)(x) = \text{ad}(ax + by)(\theta(e_1)).$$

Next,

$$\theta(\varphi(ax + by)(e_2)) = \theta(ae_1 + be_1) = (a+b)x = \text{ad}(ax + by)(-x + y) = \text{ad}(ax + by)(\theta(e_2)).$$

Thus, this shows true. To conclude, it is clear that θ is surjective, and $\mathbb{C}^2$ and L have dimension 2, so this is an isomorphism. $\square$

§7.7 Schur's Lemma

One of the best ways to understand the structure of a module for a Lie algebra is to look at the homomorphisms between it and other modules. It is natural to begin with the homomorphisms between irreducible modules.

Suppose that S and T are irreducible Lie modules and that $\theta: S \rightarrow T$ is an L -module homomorphism. Then $\text{im } \theta$ is a non-zero submodule of T , so $\text{im } \theta = T$. Similarly, $\ker \theta$ is a proper submodule of S , so $\ker \theta = 0$. It follows that θ is an isomorphism. Hence there are no non-zero homomorphisms between non-isomorphic irreducible modules. We now consider the homomorphism from an irreducible module to itself.

Lemma 7.7.1 (Schur's Lemma)

Let L be a complex Lie algebra and let S be a finite-dimensional irreducible L -module. A map $\theta: S \rightarrow S$ is an L -module homomorphism if and only if θ is a scalar multiple of the identity transformation; that is, $\theta = \lambda \text{Id}_S$ for some $\lambda \in \mathbb{C}$.

Proof. The if direction should be clear, so we work on the forward implication. Suppose that $\theta: S \rightarrow S$ is an L -module homomorphism. Then θ is, in particular, a linear map of a complex vector space, and so it must have an eigenvalue, say λ . Now $\theta - \lambda \text{Id}$ is also an L -module homomorphism. The kernel of this map contains a λ -eigenvector for θ , and so it is a non-zero submodule of S . As S is irreducible, $S = \ker(\theta - \lambda \text{Id})$; that is, $\theta = \lambda \text{Id}$. $\square$

Remark 7.7.2 — We may also rephrase this to say the following. If V, W are irreducible, then either V, W are not isomorphic, and there are no maps between them; or, $V \cong W$ and all nonzero maps $V \rightarrow W$ are isomorphisms, unique up to multiplication by elements of $\mathbb{C}^\times$.

Schur's Lemma has many applications, for example the following.

Lemma 7.7.3

Let L be a complex Lie algebra and let V be an irreducible L -module. If $z \in Z(L)$, then z acts by scalar multiplication on V ; that is, there is some $\lambda \in \mathbb{C}$ such that $z \cdot v = \lambda v$ for all $v \in V$.

Proof. The map $v \mapsto z \cdot v$ is an L -module homomorphism, for if $x \in L$ then

$$z \cdot (x \cdot v) = x \cdot (z \cdot v) + [z, x] \cdot v = x \cdot (z \cdot v)$$

since $[z, x] = 0$. Now apply Schur’s Lemma. $\square$

Corollary 7.7.4

The simple modules for an abelian Lie algebra over $\mathbb{C}$ are 1-dimensional.

Proof. Suppose that V is a simple module for the abelian Lie algebra L . By this lemma, every element of L acts by scalar multiplication on V , so any non-zero $v \in V$ spans a 1-dimensional submodule of V . As V is irreducible, this submodule must be all of V . $\square$

Remark 7.7.5 — In the previous two chapters (5 and 6), we worked with the hypothesis that a Lie algebra L was a subalgebra of $\mathfrak{gl}(V)$ for some vector space V . In other words, we were assuming that L had a faithful representation on V . Although we shall not prove it in this book, it is a theorem (*Ado’s Theorem*) that every Lie algebra has a faithful representation.

If L is a Lie algebra and $\varphi: L \rightarrow \mathfrak{gl}(V)$ is a representation (not necessarily faithful), then we can apply the results of Chapters 5 and 6 to the subalgebra $\text{im } \varphi \subseteq \mathfrak{gl}(V)$. In particular, it makes sense to talk about weights and weight spaces for L .

§7.8 Exercises

Exercise 7.8.1. Let L be the Heisenberg algebra with basis f, g, z such that $[f, g] = z$ and z is central. Show that L does not have a faithful finite dimensional irreducible representation.

Solution: Suppose that there was an injective representation $\varphi: L \hookrightarrow \mathfrak{gl}(V)$, where V is some finite dimensional vector space over $\mathbb{C}$ and L is the Heisenberg algebra. Then, by Schur’s lemma, $z \in Z(H)$, so z acts on V by scalar multiplication. However, $[f, g] = z$, so its trace must also equal to 0. This forces the scalar to be 0, which implies that the representation is not faithful. $\square$

Exercise 7.8.2. Let L be a Lie algebra over a field F . Suppose that $\varphi: L \rightarrow \mathfrak{gl}(1, F)$ is a 1-dimensional representation of L . Show that $\varphi(L') = 0$. Show that any representation of L/L' can be viewed as a representation of L on which L' acts trivially. When $F = \mathbb{C}$, show that if $L' \neq L$ then L has infinitely many non-isomorphic 1-dimensional modules, but if $L' = L$ then the only 1-dimensional representation of L is the trivial representation.

Solution: First let $x, y \in L$. Then, $\varphi([x, y]) = \varphi(x)\varphi(y) - \varphi(y)\varphi(x) = 0$, since $\varphi(x), \varphi(y) \in \mathfrak{gl}(1, F)$, so they commute.

Next, let $\rho: L/L' \rightarrow \mathfrak{gl}(V)$. Then, we may consider $\tilde{\rho}: L \rightarrow \mathfrak{gl}(V)$ defined by $\tilde{\rho}(x) = \rho(x + L')$. We want to show that the $\tilde{\rho}(L') = 0$ and that this is indeed a representation.

Note that we have the following diagram:

$$\begin{array}{ccc} L & \xrightarrow{\pi} & L/L' \\ & \searrow \tilde{\rho} & \downarrow \rho \\ & & \mathfrak{gl}(V) \end{array}$$

Then, $\tilde{\rho} = \rho \circ \pi$, and the quotient map π and ρ are Lie algebra homomorphisms, so $\tilde{\rho}$ is a Lie algebra homomorphism, so is a representation. Finally, If $z \in L'$, then indeed

$$\tilde{\rho}(z) = \rho(L') = 0,$$

so true.

Now let $F = \mathbb{C}$. If $L \neq L'$, then $L/L' \neq 0$. Now, $L \neq L'$, so let $\{x_1, \dots, x_m\}$ be a basis for L' and extend this to a basis of L : $\{x_1, \dots, x_m, y_1, \dots, y_n\}$. Then, we define a linear map $\rho_a: L \rightarrow \mathfrak{gl}(1, F)$ to be

$$\rho_a(x_i)(v) = 0, \quad \rho_a(y_j)(v) = av,$$

where $a \in \mathbb{C}$ is fixed and $v \in \mathbb{C}$. This is linear by definition, so we check it respects the bracket. Take $x, y \in L$. Then, $\rho_a([x, y]) = 0$ since $\rho_a(L') = 0$, since $L' = \text{Span}\{x_1, \dots, x_m\}$. Next, $[\rho_a(x), \rho_a(y)] = 0$, since we're taking the bracket in $\mathfrak{gl}(1, \mathbb{C}) = \mathbb{C}$, which is abelian. Thus, ρ_a is actually a representation. The claim is that this a uniquely determines an isomorphism class. Now, suppose that there is some L -module isomorphism from $\mathbb{C}$ to $\mathbb{C}$; that is, a linear map $\theta: \mathbb{C} \xrightarrow{\sim} \mathbb{C}$ such that

$$\theta(\rho_a(v)) = \rho_b(\theta(v)).$$

That is, $\theta(av) = b(\theta)(v)$. In particular, θ is not the zero map for then θ is not surjective. Thus, we have that $a\theta(v) = b\theta(v)$ if and only if $a = b$. Thus, $\rho_a: L \rightarrow \mathbb{C}$ uniquely determines a 1-dimensional representation of L .

Now let $L' = L$. Then, we define $\rho: L \rightarrow \mathbb{C}$ by

$$\rho(x) = 0.$$

In particular, since $L' = L$, L is generated by some set of commutators. Since $\rho(L') = 0$, so this is the only possibility. $\square$

Exercise 7.8.3. Let L be a solvable Lie algebra. Then, show that every irreducible representation of L is 1-dimensional.

Solution: Let $\rho: L \rightarrow \mathfrak{gl}(V)$. L is solvable, so there exists some nonzero $v \in V$ which is a simultaneous eigenvector for all $x \in L$. Then, $\rho(x)v = \lambda(x)v$, for all $x \in L$ and some $\lambda(x) \in \mathbb{C}$. Therefore, $\text{Span}\{v\}$ is L -invariant, so it is a subrepresentation of V . However, V was simple, so $V = \text{Span}\{v\}$. $\square$

8 Representations of $\mathfrak{sl}(2, \mathbb{C})$

In this chapter, we study the finite-dimensional irreducible representations of $\mathfrak{sl}(2, \mathbb{C})$. In doing this, we shall see, in a stripped-down form, many of the ideas needed to study representations of an arbitrary semisimple Lie algebra. Later we will see that representations of $\mathfrak{sl}(2, \mathbb{C})$ control a large part of the structure of all semisimple Lie algebras. We shall use the following basis of $\mathfrak{sl}(2, \mathbb{C})$:

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The structure constants are given by

$$[e, f] = h, \quad [h, e] = 2e, \quad [h, f] = -2f.$$

§8.1 The Modules V_d

We begin by constructing a family of irreducible representations of $\mathfrak{sl}(2, \mathbb{C})$.

Consider the vector space $\mathbb{C}[X, Y]$ of polynomials in two variables X, Y with complex coefficients. For each integer $d \geq 0$, let V_d be the subspace of homogeneous polynomials in X and Y of degree d . So V_0 is the 1-dimensional vector space of constant polynomials, and for $d \geq 1$, the space V_d has as a basis the monomials $X^d, X^{d-1}Y, \dots, XY^{d-1}, Y^d$. This basis shows that V_d has dimension $d + 1$ as a $\mathbb{C}$ -vector space.

We now make V_d into an $\mathfrak{sl}(2, \mathbb{C})$ -module by specifying a Lie algebra homomorphism $\varphi: \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(V_d)$. Since $\mathfrak{sl}(2, \mathbb{C})$ is linearly spanned by the matrices e, f, h , the map φ will be determined once we have specified how it acts on e, f, h . We let

$$\varphi(e) := X \frac{\partial}{\partial Y};$$

that is, $\varphi(e)$ is the linear map which first differentiates a polynomial with respect to Y and then multiplies it with X . This preserves the degrees of polynomials and so maps V_d into V_d . Similarly, we let

$$\varphi(f) := Y \frac{\partial}{\partial X}.$$

Finally, we let

$$\varphi(h) := X \frac{\partial}{\partial X} - Y \frac{\partial}{\partial Y}.$$

Notice that

$$\varphi(h)(X^a Y^b) = (a - b)X^a Y^b,$$

so h acts diagonally on V_d with respect to our chosen basis.

Theorem 8.1.1

With these definitions, $\varphi: \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(V_d)$ is a representation of $\mathfrak{sl}(2, \mathbb{C})$.

Proof. By construction, φ is linear. Thus, all we have to check is that φ preserves Lie brackets. By linearity, it is enough to check this on the basis elements of $\mathfrak{sl}(2, \mathbb{C})$, so there are just three equations we need to verify.

1. We begin by showing $[\varphi(e), \varphi(f)] = \varphi([e, f]) = \varphi(h)$. If we apply the left-hand side to a basis vector $X^a Y^b$ with $a, b \geq 1$ and $a + b = d$, we get

$$\begin{aligned} [\varphi(e), \varphi(f)](X^a Y^b) &= \varphi(e)(\varphi(f)(X^a Y^b)) - \varphi(f)(\varphi(e)(X^a Y^b)) \\ &= \varphi(e)(aX^{a-1}Y^{b+1}) - \varphi(f)(bX^{a+1}Y^{b-1}) \\ &= a(b+1)X^a Y^b - b(a+1)X^a Y^b \\ &= (a-b)X^a Y^b. \end{aligned}$$

This is the same as $\varphi(h)(X^a Y^b)$.

We check separately the action on X^d :

$$\begin{aligned} [\varphi(e), \varphi(f)](X^d) &= \varphi(e)(\varphi(f)(X^d)) - \varphi(f)(\varphi(e)(X^d)) \\ &= \varphi(e)(dX^{d-1}Y) - \varphi(f)(0) = dX^d, \end{aligned}$$

which is the same as $\varphi(h)(X^d)$. Similarly, one checks the action on Y^d , so $[\varphi(e), \varphi(f)]$ and $\varphi(h)$ agree on a basis of V_d and are the same linear map.

2. We also need $[\varphi(h), \varphi(e)] = \varphi([h, e]) = \varphi(2e) = 2\varphi(e)$. Again we can prove this by applying the maps to basis vectors of V_d . For $b \geq 1$, we get

$$\begin{aligned} [\varphi(h), \varphi(e)](X^a Y^b) &= \varphi(h)(\varphi(e)(X^a Y^b)) - \varphi(e)(\varphi(h)(X^a Y^b)) \\ &= \varphi(h)(bX^{a+1}Y^{b-1}) - \varphi(e)((a-b)X^a Y^b) \\ &= b((a+1) - (b-1))X^{a+1}Y^{b-1} - (a-b)bX^{a+1}Y^{b-1} \\ &= 2bX^{a+1}Y^{b-1}. \end{aligned}$$

This is the same as $2\varphi(e)(X^a Y^b)$. If $b = 0$ and $a = d$, then a separate verification is needed. Exercise!

3. Similarly, one can check that $[\varphi(h), \varphi(f)] = -2\varphi(f)$. Exercise!

□

§8.1.i Matrix Interpretation

It can be useful to know the matrices that correspond to the action of e, f, h on V_d ; these give the matrix representation corresponding to φ .

As usual, we take the basis $X^d, X^{d-1}Y, \dots, Y^d$ of V_d . The calculations in the proof of Theorem 8.1.3 show that the matrix of $\varphi(e)$ with respect to this basis is

$$\begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 2 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix},$$

the matrix of $\varphi(f)$ is

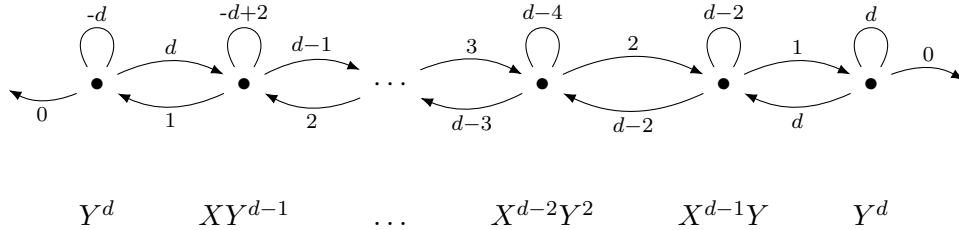
$$\begin{pmatrix} 0 & 0 & \dots & 0 & 0 \\ d & 0 & \dots & 0 & 0 \\ 0 & d-1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix},$$

and $\varphi(h)$ is diagonal:

$$\begin{pmatrix} d & 0 & \dots & 0 & 0 \\ 0 & d-2 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & -d+2 & 0 \\ 0 & 0 & \dots & 0 & -d \end{pmatrix}.$$

where the diagonal entries are the numbers $d - 2k$, where $k = 0, 1, \dots, d$. By explicitly computing the commutators of these matrices, we can give another (but equivalent) way to prove that φ is a representation of $\mathfrak{sl}(2, \mathbb{C})$.

Another way to represent the action of h, e, f is to draw a diagram like



where loops represent the action of h , arrows to the right represent the action of e , and arrows to the left represent the action of f .

§8.1.ii Irreducibility

One virtue of the diagram above is that it makes it almost obvious that the $\mathfrak{sl}(2, \mathbb{C})$ -submodule of V_d generated by any particular basis element $X^a Y^b$ contains all the basis elements and so is all of V_d .

Exercise 8.1.2. Check this assertion.

Solution: This is clear. If we take some arbitrary $X^a Y^b$, where $a + b = d$ and $a, b \geq 0$, then $\varphi(e)X^a Y^b = bX^{a+1}Y^{b-1}$, and $\varphi(f)X^a Y^b = aX^{a-1}Y^{b+1}$, so we attain all. $\square$

A possible disadvantage of our diagram is that it may blind us to the existence of the *many other* vectors in V_d , which, while linear combinations of the basis vectors, are not basis vectors themselves.

Theorem 8.1.3

The $\mathfrak{sl}(2, \mathbb{C})$ -module V_d is irreducible.

Proof. Suppose U is a non-zero $\mathfrak{sl}(2, \mathbb{C})$ -submodule of V_d . Then $h \cdot u \in U$ for all $u \in U$. Since h acts diagonalizably on V_d , it also acts diagonalizably on U , so there is an eigenvector of h which lies in U . We have seen that all eigenspaces of h on V_d are one-dimensional, and each eigenspace is spanned by some monomial $X^a Y^b$, so the submodule U must contain some monomial, and by the exercise above, U contains a basis for V_d . Hence $U = V_d$. $\square$

§8.2 Classifying the Irreducible $\mathfrak{sl}(2, \mathbb{C})$ -Modules

Lemma 8.2.1

Suppose that V is an $\mathfrak{sl}(2, \mathbb{C})$ -module and $v \in V$ is an eigenvector of h with eigenvalue λ .

- (i) Either $e \cdot v = 0$ or $e \cdot v$ is an eigenvector of h with eigenvalue $\lambda + 2$.
- (ii) Either $f \cdot v = 0$ or $f \cdot v$ is an eigenvector of h with eigenvalue $\lambda - 2$.

Proof. As V is a representation of $\mathfrak{sl}(2, \mathbb{C})$, we have

$$h \cdot (e \cdot v) = e \cdot (h \cdot v) + [h, e] \cdot v = e \cdot (\lambda v) + 2e \cdot v = (\lambda + 2)e \cdot v.$$

The calculation for $f \cdot v$ is similar. □

Lemma 8.2.2

Let V be a finite-dimensional $\mathfrak{sl}(2, \mathbb{C})$ -module. Then V contains an eigenvector w for h such that $e \cdot w = 0$.

Proof. As we work over $\mathbb{C}$, the linear map $h: V \rightarrow V$ has at least one eigenvalue and so at least one eigenvector. Let $h \cdot v = \lambda v$. Consider the vectors

$$v, e \cdot v, e^2 \cdot v, \dots$$

If they are non-zero, then by Lemma 8.2.1 they form an infinite sequence of h -eigenvectors with distinct eigenvalues. Eigenvectors with different eigenvalues are linearly independent, so V would contain infinitely many linearly independent vectors, a contradiction. Thus, there exists $k \geq 0$ such that $e^k \cdot v \neq 0$ and $e^{k+1} \cdot v = 0$. If we set $w = e^k \cdot v$, then $h \cdot w = (\lambda + 2k)w$ and $e \cdot w = 0$. □

Remark 8.2.3 — The vector w above is also called the **highest weight** vector for h .

Theorem 8.2.4

If V is a finite-dimensional irreducible $\mathfrak{sl}(2, \mathbb{C})$ -module, then V is isomorphic to one of the V_d .

Proof. By Lemma 8.2.2, V has an h -eigenvector w such that $e \cdot w = 0$. Suppose that $h \cdot w = \lambda w$. Consider the sequence of vectors

$$w, f \cdot w, f^2 \cdot w, \dots$$

By the proof of Lemma 8.2.2, there exists a $d \geq 0$ such that $f^d \cdot w \neq 0$ and $f^{d+1} \cdot w = 0$.

Step 1: We claim that the vectors $w, f \cdot w, \dots, f^d \cdot w$ form a basis for a submodule of V . They are linearly independent because, by Lemma 8.2.1, they are eigenvectors for h

with distinct eigenvalues. By construction, the span of $w, f \cdot w, \dots, f^d \cdot w$ is invariant under h and f . To show that it is invariant under e , we shall prove by induction on k that

$$e \cdot (f^k \cdot w) \in \text{Span} \{f^j \cdot w : 0 \leq j \leq k\}.$$

If $k = 0$, then we know that $e \cdot w = 0$. For the inductive step, note that

$$e \cdot (f^{k+1} \cdot w) = (fe + h) \cdot (f^k \cdot w).$$

By the inductive hypothesis, $e \cdot (f^k \cdot w)$ is in the span of the $f^j \cdot w$ for $j \leq k-1$, and therefore $fe \cdot (f^k \cdot w)$ is in the span of all $f^j \cdot w$ for $j < k$. Moreover, $hf^k \cdot w$ is a scalar multiple of $f^k \cdot w$. This gives the inductive step. Now, since V is irreducible, the submodule spanned by the $f^k \cdot w$ for $0 \leq k \leq d$ is equal to V .

Step 2: In this step, we shall show that $\lambda = d$. The matrix of h with respect to the basis $w, f \cdot w, \dots, f^d \cdot w$ is diagonal, with trace

$$\lambda + (\lambda - 2) + \dots + (\lambda - 2d) = (d+1)\lambda - d(d+1).$$

Since $[e, f] = h$, the matrix of h is equal to the commutator of the matrices of e and f , so it has trace zero and $\lambda = d$.

Step 3: To finish, we produce an explicit isomorphism $V \cong V_d$. As we have seen, V has basis $\{w, f \cdot w, \dots, f^d \cdot w\}$. Furthermore, V_d has basis

$$\{X^d, f \cdot X^d, \dots, f^d \cdot X^d\},$$

where $f^k \cdot X^d$ is a scalar multiple of $X^{d-k}Y^k$. Moreover, the eigenvalue of h on $f^k \cdot w$ is the same as the eigenvalue of h on $f^k \cdot X^d$. Clearly, to have a homomorphism, we must have a map which takes h -eigenvectors to h -eigenvectors for the same eigenvalue. So we may set

$$\psi(w) = X^d$$

and then we must define ψ by

$$\psi(f^k \cdot w) := f^k \cdot X^d.$$

This defines a vector space isomorphism, which commutes with the actions of f and h . To show that it also commutes with the action of e , we use induction on k and a method similar to Step 1. Explicitly, for $k = 0$ we have $\psi(e \cdot w) = 0$ and $e\psi(w) = e \cdot X^d = 0$. For the inductive step,

$$\psi(e f^k \cdot w) = \psi((fe + h) \cdot (f^{k-1} \cdot w)) = f \cdot \psi(e f^{k-1} \cdot w) + h \cdot \psi(f^{k-1} \cdot w)$$

using that ψ commutes with f and h . We use the inductive hypothesis to take e out and obtain that the expression can be written as

$$(fe + h) \cdot \psi(f^{k-1} \cdot w) = ef \cdot \psi(f^{k-1} \cdot w) = e \cdot \psi(f^k \cdot w).$$

□

Corollary 8.2.5

If V is a finite-dimensional representation of $\mathfrak{sl}(2, \mathbb{C})$ and $w \in V$ is an h -eigenvector such that $e \cdot w = 0$, then $h \cdot w = dw$ for some non-negative integer d and the submodule of V generated by w is isomorphic to V_d .

Proof. Step 1 in the previous proof shows that for some $d \geq 0$ the vectors $w, f \cdot w, \dots, f^d \cdot w$ span a submodule of V . Now apply Steps 2 and 3 to this submodule to get the required result. $\square$

Remark 8.2.6 — A vector v of this type considered in this corollary is called a **highest-weight vector**. If d is the associated eigenvalue of h , then d is said to be a **highest weight**.

§8.3 Weyl's Theorem

We include the following theorem for completeness, but we shall not prove it because it is too difficult.

Theorem 8.3.1 (Weyl's Theorem)

Every finite dimensional representation of a semisimple Lie algebra is completely reducible, i.e. it is a direct sum of irreducible representations.

Check appendix B.

§8.4 Exercises

Exercise 8.4.1. Find explicit isomorphisms between:

1. the trivial representation of $\mathfrak{sl}(2, \mathbb{C})$ and V_0 .
2. the natural representation of $\mathfrak{sl}(2, \mathbb{C})$ and V_1 .
3. the adjoint representation of $\mathfrak{sl}(2, \mathbb{C})$ and V_2 . Note that this is faithful.

Solution: 1. The trivial representation is $\varphi: \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(\mathbb{C})$ defined by

$$\varphi(x)(v) = 0.$$

If $\psi: \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(V_0)$ is the representation given in the text, then define $\theta: \mathbb{C} \mapsto V_0$ by $x \mapsto x$. This is a homomorphism of Lie modules, since it is clearly linear. Furthermore, for any $x \in \mathfrak{sl}(2, \mathbb{C})$ and $v \in \mathbb{C}$, then

$$\theta(\varphi(x)v) = \theta(0) = 0 = \psi(x) \cdot (v) = \psi(x) \cdot \theta(v).$$

2. Next, let $V_1 = \text{Span}\{X, Y\}$, and $\psi: \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(V_1)$ be the usual $\mathfrak{sl}(2, \mathbb{C})$ representation. Next, let $\varphi: \mathfrak{sl}(2, \mathbb{C}) \hookrightarrow \mathfrak{gl}(2, \mathbb{C})$ be given by the natural inclusion. Define $\theta: \mathbb{C}^2 \rightarrow V_1$ by $e_1 \mapsto X, e_2 \mapsto Y$. Then, we check that this is a homomorphism of $\mathfrak{sl}(2, \mathbb{C})$ -modules. Firstly, it is clearly linear. If $v = v_1 e_1 + v_2 e_2, w = w_1 e_1 + w_2 e_2$, then

$$\theta(av + bw) = (av_1 + bw_1)X + (av_2 + bw_2)Y = a\theta(v) + b\theta(w).$$

This is also bijective, since it is a linear map between vector spaces of the same dimension, and is surjective. Let e_1, e_2 be the standard basis vectors of $\mathbb{C}^2$, and $a, b \in \mathbb{C}$.

$$\theta(\varphi(e)(ae_1 + be_2)) = \theta(e_{12} \cdot (ae_1 + be_2)) = \theta(be_1) = bX.$$

On the other side, we have

$$\psi(e) \cdot \theta(ae_1 + be_2) = \psi(e) \cdot (aX + bY) = bX.$$

Thus, $\theta(\varphi(e) \cdot v) = \psi(e) \cdot \theta(v)$. The other cases follow similarly.

3. Again, we let $\psi: \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(V_2)$ be the usual representation, where $V_2 = \text{Span}\{X^2, XY, Y^2\}$. We also have the adjoint representation $\text{ad}: \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(\mathfrak{sl}(2, \mathbb{C}))$. We define $\theta: \mathfrak{sl}(2, \mathbb{C}) \rightarrow V_2$ by $e \mapsto X^2$, $f \mapsto -Y^2$, $h \mapsto -2XY$. Linearity is clear, so we check $\theta(x \cdot v) = x \cdot \theta(v)$, or that

$$\theta(\text{ad } x(v)) = \psi(x) \cdot \theta(v),$$

for all $x \in \mathfrak{sl}(2, \mathbb{C})$ and $v \in \mathfrak{sl}(2, \mathbb{C})$. We check this on the basis elements. First, if $x = v$ (and is equal to a basis element), we have the following:

$$\theta(\text{ad } x(x)) = 0.$$

Now, $\psi(x) \cdot \theta(v) = 0$ for all $v = e, f, h$. If $v = e$, then, $\psi(e) = X \frac{\partial}{\partial Y}$ and $\theta(e) = X^2$, which is zero. If $v = f$, then $\psi(f) = Y \frac{\partial}{\partial X}$ and $\theta(f) = -Y^2$, so zero. Finally, if $v = h$, then $\psi(h) = X \frac{\partial}{\partial X} - Y \frac{\partial}{\partial Y}$ and $\theta(h) = -2XY$, so $\psi(h) \cdot \theta(h) = 2XY - 2XY = 0$. Now, we consider when $x \neq v$. We have the following:

$$\begin{aligned} \theta(\text{ad } e(f)) &= \theta(h) = -2XY = X \frac{\partial}{\partial Y}(-Y^2) = \psi(e) \cdot \theta(f) \\ \theta(\text{ad } e(h)) &= \theta(-2e) = -2X^2 = X \frac{\partial}{\partial Y}(2XY) = \psi(e) \cdot \theta(h) \\ \theta(\text{ad } f(e)) &= \theta(-h) = 2XY = Y \frac{\partial}{\partial X}(X^2) = \psi(f) \cdot \theta(e) \\ \theta(\text{ad } f(h)) &= \theta(2f) = -2Y^2 = Y \frac{\partial}{\partial X}(-XY) = \psi(f) \cdot \theta(h) \\ \theta(\text{ad } h(e)) &= \theta(2e) = 2X^2 = (X \frac{\partial}{\partial X} - Y \frac{\partial}{\partial Y})(X^2) = \psi(h) \cdot \theta(e) \\ \theta(\text{ad } h(f)) &= \theta(-2f) = 2Y^2 = (X \frac{\partial}{\partial X} - Y \frac{\partial}{\partial Y})(-2Y^2) = \psi(h) \cdot \theta(f). \end{aligned}$$

Thus, true! □

Exercise 8.4.2. Show that the subalgebra of $\mathfrak{sl}(3, \mathbb{C})$ consisting of matrices of the form

$$\begin{pmatrix} * & * & 0 \\ * & * & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. We may therefore regard $\mathfrak{sl}(3, \mathbb{C})$ as a module for $\mathfrak{sl}(2, \mathbb{C})$, with the action given by $x \cdot y = [\varphi(x), y]$ for $x \in \mathfrak{sl}(2, \mathbb{C})$ and $y \in \mathfrak{sl}(3, \mathbb{C})$, where φ is the isomorphism. Show that as an $\mathfrak{sl}(2, \mathbb{C})$ -module

$$\mathfrak{sl}(3, \mathbb{C}) \cong V_2 \oplus V_1 \oplus V_1 \oplus V_0.$$

Solution: We can just send

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \mapsto \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

This isomorphism is clear. The main tool for the next part will be Corollary 8.2.5. That is, for each element in this subalgebra of $\mathfrak{sl}(3, \mathbb{C})$, we see what the action of e and h on it are. Let us define e_{ij} to be the usual matrices. Then, $e \cdot e_{12} = [e_{12}, e_{12}] = 0$ and $h \cdot e_{12} = [h', e_{12}] = 2e_{12}$, so $\langle e' \rangle \cong V_2$. We note that $f \cdot e_{12} = [f', e_{12}] = e_{22} - e_{11}$. Further, $f \cdot (e_{22} - e_{11}) = -2e_{21}$.

Next, consider e_{13} . We get that $e \cdot e_{13} = [e', e_{13}] = 0$ and $h \cdot e_{13} = [h', e_{13}] = [e_{11} - e_{22}, e_{13}] = e_{13}$. so $\langle e_{13} \rangle \cong V_1$. We shall note that $f \cdot e_{13} = [f', e_{13}] = e_{23}$.

Next, consider e_{32} . Then, $e \cdot e_{32} = [e', e_{32}] = 0$ and $h \cdot e_{32} = [h', e_{32}] = [e_{11} - e_{22}, e_{32}] = e_{32}$, so $\langle e_{32} \rangle \cong V_1$. We note that $f \cdot e_{32} = [f', e_{32}] = -e_{31}$.

Finally, we consider $e_{11} + e_{22} - 2e_{33}$. Then, $e \cdot (e_{11} + e_{22} - 2e_{33}) = -e_{12} + e_{12} = 0$. Further, $h \cdot (e_{11} + e_{22} - 2e_{33}) = [h', e_{11} + e_{22} - 2e_{33}] = 0$. Thus, $\langle e_{11} + e_{22} - 2e_{33} \rangle \cong V_0$. Thus, we have the desired isomorphism. $\square$

Exercise 8.4.3. Suppose that V is a finite-dimensional module for $\mathfrak{sl}(2, \mathbb{C})$. Show, by using Weyl’s Theorem and the classification of irreducible representations in this chapter, that V is determined up to isomorphism by the eigenvalues of h . In particular, prove that if V is the direct sum of k irreducible modules, then

$$k = \dim W_0 + \dim W_1,$$

where $W_r = \{v \in V : h \cdot v = rv\}$.

Solution: By Weyl’s theorem, it follows that V is a direct sum of irreducible representations of $\mathfrak{sl}(2, \mathbb{C})$. Further, by Theorem 8.2.4, we know that V is a direct sum of V_{d_i} s:

$$V \cong \bigoplus_{i=1}^k V_{d_i}.$$

Now, consider $W_r = \{v \in V : h \cdot v = rv\}$. The way we do this is by determining W_0 and W_1 for each V_d by seeing when d is even or odd. Before classifying, note that

$$h \cdot x^i y^j = (i - j)x^i y^j.$$

Now suppose d is even. Then, $W_0 = \text{Span}\{x^{\frac{d}{2}} y^{\frac{d}{2}}\}$, and $W_1 = 0$, since we can’t find a pair of integers i, j such that $i - j = 1$ and $i + j$ is even. Next, suppose d is odd. Then, $W_0 = 0$ now and $W_1 = \text{Span}\{x^{\frac{d+1}{2}} y^{\frac{d-1}{2}}\}$. Thus, we note that depending on the parity of d , we only get one of W_0 or W_1 to have dimension 0 and dimension 1. Therefore, if we let

$$W_0 \cong \bigoplus_{i=1}^k W_{0_{d_i}}, \quad W_1 \cong \bigoplus_{i=1}^k W_{1_{d_i}},$$

then it is apparent that

$$k = \dim(W_0) + \dim(W_1).$$

$\square$

Exercise 8.4.4. Let V be an $\mathfrak{sl}(2, \mathbb{C})$ -module, not necessarily finite-dimensional. Suppose $w \in V$ is a highest-weight vector of weight λ ; that is, $e \cdot w = 0$ and $h \cdot w = \lambda w$ for some $\lambda \in \mathbb{C}$, and $w \neq 0$. Show that

1. for $k = 1, 2, \dots$ we have $e \cdot (f^k \cdot w) = k(\lambda - k + 1)f^{k-1} \cdot w$, and
2. $e^k f^k \cdot w = (k!)^2 \binom{\lambda}{k} w$.

Deduce that if $\binom{\lambda}{k} \neq 0$ then the set of all $f^j \cdot w$ for $0 \leq j \leq k$ is linearly independent. Hence show that if V is finite-dimensional, then λ must be a non-negative integer.

Solution: 1. We approach this by induction. For $k = 1$, we have

$$e \cdot (f \cdot w) = e \cdot (\lambda - 2)w = \lambda w = 1(\lambda - 1 + 1) \cdot w.$$

Assume this holds up to some n . Then, we have

$$\begin{aligned} e \cdot (f^{n+1} \cdot w) &= (f \cdot e + [e, f]) \cdot f^n \cdot w \\ &= f \cdot (n(\lambda - n + 1)f^{n-1} \cdot w) + h \cdot f^n \cdot w \\ &= (n(\lambda - n + 1)f^n \cdot w + (f \cdot h + [h, f]) \cdot f^{n-1} \cdot w) \\ &= (n(\lambda - n + 1)f^n \cdot w + f \cdot h \cdot f^{n-1} \cdot w - 2 \cdot f^n \cdot w) \\ &\vdots \\ &= (n(\lambda - n + 1)f^n \cdot w + f^n \cdot h \cdot w - 2nf^n \cdot w) \\ &= (n(\lambda - n + 1)f^n \cdot w + f^n \cdot \lambda \cdot w - 2nf^n \cdot w) \\ &= ((n + 1)\lambda - n^2 - n)f^n \cdot w \\ &= (n + 1)(\lambda - n)f^n \cdot w \end{aligned}$$

Thus, the induction is complete.

2. Induction again. For $k = 1$, we have

$$e \cdot f \cdot w = (f \cdot e + [e, f])w = 0 + \lambda w,$$

so true. Suppose this holds up to some n . Then,

$$\begin{aligned} e^{n+1} \cdot f^{n+1} \cdot w &= e^n \cdot e \cdot f^{n+1} \cdot w \\ &= e^n \cdot (n + 1)(\lambda - n)f^n \cdot w \\ &= (n + 1)(\lambda - n)(n!)^2 \binom{\lambda}{n} w \\ &= (n + 1)^2 \binom{\lambda}{n + 1} w. \end{aligned}$$

This completes the induction. □

Exercise 8.4.5. Let M be a finite-dimensional $\mathfrak{sl}(2, \mathbb{C})$ -module. Define a linear map $c: M \rightarrow M$ by

$$c(v) = \left(ef + fe + \frac{1}{2}h^2 \right) \cdot v \quad \text{for } v \in M.$$

1. Show that c is a homomorphism of $\mathfrak{sl}(2, \mathbb{C})$ -modules. *Hint:* For example, to show that c commutes with the action of e , show that $(efe + fe^2 + \frac{1}{2}h^2e) \cdot v$ and $(e^2f + efe + \frac{1}{2}eh^2) \cdot v$ are both equal to $(2efe + \frac{1}{2}eh^2) \cdot v$.
2. By Schur's Lemma, c must act as a scalar, say λ_d , on the irreducible module V_d . Show that $\lambda_d = \frac{1}{2}d(d + 2)$, and deduce that d is determined by λ_d .
3. Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of c acting on M . Let the primary decomposition of M be

$$M = \bigoplus_{i=1}^r \ker(c - \lambda_i 1_M)^{m_i}.$$

Show that the summands are $\mathfrak{sl}(2, \mathbb{C})$ -submodules.

So, to express the module as a direct sum of simple modules, we therefore may assume that M has just one generalized eigenspace, where c has, say, eigenvalue λ .

4. Let U be an irreducible submodule of M . Suppose that U is isomorphic to V_d . Show by considering the action of c on V_d that $\lambda = \frac{1}{2}d(d+2)$ and hence that any irreducible submodule of M is isomorphic to V_d .
5. Show more generally that if N is a submodule of M , then any irreducible submodule of M/N is isomorphic to V_d .

The linear map c is known as the **Casimir operator**. The following exercise gives an indication of how it was first discovered; it will appear again in the proof of Weyl's Theorem (see Appendix B).

Solution: 1. We need to check that $c(ev) = ec(v)$, and the same for f and h . We have that

$$\begin{aligned} c(ev) &= (efe + fe^2 + \frac{1}{2}h^2e)v = \left(efe + e(fe + h) + \frac{1}{2}(he - 2e)h\right)v \\ &= (2efe + \frac{1}{2}eh^2) \cdot v \end{aligned}$$

and

$$\begin{aligned} ec(v) &= (e^2f + efe + e\frac{1}{2}h^2)v = \left(efe + (ef - h)e + \frac{1}{2}h(eh + 2e)\right)v \\ &= (2efe + \frac{1}{2}eh^2) \cdot v. \end{aligned}$$

Thus, we achieve equality. Similar results hold for f and h .

2. Lemma 8.2.2 tells us that there is a highest weight vector w such that w is an eigenvector for H and $e \cdot w = 0$.

$$c(w) = \left(ef + fe + \frac{1}{2}h^2\right)w = dw + \frac{1}{2}d^2w = \left(d + \frac{1}{2}d^2\right)w,$$

so $\lambda = d + \frac{1}{2}d^2$.

3. We need to show that for every $v \in M_i = \ker(c - \lambda_i 1_M)^{m_i}$, we have that $ev, fv, hv \in M_i$. We just verify this for ev :

$$(c - \lambda_i 1_M)^{m_i}(ev) = e(c - \lambda_i 1_M)^{m_i}(v) = 0$$

Note that we can just bracket out the e because e and c commute. The same hold for f, h , so we're done.

4. Suppose U is irreducible and isomorphic to V_d . We've shown that $\lambda = \frac{1}{2}d(d+2)$ previously, so it suffices to show that any irreducible submodule of M is isomorphic to V_d . If W is another irreducible submodule of M that is isomorphic to V_k , then $\lambda = \frac{1}{2}k(k+2)$, but then

$$\frac{1}{2}k(k+2) = \lambda = \frac{1}{2}d(d+2),$$

so $k = d$, and so $W \xrightarrow{\sim} V_d$.

5. We note that the previous part implies the isomorphism

$$M \cong V_d \oplus \cdots \oplus V_d.$$

If N is a submodule of M , then N is also isomorphic to a direct sum of V_d s. Then, M/N is, again, isomorphic to a direct sum of V_d s. Therefore, any irreducible submodule of M/N will be isomorphic to V_d . $\square$

Exercise 8.4.6. Exercise 1.7.7 gives a way to embed the real Lie algebra $\mathbb{R}_\lambda^3$ into $\mathfrak{sl}(2, \mathbb{C})$. With the given solution, we would take

$$\psi(x) = \begin{pmatrix} 0 & 1/2 \\ -1/2 & 0 \end{pmatrix}, \quad \psi(y) = \begin{pmatrix} 0 & -i/2 \\ -i/2 & 0 \end{pmatrix}, \quad \psi(z) = \begin{pmatrix} -i/2 & 0 \\ 0 & i/2 \end{pmatrix}$$

Check that $\psi(x)^2 + \psi(y)^2 + \psi(z)^2 = -3/4 I$, where I is the 2×2 identity matrix. By expressing x, y, z in terms of e, f, h , recover the description of the Casimir operator given above.

Solution: Each of these matrices squared is $-\frac{1}{4}I$. Now, we have that

$$\begin{aligned} e &= \psi(x) + i\psi(y) \\ f &= i\psi(y) - \psi(x) \\ h &= 2i\psi(z). \end{aligned}$$

We then just plug this into the Casimir operator to get that

$$ef + fe + \frac{1}{2}h^2 = \frac{3}{2}I.$$

$\square$

9 Cartan's Criteria

§9.1 Jordan Decomposition

Working over the complex numbers allows us to consider the Jordan normal form of linear transformations. We use this to define for each linear transformation x of a complex vector space V a unique *Jordan decomposition*. We illustrate what the key aspects of Jordan decomposition in the following example.

Example 9.1.1 (Motivating Example)

Consider the matrix

$$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

which is in Jordan decomposition. Then, the top 2×2 block is the generalized eigenspace for 2 and the bottom 1×1 is that for 3. A nice way to rewrite this in the following way:

$$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

The matrix on the left in the RHS is diagonal and the one on the right in the RHS is nilpotent. An important aspect is that the two matrices on the right commute.

Definition 9.1.2. The **Jordan decomposition** of x is the unique expression of x as a sum $x = d + n$ where $d: V \rightarrow V$ is diagonalizable, $n: V \rightarrow V$ is nilpotent, and d and n commute.

Remark 9.1.3 — Very often, a diagonalizable linear map of a complex vector space is also said to be **semisimple**.

Example 9.1.4

Consider the example above again. The same will be true for general Jordan blocks. Then, note that

$$A - 3 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (A - 2) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (A - 2)^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Then, we get

$$(A - 2) - (A - 2)^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = n.$$

Then, if $d = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = A - n$, so we get

$$d - 3(A - 2)^2 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Remark 9.1.5 — The takeaway from this example is that from the original matrix, we can get each original Jordan block. For instance, in the case of 1-dimensional Jordan blocks

$$\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix},$$

then $(A - \lambda_1)(A - \lambda_2) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & (\lambda_3 - \lambda_1)(\lambda_3 - \lambda_2) \end{pmatrix}$. The conclusion is that if A has eigenvalues $\lambda_1, \dots, \lambda_r$ (distinct but having multiplicities $m_1, \dots, m_r$), then not only is $\text{diag}_{m_1}(\lambda_1) \oplus \text{diag}_{m_2}(\lambda_2) \oplus \dots \oplus \text{diag}_{m_r}(\lambda_r)$ a polynomial in A , but so is

$$\text{diag}_{m_1}(\mu_1) \oplus \text{diag}_{m_2}(\mu_2) \oplus \dots \oplus \text{diag}_{m_r}(\mu_r),$$

for any complex numbers $\mu_1, \dots, \mu_r$. Here, $\text{diag}_n(\lambda)$ is a diagonal $n \times n$ matrix with entries λ . In particular, if $x = d + n$, is the Jordan decomposition of x , then $d, n, \bar{d}$ ($\bar{d}$ is complex conjugation) are all polynomials in x . This is summarized in the lemma below.

Lemma 9.1.6

Let x be a linear transformation of the complex vector space V . Then, we can write x in the Jordan decomposition $x = d + n$, where d is diagonalizable (semisimple), n is nilpotent, and d and n commute.

- (i) There is a polynomial $p(X) \in \mathbb{C}[X]$ such that $p(x) = d$.
- (ii) Fix a basis of V in which d is diagonal. Let $\bar{d}$ be the linear map whose matrix with respect to this basis is the complex conjugate of the matrix of d . There is a polynomial $q(X) \in \mathbb{C}[X]$ such that $q(x) = \bar{d}$.
- (iii) There is a polynomial $r(X) \in \mathbb{C}[X]$ such that $r(x) = n$.

Proof. Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of x . The minimal polynomial of x is then

$$m(X) = (X - \lambda_1)^{a_1} \cdots (X - \lambda_r)^{a_r},$$

where a_i is the size of the largest Jordan block with eigenvalue λ_i .

To finish, we require the lemma below (Lemma 9.1.7) to get the polynomials we seek. For (i) take $\mu_i = \lambda_i$, and for (ii) take $\mu_i = \overline{\lambda_i}$. For (iii), we have $x = d + n$, so $r(x) = x - p(x)$ works. $\square$

Lemma 9.1.7

Suppose that x has minimal polynomial

$$f(X) = (X - \lambda_1)^{a_1} \cdots (X - \lambda_r)^{a_r},$$

where the λ_i are pairwise distinct. Let the corresponding primary decomposition of V as a direct sum of generalised eigenspaces be

$$V = V_1 \oplus \cdots \oplus V_r,$$

where $V_i = \ker(x - \lambda_i 1_V)^{a_i}$. Then, given any $\mu_1, \dots, \mu_r \in \mathbb{C}$, there is a polynomial $p(X)$ such that

$$p(x) = \mu_1 1_{V_1} + \mu_2 1_{V_2} + \cdots + \mu_r 1_{V_r}.$$

Proof. Suppose we could find a polynomial $f(X) \in \mathbb{C}[X]$ such that

$$f(X) \equiv \mu_i \pmod{(X - \lambda_i)^{a_i}}.$$

Take $v \in V_i = \ker(x - \lambda_i 1_V)^{a_i}$. By hypothesis, $f(X) = \mu_i + a(X)(X - \lambda_i)^{a_i}$ for some polynomial $a(X)$. Hence

$$f(x)v = \mu_i 1_V v + a(x)(x - \lambda_i)^{a_i} v = \mu_i v,$$

as required.

To obtain the polynomial $f(X)$ in the first place, we apply the Chinese remainder theorem. The polynomials $(X - \lambda_1)^{a_1}, \dots, (X - \lambda_r)^{a_r}$ are coprime, so the Chinese

Remainder Theorem states that the map

$$\begin{aligned} \mathbb{C}[X] &\longrightarrow \bigoplus_{i=1}^r \frac{\mathbb{C}[X]}{(X - \lambda_i)^{a_i}} \\ f(X) &\longmapsto (f(X) \bmod (X - \lambda_1)^{a_1}, \dots, f(X) \bmod (X - \lambda_r)^{a_r}) \end{aligned}$$

is surjective, providing a suitable $p(X)$. □

Example 9.1.8

Let

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \lambda_1 \neq \lambda_2.$$

Then, $d = x$ and $n = 0$. Then, $\bar{d} = \frac{\bar{\lambda}_1}{\lambda_1 - \lambda_2}(x - \lambda_1) + \frac{\bar{\lambda}_2}{\lambda_1 - \lambda_2}(x - \lambda_2)$. If $\lambda_1 = \lambda_2$, then $d = \lambda_1$ so $\bar{d} = \bar{\lambda}_1$.

Now consider

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_1 \end{pmatrix}.$$

This part is an exercise.

Using Jordan decomposition, we can give a concise reinterpretation of two earlier results (see Exercise 1.7.9 and Lemma 5.1.1).

Exercise 9.1.9. Let V be a vector space, and suppose that $x \in \mathfrak{gl}(V)$ has Jordan decomposition $d + n$. Show that $\text{ad } x: \mathfrak{gl}(V) \rightarrow \mathfrak{gl}(V)$ has Jordan decomposition $\text{ad } d + \text{ad } n$.

Solution: By the uniqueness of Jordan decomposition, we need to show that $\text{ad } d$ is diagonalizable, and $\text{ad } n$ is nilpotent, and that $\text{ad } d$ and $\text{ad } n$ commute. We have that

$$[\text{ad } d, \text{ad } n] = \text{ad}([d, n]) = 0,$$

so they commute. Also, $\text{ad } n$ is nilpotent by a previous lemma. Now, we need to prove that $\text{ad } d$ is diagonalizable. We may as well assume that $d = \text{diag}(d_1, \dots, d_n)$, and we just see how this acts on the bases $E_{ij} = (\delta_{ij})$; this is exercise 1.7.9. □

Example 9.1.10

Let us return the example before lemma 9.1.6 and try to find the key polynomials. Consider

$$x = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

The characteristic polynomial is $(X - 2)^2(X - 3)$, and we are searching, first, for a polynomial $p(X) \equiv a \pmod{(X - 3)}$ and $p(X) \equiv b \pmod{(X - 2)^2}$. Note that such $p(X)$ exists by the Chinese remainder theorem. Now,

$$(x - 2)^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (x - 3) = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Now, we can write $(X - 2)^2 = (X - 3)(X - 1) + 1$. Now,

$$(1 - x)(x - 3) = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Now, we get that

$$a(1 - x)(x - 3) + b(x - 2)^2 = \begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{pmatrix},$$

so in particular setting $a = 2$ and $b = 3$, we get

$$p(x) = 2(1 - x)(x - 3) + 3(x - 2)^2 = d.$$

for the diagonal part of x . Now, if we search for the nilpotent part, we can just let $n = q(x) = x - p(x)$.

Remark 9.1.11 (Uniqueness of Jordan Decomposition) — Note that the Jordan decomposition $x = d + n$ is unique. Suppose we can also write that $x = d' + n'$ with $[d', n'] = 0$. Then, the claim is that $d = d'$ and $n = n'$.

Proof. If $[d', n'] = 0$, then $[d', x] = [d', d'] + [d', n'] = 0$. However, we saw that $d = p(x)$ for some polynomial in $\mathbb{C}[X]$, so $[d', d] = [d', p(x)] = 0$. Thus, d, d' are commuting and diagonalizable, so they are simultaneously diagonalizable. Thus, $d - d'$ is also diagonalizable. Similarly, $[n, n'] = 0$. From this, we can see that $n' - n$ is nilpotent (just raise this for some large enough power). Now, $d - d' = n' - n$, so the LHS and RHS are both diagonalizable and nilpotent, so it's zero. $\square$

§9.2 Testing for Solvability

Let V be a complex vector space and let L be a Lie subalgebra of $\mathfrak{gl}(V)$. What we work towards is Cartan's criteria, which exploits the trace map to test for solvability, but why would it be reasonable to expect solvability to be visible from the traces of the elements

of L ? The following exercise (which repeats part of Exercise 6.2.5 which you should recall) gives one indication.

Exercise 9.2.1. Suppose that $L \subseteq \mathfrak{gl}(V)$ is solvable. Use Lie’s Theorem to show that there is a basis of V in which every element of L' is represented by a strictly upper triangular matrix. Conclude that $\mathrm{tr}(xy) = 0$ for all $x \in L$ and $y \in L'$.

Solution: L is solvable, so there is a basis for which every element of L is represented by an upper triangular matrix. Now, L' is just $[L, L]$, so we get that L' is spanned by the strictly upper triangular matrices (by Exercise 4.2.11). If $x \in L$ and $y \in L'$, then we are multiplying an upper triangular matrix with one that is strictly upper triangular. Thus, it follows that the trace is zero, since every diagonal entry is zero. $\square$

Thus we have a necessary condition, in terms of traces, for L to be solvable. Remarkably, this condition is also sufficient. Before proving this, we give a small example.

Example 9.2.2

Let L be the 2-dimensional non-abelian Lie algebra with basis x, y such that $[x, y] = x$. In this basis we have

$$\mathrm{ad} x = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathrm{ad} y = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}.$$

As expected, $\mathrm{tr}(\mathrm{ad} x) = 0$.

Proposition 9.2.3

Let V be a complex vector space and let L be a Lie sub-Lie algebra of $\mathfrak{gl}(V)$. If $\mathrm{tr}(xy) = 0$ for all $x, y \in L$, then L is solvable.

Proof. We shall show that every $x \in L'$ is a nilpotent linear map. It will then follow from Engel’s Theorem (Theorem 6.1.5) that L' is nilpotent, and so, by Exercise 6.2.5, L is solvable.

Let $x \in L'$ have Jordan decomposition $x = d + n$, where d is diagonalizable, n is nilpotent, and d and n commute. We may fix a basis of V in which d is diagonal and n is strictly upper triangular. Suppose that d has diagonal entries $\lambda_1, \dots, \lambda_m$ (note that these are really the eigenvalues of x). Since our aim is to show that $d = 0$, it will suffice to show that

$$\sum_{i=1}^m \lambda_i \bar{\lambda}_i = 0.$$

The matrix of $\bar{d}$ is diagonal, with diagonal entries $\bar{\lambda}_i$ for $1 \leq i \leq m$. A simple computation shows that

$$\mathrm{tr} d \bar{x} = \sum_{i=1}^m \lambda_i \bar{\lambda}_i,$$

so it suffices to show $\mathrm{tr} d \bar{x} = 0$. Now, as $x \in L'$, we may express x as a linear combination of commutators $[y, z]$ with $y, z \in L$, so it suffices show that $\mathrm{tr}(\bar{d}[y, z]) = 0$, for all $y, z \in L$. Then, linearity of the trace map finishes it off. By the identity

$$\mathrm{tr}([a, b]c) = \mathrm{tr}(a[b, c]),$$

which follows from the fact that $\text{tr}(b(ac)) = \text{tr}((ac)b)$. Therefore, it suffices to show that

$$\text{tr}([\bar{d}, y]z) = 0.$$

This will hold by our hypothesis that $\text{tr}(ab) = 0$ for all $a, b \in L$, provided we can show that $[\bar{d}, y] \in L$. In other words, it suffices to show that $\text{ad } \bar{d}$ maps L into L . By Exercise 9.1.9, the Jordan decomposition of $\text{ad } x$ is $\text{ad } d + \text{ad } n$. Therefore, by part (ii) of Lemma 9.1.6, there is a polynomial $p(X) \in \mathbb{C}[X]$ such that $p(\text{ad } x) = \overline{\text{ad } d} = \text{ad } \bar{d}$. Now $\text{ad } x$ maps L into itself, so $p(\text{ad } x)$ does also. $\square$

To apply this proposition to an abstract Lie algebra L , we need a way to regard L as a subalgebra of some $\mathfrak{gl}(V)$. The adjoint representation of L is well-suited to this purpose, as L is solvable if and only if $\text{ad } L$ is solvable.

Theorem 9.2.4

Let L be a complex Lie algebra. Then L is solvable if and only if $\text{tr}(\text{ad } x \circ \text{ad } y) = 0$ for all $x \in L$ and all $y \in L'$.

Proof. Suppose that L is solvable. Then, $\text{ad } L \subseteq \mathfrak{gl}(L)$ is solvable, so we can apply Exercise 9.2.1 to $\text{ad } L$ and $(\text{ad } L)' = \text{ad } L'$. If $x \in L$ and $y \in L'$, then $\text{ad } x \in \text{ad } L$ and $\text{ad } y \in \text{ad } L'$, so

$$\text{tr}(\text{ad } x \circ \text{ad } y) = 0.$$

Conversely, if $\text{tr}(\text{ad } x \circ \text{ad } y) = 0$ for all $x \in L$ and all $y \in L'$, then Proposition 9.2.3 implies that $\text{ad } L' = (\text{ad } L)'$ is solvable. It follows that $\text{ad } L$ is solvable, so it follows that L is solvable (since $L/Z(L) \cong \text{ad } L$). $\square$

§9.3 The Killing Form

Definition 9.3.1. Let L be a complex Lie algebra. The **Killing form** on L is the symmetric bilinear form defined by

$$\kappa(x, y) := \text{tr}((\text{ad } x) \circ (\text{ad } y)) \quad \text{for } x, y \in L.$$

Remark 9.3.2 — The Killing form is bilinear because ad is linear, the composition of maps is bilinear, and tr is linear.

Remark 9.3.3 — It is symmetric because $\text{tr}(ab) = \text{tr}(ba)$ for linear maps a and b .

Remark 9.3.4 — Another very important property of the Killing form is its associativity, which states that for all $x, y, z \in L$ we have

$$\kappa([x, y], z) = \kappa(x, [y, z]).$$

This follows from the identity for trace mentioned in the proof of Proposition 9.2.3. An equivalent way to remember this would be

$$\kappa([y, x], z) + \kappa(x, [y, z]) = 0.$$

That is,

$$\kappa(\operatorname{ad} y(x), z) + \kappa(x, \operatorname{ad}(y)(z)) = 0.$$

Using the Killing form, we can state Theorem 9.2.4 as follows.

Theorem 9.3.5 (Cartan's First Criterion)

The complex Lie algebra L is solvable if and only if $\kappa(x, y) = 0$ for all $x \in L$ and $y \in L'$.

Example 9.3.6

Let L be the 2-dimensional non-abelian Lie algebra with basis x, y such that $[x, y] = x$. The matrices computed in Example 9.2.2 show that $\kappa(x, x) = \kappa(x, y) = \kappa(y, x) = 0$ and $\kappa(y, y) = 1$. The matrix of κ in the basis x, y is therefore

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Remark 9.3.7 — The Killing form is compatible with restriction to ideals. Suppose that L is a Lie algebra and I is an ideal of L . We write κ for the Killing form on L and κ_I for the Killing form on I , considered as a Lie algebra in its own right.

Lemma 9.3.8

Let I be an ideal of L . If $x, y \in I$, then $\kappa_I(x, y) = \kappa(x, y)$; that is, it is the restriction to I of the Killing form for L .

Proof. Take a basis for I and extend it to a basis of L . If $x \in I$, then $\operatorname{ad} x$ maps L into I , so the matrix of $\operatorname{ad} x$ in this basis is of the form

$$\begin{pmatrix} A_x & B_x \\ 0 & 0 \end{pmatrix},$$

where A_x is the matrix of $\operatorname{ad} x$ restricted to I . If $y \in I$, then $\operatorname{ad} x \circ \operatorname{ad} y$ has matrix

$$\begin{pmatrix} A_x A_y & A_x B_y \\ 0 & 0 \end{pmatrix},$$

where $A_x \circ A_y$ is the matrix of $\operatorname{ad} x \circ \operatorname{ad} y$ restricted to I . Only the block $A_x A_y$ contributes to the trace of this matrix, so

$$\kappa(x, y) = \operatorname{tr}(A_x A_y) = \kappa_I(x, y).$$

□

§9.4 Testing for Semisimplicity

Recall that a Lie algebra is said to be semisimple if its radical is zero; that is, if it has no non-zero solvable ideals. Since we can detect solvability by using the Killing form, it is perhaps not too surprising that we can also use the Killing form to decide whether or not a Lie algebra is semisimple.

§9.4.i Recap on Bilinear Forms

We begin by recalling a small part of the general theory of bilinear forms. Let β be a symmetric bilinear form on a finite-dimensional complex vector space V . Recall that if S is a subset of V , we define the **perpendicular space** to S by

$$S^\perp = \{x \in V \mid \beta(x, s) = 0 \text{ for all } s \in S\}.$$

This is a vector subspace of V . We say that β is **non-degenerate** if $V^\perp = 0$; that is, there is no non-zero vector $v \in V$ such that $\beta(v, x) = 0$ for all $x \in V$. Whether or not β is non-degenerate, we have a map

$$\begin{aligned} V &\longrightarrow V^* \\ x &\longmapsto (y \mapsto \beta(x, y)). \end{aligned}$$

Restricting this to a subspace $W \subset V$, we obtain a map

$$\begin{aligned} V &\longrightarrow W^* \\ x &\longmapsto (w \mapsto \beta(x, w)). \end{aligned}$$

The kernel of this map is $W^\perp$, so we can use the first isomorphism theorem to say that we get an injection $V/W^\perp \hookrightarrow W^*$. In the setting of finite dimensional vector spaces, we get that

$$\dim W^* = \dim W \geq \dim(V/W^\perp) = \dim V - \dim W^\perp.$$

In other words, $\dim W + \dim W^\perp \geq \dim V$.

If β is non-degenerate, then we achieve equality here: $V/W^\perp \hookrightarrow V^*$ is an injection, and $V^\perp = 0$, so $V \hookrightarrow V^*$ is an injection. Therefore, $V \rightarrow V^*$ is an isomorphism, since $\dim V = \dim V^*$. Now, the map $V^* \rightarrow W^*$ is always surjective, since $W \subset V$. Then, we may view the map $V \rightarrow W^*$ as

$$V \xrightarrow{\sim} V^* \twoheadrightarrow W^*,$$

which has kernel $W^\perp$. Thus, we get that

$$\dim W + \dim W^\perp = \dim V.$$

Remark 9.4.1 — Note that even if β is non-degenerate it is possible that $W \cap W^\perp \neq 0$ (so we don't get $V = W \oplus W^\perp$). For example, if κ is the Killing form of $\mathfrak{sl}(2, \mathbb{C})$, then $\kappa(e, e) = 0$. You are asked to compute the Killing form of $\mathfrak{sl}(2, \mathbb{C})$ in Exercise 9.4.2 below.

Exercise 9.4.2.

§9.4.ii Back to Lie Algebras and the Killing form

Now we specialize to the case where L is a Lie algebra and κ is its Killing form, so perpendicular spaces are taken with respect to κ . We begin with a simple observation which requires the associativity of κ .

Exercise 9.4.3. Suppose that I is an ideal of a Lie algebra L . Show that $I^\perp$ (with respect to κ) is an ideal of L .

Solution: $I^\perp$ is certainly a subspace, so we just need to show that this is preserved by the Lie bracket. That is, $[y, z] \in I^\perp$, where $y \in L$ and $z \in I^\perp$. Thus, we need to verify that $\kappa(x, [y, z]) = 0$, for any $x \in I$. However, we have this associativity condition on the Killing form where

$$\kappa(x, [y, z]) = \kappa([x, y], z),$$

and $[x, y] \in I$, so this is zero, since $z \in I^\perp$. $\square$

Again the converse also holds.

Theorem 9.4.4 (Cartan’s Second Criterion)

The complex Lie algebra L is semisimple if and only if the Killing form κ of L is non-degenerate.

Proof. First, the only if direction, so assume that L is semisimple. By definition, we want to show that $L^\perp = 0$. By the exercise above, $L^\perp$ is an ideal of L , so it suffices to show that $L^\perp$ is solvable. By Cartan’s first criterion, it suffices to show that if $x \in L^\perp$ and $y \in (L^\perp)'$, then, as in particular $y \in L$, we have $\kappa(x, y) = 0$. However, $\kappa(x, y) = 0$ whenever $x \in L^\perp$, $y \in L$, by definition of $L^\perp$. Hence it follows from Cartan’s First Criterion that $L^\perp$ is a solvable ideal of L . Therefore, if L is semisimple, then $L^\perp = 0$ and κ is non-degenerate.

Now for the if direction. Suppose that L is not semisimple, and so $\text{rad } L$ is non-zero. By Exercise 4.4.6, L has a non-zero abelian ideal, say I . Take a nonzero element $a \in I$, and let $x \in L$. The composite map

$$\text{ad } a \circ \text{ad } x \circ \text{ad } a$$

sends L to zero, since the image of $\text{ad } x \circ \text{ad } a$ is contained in I and I is abelian. Therefore $(\text{ad } x \circ \text{ad } a)^2 = 0$, so $\text{ad } x \circ \text{ad } a \in \mathfrak{gl}(L)$ is nilpotent; thus, has trace 0, so $\kappa(x, a) = 0$. Therefore, $I \subseteq L^\perp$, showing that κ is not non-degenerate, as $a \neq 0$ and $\kappa(x, a) = 0$ for all $x \in L$. This gives a contradiction; thus L is semisimple. $\square$

Remark 9.4.5 — It is possible that $L^\perp$ is properly contained in $\text{rad } L$. For example, Exercise 9.2.1 shows that this is the case if L is the 2-dimensional non-abelian Lie algebra.

Cartan’s Second Criterion is an extremely powerful characterization of semisimplicity. In our first application, we shall show that a semisimple Lie algebra is a direct sum of simple Lie algebras; this finally justifies the name semisimple which we have been using. The following lemma contains the main idea needed.

Lemma 9.4.6

If I is an ideal in a complex semisimple Lie algebra L , then $L = I \oplus I^\perp$. Further, if $I \neq 0$, then I is also a semisimple Lie algebra in its own right.

Proof. As usual, let κ denote the Killing form on L . Note that the Killing form of I is the restriction of the Killing form of L . If $x, y \in I \cap I^\perp$, then $\kappa(x, y) = 0$. Therefore, the restriction of κ to $I \cap I^\perp$ is identically 0. Since $I \cap I^\perp$ is an ideal of L , Cartan's First Criterion, then $I \cap I^\perp$ is solvable. In particular, since L is semisimple, then $I \cap I^\perp = 0$. Since $\dim L = \dim I + \dim I^\perp$, indeed, $L = I \oplus I^\perp$ as vector spaces. Then it follows that $L = I \oplus I^\perp$ as Lie algebras (direct sum): if $x \in I$, $y \in I^\perp$, then $[x, y] \in I \cap I^\perp$. $I \cap I^\perp = 0$, so indeed this is true.

It remains to check that the restriction of κ to I is non-degenerate. Take $x \neq 0$ in I . Then, there exists $y \in L$ such that $\kappa(x, y) \neq 0$. Write $y = y_1 + y_2$, $y_1 \in I$ and $y_2 \in I^\perp$. Then,

$$\kappa(x, y) = \kappa(x, y_1) + \kappa(x, y_2) = \kappa(x, y_1).$$

Since $\kappa(x, y_1) \neq 0$, we've shown that $\kappa|_I$ is non-degenerate; finish with Cartan's second criterion. $\square$

Corollary 9.4.7

If L is semisimple, I is an ideal, J is an ideal of I , then J is an ideal of L .

Proof. We have that $L = I \oplus I^\perp$. To check that J is an ideal of L , it suffices to show that if $x \in J$, $y \in I$ and $z \in I^\perp$, then $[x, y], [x, z] \in J$. The former is in I , by definition of J , and the second is always zero because $x \in I$, so $[x, z] = 0$. $\square$

Remark 9.4.8 — This is definitely not true if L is not semisimple.

We can now prove the following theorem.

Theorem 9.4.9

Let L be a complex Lie algebra. Then L is semisimple if and only if there are simple ideals $L_1, \dots, L_r$ of L such that $L = L_1 \oplus L_2 \oplus \dots \oplus L_r$.

Proof. We begin with the only if direction, working by induction on $\dim L$. Let I be an ideal in L of the smallest possible non-zero dimension. If $I = L$, we are done. Otherwise I is a proper simple ideal of L . (It cannot be abelian as by hypothesis L has no non-zero abelian ideals.) By the preceding lemma, $L = I \oplus I^\perp$. Also note that if $J \subset I$ is any ideal, we saw before that J is actually an ideal of L . However, by our choice of I , every ideal J of I must either be 0 or I , so I is simple. Now note that $I^\perp$ has dimension less than L , so by induction, $I^\perp$ is a direct sum of simple ideals,

$$I^\perp = L_2 \oplus \dots \oplus L_r.$$

Each L_i is also an ideal of L , as $[I, L_i] \subseteq I \cap I^\perp = 0$, so putting $L_1 = I$ we get the required decomposition.

Conversely, suppose that $L = L_1 \oplus \dots \oplus L_r$, where the L_r are simple ideals. We want to show that $\text{rad } L = 0$. Let I be a solvable ideal. Then, $[L, I] = [L_1, I] + \dots + [L_r, I]$, since we can just write elements in L as a sum of elements in $L_1, \dots, L_r$. For each $i = 1, \dots, r$,

we have $[L_i, I] \subset L_i \cap I$. In particular, it is a solvable ideal of the simple Lie algebra L_i . Then, each $[L_i, I]$ must be zero because L_i is simple. Thus,

$$[L, I] \subseteq [L_1, I] \oplus \cdots \oplus [L_r, I] = 0.$$

This shows that I is contained in $Z(L)$. But by Exercise 2.4.3,

$$Z(L) = Z(L_1) \oplus \cdots \oplus Z(L_r).$$

We know that $Z(L_1) = \cdots = Z(L_r) = 0$ as the L_i are simple ideals, so $Z(L) = 0$ and $I = 0$. $\square$

Remark 9.4.10 — Exercise 9.7.2 tells us more. In fact, these are all the ideals of L , since if we have an ideal, we can just take the intersection with L .

Corollary 9.4.11

If L is semisimple, then $L = [L, L]$.

Proof. Write $L = L_1 \oplus \cdots \oplus L_r$, with each L_i simple. Then, taking the bracket gives

$$L' = L'_1 \oplus \cdots \oplus L'_r.$$

Thus, we only need to show that if L is simple, then $L = L'$. However, L' is an ideal so it’s enough to show that this is nonzero. However, this is true because L is not abelian, which is part of the definition for L being simple. $\square$

Using very similar ideas, we can prove the following.

Lemma 9.4.12

If L is a semisimple Lie algebra and I is an ideal of L , then L/I is semisimple.

Proof. We have seen that $L = I \oplus I^\perp$, so $L/I \xrightarrow{\sim} I^\perp$, which we have seen is a semisimple Lie algebra in its own right (since $I^\perp$ is an ideal of L). $\square$

§9.5 Derivations of Semisimple Lie Algebras

Recall that the derivations of L $\text{Der} L$ is a sub-Lie-algebra of $\mathfrak{gl}(L)$, with an ideal called the inner derivations of L , $\text{IDer}(L)$, given by $\text{ad } L \subset \mathfrak{gl}(L)$. The next application of Cartan’s Second Criterion will show that the only derivations of a complex semisimple Lie algebra are those of the form $\text{ad } x$ for $x \in L$. More precisely, we have the following.

Proposition 9.5.1

If L is a finite-dimensional complex semisimple Lie algebra, then $\text{ad } L = \text{Der} L$.

Proof. We showed that for each $x \in L$ the linear map $\text{ad } x$ is a derivation of L , so ad is a Lie algebra homomorphism from L to $\text{Der } L$. Moreover, if δ is a derivation of L and $x, y \in L$, then

$$\begin{aligned} [\delta, \text{ad } x]y &= \delta[x, y] - \text{ad } x(\delta y) \\ &= [\delta x, y] + [x, \delta y] - [x, \delta y] \\ &= \text{ad } (\delta x)y. \end{aligned}$$

Thus the image of $\text{ad}: L \rightarrow \text{Der } L$ is an ideal of $\text{Der } L$. This is true for any Lie algebra.

Now we bring in our assumption that L is complex and semisimple. First, note that $\text{ad}: L \rightarrow \text{Der } L$ is one-to-one, as $\ker \text{ad} = Z(L) = 0$, so setting $M := \text{ad } L$, we get that $\text{ad}: L \rightarrow M$ is an isomorphism; particularly, M is semisimple as well.

To show that $M = \text{Der } L$, we exploit the Killing form on the Lie algebra $\text{Der } L$. Let $M^\perp$ be the orthogonal of M in $\text{Der}(L)$ under the Killing form of $\text{Der}(L)$. Assume for the sake of contradiction, that $M \neq \text{Der}(L)$. We have that $\dim M + \dim M^\perp \geq \dim \text{Der}(L)$, so

$$\dim M^\perp \geq \dim \text{Der}(L) - \dim(M) > 0.$$

That is, $\dim M^\perp \neq 0$. Take $D \in M^\perp$. We need to show that in fact $D = 0$, which will contradict the assumption that $M^\perp \neq 0$. Note that κ_M is non-degenerate by Cartan's second criterion, so $M^\perp \cap M = 0$, implying $[M, M^\perp] = 0$. Thus, for all $x \in L$, we have

$$[\text{ad } x, D] = 0,$$

since $\text{ad } x \in M$ and $D \in M^\perp$. However, the LHS is equal to $-\text{ad } D(x)$. However, $D(x)$ is in the kernel of ad , which is injective, so $D(x) = 0$. Since this holds for all $x \in L$, $D = 0$, a contradiction. This contradicts the fact that $\text{ad}(L) = M \subsetneq \text{Der}(L)$. Thus, we get equality as required. $\square$

In Exercise 9.5.2, this proposition is used to give an alternative proof that a semisimple Lie algebra is a direct sum of simple Lie algebras. Another important application occurs in the following section.

Exercise 9.5.2. Use Exercise 2.4.9 and Proposition 9.5.1 to give an alternative proof of Theorem 9.4.9 (that a complex semisimple Lie algebra is a direct sum of simple ideals).

Solution: We proceed by induction on the dimension of L . If $\dim L = 1$, there is nothing to show. Suppose the result is true up to some $\dim L - 1$. Let L be a semisimple Lie algebra and I be an ideal of smallest dimension in L . Then, by construction I is simple, so $Z(I) = 0$. Further, since I is a simple ideal, I is semisimple in its own right. Therefore, any derivation of I is actually of the form $\text{ad } x$, for some $x \in I$ by the previous proposition. Then, Exercise 2.4.9 says that

$$L = I \oplus C_L(I),$$

where $C_L(I) = \{x \in L: [x, y] = 0, \forall y \in I\}$ is the centralizer of I in L . Then, since $C_L(I)$ is necessarily of dimensions less than L , we can apply the inductive hypothesis to say that $C_L(I)$ decomposes into a direct sum of simple ideals $L_2, \dots, L_r$. Setting $L_1 := I$, we therefore have that

$$L = L_1 \oplus L_2 \oplus \dots \oplus L_r.$$

$\square$

§9.6 Abstract Jordan Decomposition

Let $x \in \mathfrak{gl}(V)$, and consider its Jordan decomposition, specified by $x = d + n$, with $[d, n] = 0$. Given a representation $\varphi: L \rightarrow \mathfrak{gl}(V)$ of a Lie algebra L , we may consider the Jordan decomposition of the linear maps $\varphi(x)$ for $x \in L$.

For a general Lie algebra there is not much that can be said about this decomposition without knowing more about the representation φ . Usually there is no relationship between the Jordan decomposition of x and that of $\varphi(x)$.

Example 9.6.1

If L is the 1-dimensional abelian Lie algebra, spanned, say by x , then we may define a representation of L on a vector space V by mapping x to any element of $\mathfrak{gl}(V)$. If θ is our representation, then the Jordan decomposition of $\theta(x)$ could be anything then.

However, representations of a complex semisimple Lie algebra are much better behaved. To demonstrate this, we use derivations to define a Jordan decomposition for elements of an arbitrary complex semisimple Lie algebra. We need the following proposition.

Definition 9.6.2. Given a derivation δ of L , for any $\lambda \in \mathbb{C}$, we write

$$L_\lambda := \{x \in L : (\delta - \lambda I_L)^n x = 0, \text{ for some } n \geq 1\}.$$

Proposition 9.6.3

Let L be a complex Lie algebra. Suppose that δ is a derivation of L with Jordan decomposition $\delta = \sigma + \nu$, where σ is diagonalizable and ν is nilpotent. Then σ and ν are also derivations of L .

Proof. For $\lambda \in \mathbb{C}$, let

$$L_\lambda = \{x \in L : (\delta - \lambda I_L)^m x = 0 \text{ for some } m \geq 1\}$$

be the generalized eigenspace of δ corresponding to λ . Note that if λ is not an eigenvalue of δ , then $L_\lambda = 0$. By the Primary Decomposition Theorem, L decomposes as a direct sum of generalized eigenspaces, $L = \bigoplus_\lambda L_\lambda$, where the sum runs over the eigenvalues of δ . Note that $\sigma: L_\lambda \rightarrow L_\lambda$ is equal to multiplication by λ . Let $x \in L_\lambda$ and $y \in L_\mu$. Since we want to show that σ is a derivation, we want to show that

$$\sigma([x, y]) = [\sigma(x), y] + [x, \sigma(y)].$$

Then,

$$[\sigma(x), y] + [x, \sigma(y)] = [\lambda x, y] + [x, \mu y] = (\lambda + \mu)[x, y].$$

Therefore, we need to prove that $[x, y] \in L_{\lambda+\mu}$. That is, we need $[L_\lambda, L_\mu] \subseteq L_{\lambda+\mu}$, which is given by Exercise 9.6.4. $\square$

Exercise 9.6.4. Let δ be a derivation of a Lie algebra L . Show that if $\lambda, \mu \in \mathbb{C}$ and $x, y \in L$, then

$$(\delta - (\lambda + \mu)1_L)^n [x, y] = \sum_{k=0}^n \binom{n}{k} [(\delta - \lambda 1_L)^k x, (\delta - \mu 1_L)^{n-k} y].$$

Hence show that if the primary decomposition of L with respect to δ is $L = \bigoplus_{\lambda} L_{\lambda}$ (as in the proof of Proposition 9.6.3), then

$$[L_{\lambda}, L_{\mu}] \subseteq L_{\lambda+\mu}.$$

Solution: We prove this formula by induction.

Given this, the containment given below is clear, by taking k and $n - k$ sufficiently large. $\square$

Theorem 9.6.5

Let L be a complex semisimple Lie algebra. Each $x \in L$ can be written uniquely as $x = d + n$, where $d, n \in L$ are such that $\text{ad } d$ is diagonalizable, and $\text{ad } n$ is nilpotent, and $[d, n] = 0$. Furthermore, if $y \in L$ commutes with x , then $[d, y] = 0$ and $[n, y] = 0$.

Proof. Let $\text{ad } x = \sigma + \nu$ where $\sigma \in \mathfrak{gl}(L)$ is diagonalizable, $\nu \in \mathfrak{gl}(L)$ is nilpotent, and $[\sigma, \nu] = 0$. By Proposition 9.6.3, we know that σ and ν are derivations of the Lie algebra L . In Proposition 9.5.1, we saw that $\text{ad } L = \text{Der } L$, so there exist $d, n \in L$ such that $\text{ad } d = \sigma$ and $\text{ad } n = \nu$. As ad is injective and

$$\text{ad } x = \sigma + \nu = \text{ad } d + \text{ad } n = \text{ad } d + n,$$

we get that $x = d + n$. Moreover, $\text{ad } [d, n] = [\text{ad } d, \text{ad } n] = 0$ so $[d, n] = 0$, since the center is trivial. The uniqueness of d and n follows from the uniqueness of the Jordan decomposition of $\text{ad } x$.

Suppose that $y \in L$ and that $\text{ad } x(y) = 0$. By Lemma 9.1.6, σ and ν may be expressed as polynomials in $\text{ad } x$. Let

$$\nu = c_0 I_L + c_1 \text{ad } x + \cdots + c_r (\text{ad } x)^r, \quad c_i \in \mathbb{C}.$$

Applying ν to y , we see that $\nu(y) = c_0 y$. But ν is nilpotent and $\nu(x) = c_0 x$, so $c_0 = 0$. Thus $\nu(y) = 0$ and so $\sigma(y) = (\text{ad } x - \nu)(y) = 0$ also. $\square$

Definition 9.6.6. We say that x has **abstract Jordan decomposition** $x = d + n$. If $n = 0$, then we say that x is **semisimple**.

Remark 9.6.7 — If $L \subseteq \mathfrak{gl}(V)$, then $\text{ad}(L) \subseteq \text{ad}(\mathfrak{gl}(V)) \neq \mathfrak{gl}(V)$, so it is not obvious that the abstract and usual Jordan decomposition agree. However, what is easy to show that if $x = d + n$, it is the usual Jordan decomposition of $x \in \mathfrak{gl}(V)$, and d, n are also contained in L , then $x = d + n$ is also the abstract Jordan decomposition.

We are now ready to prove the main result about the abstract Jordan decomposition, but before that, we introduce an exercise.

Exercise 9.6.8. Suppose that L_1 and L_2 are complex semisimple Lie algebras and that $\theta: L_1 \rightarrow L_2$ is a surjective homomorphism. Show that if $x \in L_1$ has abstract Jordan decomposition $x = d + n$, then $\theta(x) \in L_2$ has abstract Jordan decomposition $\theta(x) = \theta(d) + \theta(n)$. *Hint: Exercise 2.4.5 is relevant.*

Solution: We have that $\theta(x) = \theta(d + n) = \theta(d) + \theta(n)$. If we can show that this satisfies the desired axioms, the uniqueness of Jordan decomposition then shows that $\theta(x)$ has this Jordan decomposition. First, note that $[\theta(d), \theta(n)] = \theta[d, n] = 0$, so we have commutativity. Then, Exercise 2.4.5 tells us that $\theta(d)$ is semisimple; that is, $\text{ad}\theta(d)$ is diagonalizable. We ask if $\text{ad}\theta(n)$ is nilpotent. Suppose that $(\text{ad } n)^k = 0$ for some k . Let $y \in L_2$. Then, via surjectivity, we can write $y = \theta(y')$, for some $y' \in L_1$. Then, we have that

$$\text{ad}\theta(n)(y) = [\theta(n), \theta(y')] = \theta(\text{ad } n(y')).$$

Then, note that

$$\text{ad}\theta(n)^k(y) = [\theta(n), [\theta(n), \dots, [\theta(n), \theta(y')], \dots]] = \theta([\text{ad } n^k(y')]),$$

so $\text{ad } \theta$ is nilpotent because $\text{ad } n$ is nilpotent. This concludes the proof. $\square$

Theorem 9.6.9

Let L be a semisimple Lie algebra and let $\theta: L \rightarrow \mathfrak{gl}(V)$ be a representation of L . Suppose that $x \in L$ has abstract Jordan decomposition $x = d + n$. Then the Jordan decomposition of $\theta(x) \in \mathfrak{gl}(V)$ is $\theta(x) = \theta(d) + \theta(n)$.

Proof. Not present due to error in book. $\square$

The last theorem is a very powerful result, which we shall apply several times in the next chapter. For another application, see Exercise 9.6.10 below.

Exercise 9.6.10. Suppose that M is an $\mathfrak{sl}(2, \mathbb{C})$ -module. Use the abstract Jordan decomposition to show that M decomposes as a direct sum of h -eigenspaces. Hence use Exercise 8.4.5 to show that M is completely reducible.

Solution: Not done because the key theorem is not proven. $\square$

§9.7 Exercises

Exercise 9.7.1. Let $L = \mathfrak{gl}(n, \mathbb{C})$. Show that the Killing form of L is given by

$$\kappa(a, b) = 2n \text{tr}(ab) - 2(\text{tr } a)(\text{tr } b).$$

For instance, start with $(\text{ad } b)e_{rs}$, apply $\text{ad } a$, and then express the result in terms of the basis and find the coefficient of e_{rs} . Hence prove that if $n \geq 2$ then $\mathfrak{sl}(n, \mathbb{C})$ is semisimple.

Solution: The e_{rs} form a basis of $\mathfrak{gl}(n, \mathbb{C})$, as we know. Thus,

$$\kappa(a, b) = \sum_{r,s} (\text{coefficient of } e_{rs} \text{ in } (\text{ad}(a)\text{ad}(b))(e_{rs})).$$

Now

$$\text{ad}(b)(e_{rs}) = [b, e_{rs}] = be_{rs} - e_{rs}b,$$

so

$$(\text{ad}(a)\text{ad}(b))(e_{rs}) = a(be_{rs} - e_{rs}b) - (be_{rs} - e_{rs}b)a = abe_{rs} - ae_{rs}b - be_{rs}a + e_{rs}ba.$$

For any matrix $X = (x_{ij})$, one has

$$Xe_{rs} = \sum_i x_{ir} e_{is}, \quad e_{rs}X = \sum_j x_{sj} e_{rj}.$$

Thus, the coefficient of e_{rs} in Xe_{rs} is x_{rr} , and the coefficient of e_{rs} in $e_{rs}X$ is x_{ss} . Therefore:

$$\begin{aligned} \text{coeff}_{e_{rs}}(ae_{rs}) &= (ab)_{rr}, \\ \text{coeff}_{e_{rs}}(ae_{rs}b) &= a_{rr}b_{ss}, \\ \text{coeff}_{e_{rs}}(be_{rs}a) &= b_{rr}a_{ss}, \\ \text{coeff}_{e_{rs}}(e_{rs}ba) &= (ba)_{ss}. \end{aligned}$$

Thus,

$$\kappa(a, b) = \sum_{r,s} ((ab)_{rr} - a_{rr}b_{ss} - b_{rr}a_{ss} + (ba)_{ss}).$$

We simplify each term:

$$\sum_{r,s} (ab)_{rr} = n \text{tr}(ab), \quad \sum_{r,s} (ba)_{ss} = n \text{tr}(ba) = n \text{tr}(ab),$$

and

$$\sum_{r,s} a_{rr}b_{ss} = (\text{tra})(\text{tr}b), \quad \sum_{r,s} b_{rr}a_{ss} = (\text{tra})(\text{tr}b).$$

Hence,

$$\kappa(a, b) = 2n \text{tr}(ab) - 2(\text{tra})(\text{tr}b).$$

We move on to the semisimplicity of $\mathfrak{sl}(n, \mathbb{C})$. If $a, b \in \mathfrak{sl}(n, \mathbb{C})$, then $\text{tra} = \text{tr}b = 0$, so

$$\kappa(a, b) = 2n \text{tr}(ab).$$

Since we want to show that $\mathfrak{sl}(n, \mathbb{C})$ is semisimple for $n \geq 2$, we show this is nondegenerate. For this we show that for every element in $\mathfrak{sl}(n, \mathbb{C})$, there is an element which doesn't let it vanish. Let $0 \neq a \in \mathfrak{sl}(n, \mathbb{C})$. If some off-diagonal entry $a_{ij} \neq 0$ with $i \neq j$, take $b = e_{ji} \in \mathfrak{sl}(n, \mathbb{C})$. Then

$$\text{tr}(ab) = a_{ij} \neq 0.$$

Otherwise, a is diagonal with trace 0, so there exist $i \neq j$ such that $a_{ii} \neq a_{jj}$. Take $b = e_{ii} - e_{jj} \in \mathfrak{sl}(n, \mathbb{C})$. Then

$$\text{tr}(ab) = a_{ii} - a_{jj} \neq 0.$$

We have verified that there are no nonzero matrices a in $\mathfrak{sl}(n, \mathbb{C})$ such that $\kappa(a, b)$ vanishes for all $b \in \mathfrak{sl}(n, \mathbb{C})$, so it's nondegenerate. Cartan's criterion then implies that $\mathfrak{sl}(n, \mathbb{C})$ is semisimple, for all $n \geq 2$. $\square$

Exercise 9.7.2.

1. Show that if L a semisimple Lie algebra, then $L' = L$.
2. Suppose that L is the direct sum of simple ideals $L = L_1 \oplus \cdots \oplus L_k$. Show that if I is a simple ideal of L , then I is equal to one of the L_i . *Hint: Consider the ideal $[I, L]$.*
3. If $L' = L$, must L be semisimple?

Solution: 1. We have $L = L_1 \oplus \cdots \oplus L_k$, where each L_i are simple ideals. Thus, we have that

$$L' = L'_1 \oplus \cdots \oplus L'_k,$$

so it suffices to check the assertion for simple Lie algebras. However, $L' = L$ for simple Lie algebras L , since L is necessarily not abelian.

2. Let I be a simple ideal. Suppose that $I \neq L_i$, for all i . We note that

$$[I, L] = [I, L_1] \oplus \cdots \oplus [I, L_k].$$

Then, we note that $[I, L_i] \subseteq I$, so I is an ideal of L_i , for all i . However, since L_i are simple, I must either equal L_i or be 0. I cannot be the zero ideal, since it would not be simple then, so it must be equal to one of the L_i .

3. If $L' = L$, then we get that

$$L^{(n)} = L,$$

so L is not solvable. The answer is no..., but we need to construct an example where L is not semisimple. We do this by considering semidirect products (recall 3.3.iii). The specific construction we consider is the semidirect product $\mathfrak{sl}(2, \mathbb{C}) \ltimes A_2$, where A_2 is the two-dimensional abelian Lie algebra. A semidirect product comes with a Lie algebra homomorphism $\iota: \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{Der}(A_2) = \mathfrak{gl}(A_2)$, which we can view as a representation of $\mathfrak{sl}(2, \mathbb{C})$. We fix a basis of A_2 , $\text{Span}\{x, y\}$. Then, we define the action of $\mathfrak{sl}(2, \mathbb{C})$ on $\mathfrak{gl}(A_2)$ as

$$\begin{aligned} \iota: e &\mapsto Id \\ \iota: f &\mapsto \iota(f)(x) = x \\ &\quad \iota(f)(y) = 0 \\ \iota: h &\mapsto \iota(h)(x) = 0 \\ &\quad \iota(h)(y) = y. \end{aligned}$$

Then, recall that $[(s_1, x_1), (s_2, x_2)] := ([s_1, s_2], [x_1, x_2] + \iota(s_1)(x_2) - \iota(s_2)(x_1))$. We just need to check that $L' = L$, and that L is not solvable. Indeed, L is not solvable because A_2 is an ideal of our semidirect product, and so $A_2 \subseteq \text{rad } L$, which is therefore not zero. It suffices to check that $L' = L$ now. $\square$

Exercise 9.7.3. Suppose that L is a Lie algebra over $\mathbb{C}$ and that β is a symmetric, associative bilinear form of L . Show that β induces a linear map

$$\theta: L \rightarrow L^*, \quad \theta(x) = \beta(x, -).$$

Viewing both L and L^* as L -modules, show that θ is an L -module homomorphism. Deduce that if β is nondegenerate, then L and L^* are isomorphic as L -modules.

Solution: We want to show that $\theta(x \cdot y) = x \cdot \theta(y)$. Let, $x, y, z \in L$. Also recall that L^* is made into an L -module by

$$(x \cdot \theta(y))(z) = -\theta(y)(x \cdot z).$$

Then, we have:

$$\begin{aligned}
 \theta(x \cdot y)(z) &= \beta([x, y], z) \\
 &= \beta(x, [y, z]) \\
 &= \beta([y, z], x) \\
 &= -\beta(y, [x, z]) \\
 &= -\theta(y)(x \cdot z) \\
 &= (x \cdot \theta(y))(z).
 \end{aligned}$$

If β is nondegenerate, then the kernel of $\beta(x, \cdot)$ is zero. In other words, the kernel of θ is zero, and so injective. $\dim L = \dim L^*$, so we get surjectivity, so we get the isomorphism. $\square$

Exercise 9.7.4. Let L be a simple Lie algebra over $\mathbb{C}$ with Killing form κ . Use Exercise 9.7.3 to show that if β is any other symmetric, associative, non-degenerate bilinear form on L , then there exists $0 \neq \lambda \in \mathbb{C}$ such that $\kappa = \lambda\beta$.

Solution: Note that if L is simple, then κ is nondegenerate. It is also symmetric and associative by definition. Thus, θ_κ and θ_β , defined with their appropriate bilinear forms are isomorphisms between L and L^* . Then, in particular, we may let

$$\theta_\kappa \theta_\beta^{-1}: L \xrightarrow{\sim} L.$$

This is certainly not the zero map, so by Schur's lemma, we have that $\theta_\kappa \theta_\beta^{-1} = \lambda 1_L$. Then, we can compose with θ_β to get that $\theta_\kappa = \lambda \theta_\beta$. However, this is just saying that $\kappa = \lambda\beta$. $\square$

10 The Root Space Decomposition

We are now ready to start classifying complex semisimple Lie algebras. So far we have proved the simplicity of only one family of Lie algebras, namely the algebras $\mathfrak{sl}(n, \mathbb{C})$ for $n \geq 2$ (see Exercise 9.7.1). There is, however, a strong sense in which their behavior is typical of all complex semisimple Lie algebras. We therefore begin by looking at the structures of $\mathfrak{sl}(2, \mathbb{C})$ and $\mathfrak{sl}(3, \mathbb{C})$ in the belief that this will motivate the strategy adopted in this chapter.

Previously, we also proved that $\mathfrak{sl}(2, \mathbb{C})$ was the unique 3-dimensional semisimple complex Lie algebra by proceeding as follows:

1. We first showed that if L was a 3-dimensional Lie algebra such that $L = L'$, then there was some $h \in L$ such that $\text{ad } h$ was diagonalizable.
2. We then took a basis of L consisting of eigenvectors for $\text{ad } h$ and by finding the structure constants with respect to this basis showed that L was isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

In the case of $\mathfrak{sl}(3, \mathbb{C})$, a suitable replacement for the element $h \in \mathfrak{sl}(2, \mathbb{C})$ is the 2-dimensional subalgebra H of diagonal matrices in $\mathfrak{sl}(3, \mathbb{C})$. One can see directly that $\mathfrak{sl}(3, \mathbb{C})$ decomposes into a direct sum of common eigenspaces for the elements of $\text{ad } H$. Suppose $h \in H$ has diagonal entries a_1, a_2, a_3 :

$$h = \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix}.$$

Then

$$[h, e_{ij}] = (a_i - a_j)e_{ij},$$

so the elements e_{ij} for $i \neq j$ are common eigenvectors for the elements of $\text{ad } h$. Further, $\text{ad } h$ is diagonalizable. This holds more generally, for $\mathfrak{sl}(n, \mathbb{C})$, so the diagonal matrices in $\mathfrak{sl}(n, \mathbb{C})$ are an abelian subalgebra, all of whose elements are semisimple (nilpotent parts zero). Moreover, as H is abelian, H is contained in the kernel of every element of $\text{ad } H$.

It will be helpful to express this decomposition using the language of weights and weight spaces introduced in Chapter 5. Define $\varepsilon_i: H \rightarrow \mathbb{C}$ by $\varepsilon_i(h) = a_i$. We have

$$(\text{ad } h)e_{ij} = (\varepsilon_i - \varepsilon_j)(h)e_{ij}.$$

Here $\varepsilon_i - \varepsilon_j$ is a weight and e_{ij} is in its associated weight space. In fact one can check that if L_{ij} is the weight space for $\varepsilon_i - \varepsilon_j$, that is

$$L_{ij} = \{x \in \mathfrak{sl}(3, \mathbb{C}) : (\text{ad } h)x = (\varepsilon_i - \varepsilon_j)(h)x \text{ for all } h \in H\},$$

then we have $L_{ij} = \text{Span}\{e_{ij}\}$ for $i \neq j$. Hence there is a direct sum decomposition

$$\mathfrak{sl}(3, \mathbb{C}) = H \oplus \bigoplus_{i \neq j} L_{ij}.$$

The existence of this decomposition can be seen in a more abstract way. Let L be a complex semisimple Lie algebra and let H be an abelian subalgebra of L consisting of semisimple elements. By definition, $\text{ad } h$ is diagonalizable for every $h \in H$. Moreover, as commuting linear maps may be simultaneously diagonalized, H acts diagonalizably on L in the adjoint representation. We may therefore decompose L into a direct sum of weight spaces for the adjoint action of H .

Our strategy is therefore:

1. to find an abelian Lie subalgebra H of L that consists entirely of semisimple elements; and
2. to decompose L into weight spaces for the action of $\text{ad } H$ and then exploit this decomposition to determine information about the structure constants of L .

In the following section, we identify the desirable properties of the subalgebra H and prove some preliminary results about the decomposition. We then show that subalgebras H with these desirable properties always exist and complete step (2).

§10.1 The Weight Spaces L_α

Suppose that L is a complex semisimple Lie algebra containing an abelian subalgebra H consisting of semisimple elements. What information does this give us about L ?

We have seen that L has a basis of common eigenvectors for the elements of $\text{ad } H$. Given a common eigenvector $x \in L$, the eigenvalues are given by the associated weight, $\alpha: H \rightarrow \mathbb{C}$, defined by

$$(\text{ad } h)x = \alpha(h)x \text{ for all } h \in H.$$

Weights are elements of the dual space H^* . For each $\alpha \in H^*$, let

$$L_\alpha := \{x \in L : [h, x] = \alpha(h)x \text{ for all } h \in H\}$$

denote the corresponding **weight space**. One of these weight spaces is the zero weight space:

$$L_0 = \{z \in L : [h, z] = 0 \text{ for all } h \in H\}.$$

This is the same as the centralizer of H in L , $C_L(H)$. As H is abelian, we have $H \subseteq L_0$.

Let Φ denote the set of non-zero $\alpha \in H^*$ for which L_α is non-zero. We can write the decomposition of L into weight spaces for H as

$$L = L_0 \oplus \bigoplus_{\alpha \in \Phi} L_\alpha. \quad (\star)$$

Since L is finite-dimensional, this implies that Φ is finite.

Lemma 10.1.1

Suppose that $\alpha, \beta \in H^*$. Then

- (i) $[L_\alpha, L_\beta] \subseteq L_{\alpha+\beta}$.
- (ii) If $\alpha + \beta \neq 0$, then $\kappa(L_\alpha, L_\beta) = 0$.
- (iii) The restriction of κ to L_0 is non-degenerate.

Proof. (i) Take $x \in L_\alpha$ and $y \in L_\beta$. We must show that $[x, y]$, if non-zero, is an eigenvector for each $h \in H$, with eigenvalue $\alpha(h) + \beta(h)$. Using the Jacobi identity we get

$$\begin{aligned} [h, [x, y]] &= [[h, x], y] + [x, [h, y]] = [\alpha(h)x, y] + [x, \beta(h)y] \\ &= \alpha(h)[x, y] + \beta(h)[x, y] \\ &= (\alpha + \beta)(h)[x, y]. \end{aligned}$$

(ii) Since $\alpha + \beta \neq 0$, there is some $h \in H$ such that $(\alpha + \beta)(h) \neq 0$. Now, for any $x \in L_\alpha$ and $y \in L_\beta$, we have, using the associativity of the Killing form,

$$\alpha(h)\kappa(x, y) = \kappa([h, x], y) = -\kappa([x, h], y) = -\kappa(x, [h, y]) = -\beta(h)\kappa(x, y),$$

and hence

$$(\alpha + \beta)(h)\kappa(x, y) = 0.$$

Since by assumption $(\alpha + \beta)(h) \neq 0$, we must have $\kappa(x, y) = 0$.

(iii) By (ii), we know that $\kappa(L_0, L_\alpha) = 0$, for all $\alpha \neq 0$. Suppose $x \in L_0$ such that $\kappa(x, y) = 0$, for all $y \in L_0$. It follows that $\kappa(x, y) = 0$, for all $y \in L$, since the L_α span L and so we can write y using a basis of elements from the L_α . Since κ is non-degenerate (because L is semisimple), it follows that $x = 0$, as required. $\square$

Exercise 10.1.2. Show that if $x \in L_\alpha$ where $\alpha \neq 0$, then $\text{ad } x$ is nilpotent, i.e. x is nilpotent.

Solution: We have that $[L_\alpha, L_\beta] \subseteq L_{\alpha+\beta}$, so equivalently $\text{ad } x(L_\beta) \subseteq L_{\alpha+\beta}$. Thus,

$$(\text{ad } x)^2 \subseteq \text{ad } x(L_{\alpha+\beta}) \subseteq L_{2\alpha+\beta}.$$

Continuing, we get

$$(\text{ad } x)^n(L_\beta) \subseteq L_{n\alpha+\beta}.$$

$L_\gamma \neq 0$ for only finitely many γ , since $(\text{ad } x)^n(L_\beta) = 0$ for n sufficiently large (because L is finite dimensional). Therefore, $\text{ad } x$ is nilpotent. $\square$

If H is small, then the decomposition $(\star)$ is likely to be rather coarse, with few non-zero weight spaces other than L_0 . Furthermore, if H is properly contained in L_0 , then we get little information about how the elements in L_0 that are not in H act on L . This is illustrated by the following exercise.

Exercise 10.1.3. Let $L = \mathfrak{sl}(n, \mathbb{C})$, where $n \geq 2$, and let $H = \text{Span}\{h\}$, where $h = e_{11} - e_{22}$. Find $L_0 = C_L(H)$, and determine the direct sum decomposition $(\star)$ with respect to H .

Solution: We first determine $C_L(H)$.

$$C_L(H) = \text{Span} \{ \{E_{i,i} - E_{i+1,i+1} : 0 \leq i \leq n-1\} \cup \{E_{ij} : 3 \leq i \neq j\} \}.$$

We calculate the other weight-spaces; we distinguish depending on the eigenvalue. Let $j \geq 2$. Then,

$$[h, E_{1j}] = E_{1j}, \quad [h, E_{j1}] = -E_{j1}.$$

We also calculate that

$$[h, E_{12}] = 2E_{12}, \quad [h, E_{21}] = -2E_{21}.$$

Therefore, we have the following weight-space decomposition:

$$\begin{aligned} L &= L_0 \oplus L_1 \oplus L_{-1} \oplus L_2 \oplus L_{-2} \\ &= \left(\bigoplus_{i=1}^{n-1} \mathbb{C}(E_{ii} - E_{i+1,i+1}) \right) \oplus \left(\bigoplus_{3 \leq i \neq j} \mathbb{C}E_{ij} \right) \oplus \left(\bigoplus_{i=3}^n \mathbb{C}E_{1j} \right) \\ &\quad \oplus \left(\bigoplus_{i=1}^n \mathbb{C}E_{j1} \right) \oplus \mathbb{C}E_{12} \oplus \mathbb{C}E_{21}. \end{aligned}$$

□

We conclude that for the decomposition (★) of L into weight spaces to be as useful as possible, H should be as large as possible, and ideally we would have $H = L_0 = C_L(H)$.

Definition 10.1.4. A Lie subalgebra H of a Lie algebra L is said to be a **Cartan subalgebra** (or simply **CSA**) if

- H is abelian
- every element $h \in H$ is semisimple, i.e., $\text{ad } h$ is diagonalizable
- H is maximal subject to the first two properties

Example 10.1.5

It is not difficult to check that the diagonal matrices satisfy this.

Note that we do not assume L is semisimple in this definition. For example, the subalgebra H of $\mathfrak{sl}(3, \mathbb{C})$ considered in the introduction to this chapter is a Cartan subalgebra of $\mathfrak{sl}(3, \mathbb{C})$. One straightforward way to see this is to show that $C_{\mathfrak{sl}(3, \mathbb{C})}(H) = H$; thus H is not contained in any larger abelian subalgebra of $\mathfrak{sl}(3, \mathbb{C})$.

Remark 10.1.6 — We remark that some texts use a ‘maximal toral subalgebra’ in place of what we have called a Cartan subalgebra. The connection is discussed at the end of Appendix C, where we establish that the two types of algebras are the same.

§10.2 Cartan Subalgebras

Proposition 10.2.1

If L is semisimple, then a nonzero Cartan subalgebra exists.

Proof. If not, then there are no semisimple $x \in L$. Then, by Theorem 9.6.5, any $x \in L$ can be written as $x = d + n$, where $\text{ad } d$ is diagonalizable, $\text{ad } n$ is nilpotent, and $d, n \in L$. d , by definition, is semisimple. Then, d must be zero, so $\text{ad } x$ would be nilpotent for all $x \in L$. Then, L is solvable by Engel’s theorem, which is a contradiction. □

We shall now show that if H is a Cartan subalgebra then $H = C_L(H)$. The proof of this statement is slightly technical, so the reader may prefer to defer or skip some of the details.

Lemma 10.2.2

Let H be a Cartan subalgebra of a complex semisimple Lie algebra L . Suppose that $h \in H$ is such that the dimension of $C_L(h)$ is minimal. Then every $s \in H$ is central in $C_L(h)$, and so $C_L(h) \subseteq C_L(s)$. Hence $C_L(h) = C_L(H)$.

Proof. We shall show that if s is not central in $C_L(h)$, then there is a linear combination of s and h whose centralizer has smaller dimension than $C_L(h)$.

First we construct a suitable basis for L . We start by taking a basis of $C_L(h) \cap C_L(s)$, $\{c_1, \dots, c_n\}$, say. As s is semisimple and $s \in C_L(h)$, $\text{ad } s$ acts diagonalizably on $C_L(h)$. We may therefore extend this basis to a basis of $C_L(h)$ consisting of $\text{ad } s$ eigenvectors, say by adjoining $\{x_1, \dots, x_p\}$. Similarly we may extend $\{c_1, \dots, c_n\}$ to a basis of $C_L(s)$ consisting of $\text{ad } h$ eigenvectors, say by adjoining $\{y_1, \dots, y_q\}$. We leave it to the reader to check that

$$\{c_1, \dots, c_n, x_1, \dots, x_p, y_1, \dots, y_q\}$$

is a basis of $C_L(h) + C_L(s)$. Finally, as $\text{ad } h$ and $\text{ad } s$ commute and act diagonalizably on L , we may extend this basis to a basis of L by adjoining simultaneous eigenvectors for $\text{ad } h$ and $\text{ad } s$, say $\{w_1, \dots, w_r\}$.

Note that if $[s, x_j] = 0$ then $x_j \in C_L(s) \cap C_L(h)$, a contradiction. Similarly, one can check that $[h, y_k] \neq 0$. Let $[h, w_l] = \theta_l w_l$ and $[s, w_l] = \sigma_l w_l$. Again we have $\theta_l, \sigma_l \neq 0$ for $1 \leq l \leq r$. The following table summarizes the eigenvalues of $\text{ad } s$, $\text{ad } h$, and $\text{ad } (s + \lambda h)$, where $\lambda \neq 0$:

	c_i	x_j	y_k	w_l
$\text{ad } s$	0	$\neq 0$	0	σ_l
$\text{ad } h$	0	0	$\neq 0$	θ_l
$\text{ad } (s + \lambda h)$	0	$\neq 0$	$\neq 0$	$\sigma_l + \lambda \theta_l$

Thus, if we choose λ so that $\lambda \neq 0$ and $\lambda \neq -\frac{\sigma_l}{\theta_l}$ for any l , then we will have

$$C_L(s + \lambda h) = C_L(s) \cap C_L(h).$$

By hypothesis, $C_L(h) \not\subseteq C_L(s)$, so this subspace is of smaller dimension than $C_L(h)$; this contradicts the choice of h .

Now, since $C_L(H)$ is the intersection of the $C_L(s)$ for $s \in H$, it follows that $C_L(h) \subseteq C_L(H)$. The other inclusion is obvious, so we have proved that $C_L(h) = C_L(H)$. $\square$

Theorem 10.2.3

If H is a Cartan subalgebra of a complex semisimple Lie algebra L and $h \in H$ is such that $C_L(h) = C_L(H)$, then $C_L(h) = H$. Hence H is self-centralizing.

Proof. Since H is abelian, H is certainly contained in $C_L(h)$. Suppose $x \in C_L(h)$ has abstract Jordan decomposition $x = s + n$. As x commutes with h , Theorem 9.6.5 implies that both s and n lie in $C_L(h)$, so we must show that $s \in H$ and $n = 0$.

We almost know already that $s \in H$. Namely, since $C_L(h) = C_L(H)$, we have that s commutes with every element of H and therefore $H + \text{Span}\{s\}$ is an abelian subalgebra

of L consisting of semisimple elements. It contains the Cartan subalgebra H and hence by maximality $s \in H$. To show that the only nilpotent element in $C_L(H)$ is 0 takes slightly more work.

Step 1: $C_L(h)$ is nilpotent.

Take $x \in C_L(h)$ with $x = s + n$ as above. Since $s \in H$, it must be central in $C_L(h)$, so, regarded as linear maps from $C_L(h)$ to itself, we have $\text{ad } x = \text{ad } n$. Thus for every $x \in C_L(h)$, $\text{ad } x: C_L(h) \rightarrow C_L(h)$ is nilpotent. It now follows from the second version of Engel's Theorem (Theorem 6.1.6) that $C_L(h)$ is a nilpotent Lie algebra.

Step 2: Every element in $C_L(h)$ is semisimple.

Let $x \in C_L(h)$ have abstract Jordan decomposition $x = s + n$ as above. As $C_L(h)$ is nilpotent, it is certainly solvable, so by Lie's Theorem (Theorem 6.2.1) there is a basis of L in which the maps $\text{ad } x$ for $x \in C_L(h)$ are represented by upper triangular matrices. As $\text{ad } n: L \rightarrow L$ is nilpotent, its matrix must be strictly upper triangular. Therefore

$$\kappa(n, z) = \text{tr}(\text{ad } n \circ \text{ad } z) = 0$$

for all $z \in C_L(h)$. By Lemma 10.1.1, the restriction of κ to $C_L(H)$ is non-degenerate, so we deduce $n = 0$, as required, so $x = s$. $\square$

Corollary 10.2.4

$L_0 = H$, if H is CSA.

Proof. This is because $H = C_L(H) = L_0$. $\square$

§10.3 Definition of the Root Space Decomposition

Let H be a Cartan subalgebra of our semisimple Lie algebra L . As $H = L_0$, the direct sum decomposition of L into weight spaces for H may be written as

$$L = H \oplus \bigoplus_{\alpha \in H^*} L_\alpha,$$

where

$$L_\alpha = \{v \in L: [h, v] = \lambda(h)v, \forall h \in H\}$$

are called the **weight spaces**. We note that if $h_1, h_2 \in H$, then

$$[\text{ad } h_1, \text{ad } h_2] = \text{ad}([h_1, h_2]) = 0.$$

Note that the $0 \neq \alpha \in H^*$ where L_α is nonzero is finite, since L is finite dimensional (we denote this as the set Φ).

Definition 10.3.1. If $\alpha \in \Phi$, then we say that α is a **root** of L and L_α is the associated **root space**, so

$$\Phi = \{\alpha \in H^*: L_\alpha \neq 0, \alpha \neq 0\}$$

is the set of roots. The direct sum decomposition above is the **root space decomposition**. It should be noted that the roots and root spaces depend on the choice of Cartan subalgebra H .

Remark 10.3.2 — Using this terminology, we can write the above as

$$L = H \oplus \bigoplus_{\alpha \in \Phi} L_{\alpha}.$$

Example 10.3.3

If $L = \mathfrak{sl}(2, \mathbb{C})$, then we take $H = \left\{ \begin{pmatrix} \lambda & 0 \\ 0 & -\lambda \end{pmatrix} : \lambda \in \mathbb{C} \right\}$. Then, the roots are α and $-\alpha$, where

$$\alpha \left(\begin{pmatrix} \lambda & 0 \\ 0 & -\lambda \end{pmatrix} \right) = 2\lambda, \quad -\alpha \left(\begin{pmatrix} \lambda & 0 \\ 0 & -\lambda \end{pmatrix} \right) = -2\lambda.$$

This is because

$$L_{\alpha} = \mathbb{C} \cdot \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad L_{-\alpha} = \mathbb{C} \cdot \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

§10.4 Subalgebras Isomorphic to $\mathfrak{sl}(2, \mathbb{C})$

We shall now associate to each root $\alpha \in \Phi$ a Lie subalgebra of L isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. These subalgebras will enable us to apply the results in previous chapters (Ch. 8) on representations of $\mathfrak{sl}(2, \mathbb{C})$ to deduce several strong results on the structure of L . Later chapters give many examples of the theory we develop in the next three sections. See also Exercise 10.4.1 for a more immediate example.

Exercise 10.4.1. Let $L = \mathfrak{sl}(3, \mathbb{C})$. With the same notation as in the introduction, let $\alpha := \varepsilon_1 - \varepsilon_2$ and $\beta := \varepsilon_2 - \varepsilon_3$. Show that the set of roots is

$$\Phi = \{\pm\alpha, \pm\beta, \pm(\alpha + \beta)\}.$$

Show that the angle between the roots α and β is $2\pi/3$ and verify some of the results in subsections 5 and 6 for $\mathfrak{sl}(3, \mathbb{C})$.

Solution: H is the diagonal elements. We first show that $\alpha, \beta, \alpha + \beta$ are roots. Let $h = \text{diag}(a_1, a_2, a_3)$

$$[h, E_{12}] = (a_1 - a_2)E_{12} = \alpha(h)E_{12}.$$

Analogously, $[h, E_{23}] = \beta(h)E_{23}$ and $[h, E_{13}] = (\alpha + \beta)(h)E_{13}$. Thus, $\alpha, \beta, \alpha + \beta$ are roots. Doing the same but with E_{21}, E_{32}, E_{31} shows that $-\alpha, -\beta, -(\alpha + \beta)$ are roots. Since $\dim H = 2$, and the sum of $\dim L_{\alpha} = 6$, we are done, since $\dim \mathfrak{sl}(3, \mathbb{C}) = 8$.

Now, we calculate the angle between α and β by restricting the standard inner product on $\mathbb{R}^3$ with basis $\varepsilon_1, \varepsilon_2, \varepsilon_3$ to the hyperplane $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$. We get that

$$(\alpha, \alpha) = (\beta, \beta) = 2, \quad (\alpha, \beta) = -1.$$

Therefore,

$$\cos \theta = \frac{(\alpha, \beta)}{\sqrt{(\alpha, \alpha)(\beta, \beta)}} = \frac{-1}{2},$$

so $\theta = \frac{2\pi}{3}$ as desired. □

Lemma 10.4.2

If α is a root of L , then so is $-\alpha$, and for each nonzero x in L_α , there exists $y \in L_{-\alpha}$, such that $\text{Span}\{x, y, [x, y]\}$ is a Lie subalgebra of L isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

Proof. First we claim that there is some $y \in L_{-\alpha}$ such that $\kappa(x, y) \neq 0$ and $[x, y] \neq 0$. By Lemma 10.1.1, if $\alpha + \beta \neq 0$, then $[L_\alpha, L_\beta] = 0$. Thus, $\kappa(x, z) = 0$ for all $z \in L_\beta$, $\beta \neq -\alpha$. Since κ is non-degenerate, there is some $z \in L$ such that $\kappa(x, z) \neq 0$. Writing z using the root space decomposition, it follows then that there exists $y \in L_{-\alpha}$ with $\kappa(x, y) \neq 0$. Then, $[x, y] \in L_{\alpha+(-\alpha)} = L_0 = H$, since $[L_\alpha, L_\beta] \subseteq L_{\alpha+\beta}$. We claim that $[x, y] \neq 0$. To see this, since $\alpha \neq 0$, there exist $t \in H$ such that $\alpha(t) \neq 0$. For this t , we have

$$\kappa(t, [x, y]) = \kappa([t, x], y) = \kappa(\alpha(t)x, y) = \alpha(t)\kappa(x, y) \neq 0$$

and so $[x, y] \neq 0$.

Let $S := \text{Span}\{x, y, [x, y]\}$. We check that this defines a subalgebra. By Lemma 10.1.1, $[x, y]$ lies in $L_0 = H$, so we have

$$[[x, y], x] = \alpha([x, y])x, \quad [[x, y], y] = -\alpha([x, y])y.$$

This shows that S is a Lie subalgebra of L .

It remains to show that S is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. We claim that $\alpha([x, y]) \neq 0$. If this were true, then it follows that S is a 3-dimensional Lie algebra with $S' = S$. This implies that $S \cong \mathfrak{sl}(2, \mathbb{C})$ by the classification of three dimensional Lie algebras. Alternatively, rescale y so that $\alpha([x, y]) = 2$, so that we get an explicit isomorphism $S \xrightarrow{\sim} \mathfrak{sl}(2, \mathbb{C})$ with

$$x \mapsto e, \quad y \mapsto f, \quad [x, y] \mapsto h.$$

We just have to check our claim that $\alpha([x, y]) \neq 0$. If not, then $[[x, y], x] = \alpha([x, y])x = 0$; similarly $[[x, y], y] = -\alpha([x, y])y = 0$, so $[x, y] \in Z(S)$. Then, S is solvable. Then, the adjoint representation $\text{ad } S$ is solvable, by Lemma 4.1.17. Since $\text{ad } S$ is a Lie subalgebra of $\mathfrak{gl}(L)$, Lie’s theorem tells us that there exists a basis such that for each $z \in S$, $\text{ad } z$ is conjugate to an upper triangular matrix. However, $\text{ad } x$ and $\text{ad } y$ both commute with $\text{ad } [x, y] \in S$ (by the Jacobi identity), so $\text{ad } [x, y]$ is nilpotent (by Proposition 5.4.1). Further, $[x, y] \in H$, so is automatically semisimple. $\text{ad } [x, y]$ is therefore simultaneously nilpotent and semisimple, implying (by checking traces for instance), that $\text{ad } [x, y]$ is zero. This implies that $[x, y] \in Z(L)$, but this contradicts $Z(L) = 0$ due to the semisimplicity of L . Therefore, $\alpha([x, y]) \neq 0$, and $S \cong \mathfrak{sl}(2, \mathbb{C})$ as required. $\square$

Remark 10.4.3 — Using this lemma, we may associate to each $\alpha \in \Phi$ a subalgebra $\mathfrak{sl}(\alpha)$ of L isomorphic to $\mathfrak{sl}(2, \mathbb{C})$ (S in the proof).

The following exercise gives a standard basis for this Lie algebra.

Exercise 10.4.4. Show that for each $\alpha \in \Phi$, $\mathfrak{sl}(\alpha)$ has a basis $\{e_\alpha, f_\alpha, h_\alpha\}$ such that

1. $e_\alpha \in L_\alpha$, $f_\alpha \in L_{-\alpha}$, $[e_\alpha, f_\alpha] = h_\alpha \in H$, and $\alpha(h_\alpha) = 2$.
2. The map $\theta: \mathfrak{sl}(\alpha) \rightarrow \mathfrak{sl}(2, \mathbb{C})$ defined by $\theta(e_\alpha) = e$, $\theta(f_\alpha) = f$, and $\theta(h_\alpha) = h$ is a Lie algebra isomorphism.

Hint: With the notation used in the statement of the lemma, one can take $e_\alpha = x$ and $f_\alpha = \lambda y$ for a suitable choice of $\lambda \in \mathbb{C}$.

Solution: By the lemma above, we can just let $e_\alpha = x, f_\alpha = 2y$, so that $h_\alpha = 2h$ and $\alpha(h_\alpha) = 2$. θ is then an isomorphism since the structure constants are equal. $\square$

Remark 10.4.5 — We will see that L_α is 1-dimensional.

Exercise 10.4.6. Show that a Lie algebra L is solvable if and only if every subalgebra of L generated by two elements is solvable.

Solution: The forward direction is clear, since every subalgebra of a solvable Lie algebra is solvable. Conversely, suppose that L is not solvable. Then, $L/\text{rad } L$ is semisimple. Therefore, by the above, there is a copy of $\mathfrak{sl}(2, \mathbb{C})$ living in $L/\text{rad } L$. Let $x, y \in L$ such that under the map $L \rightarrow L/\text{rad } L$, it maps to the canonical generators e, f of $\mathfrak{sl}(2, \mathbb{C})$. Let K be a subalgebra of L generated by x, y . Then K surjects onto this copy of $\mathfrak{sl}(2, \mathbb{C})$, so it can't be solvable. $\square$

§10.5 Root Strings and Eigenvalues

We can use the Killing form to define an isomorphism between H and H^* as follows. Given $h \in H$, let θ_h denote the map $\theta_h \in H^*$ defined by

$$\theta_h(k) := \kappa(h, k) \text{ for all } k \in H.$$

By Lemma 10.1.1 the Killing form is non-degenerate on restriction to H , so the map $h \mapsto \theta_h$ is an isomorphism between H and H^* (since non-degeneracy is equivalent to this map being injective). This was also in Exercise 9.7.3. In particular, associated to each root $\alpha \in \Phi$ there is a unique element $t_\alpha \in H$ such that

$$\kappa(t_\alpha, h) = \alpha(h) \text{ for all } h \in H.$$

Note that this t_α is unique because if there was some other such t'_α with the same property, then $\kappa(t_\alpha - t'_\alpha, h) = 0$, but then non-degeneracy of $\kappa|_H$ implies that $t_\alpha = t'_\alpha$. One very useful property of this correspondence is the following lemma.

Lemma 10.5.1

Let $\alpha \in \Phi$. If $x \in L_\alpha$ and $y \in L_{-\alpha}$, then $[x, y] = \kappa(x, y)t_\alpha$. In particular, $h_\alpha = [e_\alpha, f_\alpha] \in \text{Span}\{t_\alpha\}$.

Proof. For $h \in H$, we have

$$\kappa(h, [x, y]) = \kappa([h, x], y) = \alpha(h)\kappa(x, y) = \kappa(t_\alpha, h)\kappa(x, y).$$

Now we view $\kappa(x, y)$ as a scalar and rewrite the right-hand side to get

$$\kappa(h, [x, y]) = \kappa(h, \kappa(x, y)t_\alpha).$$

This shows that $[x, y] - \kappa(x, y)t_\alpha$ is perpendicular to all $h \in H$, and hence it is zero as κ restricted to H is non-degenerate. $\square$

We are now in a position to apply the results of Chapter 8 on the representation theory of $\mathfrak{sl}(2, \mathbb{C})$. Let α be a root. We may regard L as an $\mathfrak{sl}(\alpha)$ -module via restriction of the adjoint representation. Thus, if $a \in \mathfrak{sl}(\alpha)$ and $y \in L$, then the action is given by

$$a \cdot y = (\text{ad } a)y = [a, y].$$

Note that the $\mathfrak{sl}(\alpha)$ -submodules of L are precisely the vector subspaces M of L such that $[s, m] \in M$ for all $s \in \mathfrak{sl}(\alpha)$ and $m \in M$. Of course, it is enough to check this when s is one of the standard basis elements $h_\alpha, e_\alpha, f_\alpha$. We shall also need the following lemma.

Lemma 10.5.2

If M is an $\mathfrak{sl}(\alpha)$ -submodule of L , then the eigenvalues of h_α acting on M are integers.

Proof. This relies on Weyl’s theorem. If M is a $\mathfrak{sl}(2, \mathbb{C})$ -module, then it can be decomposed into a direct sum of irreducible $\mathfrak{sl}(2, \mathbb{C})$ -modules. Each irreducible $\mathfrak{sl}(2, \mathbb{C})$ -module is isomorphic to V_d (from chapter 8), in which h acted with eigenvalues $d, d-2, \dots, -d$. $\square$

Example 10.5.3

If you did Exercise 8.4.2, then you will have seen how $\mathfrak{sl}(3, \mathbb{C})$ decomposes as an $\mathfrak{sl}(\alpha)$ -module where $\alpha = \varepsilon_1 - \varepsilon_2$ is a root of the Cartan subalgebra of $\mathfrak{sl}(3, \mathbb{C})$ consisting of all diagonal matrices.

Example 10.5.4

Let $U = H + \mathfrak{sl}(\alpha)$. Let $K = \ker \alpha \subseteq H$. By the rank-nullity formula, $\dim K = \dim H - 1$. (We know that $\dim \text{im } \alpha = 1$ as $\alpha(h_\alpha) \neq 0$.) As H is abelian, $[h_\alpha, x] = 0$ for all $x \in K$. Moreover, if $x \in K$, then

$$[e_\alpha, x] = -[x, e_\alpha] = -\alpha(x)e_\alpha = 0$$

and similarly $[f_\alpha, x] = 0$. Thus every element of $\mathfrak{sl}(\alpha)$ acts trivially on K . It follows that $U = K \oplus \mathfrak{sl}(\alpha)$ is a decomposition of U into $\mathfrak{sl}(\alpha)$ -modules. By Exercise 8.4.1, the adjoint representation of $\mathfrak{sl}(\alpha)$ is isomorphic to V_2 , so U is isomorphic to the direct sum of $\dim H - 1$ copies of the trivial representation V_0 , and one copy of the adjoint representation V_2 .

Example 10.5.5

If $\beta \in \Phi$ or $\beta = 0$, let

$$M := \bigoplus_c L_{\beta + c\alpha},$$

where the sum is over all $c \in \mathbb{C}$ such that $\beta + c\alpha \in \Phi$. It follows from Lemma 10.1.1 that M is an $\mathfrak{sl}(\alpha)$ -submodule of L . This module is said to be the α -root string through β . Analyzing these modules will give the main results of this section.

Proposition 10.5.6

Let $\alpha \in \Phi$. The root spaces $L_{\pm\alpha}$ are 1-dimensional. Moreover, the only multiples of α which lie in Φ are $\pm\alpha$.

Proof. Let $c\alpha$ be a root. Then, $\alpha(h_\alpha) = 2$, since $[h_\alpha, e_\alpha] = 2e_\alpha$. It follows that $c\alpha(h_\alpha) = 2c$, so $2c$ is an eigenvalue of h_α acting on L , so $2c \in \mathbb{Z}$ by Lemma 10.5.2. As the eigenvalues of h_α are integral, either $c \in \mathbb{Z}$ or $c \in \mathbb{Z} + \frac{1}{2}$. To rule out the unwanted values for c , we consider the root string module

$$M := H \oplus \bigoplus_{c\alpha \in \Phi} L_{c\alpha}.$$

This is an $\mathfrak{sl}(\alpha)$ -module and we want to show that it equals $H \oplus L_{-\alpha} \oplus L_\alpha$. Note that

$$[e_\alpha, L_{c\alpha}] \subset L_{(c+2)\alpha}, \quad [f_\alpha, L_{c\alpha}] \subset L_{(c-2)\alpha}, \quad [h_\alpha, L_{c\alpha}] \subset L_{c\alpha}.$$

Let $K = \ker \alpha \subseteq H$. Since $\alpha \in H^*$ and nonzero, it follows that K is an abelian Lie algebra of dimension 1 less than H . If $k \in K$, then $[k, e_\alpha] = \alpha(k)e_\alpha = 0$. Similarly, $[k, f_\alpha] = 0$ and $[k, h_\alpha] = 0$, so $\mathfrak{sl}(\alpha)$ acts by 0 on K . By Example 10.5.4, $K \oplus \mathfrak{sl}(\alpha)$ is an $\mathfrak{sl}(\alpha)$ -submodule of M . By Weyl's Theorem, modules for $\mathfrak{sl}(\alpha)$ are completely reducible, so we may write

$$M = K \oplus \mathfrak{sl}(\alpha) \oplus W.$$

We want to show that $W = 0$, for then it is clear that $L_{\pm\alpha}$ are 1-dimensional, and that all other multiples of c are not possible.

By Weyl's theorem, W is isomorphic to a direct sum of V_d , for some $d \geq 0$. Thus, if $W \neq 0$, choose a submodule V_d of W . Then, the eigenvalues of h_α on V_d are $d, d-2, \dots, 2-d, -d$. These are either all even or odd. If d is even then, 0 is an eigenvalue of h_α on V_d , so on W . The zero-eigenspace of h_α on M is H , so $W \cap H \neq 0$, but this is a contradiction, since H is already contained in $K \oplus \mathfrak{sl}(\alpha)$. Therefore, d must be odd.

Before considering the case where d is odd, we pursue another consequence of this argument. Suppose that $2\alpha \in \Phi$. Then h_α has $2\alpha(h_\alpha) = 4$ as an eigenvalue. As the eigenvalues of h_α on $K \oplus \mathfrak{sl}(\alpha)$ are 0 and ± 2 , the only way this could happen is if W contains an irreducible submodule V_d with d even, which we just saw is impossible.

Now suppose that d is odd. Then h_α has 1 as an eigenvalue on V_d , so on W . Therefore, $L_{c\alpha} \neq 0$, with $2c = 1$, so $c = \frac{1}{2}$. Thus, $\frac{1}{2}\alpha \in \Phi$. However, then $\frac{1}{2}\alpha$ and α are both in Φ , this contradicts the previous paragraph. Since we've covered when d is even and odd, then $W = 0$, as required. $\square$

Proposition 10.5.7

Suppose that $\alpha, \beta \in \Phi$ and $\beta \neq \pm\alpha$.

- (i) $\beta(h_\alpha) \in \mathbb{Z}$.
- (ii) There are integers $r, q \geq 0$ such that if $k \in \mathbb{Z}$, then $\beta + k\alpha \in \Phi$ if and only if $-r \leq k \leq q$. Moreover, $r - q = \beta(h_\alpha)$.
- (iii) If $\alpha + \beta \in \Phi$, then $[e_\alpha, e_\beta]$ is a non-zero scalar multiple of $e_{\alpha+\beta}$.
- (iv) $\beta - \beta(h_\alpha)\alpha \in \Phi$.

Proof. Let $M := \bigoplus_k L_{\beta+k\alpha}$ be the root string of α through β . To prove (i), we note that $\beta(h_\alpha)$ is the eigenvalue of h_α acting on L_β , and so it lies in $\mathbb{Z}$.

We know from the previous proposition that $\dim L_{\beta+k\alpha} = 1$ whenever $\beta + k\alpha$ is a root, so the eigenspaces of $\text{ad}h_\alpha$ on $M = \bigoplus_k L_{\beta+k\alpha}$ are all 1-dimensional and, since $(\beta + k\alpha)h_\alpha = \beta(h_\alpha) + 2k$, the eigenvalues of $\text{ad}h_\alpha$ on M are either all even or all odd. Therefore, M is an irreducible $\mathfrak{sl}(\alpha)$ -module. Then, we can suppose that $M \cong V_d$. On V_d , the element h_α acts diagonally with eigenvalues

$$\{d, d-2, \dots, -d\},$$

whereas on M , h_α acts diagonally with eigenvalues

$$\{\beta(h_\alpha) + 2k : \beta + k\alpha \in \Phi\}.$$

Equating these sets shows that if we define q and r by $q = \frac{1}{2}(d - \beta(h_\alpha))$ and $r = \frac{1}{2}(d + \beta(h_\alpha))$, then (ii) will hold.

Suppose $v \in L_\beta$, so v belongs to the h_α -eigenspace where h_α acts as $\beta(h_\alpha)$. If $(\text{ad}e_\alpha)e_\beta = 0$, then e_β is a highest-weight vector in the irreducible $\mathfrak{sl}(\alpha)$ -representation $M \cong V_d$, with highest weight $\beta(h_\alpha)$. If $\alpha + \beta$ is a root, then h_α acts on the associated root space by $(\alpha + \beta)h_\alpha = \beta(h_\alpha) + 2$. Therefore e_β is not in the highest weight space of the irreducible representation M , and so $(\text{ad}e_\alpha)e_\beta \neq 0$. This proves (iii).

Finally, (iv) follows from part (ii) as

$$\beta - \beta(h_\alpha)\alpha = \beta - (r - q)\alpha$$

and $-r \leq -r + q \leq q$. □

We now have a good idea about the structure constant of L (with respect to a basis given by the root space decomposition). The action of H on the root spaces of L is determined by the roots. Part (iii) of the previous proposition shows that (up to scalar factors) the set of roots also determines the brackets $[e_\alpha, e_\beta]$ for roots $\alpha \neq \pm\beta$. Lastly, by construction, $[e_\alpha, e_{-\alpha}]$ is in the span of $[e_\alpha, f_\alpha] = h_\alpha$. The reader keen to see a complete answer should read about Chevalley's Theorem in chapter 15.

§10.6 Cartan Subalgebras as Inner-Product Spaces

We conclude this chapter by showing that the roots of L all lie in a real vector subspace of H^* and that the Killing form induces an inner product on this space.

Lemma 10.6.1

- (i) If $h \in H$ and $h \neq 0$, then there exists a root $\alpha \in \Phi$ such that $\alpha(h) \neq 0$.
- (ii) The set Φ of roots spans H^* .

Proof. Suppose that $\alpha(h) = 0$ for all roots α . Then we have $[h, x] = \alpha(h)x = 0$ for all $x \in L_\alpha$ and for all roots α . Since H is abelian, it follows from the root space decomposition that $h \in Z(L)$, which is zero as L is semisimple.

In a sense, (ii) is just a reformulation of (i) in the language of linear algebra. Let $W \subset H^*$ denote the span of Φ . Suppose that W is a proper subspace of H^* . Then the annihilator of W in H ,

$$W^\circ := \{h \in H : \delta(h) = 0 \text{ for all } \delta \in W\},$$

has dimension $\dim H - \dim W > 0$. (See Appendix A.) Therefore there is some non-zero $h \in H$ such that $\delta(h) = 0$ for all $\delta \in W$, so in particular $\alpha(h) = 0$ for all $\alpha \in \Phi$, in contradiction to part (i). $\square$

Remark 10.6.2 — Note that this needn't be a basis, and it generally is not true. For instance, even in $\mathfrak{sl}(2, \mathbb{C})$, H^* is 1-dimensional, but we have $\alpha, -\alpha \in H^*$ already.

In the previous section, we found that the elements h_α and $h_{-\alpha}$ spanned the same 1-dimensional subspace of L . More precisely, we have the following.

Lemma 10.6.3

For each $\alpha \in \Phi$, we have

$$t_\alpha = \frac{h_\alpha}{\kappa(e_\alpha, f_\alpha)} \quad \text{and} \quad h_\alpha = \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)},$$

and

$$\kappa(t_\alpha, t_\alpha)\kappa(h_\alpha, h_\alpha) = 4.$$

Proof. We have that $h_\alpha = [e_\alpha, f_\alpha] = \kappa(e_\alpha, f_\alpha)t_\alpha$, which gives the first statement. Next, we have

$$\kappa(t_\alpha, h_\alpha) = \alpha(h_\alpha) = 2.$$

However,

$$\kappa(t_\alpha, h_\alpha) = \kappa(t_\alpha, \kappa(e_\alpha, f_\alpha)t_\alpha) = \kappa(e_\alpha, f_\alpha)\kappa(t_\alpha, t_\alpha).$$

Therefore,

$$2 = \kappa(e_\alpha, f_\alpha) \cdot \kappa(t_\alpha, t_\alpha),$$

so

$$h_\alpha = \kappa(e_\alpha, f_\alpha)t_\alpha = \frac{2}{\kappa(t_\alpha, t_\alpha)}t_\alpha.$$

This gives the second formula. For the third, we have that

$$\kappa(h_\alpha, h_\alpha) = \frac{4}{\kappa(t_\alpha, t_\alpha)^2}\kappa(t_\alpha, t_\alpha).$$

$\square$

Corollary 10.6.4

If α and β are roots, then $\kappa(h_\alpha, h_\beta) \in \mathbb{Z}$ and $\kappa(t_\alpha, t_\beta) \in \mathbb{Q}$.

Proof. We have $\text{tr}(\text{ad}h_\alpha \circ \text{ad}h_\beta)$. Consider the basis for L given by a basis for H , and the e_α for $\alpha \in \Phi$. Thus, we get

$$\kappa(h_\alpha, h_\beta) = \sum_{\gamma \in \Phi} \gamma(h_\alpha) \gamma(h_\beta).$$

Note that the RHS are just the eigenvalues of $\text{ad}h_\alpha$ (or $\text{ad}h_\beta$) on e_γ . Since the eigenvalues of h_α and h_β are integers, this shows that $\kappa(h_\alpha, h_\beta) \in \mathbb{Z}$.

We now use the previous lemma to get

$$\begin{aligned} \kappa(t_\alpha, t_\beta) &= \kappa\left(\frac{\kappa(t_\alpha, t_\alpha)h_\alpha}{2}, \frac{\kappa(t_\beta, t_\beta)h_\beta}{2}\right) \\ &= \frac{\kappa(t_\alpha, t_\alpha)\kappa(t_\beta, t_\beta)}{4} \kappa(h_\alpha, h_\beta) \\ &= \frac{4\kappa(h_\alpha, h_\beta)}{\kappa(h_\alpha, h_\alpha)\kappa(h_\beta, h_\beta)} \in \mathbb{Q}. \end{aligned}$$

□

We can translate the Killing form on H to obtain a non-degenerate symmetric bilinear form on H^* , denoted $(\cdot, \cdot): H^* \times H^* \rightarrow \mathbb{C}$. This form can be defined by

$$(x, y) = \kappa(t_x, t_y),$$

where t_x and t_y are the elements of H corresponding to x and y under the isomorphism $H \cong H^*$ induced by κ : $\kappa(t_x, h) = x(h)$. In particular, if α and β are roots, then

$$(\alpha, \beta) = \kappa(t_\alpha, t_\beta) \in \mathbb{Q}.$$

Remark 10.6.5 — We verify here that this pairing is indeed symmetric and nondegenerate.

$$(x, y) = \kappa(t_x, t_y) = x(t_y) = y(t_x) = \kappa(t_y, t_x) = (y, x)$$

Further, non-degeneracy follows from the fact that κ is nondegenerate: if $(x, y) = 0$ for all y , then $\kappa(t_x, t_y) = 0$, for all y , so in particular, $\alpha(t_x) = 0$ for all α . This implies that $t_x = 0$, i.e. $x = 0$. The bilinearity is much easier.

Exercise 10.6.6. Show that $\beta(h_\alpha) = \frac{2(\beta, \alpha)}{(\alpha, \alpha)}$. Later, we define $\frac{2(\beta, \alpha)}{(\alpha, \alpha)} = \langle \beta, \alpha \rangle$.

Solution: We note that

$$\beta(h_\alpha) = \kappa(t_\beta, h_\alpha) = \kappa\left(t_\beta, \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}\right) = 2 \frac{\kappa(t_\beta, t_\alpha)}{\kappa(t_\alpha, t_\alpha)} = 2 \frac{(\beta, \alpha)}{(\alpha, \alpha)}.$$

□

We saw in Lemma 10.6.1 that the roots of L span H^* , so H^* has a vector space basis consisting of roots, say $\{\alpha_1, \alpha_2, \dots, \alpha_\ell\}$, with each $\alpha_i \in \Phi$. We can now prove that something stronger is true as follows.

Lemma 10.6.7

If β is a root, then β is a linear combination of the α_i with coefficients in $\mathbb{Q}$.

Proof. Certainly we may write $\beta = \sum_{i=1}^{\ell} c_i \alpha_i$ with coefficients $c_i \in \mathbb{C}$. For each j with $1 \leq j \leq \ell$, we have

$$(\beta, \alpha_j) = \sum_{i=1}^{\ell} (\alpha_i, \alpha_j) c_i,$$

and $(\alpha_i, \alpha_j) \in \mathbb{Q}$ because $(\alpha_i, \alpha_j) = \kappa(t_{\alpha_i}, t_{\alpha_j})$. We can write these equations in matrix form as

$$\begin{pmatrix} (\beta, \alpha_1) \\ \vdots \\ (\beta, \alpha_{\ell}) \end{pmatrix} = \begin{pmatrix} (\alpha_1, \alpha_1) & \dots & (\alpha_{\ell}, \alpha_1) \\ \vdots & \ddots & \vdots \\ (\alpha_1, \alpha_{\ell}) & \dots & (\alpha_{\ell}, \alpha_{\ell}) \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_{\ell} \end{pmatrix}.$$

The matrix is the matrix of the non-degenerate bilinear form $(\cdot, \cdot)$ with respect to the chosen basis of roots $\{\alpha_1, \dots, \alpha_{\ell}\}$, so it's invertible. The entries are in $\mathbb{Q}$, so it has an inverse with entries in $\mathbb{Q}$. Thus, the c_i are rational. $\square$

By this lemma, the *real* subspace of H^* spanned by the roots $\alpha_1, \dots, \alpha_{\ell}$ contains all the roots of Φ and so does not depend on our particular choice of basis. Let E denote this subspace.

Proposition 10.6.8

The form $(\cdot, \cdot)$ is a real-valued inner product on $E = \text{Span}_{\mathbb{R}}\{\alpha_1, \dots, \alpha_{\ell}\}$. That is, it is positive definite symmetric bilinear.

Proof. Since $(\cdot, \cdot)$ is a symmetric bilinear form, we only need to check that the restriction of $(\cdot, \cdot)$ to E is positive definite. Let $\theta \in E$ correspond to $t_{\theta} \in H$. Using the root space decomposition and the fact that $(\text{ad } t_{\theta})e_{\beta} = \beta(t_{\theta})e_{\beta}$, we get

$$(\theta, \theta) = \kappa(t_{\theta}, t_{\theta}) = \sum_{\beta \in \Phi} \beta(t_{\theta})^2 = \sum_{\beta \in \Phi} \kappa(t_{\beta}, t_{\theta})^2 = \sum_{\beta \in \Phi} (\beta, \theta)^2.$$

As (β, θ) is real, the right-hand side is real and non-negative. Moreover, if $(\theta, \theta) = 0$, then $\beta(t_{\theta}) = 0$ for all roots β , so by Lemma 10.6.1, $\theta = 0$. $\square$

§10.7 Exercises

Exercise 10.7.1. Suppose that L is a complex semisimple Lie algebra with Cartan subalgebra H and root system Φ . Use the results of this chapter to prove that

$$\dim L = \dim H + |\Phi|.$$

Hence show that there are no semisimple Lie algebras of dimensions 4, 5, or 7.

Solution: L can be written as

$$L = H \oplus \bigoplus_{\alpha \in \Phi} L_{\alpha},$$

and we know that $\dim L_{\alpha} = 1$, so $\dim L = \dim H + |\Phi|$.

First, there are no semisimple Lie algebras of dimension 4, because then the possibilities of $\dim H$ are 0, 2, 4. $\dim H \neq 4$, for then L is abelian and $\dim H \neq 0$ because a nonzero Cartan subalgebra always exists. Thus, $\dim H = 2$. Since $\text{Span } \Phi = H^*$ and $\dim H^* = 2$, it’s not possible for Φ to contain just one set of linearly independent roots.

Next, dimension 5. It follows that $|\Phi| = 2$ or 4. If $|\Phi| = 2$, then $\dim H^* = 3$, so not possible because one root cannot span a space of dimension 3. If $|\Phi| = 4$, then $\dim H^* = \dim H = 1$. However, rank one root systems only have one scalar multiple, but we had two non-scalar multiple roots. A contradiction.

Finally, dimension 7. Here, Φ has size 2, 4, 6. If $|\Phi| = 2$, then $\dim H^* = 5$, so not enough. Similarly, if $|\Phi| = 4$, then $\dim H^* = 3$, so again not enough. Finally, if $|\Phi| = 6$, then $\dim H^* = 1$, so Φ can’t span H^* because the dimension is too small. $\square$

Exercise 10.7.2. Show that the set of diagonal matrices in $\mathfrak{so}(4, \mathbb{C})$ forms a Cartan subalgebra of $\mathfrak{so}(4, \mathbb{C})$ and determine the corresponding root space decomposition. Hence show that

$$\mathfrak{so}(4, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C}).$$

Solution: Recall that

$$\mathfrak{so}(4, \mathbb{C}) = \mathfrak{gl}_S(n, \mathbb{C}) = \{x \in \mathfrak{gl}(n, \mathbb{C}) : x^T S = -Sx\},$$

where

$$S = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}$$

Given $x = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we find conditions on a, b, c, d . We get that

$$\begin{pmatrix} c^T & a^T \\ d^T & b^T \end{pmatrix} = \begin{pmatrix} -c & -d \\ -a & -b \end{pmatrix}.$$

Therefore,

$$\mathfrak{so}(4, \mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a = -d^T, b = -b^T, c = -c^T \right\}.$$

We can therefore calculate that $\dim \mathfrak{so}(4, \mathbb{C}) = 6$. The set of diagonal matrices in $\mathfrak{so}(4, \mathbb{C})$ are matrices of the form $h = \text{diag}(a, b, -a, -b)$. This definitely forms an abelian algebra, and $\text{ad } h$ certainly contains semisimple elements. Call this set H for now. This subalgebra is of dimension 2. If H is of bigger dimension, then $\dim H$ must be 4, so $|\Phi| = 2$, so Φ cannot span H^* for dimension reasons. Therefore, H is maximal and so a Cartan subalgebra of $\mathfrak{so}(4, \mathbb{C})$. Let

$$\varepsilon_1 : (a, b, -a, -b) \mapsto a, \quad \varepsilon_2 : (a, b, -a, -b) \mapsto b.$$

Then, we can find that each matrix corresponding to a ‘free variable’ is an eigenvector, and the roots are

$$\{\pm(\varepsilon_1 - \varepsilon_2), \pm(\varepsilon_1 + \varepsilon_2)\}.$$

Setting $\alpha = \varepsilon_1 - \varepsilon_2$ and $\beta = \varepsilon_1 + \varepsilon_2$. Then, for instance

$$L_\beta = \text{Span}\{E_{14} - E_{23} =: x_\beta\}, L_{-\beta} = \text{Span}\{E_{41} - E_{32} =: x_{-\beta}\}.$$

The element h_β is then the bracket of these two:

$$h_\beta := [E_{14} - E_{23}, E_{41} - E_{32}] = E_{11} + E_{22} - E_{33} - E_{44}.$$

Similarly, we can see that

$$L_\alpha = \text{Span}\{E_{12} - E_{43} =: x_\alpha\}, L_{-\alpha} = \text{Span}\{E_{21} - E_{34} =: x_{-\alpha}\}.$$

Thus,

$$h_\alpha := [E_{12} - E_{43}, E_{21} - E_{34}] = E_{11} - E_{22} - E_{33} + E_{44}.$$

This provides the decomposition then

$$L = \mathfrak{sl}(\alpha) \oplus \mathfrak{sl}(\beta).$$

Indeed, the intersection is trivial, and we can calculate that the structure constants line up with those of $\mathfrak{sl}(\alpha)$. $\square$

Exercise 10.7.3. In Lemma 10.4.2, we defined for each $\alpha \in \Phi$ a subalgebra of L isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. In the main step in the proof, we showed that if $x \in L_\alpha$ and $y \in L_{-\alpha}$, and $h = [x, y] \neq 0$, then $\alpha(h) \neq 0$. Here is an alternative proof of this using root string modules.

Suppose that $\alpha(h) = 0$. Let $\beta \in \Phi$ be any root. Let M be the α -root string module through β ,

$$M = \bigoplus_c L_{\beta+c\alpha}.$$

By considering the trace of h on M , show that $\beta(h) = 0$ and hence get a contradiction.

Solution: First note that there should only be finitely many such $L_{\beta+c\alpha}$ because there are finitely many roots. Consider the action of $\text{ad } h$ on M and focus on the trace. Then,

$$\text{tr}(\text{ad } h) = \sum_c (\beta + c\alpha)(h) = k \cdot \beta(h),$$

where k is some finite number. We now analyze the trace of h on M . By Weyl's theorem, M decomposes into a direct sum of irreducible submodules V_d s, and the trace of the action of h on each irreducible submodule is 0, since the eigenvalues of h on such a submodule ranges from $d, d-2, \dots, -d$. Therefore, the action of h on M is zero. Thus, $0 = k \cdot \beta(h)$, so $\beta(h) = 0$. This is a contradiction, for then we found an element h which vanishes on all roots in Φ . $\square$

Exercise 10.7.4. Let L be a complex semisimple Lie algebra with Cartan subalgebra H and root space Φ . Let $\alpha \in \Phi$ and let

$$\mathfrak{sl}(\alpha) = \text{Span}\{e_\alpha, f_\alpha, h_\alpha\}$$

be the corresponding subalgebra constructed in this chapter. Show that this subalgebra is unique up to

1. scaling basis elements as $ce_\alpha, c^{-1}f_\alpha, h_\alpha$ for non-zero $c \in \mathbb{C}$; and
2. swapping e_α and f_α and then replacing h_α with $-h_\alpha$.

Solution: This just amounts to checking that the structure constants remain the same with these changes. $\square$

Exercise 10.7.5. Let L be a semisimple Lie algebra and let Φ be its set of roots. Let $\alpha \in \Phi$ and let

$$N := \text{Span}\{f_\alpha\} \oplus \text{Span}\{h_\alpha\} \oplus L_\alpha \oplus L_{2\alpha} \oplus \cdots.$$

Show that N is an $\mathfrak{sl}(\alpha)$ -submodule of L . By considering the trace of $h_\alpha: N \rightarrow N$, give an alternative proof of Proposition 10.5.6.

Solution: This is an $\mathfrak{sl}(\alpha)$ -submodule, since the action of $e_\alpha, f_\alpha, h_\alpha$ keeps N closed. This is clear for e_α and h_α , but we should check for f_α . We have that

$$[f_\alpha, L_{m\alpha}] \subseteq [L_{-\alpha}, L_{m\alpha}] \subseteq L_{(m-1)\alpha},$$

so we have closure. Particularly, for $m = 1$, we have that $[L_{-\alpha}, L_\alpha] \subseteq \text{Span}\{t_\alpha\} = \text{Span}\{h_\alpha\}$, so we are fine.

Now consider the trace of the action of h_α . We first note that the action of h_α on N must have trace zero, since

$$\text{tr}(\text{ad } h_\alpha) = \text{tr}(\text{ad } [e_\alpha, f_\alpha]) = \text{tr}(\text{ad } e_\alpha, \text{ad } f_\alpha - \text{ad } f_\alpha \text{ad } e_\alpha) = 0.$$

Next, the trace of $\text{ad } h$ is equal to $-2 + 0 + 2 \cdot \sum_{m>0} m \cdot \dim(L_{m\alpha})$. Therefore, we must have that $\dim L_\alpha = 1$ and m to not exceed 1. This proves that the only positive integral multiple of α in Φ is α . We can do the same for negatives. To show that these are the only possible multiples, one can just proceed from the start of Proposition 10.5.6 and run through the same argument. $\square$

11 Root Systems

§11.1 Definitions

Let E be a finite dimensional real vector space endowed with an inner product $(\cdot, \cdot)$; that is, symmetric, bilinear, positive definite. In our interest, we will set $E = \mathbb{R}^n$, so that the inner product will be isomorphic to the usual dot product $(v, w) = v \cdot w$. Given a nonzero vector $v \in E$, we let s_v be the reflection in the hyperplane normal to v , so that $s_v(v) = -v$. Further, s_v fixes all elements y such that $(y, v) = 0$. We can then provide an explicit formula for s_v :

$$s_v(x) = x - \frac{2(x, v)}{(v, v)}v, \quad \forall x \in E$$

Further, s_v preserves the inner product:

$$(s_v(x), s_v(y)) = (x, y) \quad \forall x, y \in E$$

Notation 11.1.1. We denote $\langle x, v \rangle := \frac{2(x, v)}{(v, v)}$.

Remark 11.1.2 — This is not symmetric, and is only linear in the first variable, x .

With this notation, we can now define root systems.

Definition 11.1.3. Let E be a real inner-product space. A **root system** R in E is a subset $R \subset E$ of **roots** if it satisfies the following axioms.

- (R1) R is finite, it spans E , and $0 \notin R$.
- (R2) If $\alpha \in R$, then the only scalar multiples of α in R are $\pm\alpha$.
- (R3) If $\alpha \in R$, then the reflection s_α permutes the elements of R .
- (R4) If $\alpha, \beta \in R$, then $\langle \beta, \alpha \rangle \in \mathbb{Z}$.

Remark 11.1.4 — We don't need to really specify E at the beginning, since R spans E .

Example 11.1.5 (The Root Space Decomposition)

Let L be a complex semisimple Lie algebra, and suppose that Φ be the set of roots of L with respect to some fixed Cartan subalgebra H . Let E denote the real span of Φ , $E = \mathbb{R} \cdot \Phi$. By Proposition 10.6.8, the Killing form on L restricts to an inner product on E . The claim is that Φ is a root system.

We can use the results of the previous chapter to show that Φ is a root system in E . By definition, $0 \notin \Phi$ and, as we observed early on, Φ is finite. We showed that the (R2) holds in Proposition 10.5.6. To show that the third holds, we note that if $\alpha, \beta \in \Phi$ then

$$s_\alpha(\beta) = \beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)}\alpha = \beta - \beta(h_\alpha)\alpha,$$

which lies in Φ by Proposition 10.5.7. To get the second equality above, we used the identity of Exercise 10.6.6, which may be proved as follows:

$$\beta(h_\alpha) = \kappa(t_\beta, h_\alpha) = \kappa\left(t_\beta, \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}\right) = \frac{2(\beta, \alpha)}{(\alpha, \alpha)} = \langle \beta, \alpha \rangle.$$

As the eigenvalues of h_α are integers, this identity also establishes fourth property.

Exercise 11.1.6. We work in $\mathbb{R}^{\ell+1}$, with the Euclidean inner product. Let ε_i be the vector in $\mathbb{R}^{\ell+1}$ with i -th entry 1 and all other entries zero. Define

$$R := \{\pm(\varepsilon_i - \varepsilon_j) : 1 \leq i < j \leq \ell + 1\}$$

and let $E = \text{Span } R = \{\sum \alpha_i \varepsilon_i : \sum \alpha_i = 0\}$. Show that R is a root system in E .

Solution: R is clearly finite, and E is defined as $\text{Span } R$. We just need to show that $0 \notin R$, but this is also clear. Next, we need to show that if $r \in R$, then the only multiples of r are $\pm r$. This is also very clear. Next, we check that the reflection permutes elements of R .

$$\begin{aligned} s_{\pm(\varepsilon_i - \varepsilon_j)}(\pm(\varepsilon_k - \varepsilon_l)) &= \pm(\varepsilon_k - \varepsilon_l) - 2 \frac{(\pm(\varepsilon_k - \varepsilon_l), \pm(\varepsilon_i - \varepsilon_j))}{(\pm(\varepsilon_i - \varepsilon_j), \pm(\varepsilon_i - \varepsilon_j))} (\pm(\varepsilon_i - \varepsilon_j)) \\ &= \pm(\varepsilon_k - \varepsilon_l) - (\pm(\varepsilon_k - \varepsilon_l), \pm(\varepsilon_i - \varepsilon_j)) \cdot (\pm(\varepsilon_i - \varepsilon_j)). \end{aligned}$$

This is true...CHECK. The final condition we check is that $\langle \pm(\varepsilon_i - \varepsilon_j), \pm(\varepsilon_k - \varepsilon_l) \rangle \in \mathbb{Z}$. This just suffices to check that

$$\frac{(\pm(\varepsilon_k - \varepsilon_l), \pm(\varepsilon_i - \varepsilon_j))}{(\pm(\varepsilon_i - \varepsilon_j), \pm(\varepsilon_i - \varepsilon_j))} \in \mathbb{Z}.$$

However, this is just

$$(\pm(\varepsilon_k - \varepsilon_l), \pm(\varepsilon_i - \varepsilon_j)),$$

which is definitely in $\mathbb{Z}$. □

Remark 11.1.7 — We shall see that our axioms isolate all the essential properties of roots of Lie algebras. For this reason, there is no need in this chapter to keep the full body of theory we have developed in mind—doing so would burden us with extraneous notions while needlessly reducing the applicability of our arguments. In any case, we shall see later that every root system is the set of roots of a complex

semisimple Lie algebra, so our problem is no more general than is necessary.

§11.2 First Steps in the Classification

The following lemma gives the first indication that the axioms for root systems are quite restrictive.

Lemma 11.2.1 (Finiteness Lemma)

Suppose that R is a root system in the real inner-product space E . Let $\alpha, \beta \in R$ with $\beta \neq \pm\alpha$. Then

$$\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle \in \{0, 1, 2, 3\}.$$

Proof. By (R4), the product is an integer, so we must establish the bounds. For any non-zero $v, w \in E$, the angle θ between v and w is such that

$$(v, w)^2 = (v, v)(w, w) \cos^2 \theta.$$

This gives

$$\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle = 4 \cos^2 \theta \leq 4.$$

Suppose we have $\cos^2 \theta = 1$. Then θ is an integer multiple of π and so α and β are linearly dependent, contrary to our assumption. Thus, the inequality is strict. $\square$

We now use this lemma to show that there are only a few possibilities for the integers $\langle \alpha, \beta \rangle$. Take two roots α, β in a root system R with $\alpha \neq \pm\beta$. We may choose the labeling so that $(\beta, \beta) \geq (\alpha, \alpha)$ and hence

$$|\langle \beta, \alpha \rangle| = \frac{2|(\beta, \alpha)|}{(\alpha, \alpha)} \geq \frac{2|(\alpha, \beta)|}{(\beta, \beta)} = |\langle \alpha, \beta \rangle|.$$

By the Finiteness Lemma, the possibilities are:

$\langle \alpha, \beta \rangle$	$\langle \beta, \alpha \rangle$	θ	$\frac{(\beta, \beta)}{(\alpha, \alpha)}$
0	0	$\pi/2$	undetermined
1	1	$\pi/3$	1
-1	-1	$2\pi/3$	1
1	2	$\pi/4$	2
-1	-2	$3\pi/4$	2
1	3	$\pi/6$	3
-1	-3	$5\pi/6$	3

Example 11.2.2

Consider when $E = \mathbb{R}$. There is only one possibility for a root system R then: $R = \{\alpha, -\alpha\}$, where $\alpha \neq 0$.

Remark 11.2.3 — There's a bijection between semisimple Lie algebras and root systems, and this particular example comes from $\mathfrak{sl}(\alpha)$.

Given roots α and β , we would like to know when their sum and difference lie in R . Our table gives some information about this question.

Proposition 11.2.4

Let $\alpha, \beta \in R$.

- (iv) If the angle between α and β is strictly obtuse ($\frac{\pi}{2} < \theta < \pi$), then $\alpha + \beta \in R$.
- (iv) If the angle between α and β is strictly acute ($0 < \theta < \frac{\pi}{2}$), then $\alpha - \beta \in R$ and $\beta - \alpha \in R$, since R is preserved by multiplication by -1 .

Proof. In either case, we may assume that $(\beta, \beta) \geq (\alpha, \alpha)$. By (R3), we know that

$$s_\beta(\alpha) = \alpha - \langle \alpha, \beta \rangle \beta$$

lies in R . If θ is strictly obtuse, the table shows that $\langle \alpha, \beta \rangle = -1$. If θ is strictly acute, then $\langle \alpha, \beta \rangle = 1$. Then, using the equation above, this shows the result. $\square$

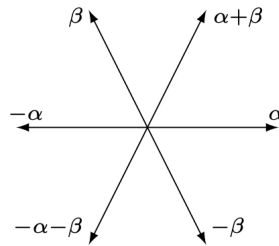
Definition 11.2.5. The **rank** of a root system is the dimension of E .

Example 11.2.6 (Root Systems of Rank 2)

Let $E = \mathbb{R}^2$ with the Euclidean inner product. We shall find all root systems R contained in E .

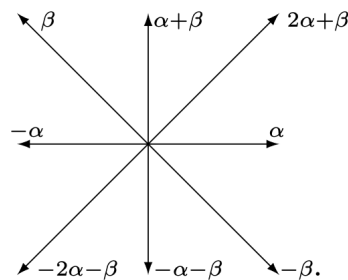
Solution: Take a root α of the shortest possible length. Since R spans E , it must contain some root $\beta \neq \pm\alpha$. By considering $-\beta$ if necessary, we may assume that β makes an obtuse angle with α . Moreover, we may assume that this angle, say θ , is as large as possible.

1. Suppose that $\theta = 2\pi/3$. Using Proposition 11.2.4, we find that R contains the six roots shown below.



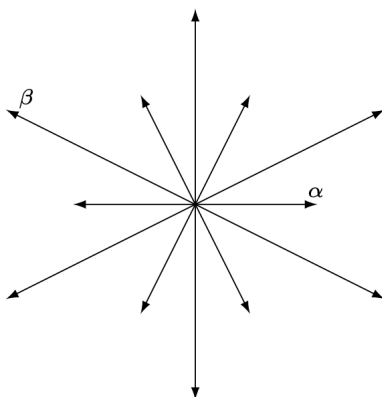
One can check that this set is closed under the action of the reflections $s_\alpha, s_\beta, s_{\alpha+\beta}$. As $s_{-\alpha} = s_\alpha$, and so on, this is sufficient to verify (R3). We have therefore found a root system in E . This root system is said to have **type** A_2 . (The 2 refers to the dimension of the underlying space.)

2. Suppose that $\theta = 3\pi/4$. Proposition 11.2.4 shows that $\alpha + \beta$ is a root, and applying s_α to β shows that $2\alpha + \beta$ is a root, so R must contain



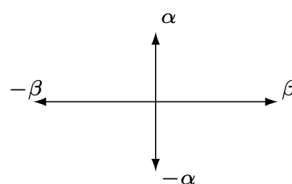
This root system is said to have **type** B_2 . A further root would make an angle of at most $\pi/8$ with one of the existing roots, so this must be all of R .

3. Suppose that $\theta = 5\pi/6$. We leave it to the reader to show that R must be



and to determine the correct labels for the remaining roots. This root system is said to have **type** G_2 .

4. Suppose that β is perpendicular to α . This gives us the root system of **type** $A_1 \times A_1$.



Here, as $(\alpha, \beta) = 0$, the reflection s_α fixes the roots $\pm\beta$ lying in the space perpendicular to α , so there is no interaction between the roots $\pm\alpha$ and $\pm\beta$. In particular, knowing the length of α tells us nothing about the length of β . This corresponds to the Lie algebra $\mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$.

□

Remark 11.2.7 — In constructing the diagrams above, we've chosen α and β to create the largest angle possible (less than π). This is because if they make an obtuse angle, then $\alpha + \beta$ is in the root system. The reason we do this is because we will show that all the roots can have all positive integer (or all negative) coefficients.

These considerations suggest the following definition.

Definition 11.2.8. The root system R is **irreducible** if R cannot be expressed as a disjoint union of two non-empty subsets $R_1 \cup R_2$ such that $(\alpha, \beta) = 0$ for all $\alpha \in R_1$ and $\beta \in R_2$.

Note that if such a decomposition exists, then R_1 and R_2 are root systems in their respective spans. The next lemma tells us that it will be enough to classify the irreducible root systems.

Lemma 11.2.9

Let R be a root system in the real vector space E . We may write R as a disjoint union

$$R = R_1 \cup R_2 \cup \cdots \cup R_k,$$

where each R_i is an irreducible root system in the space $E_i := \text{Span}_{\mathbb{R}}(R_i)$, and $E = E_1 \oplus \cdots \oplus E_k$.

Proof. Define an equivalence relation $\sim$ on R by letting $\alpha \sim \beta$ if there exist $\gamma_1, \gamma_2, \dots, \gamma_s$ in R with $\alpha = \gamma_1$ and $\beta = \gamma_s$ such that $(\gamma_i, \gamma_{i+1}) \neq 0$ for $1 \leq i < s$. Let the R_i be the equivalence classes for this relation. It is clear that they satisfy axioms (R1), (R2), and (R4); you are asked to check (R3) in the following exercise. That each R_i is irreducible follows immediately from the construction.

As every root appears in some E_i , the sum of the E_i spans E . Suppose that $v_1 + \cdots + v_k = 0$, where $v_i \in E_i$. Taking inner products with v_j , we get

$$0 = (v_1, v_j) + \cdots + (v_j, v_j) + \cdots + (v_k, v_j) = (v_j, v_j)$$

so each $v_j = 0$. Hence $E = E_1 \oplus \cdots \oplus E_k$. $\square$

Exercise 11.2.10. Show that if $(\alpha, \beta) \neq 0$, then $(\alpha, s_\alpha(\beta)) \neq 0$. Deduce that the equivalence classes defined in the proof of the lemma satisfy (R3).

Solution: We have that

$$\begin{aligned} (\alpha, s_\alpha(\beta)) &= (\alpha, \beta - 2 \frac{(\beta, \alpha)}{(\alpha, \alpha)} \alpha) = (\alpha, \beta) - 2(\alpha, \frac{(\beta, \alpha)}{(\alpha, \alpha)} \alpha) \\ &= (\alpha, \beta) - 2 \frac{(\beta, \alpha)}{(\alpha, \alpha)} (\alpha, \alpha) = (\alpha, \beta) - 2(\alpha, \beta) = -(\alpha, \beta). \end{aligned}$$

Now, we need to show that the equivalence class defined satisfies the third axiom, which is that for any $\alpha, \beta \in R_i$, $s_\alpha(\beta) \in R_i$. Now, since $\alpha, \beta \in R_i$, $\alpha \sim \beta$, so there exists some $\gamma_1, \dots, \gamma_k$ with $\gamma_1 = \alpha$ and $\gamma_k = \beta$ such that $(\gamma_i, \gamma_{i+1}) \neq 0$. A key property of s_α is that it preserves the inner product:

$$0 \neq (\gamma_i, \gamma_{i+1}) = (s_\alpha(\gamma_i), s_\alpha(\gamma_{i+1})).$$

Noting that $\gamma_1 = \alpha$ and that $s_\alpha(\alpha) = \alpha$ we have that

$$(\alpha, s_\alpha(\gamma_2)) \neq 0, \dots, (s_\alpha(\gamma_{k-1}), s_\alpha(\beta)) \neq 0.$$

Therefore, we have a new chain, implying that $\alpha \sim s_\alpha(\beta)$. Therefore, $s_\alpha(\beta) \in R_i$. $\square$

§11.3 Bases for Root Systems

Let R be a root system in the real inner-product space E . Because R spans E , any maximal linearly independent subset of R is a vector space basis for E . Proposition 11.5 suggests that it might be convenient if we could find such a subset where every pair of elements made an obtuse angle. In fact, we can ask for something stronger, as in the following.

Definition 11.3.1. A subset B of R is a **base** for the root system R if

(B1) B is a vector space basis for E , and

(B2) every $\beta \in R$ can be written as

$$\beta = \sum_{\alpha \in B} k_{\alpha} \alpha$$

with $k_{\alpha} \in \mathbb{Z}$, where all the non-zero coefficients k_{α} have the same sign.

Exercise 11.3.2. Show that if B is a base for a root system, then the angle between any two distinct elements of B is obtuse (that is, at least $\frac{\pi}{2}$).

Solution: Suppose that there exist two elements $\alpha, \beta \in B \subseteq R$ such that the angle between them is acute. Then, $\alpha - \beta \in R$, but their signs differ. This contradicts that every element in R can be written using the elements of B using the same signed coefficients. $\square$

Definition 11.3.3. We say that a root $\beta \in R$ is **positive** with respect to B if the coefficients given in (B2) are positive, and otherwise it is **negative** with respect to B .

Exercise 11.3.4. Let

$$R = \{\pm(\varepsilon_i - \varepsilon_j) : 1 \leq i < j \leq \ell + 1\}$$

be the root system in Exercise 11.1.6. Let $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$ for $1 \leq i \leq \ell$. Show that

$$B = \{\alpha_1, \dots, \alpha_{\ell}\}$$

is a base for R and find the positive roots.

Solution: We first show that B is a basis for the vector space E . We do this by first showing that E spans R . This is clear:

$$\pm(\varepsilon_i - \varepsilon_j) = \pm(\alpha_i + \alpha_{i+1} + \dots + \alpha_{j-1}).$$

Thus, B spans E . To show that this is a linearly independent set, suppose that

$$\lambda_1 \alpha_1 + \dots + \lambda_{\ell} \alpha_{\ell} = 0.$$

Then, by focusing on each entry, it follows that $\lambda_1 = \dots = \lambda_{\ell} = 0$. Thus, B is a vector space basis for E . We now must show that every element in R is either positive or negative. However, this is clear from our formula above, so we are done. $\square$

A natural way to label the elements of R as positive or negative is to fix a hyperplane of codimension 1 in E which does not contain any element of R and then to label the roots of one side of the hyperplane as positive and those on the other side as negative. Suppose that R has a base B compatible with this labeling. Then the elements of B must lie on the positive side of the hyperplane.

Theorem 11.3.5

Every root system has a base.

Proof. Let R be a root system in the real inner-product space E . We may assume that E has dimension at least 2, as the case $\dim E = 1$ is obvious. We may choose a vector $z \in E$ which does not lie in the perpendicular space of any of the roots; such a z exists, since $(z, x) = z \cdot x = \sum z_i x_i$, which is a hyperplane. We shall say that α is positive if $(z, \alpha) > 0$ and negative if $(z, \alpha) < 0$. Let R^+ be the set of $\alpha \in R$ which lie on the positive side of z . Let

$$B := \{\alpha \in R^+ : \alpha \text{ is not the sum of two elements in } R^+\}.$$

We claim that B is a base for R .

We first show that (B2) holds. If $\beta \in R$, then either $\beta \in R^+$ or $-\beta \in R^+$, so it is sufficient to prove that every $\beta \in R^+$ can be expressed as

$$\beta = \sum_{\alpha \in B} k_\alpha \alpha$$

for some $k_\alpha \in \mathbb{Z}$ with each $k_\alpha \geq 0$. Suppose this doesn’t hold. Then, amongst these ‘bad’ roots, we may choose an element $\beta \in R^+$ such that the inner product (z, β) is minimal. As $\beta \notin B$ and is positive, it must be a sum of two roots. Thus, there exist $\beta_1, \beta_2 \in R^+$ such that $\beta = \beta_1 + \beta_2$. By linearity,

$$(z, \beta) = (z, \beta_1) + (z, \beta_2)$$

is the sum of two positive numbers, and therefore

$$0 < (z, \beta_i) < (z, \beta)$$

for $i = 1, 2$. By minimality of (z, β) , it follows that β_i can be written as a positive integral linear combination, giving a contradiction.

It remains to show that B is linearly independent. First note that if α and β are distinct elements of B , then by Exercise 11.3.2 the angle between them must be obtuse. Suppose that $\sum_{\alpha \in B} r_\alpha \alpha = 0$, where $r_\alpha \in \mathbb{R}$. Collecting all the terms with positive coefficients to one side gives an element

$$x := \sum_{\alpha: r_\alpha > 0} r_\alpha \alpha = \sum_{\beta: r_\beta < 0} (-r_\beta) \beta.$$

Hence

$$(x, x) = \sum_{\alpha: r_\alpha > 0} \sum_{\beta: r_\beta < 0} r_\alpha (-r_\beta) (\alpha, \beta) \leq 0.$$

Note that for the final inequality, we have $(\alpha, \beta) \leq 0$, since if $(\alpha, \beta) > 0$, then $\alpha - \beta, \beta - \alpha \in R$ (since the angle they create are acute). Then, we can write $\alpha = \beta + (\alpha - \beta)$ and $\beta = \alpha + (\beta - \alpha)$, but one of these are a sum of two positive roots, which contradict $\alpha, \beta \in B$. It follows that $(x, x) \leq 0$, so $x = 0$. Therefore

$$0 = (x, z) = \sum_{\alpha: r_\alpha > 0} r_\alpha (\alpha, z) = \sum_{\beta: r_\beta < 0} (-r_\beta) (\beta, z).$$

This is a sum of positives, so we must have that $r_\alpha = 0$ for all α . □

Definition 11.3.6. Let R^+ denote the set of all positive roots in a root system R with respect to a base B , and let R^- be the set of all negative roots. Then $R = R^+ \sqcup R^-$, a disjoint union. The set B is contained in R^+ ; the elements of B are called **simple roots**. The reflections s_α for $\alpha \in B$ are known as **simple reflections**.

Remark 11.3.7 — A root system R will usually have many possible bases. For example, if B is a base then so is $\{-\alpha : \alpha \in B\}$. In particular, the terms ‘positive’ and ‘negative’ roots are always taken with reference to a fixed base B .

Exercise 11.3.8. Let R be a root system with a base B . Take any $\gamma \in R$. Show that the set $s_\gamma(B) = \{s_\gamma(\alpha) : \alpha \in B\}$ is also a base of R .

Solution: We need to show that $s_\gamma(B)$ first serves as a basis for $\text{Span } R$ as a vector space. This follows from the fact that s_γ is a linear isomorphism of E . Next, we show that every element in R can be written using either all positive coefficients or negative. Suppose we can write $r \in R$. Then, s_γ permutes R , so there exists some element $r' \in R$ such that $s_\gamma(r') = r$ and

$$r' = \sum_{\beta \in B} d_\beta \beta,$$

with every d_β being of the same sign. Then, we have that

$$r = s_\gamma(r') = \sum_{\beta \in B} d_\beta s_\gamma(\beta),$$

so we are done. □

Remark 11.3.9 — There is a converse to this exercise, which is explained in the next section.

§11.4 The Weyl Group of a Root System

For each root $\alpha \in R$, we have defined a reflection s_α which acts as an invertible linear map on E . We may therefore consider the group of invertible linear transformations of E generated by the reflections s_α for $\alpha \in R$.

Definition 11.4.1. This is known as the **Weyl group** of R and is denoted by W or $W(R)$.

Lemma 11.4.2

The Weyl group W associated to R is finite.

Proof. By axiom (R3), the elements of W permute R , so there is a group homomorphism from W into the group of all permutations of R , which is a finite group because R is finite. We claim that this homomorphism is injective, and so W is finite.

Suppose that $g \in W$ is in the kernel of this homomorphism. Then, by definition, g fixes all the roots in R . But E is spanned by the roots, so g fixes all elements in a basis of E , and so g must be the identity map. □

§11.5 Recovering the Roots

Suppose that we are given a base B for a root system R . We shall show that this alone gives us enough information to recover R . To do this, we use the Weyl group and prove

that every root β is of the form $\beta = g(\alpha)$ for some $\alpha \in B$ and some g in the subgroup $W_0 := \langle s_\gamma : \gamma \in B \rangle$ of W . (We shall also see that $W = W_0$.) Thus, if we repeatedly apply reflections in the simple roots, we will eventually recover the full root system.

Some evidence for this statement is given by Example 11.2.6: in each case we started with a pair of roots $\{\alpha, \beta\}$, and knowing only the positions of α and β , we used repeated reflections to construct the unique root system containing these roots as a base.

Lemma 11.5.1

If $\alpha \in B$, then the reflection s_α permutes the set of positive roots other than α .

Proof. Suppose that $\beta \in R^+$ and $\beta \neq \alpha$. We know that

$$\beta = \sum_{\gamma \in B} k_\gamma \gamma \quad \text{for some } k_\gamma \geq 0.$$

Since $\beta \neq \alpha$ and $\beta \in R$, there is some $\gamma \in B$ with $k_\gamma \neq 0$ and $\gamma \neq \alpha$. We know that $s_\alpha(\beta) \in R$; and from

$$s_\alpha(\beta) = \beta - \langle \beta, \alpha \rangle \alpha$$

we see that the coefficient of γ in $s_\alpha(\beta)$ is k_γ , which is positive. As all the non-zero coefficients in the expression of $s_\alpha(\beta)$ as a linear combination of base elements must have the same sign, this tells us that $s_\alpha(\beta)$ lies in R^+ . $\square$

Proposition 11.5.2

Suppose that $\beta \in R$. There exists $g \in W_0$ and $\alpha \in B$ such that $\beta = g(\alpha)$.

Proof. Suppose first of all that $\beta \in R^+$ and that $\beta = \sum_{\gamma \in B} k_\gamma \gamma$ with $k_\gamma \in \mathbb{Z}$, $k_\gamma \geq 0$. We shall proceed by induction on the height of β defined by

$$\text{ht}(\beta) = \sum_{\gamma \in B} k_\gamma.$$

If $\text{ht}(\beta) = 1$, then $\beta \in B$, so we may take $\alpha = \beta$ and let g be the identity map. For the inductive step, suppose that $\text{ht}(\beta) = n \geq 2$. By axiom (R2), at least two of the k_γ are strictly positive.

We claim that there is some $\gamma \in B$ such that $\langle \beta, \gamma \rangle > 0$. If not, then $\langle \beta, \gamma \rangle \leq 0$ for all $\gamma \in B$ and so

$$(\beta, \beta) = \sum_{\gamma} k_\gamma (\beta, \gamma) \leq 0,$$

which is a contradiction because $\beta \neq 0$. We may therefore choose some $\gamma \in B$ with $\langle \beta, \gamma \rangle > 0$. Then $\langle \beta, \gamma \rangle > 0$ and so

$$\text{ht}(s_\gamma(\beta)) = \text{ht}(\beta) - \langle \beta, \gamma \rangle < \text{ht}(\beta).$$

(We have $s_\gamma(\beta) \in R^+$ by the previous lemma.) The inductive hypothesis now implies that there exists $\alpha \in B$ and $h \in W_0$ such that $s_\gamma(\beta) = h(\alpha)$. Hence $\beta = s_\gamma(h(\alpha))$ because $s_\gamma^2 = \text{id}$. Therefore, we may take $g = s_\gamma h$, which lies in W_0 .

Now suppose that $\beta \in R^-$, so $-\beta \in R^+$. By the first part, $-\beta = g(\alpha)$ for some $g \in W_0$ and $\alpha \in B$. By linearity of g , we get $\beta = g(-\alpha) = g(s_\alpha(\alpha))$, where $gs_\alpha \in W_0$. $\square$

We end this section by proving that a base for a root system determines its full Weyl group. We need the following straightforward result.

Exercise 11.5.3. Suppose that α is a root and that $g \in W$. Show that $gs_\alpha g^{-1} = s_{g\alpha}$.

Solution: Let $\beta \in R$ be arbitrary. The LHS gives as follows:

$$gs_\alpha g^{-1}(\beta) = g(g^{-1}\beta - \langle g^{-1}\beta, \alpha \rangle \alpha) = \beta - \langle \beta, g\alpha \rangle g\alpha.$$

In the above, we used that $\langle g^{-1}\beta, \alpha \rangle = \langle \beta, g\alpha \rangle$ because $(gu, gv) = (u, v)$, since $g \in W$, which is the group generated by reflections. The RHS gives

$$s_{g\alpha}(\beta) = \beta - \langle \beta, g\alpha \rangle g\alpha.$$

Thus, the two sides agree and so we are done. $\square$

Lemma 11.5.4

We have $W_0 = W$; that is, W is generated by the s_α for $\alpha \in B$.

Proof. By definition, W is generated by the reflections s_β for $\beta \in R$, so it is sufficient to prove that $s_\beta \in W_0$ for any $\beta \in R$. By Proposition 11.5.2, we know that given β there is some $g \in W_0$ and $\alpha \in B$ such that $\beta = g(\alpha)$. Now the reflection s_β is equal to $gs_\alpha g^{-1} \in W_0$ by the previous exercise. $\square$

§11.6 Cartan Matrices and Dynkin Diagrams

Although in general a root system can have many different bases, the following theorem shows that from a geometric point of view they are all essentially the same. As the proof of this theorem is slightly technical, we postpone it to Appendix D.

Theorem 11.6.1

Let R be a root system and suppose that B and B' are two bases of R . Then there exists an element g in the Weyl group $W(R)$ such that $B' = g(B) = \{g(\alpha) : \alpha \in B\}$. $\square$

Remark 11.6.2 — The main point of all of the above is that everything we do is independent (up to isomorphism) of the choice of a base.

Definition 11.6.3. Let B be a base in a root system R . Fix an order on the elements of B , say $(\alpha_1, \dots, \alpha_\ell)$. The **Cartan matrix** of R is defined to be the $\ell \times \ell$ matrix with ij -th entry $(\langle \alpha_i, \alpha_j \rangle)_{ij}$.

Remark 11.6.4 — Note that this matrix is unique to B up to reordering. We can say more. This is independent of B , in the sense that any other basis B' is of the form $g(B) = \{g(\alpha_1), \dots, g(\alpha_\ell)\}$ for some $g \in W$, and $\langle g(\alpha_i), g(\alpha_j) \rangle = \langle \alpha_i, \alpha_j \rangle$.

By the remark above, it follows from Theorem 11.6.1 that the Cartan matrix depends only on the ordering adopted with our chosen base B and not on the base itself. Note that by (R4) the entries of the Cartan matrix are integers.

Example 11.6.5

Consider the example A_2 . Then, the Cartan matrix with respect to the base α_1, α_2 is

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

Example 11.6.6

Let R be as in Exercise 11.3.4. Calculation shows that the Cartan matrix with respect to the ordered base $(\alpha_1, \dots, \alpha_\ell)$ is

$$\begin{pmatrix} 2 & -1 & 0 & \dots & 0 & 0 \\ -1 & 2 & -1 & \dots & 0 & 0 \\ 0 & -1 & 2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 2 & -1 \\ 0 & 0 & 0 & \dots & -1 & 2 \end{pmatrix}.$$

Example 11.6.7

Let R be the root system which we have drawn in Example 11.2.6 (2); that is, the root system with type B_2 . This has ordered base (α, β) , and the corresponding Cartan matrix is

$$C = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}.$$

Remark 11.6.8 — In general, we should get diagonal entries 2, and negative entries on the non-diagonal, since if $\alpha_i, \alpha_j \in B$ ($i \neq j$), then $\langle \alpha_i, \alpha_j \rangle \leq 0$. Otherwise, $\alpha_i - \alpha_j$ or $\alpha_j - \alpha_i \in R$.

Definition 11.6.9. Let $d_{\alpha, \beta} := \langle \alpha, \beta \rangle \langle \beta, \alpha \rangle \in \{0, 1, 2, 3\}$. The **Dynkin diagram** of R is the graph whose vertices are the $\alpha_1, \dots, \alpha_\ell$ and the edges are given by drawing $d_{\alpha_i \alpha_j}$ lines between α_i and α_j for all $\alpha_i \neq \alpha_j$. By Theorem 11.6.1, the Dynkin diagram of R is independent of the choice of base. If $d_{\alpha_i \alpha_j} > 0$, we label the lines with an arrow from the longer of α_i, α_j to the shorter; that is, from the bigger of $(\alpha_i, \alpha_i), (\alpha_j, \alpha_j)$.

Definition 11.6.10. The graph with the same vertices and edges, but without the arrows, is known as the **Coxeter graph** of R .

Example 11.6.11

Using the base given in Exercise 11.5.3, the Dynkin diagram of the root system introduced in Exercise 11.1.6 is

$$\begin{array}{ccccccc} \alpha_1 & & \alpha_2 & & & & \alpha_{\ell-2} & & \alpha_{\ell-1} \\ \circ & \text{---} & \circ & & \dots & & \circ & \text{---} & \circ \end{array}$$

Example 11.6.12

The Dynkin diagram for the root system in Example 11.2.6(2) is



Exercise 11.6.13. Show that a root system is irreducible if and only if its Dynkin diagram is connected; that is, given any two vertices, there is a path joining them.

Solution. This follows from the remark that we have an edge between two vertices precisely when two roots are not orthogonal to each other. $\square$

Remark 11.6.14 — The data of the Cartan matrix and the data of a Dynkin diagram are exactly the same, since $\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle$ are in $\{0, 1, 2, 3\}$ and the each of the terms in the product are integers. These must both be negative, and the entry is determined by the arrow. In fact, more is true: The next section shows that a root system is essentially determined by its Dynkin diagram.

§11.7 Isomorphisms of Root Systems

Definition 11.7.1. Let R and R' be two root systems in real inner-product spaces E and E' , respectively. We say that R and R' are **isomorphic** if there is a vector space isomorphism $\varphi: E \rightarrow E'$ such that

1. $\varphi(R) = R'$, and
2. for any two roots $\alpha, \beta \in R$, $\langle \alpha, \beta \rangle = \langle \varphi(\alpha), \varphi(\beta) \rangle$.

Recall that if θ is the angle between roots α and β , then $4\cos^2\theta = \langle \alpha, \beta \rangle \langle \beta, \alpha \rangle$, so condition (b) says that φ should preserve angles between root vectors. For irreducible root systems, a stronger geometric characterization is possible—see Exercise ?? at the end of this chapter.

Example 11.7.2

Let R be a root system in the inner-product space E . We used that the reflection maps s_α for $\alpha \in R$ are isomorphisms (from R to itself) when we defined the Cartan matrix of a root system.

Example 11.7.3

An example of an isomorphism that is not distance preserving is given by scaling: For any non-zero $c \in \mathbb{C}$, the set $cR := \{c\alpha : \alpha \in R\}$ is a root system in E , and the map $\nu \mapsto c\nu$ induces an isomorphism between R and cR .

It follows immediately from the definition of isomorphism that isomorphic root systems have the same Dynkin diagram. We now prove that the converse holds.

Proposition 11.7.4

Let R and R' be root systems in the real vector spaces E and E' , respectively. Then, the Dynkin diagrams of R and R' are the same if and only if the root systems are isomorphic.

Proof. The converse is immediate by definition, as noted above. We may choose bases $B = \{\alpha_1, \dots, \alpha_\ell\}$ in R and $B' = \{\alpha'_1, \dots, \alpha'_\ell\}$ in R' so that for all i, j one has

$$\langle \alpha_i, \alpha_j \rangle = \langle \alpha'_i, \alpha'_j \rangle.$$

Let $\varphi: E \rightarrow E'$ be the linear map which maps α_i to α'_i . We first show the second condition.

Note that if $\alpha \in R$ then we can write $\alpha = \sum k_i \alpha_i$ for some $k_i \in \mathbb{Z}$. Since $\langle \cdot, \cdot \rangle$ is linear in its first component we have

$$\langle \varphi(\alpha), \varphi(\alpha_j) \rangle = \langle \alpha, \alpha_j \rangle. \quad (11.1)$$

This shows that the desired condition holds when the second entry is in B . Now observe that for any $\alpha_i, \alpha_j \in B$ we have

$$\varphi(s_{\alpha_i}(\alpha_j)) = \varphi(\alpha_j - \langle \alpha_j, \alpha_i \rangle \alpha_i) = \varphi(\alpha_j) - \langle \alpha_j, \alpha_i \rangle \varphi(\alpha_i) = \alpha'_j - \langle \alpha'_j, \alpha'_i \rangle \alpha'_i = s_{\alpha'_i}(\alpha'_j).$$

By linearity of φ it follows that

$$\varphi(s_{\alpha_i}(v)) = s_{\alpha'_i}(\varphi(v)) \quad (11.2)$$

for all $v \in E$. Now, we must deal how to get around the non-linearity of $\langle \cdot, \cdot \rangle$ in the second component. We will use the Weyl group for this. Let $\beta \in R$. By Proposition 11.5.2 there exist $\alpha_{i_1}, \dots, \alpha_{i_r} \in B$ and $\alpha_j \in B$ such that

$$s_{\alpha_{i_1}} \cdots s_{\alpha_{i_r}}(\alpha_j) = \beta.$$

By induction on r it follows from (11.2) that

$$\varphi(\beta) = \varphi(s_{\alpha_{i_1}} \cdots s_{\alpha_{i_r}}(\alpha_j)) = s_{\alpha'_{i_1}} \cdots s_{\alpha'_{i_r}}(\alpha'_j).$$

Hence for any $\alpha \in R$ we have

$$\langle \varphi(\alpha), \varphi(\beta) \rangle = \langle \varphi(\alpha), s_{\alpha'_{i_1}} \cdots s_{\alpha'_{i_r}}(\alpha'_j) \rangle.$$

For any $u, v \in E$ and $\alpha_j \in E$ we have

$$\langle u, s_{\alpha_j}(v) \rangle = \frac{2(u, s_{\alpha_j}(v))}{(s_{\alpha_j}(v), s_{\alpha_j}(v))} = \frac{2(s_{\alpha_j}(u), v)}{(v, v)} = \langle s_{\alpha_j}(u), v \rangle.$$

Hence

$$\langle \varphi(\alpha), \varphi(\beta) \rangle = \langle s_{\alpha'_{i_r}} \cdots s_{\alpha'_{i_1}}(\varphi(\alpha)), \alpha'_j \rangle.$$

It now follows from (11.2) that

$$\langle \varphi(\alpha), \varphi(\beta) \rangle = \langle \varphi(s_{\alpha_{i_r}} \cdots s_{\alpha_{i_1}}(\alpha)), \alpha'_j \rangle.$$

By (11.1) we get

$$\langle \varphi(\alpha), \varphi(\beta) \rangle = \langle s_{\alpha_{i_r}} \cdots s_{\alpha_{i_1}}(\alpha), \alpha_j \rangle.$$

By moving the reflections back to the right-hand side we get

$$\langle \varphi(\alpha), \varphi(\beta) \rangle = \langle \alpha, s_{\alpha_{i_1}} \cdots s_{\alpha_{i_r}}(\alpha_j) \rangle = \langle \alpha, \beta \rangle,$$

as required.

Next, we must show that $\varphi(R) = R'$. Let $v \in E$ and $\alpha \in B$. We have

$$\varphi(s_\alpha(v)) = \varphi(v - \langle v, \alpha \rangle \alpha) = \varphi(v) - \langle v, \alpha \rangle \alpha'.$$

We claim that $\langle v, \alpha_i \rangle = \langle \varphi(v), \alpha'_i \rangle$. To show this, express v as a linear combination of $\alpha_1, \dots, \alpha_n$ and then use that $\langle -, - \rangle$ is linear in its first component. Therefore the last equation may be written as

$$\varphi(s_{\alpha_i}(v)) = s_{\alpha'_i}(\varphi(v)).$$

By Lemma 11.5.4, the simple reflections s_{α_i} generate the Weyl group of R . Hence the image under φ of the orbit of $v \in E$ under the Weyl group of R is contained in the orbit of $\varphi(v)$ under the Weyl group of R' . Now Proposition 11.5.2 tells us that $\{g(\alpha) : g \in W_0, \alpha \in B\} = R$ so, since $\varphi(B) = B'$, we must have $\varphi(R) \subseteq R'$.

The same argument may be applied to the inverse of φ to show that $\varphi^{-1}(R') \subseteq R$. Hence $\varphi(R) = R'$, as required. $\square$

§11.8 Lie Algebras, Root Systems, and Dynkin Diagrams

Theorem 11.8.1

There is a natural bijection between isomorphism classes of:

- (i) Semisimple Lie algebras over $\mathbb{C}$
- (ii) Root systems
- (iii) Dynkin diagrams

Remark 11.8.2 — We shall note that this induces bijections between:

- (i) Simple Lie algebras (over $\mathbb{C}$)
- (ii) Irreducible root systems
- (iii) Connected Dynkin diagrams

We note that these bijections are compatible with:

- direct sums of Lie algebras
- disjoint unions of root systems
- disjoint unions of Dynkin diagrams.

§11.9 Exercises

Exercise 11.9.1. Suppose that R is a root system in the real inner-product space E . Show that

$$\tilde{R} := \left\{ \frac{2\alpha}{(\alpha, \alpha)} : \alpha \in R \right\}$$

is also a root system in E . Show that the Cartan matrix of $\tilde{R}$ is the transpose of the Cartan matrix of R (when each is taken with respect to suitable ordering of the roots) and that the Weyl groups of R and $\tilde{R}$ are isomorphic. One says $\tilde{R}$ is the **dual root system** to R .

Solution: First, $\tilde{R}$ is finite because R is finite. Further, since the only multiples of α in R are $\pm\alpha$, it follows that the only multiples of $\frac{2\alpha}{(\alpha, \alpha)}$ are $\frac{2\pm\alpha}{(\pm\alpha, \pm\alpha)} = \pm \frac{2\alpha}{(\alpha, \alpha)}$. We next check that $s_{\frac{2\alpha}{(\alpha, \alpha)}}$ permutes the roots of $\tilde{R}$, where $\frac{2\alpha}{(\alpha, \alpha)} \in \tilde{R}$. Let $\frac{2\beta}{(\beta, \beta)} \in \tilde{R}$. Then we have as follows:

$$\begin{aligned} s_{\frac{2\alpha}{(\alpha, \alpha)}} \left(\frac{2\beta}{(\beta, \beta)} \right) &= \frac{2\beta}{(\beta, \beta)} - \left\langle \frac{2\beta}{(\beta, \beta)}, \frac{2\alpha}{(\alpha, \alpha)} \right\rangle \frac{2\alpha}{(\alpha, \alpha)} \\ &= \frac{2\beta}{(\beta, \beta)} - \frac{2 \left(\frac{2\beta}{(\beta, \beta)}, \frac{2\alpha}{(\alpha, \alpha)} \right)}{\left(\frac{2\alpha}{(\alpha, \alpha)}, \frac{2\alpha}{(\alpha, \alpha)} \right)} \frac{2\alpha}{(\alpha, \alpha)} \\ &= \frac{2\beta}{(\beta, \beta)} - \frac{4(\beta, \alpha)}{(\beta, \beta)} \alpha \\ &= \frac{2}{(\beta, \beta)} \left(\beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)} \alpha \right) \\ &= \frac{2}{(\beta, \beta)} \cdot s_\alpha(\beta) \\ &= \frac{2s_\alpha(\beta)}{(s_\alpha(\beta), s_\alpha(\beta))}. \end{aligned}$$

Since $s_\alpha(\beta) \in R$, it follows that this is in $\tilde{R}$. Next, we need to show that integrality condition. We have as follows:

$$\begin{aligned} \left\langle \frac{2\beta}{(\beta, \beta)}, \frac{2\alpha}{(\alpha, \alpha)} \right\rangle &= \frac{2 \left(\frac{2\beta}{(\beta, \beta)}, \frac{2\alpha}{(\alpha, \alpha)} \right)}{\left(\frac{2\alpha}{(\alpha, \alpha)}, \frac{2\alpha}{(\alpha, \alpha)} \right)} \\ &= \frac{2(\beta, \alpha)}{(\beta, \beta)} \\ &= \frac{2(\alpha, \beta)}{(\beta, \beta)} \\ &= \langle \alpha, \beta \rangle \end{aligned}$$

This last term is in $\mathbb{Z}$ so we have verified true.

Next, let $B = \{\alpha_1, \dots, \alpha_\ell\}$ be a base for R . Then $\tilde{B} = \left\{ \frac{2\alpha_i}{(\alpha_i, \alpha_i)} : \alpha_i \in B \right\}$ is a base for $\tilde{R}$. We calculate each product of pairs. For any i, j , we have that

$$\left\langle \frac{2\alpha_i}{(\alpha_i, \alpha_i)}, \frac{2\alpha_j}{(\alpha_j, \alpha_j)} \right\rangle = \langle \alpha_j, \alpha_i \rangle$$

from the above, so the Cartan matrix is given by $((\langle \alpha_j, \alpha_i \rangle))_{ij}$. This is exactly the transpose of the Cartan matrix of R , because

$$(\langle \alpha_i, \alpha_j \rangle)_{ij}^T = (\langle \alpha_j, \alpha_i \rangle)_{ij},$$

which is the Cartan matrix of $\tilde{R}$.

To conclude this exercise, we want to show that the $W(R)$ and $W(\tilde{R})$ are isomorphic. If we can construct an isomorphism of the reflections, then we are done, since the reflections of R generate $W(R)$. The map we consider is essentially given from the first part. We can see that

$$s_\alpha(v) = s_{\frac{2\alpha}{\langle\alpha,\alpha\rangle}}(v)$$

from the calculation above! Therefore, we can just map every reflection to itself, so the group they generate are equal. $\square$

Exercise 11.9.2. Suppose that R is a root system in E , that R' is a root system in E' , and that $\varphi: E \rightarrow E'$ is a linear map which induces an isomorphism of root systems. Show that for $\alpha \in R$ one then has

$$s_\alpha = \varphi^{-1} \circ s_{\varphi(\alpha)} \circ \varphi.$$

Prove that the Weyl group associated to R is isomorphic to the Weyl group associated to R' . (If you know what it means, prove that the pairs $(R, W(R))$ and $(R', W(R'))$ are isomorphic as G -spaces.)

Solution: Let $v \in E$. On the LHS, we have $s_\alpha(v) = v - \langle v, \alpha \rangle \alpha$. On the RHS, we have as follows:

$$(\varphi^{-1} \circ s_{\varphi(\alpha)} \circ \varphi)(v) = \varphi^{-1}(\varphi(v) - \langle \varphi(v), \varphi(\alpha) \rangle \varphi(\alpha)) = v - \langle v, \alpha \rangle \alpha.$$

Note that in the above, we used that $\langle \varphi(v), \varphi(\alpha) \rangle = \langle v, \alpha \rangle$, which is a defining property of an isomorphism between root systems.

This relation induces an isomorphism of Weyl groups. In particular, the map $\Psi := \varphi^{-1} \circ - \circ \varphi: W(R') \rightarrow W(R)$ is an isomorphism of groups. It is clearly a group homomorphism, and it is also bijective because φ is invertible; let Φ be the inverse.

We show the isomorphism of G -spaces. Since Φ is a group isomorphism, it is enough to show that it respects the group actions. Let $W(R)$ act on R' via Φ : $g \cdot w = \Phi(g)w$, where $g \in W(R)$, $w \in R'$. Note that we have $\varphi \circ s_\alpha = s_{\varphi(\alpha)} \circ \varphi$. Therefore,

$$\varphi(g(v)) = \Phi(g)\varphi(v) = g \cdot \varphi(v).$$

Thus, we are done. $\square$

12 The Classical Lie Algebras

Our aim in this chapter is to study the classical Lie algebras $\mathfrak{sl}(n, \mathbb{C})$, $\mathfrak{so}(n, \mathbb{C})$, and $\mathfrak{sp}(n, \mathbb{C})$ for $n \geq 2$. The main result we prove is the following theorem:

Theorem 12.0.1

If L is a classical Lie algebra other than $\mathfrak{so}(2, \mathbb{C})$ and $\mathfrak{so}(4, \mathbb{C})$, then L is simple.

§12.1 General Strategy

Let L be a classical Lie algebra. In each case, it follows that L has a large subalgebra H of diagonal matrices. The maps $\text{ad } h$ for $h \in H$ are diagonalizable, as was first seen in Exercise 1.7.9, so H consists of semisimple elements.

We can immediately say a bit more about the action of H . The subspace $L \cap \text{Span}\{e_{ij} : i \neq j\}$ of off-diagonal matrices in L is also invariant under $\text{ad } h$ for $h \in H$ and hence the action of $\text{ad } H$ on this space is diagonalizable. Let

$$L \cap \text{Span}\{e_{ij} : i \neq j\} = \bigoplus_{\alpha \in \Phi} L_{\alpha},$$

where for $\alpha \in H^*$, L_{α} is the α -eigenspace of H on the off-diagonal part of L and

$$\Phi = \{\alpha \in H^* : \alpha \neq 0, L_{\alpha} \neq 0\}.$$

This gives us the decomposition

$$L = H \oplus \bigoplus_{\alpha \in \Phi} L_{\alpha}, \quad (12.1)$$

which looks very much like a root space decomposition. We shall first show that $H = L_0$, from which it will follow that H is a Cartan subalgebra of L .

Lemma 12.1.1

Let $L \subseteq \mathfrak{gl}(n, \mathbb{C})$ and H be as in (12.1) above, so that for all $\alpha \in \Phi$, there exists some $h \in H$ such that $\alpha(h) \neq 0$. Then H is a Cartan subalgebra of L .

Proof. We know already that H is abelian and that all the elements of H are semisimple. It remains to show that H is maximal with these properties. Suppose that $x \in L$ and that $[H, x] = 0$. (Equivalently, $x \in L_0$.) Using the direct sum decomposition (12.1), we may write $x = x_h + \sum_{\alpha \in \Phi} c_{\alpha} x_{\alpha}$, where $x_{\alpha} \in L_{\alpha}$, $c_{\alpha} \in \mathbb{C}$, and $x_h \in H$. For all $h \in H$, we have

$$0 = [h, x] = \sum_{\alpha} c_{\alpha} \alpha(h) x_{\alpha}.$$

By the hypothesis, for every $\alpha \in \Phi$ there is some $h \in H$ such that $\alpha(h) \neq 0$, so $c_{\alpha} = 0$ for each α and hence $x \in H$. $\square$

Proposition 12.1.2

Let L be a complex Lie algebra with Cartan subalgebra H . Let

$$L = H \oplus \bigoplus_{\alpha \in \Phi} L_{\alpha}$$

be the direct sum decomposition of L into simultaneous eigenspaces for the elements of $\text{ad } H$, where Φ is the set of non-zero $\alpha \in H^*$ such that $L_{\alpha} \neq 0$. (So we assume that $H = L_0$.) Suppose that the following conditions hold:

- (i) For each $0 \neq h \in H$, there is some $\alpha \in \Phi$ such that $\alpha(h) \neq 0$.
- (ii) For each $\alpha \in \Phi$, the space L_{α} is 1-dimensional.
- (iii) If $\alpha \in \Phi$, then $-\alpha \in \Phi$, and if L_{α} is spanned by x_{α} , then $[[x_{\alpha}, x_{-\alpha}], x_{\alpha}] \neq 0$.

Then L is semisimple, H is a Cartan subalgebra, and the α are the roots.

Proof. By Exercise 4.4.6, it is enough to show that L has no non-zero abelian ideals. Let A be an abelian ideal of L . By hypothesis, H acts diagonalizably on L and $[H, A] \subseteq A$, so H also acts diagonalizably on A . We can therefore decompose A as

$$A = (A \cap H) \oplus \bigoplus_{\alpha \in \Phi} (A \cap L_{\alpha}).$$

Suppose for a contradiction that $A \cap L_{\alpha} \neq 0$ for some $\alpha \in \Phi$. Then, because L_{α} is 1-dimensional, we must have $L_{\alpha} \subseteq A$. Since A is an ideal, this implies that $[L_{\alpha}, L_{-\alpha}] \subseteq A$, so A contains an element h of the form $h = [x_{\alpha}, x_{-\alpha}]$, where x_{α} spans L_{α} and $x_{-\alpha}$ spans $L_{-\alpha}$. Since A is abelian and both x_{α} and h are known to lie in A , we deduce that $[h, x_{\alpha}] = 0$. However, condition (iii) says that $[h, x_{\alpha}] \neq 0$, a contradiction.

We have therefore proved that $A = A \cap H$; that is, $A \subseteq H$. If A contains some non-zero element h , then, by condition (i), we know that there is some $\alpha \in \Phi$ such that $\alpha(h) \neq 0$. But then $[h, x_{\alpha}] = \alpha(h)x_{\alpha} \in L_{\alpha}$ and also $[h, x_{\alpha}] \in A$, so $x_{\alpha} \in L_{\alpha} \cap A$, which contradicts the previous paragraph. Therefore $A = 0$. $\square$

Note that since $[L_{\alpha}, L_{-\alpha}] \subseteq L_0 = H$, condition (iii) holds if and only if $\alpha([L_{\alpha}, L_{-\alpha}]) \neq 0$. Therefore, to show that condition (iii) holds, it is enough to verify that $[[L_{\alpha}, L_{-\alpha}], L_{\alpha}] \neq 0$ for one member of each pair of roots $\pm\alpha$; this will help to reduce the amount of calculation required.

Having found a Cartan subalgebra of L and shown that L is semisimple, we will then attempt to identify the root system. We must find a base for Φ , and then for β, γ in the base we must find the Cartan number $\langle \beta, \gamma \rangle$. To do this, we shall use the identity

$$\langle \beta, \gamma \rangle = \beta(h_{\gamma}),$$

where h_{γ} is part of the standard basis of the subalgebra $\mathfrak{sl}(\gamma)$ associated to the root γ . To find h_{γ} will be an easy calculation for which we can use the work done in checking condition (iii) of Proposition 12.1.2.

Now, to show that L is simple, it is enough, by the following proposition, to show that Φ is irreducible, or equivalently (by Exercise 11.6.13) that the Dynkin diagram of Φ is connected.

Proposition 12.1.3

Let L be a complex semisimple Lie algebra with Cartan subalgebra H and root system Φ . If Φ is irreducible, then L is simple.

Proof. By the root space decomposition, we may write L as

$$L = H \oplus \bigoplus_{\alpha \in \Phi} L_{\alpha}.$$

Suppose that L has a proper non-zero ideal I . Since H consists of semisimple elements, it acts diagonalizably on I , and so I has a basis of common eigenvectors for the elements of $\text{ad } H$. As we know that each root space L_{α} is 1-dimensional, this implies that

$$I = H_1 \oplus \bigoplus_{\alpha \in \Phi_1} L_{\alpha}$$

for some subspace H_1 of $H = L_0$ and some subset Φ_1 of Φ . Similarly, we have

$$I^{\perp} = H_2 \oplus \bigoplus_{\alpha \in \Phi_2} L_{\alpha},$$

where $I^{\perp}$ is the perpendicular space to I with respect to the Killing form. As $I \oplus I^{\perp} = L$, we must have $H_1 \oplus H_2 = H$, $\Phi_1 \cap \Phi_2 = \emptyset$, and $\Phi_1 \cup \Phi_2 = \Phi$.

If Φ_2 is empty, then $L_{\alpha} \subseteq I$ for all $\alpha \in \Phi$. As L is generated by its root spaces, this implies that $I = L$, a contradiction. Similarly, Φ_1 is non-empty. Now, given $\alpha \in \Phi_1$ and $\beta \in \Phi_2$, we have

$$\langle \alpha, \beta \rangle = \alpha(h_{\beta}) = 0$$

as $\alpha(h_{\beta})e_{\alpha} = [h_{\beta}, e_{\alpha}] \in I^{\perp} \cap I = 0$, so $\langle \alpha, \beta \rangle = 0$ for all $\alpha \in \Phi_1$ and $\beta \in \Phi_2$, which shows that Φ is reducible. $\square$

In summary, our program is:

1. Find the subalgebra H of diagonal matrices in L and determine the decomposition (12.1). This will show directly that conditions (i) and (ii) of Proposition 12.1.2 hold.
2. Check that $[[L_{\alpha}, L_{-\alpha}], L_{\alpha}] \neq 0$ for each root $\alpha \in \Phi$.

By Lemma 12.1.1 and Proposition 12.1.2, we now know that L is semisimple and that H is a Cartan subalgebra of L .

3. Find a base for Φ .
4. For γ, β in the base, find h_{γ} and e_{β} and hence $\langle \beta, \gamma \rangle = \beta(h_{\gamma})$. This will determine the Dynkin diagram of our root system, from which we can verify that Φ is irreducible and L is simple.

§12.2 $\mathfrak{sl}(\ell + 1, \mathbb{C})$

For this Lie algebra, most of the work has already been done.

1. We saw that the root space decomposition of $L = \mathfrak{sl}(\ell + 1, \mathbb{C})$ is

$$L = H \oplus \bigoplus_{i \neq j} L_{\varepsilon_i - \varepsilon_j},$$

where $\varepsilon_i(h)$ is the i -th entry of h and the root space $L_{\varepsilon_i - \varepsilon_j}$ is spanned by e_{ij} . Thus $\Phi = \{\pm(\varepsilon_i - \varepsilon_j) : 1 \leq i < j \leq \ell + 1\}$.

2. If $i < j$, then $[e_{ij}, e_{ji}] = e_{ii} - e_{jj} = h_{ij}$ and $[h_{ij}, e_{ij}] = 2e_{ij} \neq 0$.
3. We know from Exercise 11.3.4 that the root system Φ has as a base $\{\alpha_i : 1 \leq i \leq \ell\}$, where $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$.
4. From (2) we see that standard basis elements for the subalgebras $\mathfrak{sl}(\alpha_i)$ can be taken as $e_{\alpha_i} = e_{i,i+1}$, $f_{\alpha_i} = e_{i+1,i}$, $h_{\alpha_i} = e_{ii} - e_{i+1,i+1}$. Calculation shows that

$$\langle \alpha_i, \alpha_j \rangle = \alpha_i(h_{\alpha_j}) = \begin{cases} 2 & i = j \\ -1 & |i - j| = 1 \\ 0 & \text{otherwise,} \end{cases}$$

so the Cartan matrix of Φ is as calculated in Example 11.6.6 and the Dynkin diagram is

$$\begin{array}{ccccccc} \alpha_1 & & \alpha_2 & & \dots & & \alpha_{\ell-1} & & \alpha_\ell \\ \circ & \text{---} & \circ & & & & \circ & \text{---} & \circ \end{array}$$

This diagram is connected, so L is simple. We say that the root system of $\mathfrak{sl}(\ell + 1, \mathbb{C})$ has type A_ℓ .

§12.3 $\mathfrak{so}(2\ell + 1, \mathbb{C})$

Let $L = \mathfrak{gl}_S(2\ell + 1, \mathbb{C})$ for $\ell \geq 1$, where

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_\ell \\ 0 & I_\ell & 0 \end{pmatrix}.$$

Recall that this means

$$L = \{x \in \mathfrak{gl}(2\ell + 1, \mathbb{C}) : x^t S = -Sx\}.$$

We write elements of L as block matrices, of shapes adapted to the blocks of S . Calculation shows, using Exercise 2.4.8, that

$$L = \left\{ \begin{pmatrix} 0 & c^t & -b^t \\ b & m & p \\ -c & q & -m^t \end{pmatrix} : p = -p^t \text{ and } q = -q^t \right\}.$$

As usual, let H be the set of diagonal matrices in L . It will be convenient to label the matrix entries from 0 to 2ℓ . Let $h \in H$ have diagonal entries $0, a_1, \dots, a_\ell, -a_1, \dots, -a_\ell$, so with our numbering convention,

$$h = \sum_{i=1}^{\ell} a_i (e_{ii} - e_{i+\ell, i+\ell}).$$

- 1a. We start by finding the root spaces for H . Consider the subspace of L spanned by matrices whose non-zero entries occur only in the positions labeled by b and c . This subspace has as a basis $b_i = e_{i,0} - e_{0,\ell+i}$ and $c_i = e_{0,i} - e_{\ell+i,0}$ for $1 \leq i \leq \ell$. (Note that b_i and c_i are matrices, not scalars!) We calculate that

$$[h, b_i] = a_i b_i, \quad [h, c_i] = -a_i c_i.$$

- 1b. We extend to a basis of L by the matrices

$$\begin{aligned} m_{ij} &= e_{ij} - e_{\ell+j,\ell+i} \quad \text{for } 1 \leq i \neq j \leq \ell, \\ p_{ij} &= e_{i,\ell+j} - e_{j,\ell+i} \quad \text{for } 1 \leq i < j \leq \ell, \\ q_{ji} &= p_{ij}^t = e_{\ell+j,i} - e_{\ell+i,j} \quad \text{for } 1 \leq i < j \leq \ell. \end{aligned}$$

Again we are fortunate that the obvious basis elements are in fact simultaneous eigenvectors for the action of H . Calculation shows that

$$[h, m_{ij}] = (a_i - a_j)m_{ij}, \quad [h, p_{ij}] = (a_i + a_j)p_{ij}, \quad [h, q_{ji}] = -(a_i + a_j)q_{ji}.$$

We can now list the roots. For $1 \leq i \leq \ell$, let $\varepsilon_i \in H^*$ be the map sending h to a_i , its entry in position i ; $\varepsilon_i(h) = a_i$.

root	ε_i	$-\varepsilon_i$	$\varepsilon_i - \varepsilon_j$	$\varepsilon_i + \varepsilon_j$	$-(\varepsilon_i + \varepsilon_j)$
eigenvector	b_i	c_i	$m_{ij} \ (i \neq j)$	$p_{ij} \ (i < j)$	$q_{ji} \ (i < j)$

2. We check that $[h, x_\alpha] \neq 0$, where $h = [x_\alpha, x_{-\alpha}]$.

- 2a. For $\alpha = \varepsilon_i$, we have

$$h_i := [b_i, c_i] = e_{ii} - e_{\ell+i,\ell+i}$$

and, by (1a), $[h_i, b_i] = b_i$.

- 2b. For $\alpha = \varepsilon_i - \varepsilon_j$ and $i < j$, we have

$$h_{ij} := [m_{ij}, m_{ji}] = (e_{ii} - e_{\ell+i,\ell+i}) - (e_{jj} - e_{\ell+j,\ell+j})$$

and, by (1b), $[h_{ij}, m_{ij}] = 2m_{ij}$.

- 2c. Finally, for $\alpha = \varepsilon_i + \varepsilon_j$, for $i < j$, we have

$$h_{ij} := [p_{ij}, q_{ji}] = (e_{ii} - e_{\ell+i,\ell+i}) + (e_{jj} - e_{\ell+j,\ell+j})$$

and, by (1b), $[h_{ij}, p_{ij}] = 2p_{ij}$.

3. We claim that a base for our root system is given by

$$B = \{\alpha_i : 1 \leq i < \ell\} \cup \{\beta_\ell\},$$

where $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$ and $\beta_\ell = \varepsilon_\ell$. To see this, note that when $1 \leq i < \ell$,

$$\varepsilon_i = \alpha_i + \alpha_{i+1} + \cdots + \alpha_{\ell-1} + \beta_\ell,$$

and that when $1 \leq i < j \leq \ell$,

$$\varepsilon_i - \varepsilon_j = \alpha_i + \alpha_{i+1} + \cdots + \alpha_{j-1},$$

$$\varepsilon_i + \varepsilon_j = \alpha_i + \cdots + \alpha_{j-1} + 2(\alpha_j + \alpha_{j+1} + \cdots + \alpha_{\ell-1} + \beta_\ell).$$

Going through the table of roots shows that if $\gamma \in \Phi$ then either γ or $-\gamma$ appears above as a non-negative linear combination of elements of B . Since B has ℓ elements and $\ell = \dim H$, this is enough to show that B is a base of Φ .

4. We now determine the Cartan matrix. For $i < \ell$, we take $e_{\alpha_i} = m_{i,i+1}$, and then $h_{\alpha_i} = h_{i,i+1}$ follows from (2b). We take $e_{\beta_\ell} = b_\ell$, and then from (2a) we see that $h_{\beta_\ell} = 2(e_{\ell,\ell} - e_{2\ell,2\ell})$.

For $1 \leq i, j < \ell$, we calculate that

$$[h_{\alpha_j}, e_{\alpha_i}] = \begin{cases} 2e_{\alpha_j} & i = j \\ -e_{\alpha_j} & |i - j| = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Hence

$$\langle \alpha_i, \alpha_j \rangle = \begin{cases} 2 & i = j, \\ -1 & |i - j| = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Similarly, by calculating $[h_{\beta_\ell}, e_{\alpha_i}]$ and $[h_{\alpha_i}, e_{\beta_\ell}]$, we find that

$$\langle \alpha_i, \beta_\ell \rangle = \begin{cases} -2 & i = \ell - 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$\langle \beta_\ell, \alpha_i \rangle = \begin{cases} -1 & i = \ell - 1, \\ 0 & \text{otherwise.} \end{cases}$$

This shows that the Dynkin diagram of Φ is

$$\begin{array}{ccccccc} \alpha_1 & & \alpha_2 & & \dots & & \alpha_{\ell-2} & \alpha_{\ell-1} & \beta_\ell \\ \circ & \text{---} & \circ & \text{---} & & \text{---} & \circ & \circ & \circ \end{array}$$

As the Dynkin diagram is connected, Φ is irreducible and so L is simple. The root system of $\mathfrak{so}(2\ell + 1, \mathbb{C})$ is said to have **type** B_ℓ .

§12.4 $\mathfrak{so}(2\ell, \mathbb{C})$

Let $L = \mathfrak{gl}_S(2\ell, \mathbb{C})$, where

$$S = \begin{pmatrix} 0 & I_\ell \\ I_\ell & 0 \end{pmatrix}.$$

We write elements of L as block matrices, of shapes adapted to the blocks of S . Calculation shows that

$$L = \left\{ \begin{pmatrix} m & p \\ q & -m^t \end{pmatrix} : p = -p^t \text{ and } q = -q^t \right\}.$$

We see that if $\ell = 1$ then the Lie algebra is 1-dimensional, and so, by definition, not simple or semisimple. For this reason, we assumed in the statement of Theorem 12.0.1 that $\ell \geq 2$.

As usual, we let H be the set of diagonal matrices in L . We label the matrix entries from 1 up to 2ℓ . This means that we can use the calculations already done for $\mathfrak{so}(2\ell + 1, \mathbb{C})$ by ignoring the row and column of matrices labeled by 0.

1. All the work needed to find the root spaces in $\mathfrak{so}(2\ell, \mathbb{C})$ is done for us by step (1b) for $\mathfrak{so}(2\ell + 1, \mathbb{C})$. Taking the notation from this part, we get the following roots:

root	$\varepsilon_i - \varepsilon_j$	$\varepsilon_i + \varepsilon_j$	$-(\varepsilon_i + \varepsilon_j)$
eigenvector	$m_{ij} \ (i \neq j)$	$p_{ij} \ (i < j)$	$q_{ji} \ (i < j)$

2. The work already done in steps (2b) and (2c) for $\mathfrak{sl}(2\ell + 1, \mathbb{C})$ shows that

$$[[L_\alpha, L_{-\alpha}], L_\alpha] \neq 0$$

for each root α .

3. We claim that a base for our root system is given by

$$B = \{\alpha_i : 1 \leq i < \ell\} \cup \{\beta_\ell\},$$

where $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$ and $\beta_\ell = \varepsilon_{\ell-1} + \varepsilon_\ell$. To see this, note that when $1 \leq i < j \leq \ell$,

$$\varepsilon_i - \varepsilon_j = \alpha_i + \alpha_{i+1} + \cdots + \alpha_{j-1},$$

$$\varepsilon_i + \varepsilon_j = (\alpha_i + \alpha_{i+1} + \cdots + \alpha_{\ell-2}) + (\alpha_j + \alpha_{j+1} + \cdots + \alpha_{\ell-1} + \beta_\ell).$$

This shows that if $\gamma \in \Phi$, then either γ or $-\gamma$ is a non-negative linear combination of elements of B with integer coefficients, so B is a base for our root system.

4. We calculate the Cartan integers. The work already done for $\mathfrak{so}(2\ell + 1, \mathbb{C})$ gives us the Cartan numbers $\langle \alpha_i, \alpha_j \rangle$ for $i, j < \ell$. For the remaining ones, we take $e_{\beta_\ell} = e_{\ell-1, \ell}$ and then we find from step (2c) for $\mathfrak{so}(2\ell + 1, \mathbb{C})$ that

$$h_{\beta_\ell} = (e_{\ell-1, \ell-1} - e_{2\ell-1, 2\ell-1}) + (e_{\ell, \ell} - e_{2\ell, 2\ell}).$$

Hence

$$\langle \alpha_j, \beta_\ell \rangle = \begin{cases} -1 & j = \ell - 2 \\ 0 & \text{otherwise,} \end{cases} \quad \langle \beta_\ell, \alpha_j \rangle = \begin{cases} -1 & j = \ell - 2 \\ 0 & \text{otherwise.} \end{cases}$$

If $\ell = 2$, then the base has only the two orthogonal roots α_1 and β_2 , so in this case, Φ is reducible. In fact, $\mathfrak{so}(4, \mathbb{C})$ is isomorphic to $\mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$, as you were asked to prove in Exercise 10.7.2. This explains the other Lie algebra excluded from the statement of Theorem 12.0.1.

If $\ell \geq 3$, then our calculation shows that the Dynkin diagram of Φ is



As this diagram is connected, the Lie algebra is simple. When $\ell = 3$, the Dynkin diagram is the same as that of A_3 , the root system of $\mathfrak{sl}(4, \mathbb{C})$, so we might expect that $\mathfrak{so}(6, \mathbb{C})$ should be isomorphic to $\mathfrak{sl}(4, \mathbb{C})$. This is indeed the case; see Exercise ???. For $\ell \geq 4$, the root system of $\mathfrak{so}(2\ell, \mathbb{C})$ is said to have **type** D_ℓ .

§12.5 $\mathfrak{sp}(2\ell, \mathbb{C})$

Let $L = \mathfrak{gl}_S(2\ell, \mathbb{C})$, where S is the matrix

$$S = \begin{pmatrix} 0 & I_\ell \\ -I_\ell & 0 \end{pmatrix}.$$

We write elements of L as block matrices, of shapes adapted to the blocks of S . Calculation shows that

$$L = \left\{ \begin{pmatrix} m & p \\ q & -m^t \end{pmatrix} : p = p^t \text{ and } q = q^t \right\}.$$

We see that when $\ell = 1$, this is the same Lie algebra as $\mathfrak{sl}(2, \mathbb{C})$. In what follows, we shall assume that $\ell \geq 2$.

Let H be the set of diagonal matrices in L . We label the matrix entries in the usual way from 1 to 2ℓ . Let $h \in H$ have diagonal entries $a_1, \dots, a_\ell, -a_1, \dots, -a_\ell$, that is,

$$h = \sum_{i=1}^{\ell} a_i (e_{ii} - e_{i+\ell, i+\ell}).$$

1. We take the following basis for the root spaces of L :

$$\begin{aligned} m_{ij} &= e_{ij} - e_{\ell+j, \ell+i} & \text{for } 1 \leq i \neq j \leq \ell, \\ p_{ij} &= e_{i, \ell+j} + e_{j, \ell+i} & \text{for } 1 \leq i < j \leq \ell, & \quad p_{ii} = e_{i, \ell+i} & \text{for } 1 \leq i \leq \ell, \\ q_{ji} &= p_{ij}^t = e_{\ell+j, i} + e_{\ell+i, j} & \text{for } 1 \leq i < j \leq \ell, & \quad q_{ii} = e_{\ell+i, i} & \text{for } 1 \leq i \leq \ell. \end{aligned}$$

Calculation shows that

$$\begin{aligned} [h, m_{ij}] &= (a_i - a_j) m_{ij}, \\ [h, p_{ij}] &= (a_i + a_j) p_{ij}, \\ [h, q_{ji}] &= -(a_i + a_j) q_{ji}. \end{aligned}$$

Notice that for p_{ij} and q_{ji} it is allowed that $i = j$, and in these cases we get the eigenvalues $2a_i$ and $-2a_i$, respectively.

We can now list the roots. Write ε_i for the element in H^* sending h to a_i .

root	$\varepsilon_i - \varepsilon_j$	$\varepsilon_i + \varepsilon_j$	$-(\varepsilon_i + \varepsilon_j)$	$2\varepsilon_i$	$-2\varepsilon_i$
eigenvector	$m_{ij} \ (i \neq j)$	$p_{ij} \ (i < j)$	$q_{ji} \ (i < j)$	p_{ii}	q_{ii}

2. For each root α , we must check that $[h, x_\alpha] \neq 0$, where $h = [x_\alpha, x_{-\alpha}]$. When $\alpha = \varepsilon_i - \varepsilon_j$, this has been done in step (2b) for $\mathfrak{so}(2\ell + 1, \mathbb{C})$. If $\alpha = \varepsilon_i + \varepsilon_j$, then $x_\alpha = p_{ij}$ and $x_{-\alpha} = q_{ji}$ and

$$h = (e_{ii} - e_{\ell+i, \ell+i}) + (e_{jj} - e_{\ell+j, \ell+j})$$

if $i \neq j$, and $h = e_{ii} - e_{\ell+i, \ell+i}$ if $i = j$. Hence $[h, x_\alpha] = 2x_\alpha$ in both cases.

3. Let $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$ for $1 \leq i \leq \ell - 1$ as before, and let $\beta_\ell = 2\varepsilon_\ell$. We claim that $\{\alpha_1, \dots, \alpha_{\ell-1}, \beta_\ell\}$ is a base for Φ . By the same argument as used before, this follows once we observe that for $1 \leq i < j \leq \ell$

$$\begin{aligned} \varepsilon_i - \varepsilon_j &= \alpha_i + \alpha_{i+1} + \dots + \alpha_{j-1}, \\ \varepsilon_i + \varepsilon_j &= \alpha_i + \alpha_{i+1} + \dots + \alpha_{\ell-1} + 2(\alpha_j + \dots + \alpha_{\ell-1}) + \beta_\ell, \\ 2\varepsilon_i &= 2(\alpha_i + \alpha_{i+1} + \dots + \alpha_{\ell-1}) + \beta_\ell. \end{aligned}$$

4. We calculate the Cartan integers. The numbers $\langle \alpha_i, \alpha_j \rangle$ are already known. Take $e_{\beta_\ell} = p_{\ell\ell}$, then we find that $h_{\beta_\ell} = e_{\ell, \ell} - e_{2\ell, 2\ell}$ and so

$$\begin{aligned} \langle \alpha_i, \beta_\ell \rangle &= \begin{cases} -1 & i = \ell - 1 \\ 0 & \text{otherwise,} \end{cases} \\ \langle \beta_\ell, \alpha_j \rangle &= \begin{cases} -2 & j = \ell - 1 \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

The Dynkin diagram of Φ is



which is connected, so L is simple. The root system of $\mathfrak{sp}(2\ell, \mathbb{C})$ is said to have type C_ℓ . Since the root systems C_2 and B_2 have the same Dynkin diagram, we might expect that the Lie algebras $\mathfrak{sp}(4, \mathbb{C})$ and $\mathfrak{so}(5, \mathbb{C})$ would be isomorphic. This is the case, see Exercise ??.

§12.6 Killing Forms of the Classical Lie Algebras

Now that we know that (with two exceptions) the classical Lie algebras are simple, we can use some of our earlier work to compute their Killing forms. We shall see that they can all be given by a simple closed formula.

Lemma 12.6.1

Let $L \subseteq \mathfrak{gl}(n, \mathbb{C})$ be a simple classical Lie algebra. Let $\beta: L \times L \rightarrow \mathbb{C}$ be the symmetric bilinear form

$$\beta(x, y) := \text{tr}(xy).$$

Then β is non-degenerate.

Proof. Let $J = \{x \in L : \beta(x, y) = 0 \text{ for all } y \in L\}$. It follows from the associative property of trace, as in Exercise 9.4.3, that J is an ideal of L . Since L is simple, and clearly β is not identically zero, we must have $J = 0$. Therefore β is non-degenerate. $\square$

In Exercise 9.7.4, we showed that any two non-degenerate symmetric associative bilinear forms on a simple Lie algebra are scalar multiples of one another. Hence, by Cartan's Second Criterion (Theorem 9.4.4), if κ is the Killing form on L , then $\kappa = \lambda\beta$ for some non-zero scalar $\lambda \in \mathbb{C}$. To determine the scalar λ , we use the root space decomposition to compute $\kappa(h, h')$ for $h, h' \in H$.

Example 12.6.2

For example, for $\mathfrak{sl}(\ell + 1, \mathbb{C})$ let $h \in H$, with diagonal entries $a_1, \dots, a_{\ell+1}$, and similarly let $h' \in H$ with diagonal entries $a'_1, \dots, a'_{\ell+1}$. Then, using the root space decomposition given in step (1) of Section 12.2, we get

$$\kappa(h, h') = \sum_{\alpha \in \Phi} \alpha(h)\alpha(h') = 2 \sum_{i < j} (a_i - a_j)(a'_i - a'_j).$$

Putting $h = h'$, and $a_1 = 1$, $a_2 = -1$ and all other entries zero, we get $\kappa(h, h) = 8 + 4(\ell - 1) = 4(\ell + 1)$. Since $\text{tr } h^2 = 2$, this implies that $\lambda = 2(\ell + 1)$.

For the remaining three families, see Exercise 12.8.3 below. We get $\kappa(x, y) = \lambda \text{tr}(xy)$, where

$$\lambda = \begin{cases} 2(\ell + 1) & L = \mathfrak{sl}(\ell + 1, \mathbb{C}) \\ 2\ell - 1 & L = \mathfrak{so}(2\ell + 1, \mathbb{C}) \\ 2(\ell + 1) & L = \mathfrak{sp}(2\ell, \mathbb{C}) \\ 2(\ell - 1) & L = \mathfrak{so}(2\ell, \mathbb{C}). \end{cases}$$

§12.7 Root Systems and Isomorphisms

Let L be a complex semisimple Lie algebra. We have seen how to define the root system associated to a Cartan subalgebra of L . Could two different Cartan subalgebras of L give different root systems? The following theorem shows that the answer is no.

Theorem 12.7.1

if L is a semisimple Lie algebra, and Φ_1, Φ_2 are the root systems associated to two different Cartan subalgebras, then $\Phi_1 \cong \Phi_2$.

Proof. In Appendix C. □

Remark 12.7.2 — Recall Proposition 11.7.4, which states that two root systems with the same Dynkin diagrams are isomorphic.

Suppose now that L_1 and L_2 are complex semisimple Lie algebras that have non-isomorphic root systems (with respect to some Cartan subalgebras). Then, by the theorem, L_1 and L_2 cannot be isomorphic. Thus we can use root systems to rule out isomorphisms between the classical Lie algebras. This does most of the work needed to prove the following proposition.

Proposition 12.7.3

The only isomorphisms between classical Lie algebras are:

- (i) $\mathfrak{so}(3, \mathbb{C}) \cong \mathfrak{sp}(2, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C})$; root systems of type A_1 ,
- (ii) $\mathfrak{so}(4, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$; root systems of type $A_1 \times A_1$,
- (iii) $\mathfrak{so}(5, \mathbb{C}) \cong \mathfrak{sp}(4, \mathbb{C})$; root systems of types B_2 and C_2 ,
- (iv) $\mathfrak{so}(6, \mathbb{C}) \cong \mathfrak{sl}(4, \mathbb{C})$; root systems of types D_3 and A_3 .

Note that we have not yet proved the existence of all these isomorphisms. However, we have already seen the first two (see Exercises 1.7.7 and 10.7.2). The third isomorphism appears in Exercise 12.8.2 below and the last is discussed in Chapter 15. We are therefore led to conjecture that the converse of Theorem 12.7.1 also holds; that is, if two complex semisimple Lie algebras have isomorphic root systems, then they are isomorphic as Lie algebras.

We shall see in Chapter 14 that this is a corollary of Serre’s Theorem. Thus isomorphisms of root systems precisely reflect isomorphisms of complex semisimple Lie algebras. To classify the complex semisimple Lie algebras, we should therefore first classify root systems. This is the subject of the next chapter.

§12.8 Exercises

Exercise 12.8.1. Show that the dimensions of the classical Lie algebras are as follows:

$$\begin{aligned}\dim \mathfrak{sl}(\ell + 1, \mathbb{C}) &= \ell^2 + 2\ell, \\ \dim \mathfrak{so}(2\ell + 1, \mathbb{C}) &= 2\ell^2 + \ell, \\ \dim \mathfrak{sp}(2\ell, \mathbb{C}) &= 2\ell^2 + \ell, \\ \dim \mathfrak{so}(2\ell, \mathbb{C}) &= 2\ell^2 - \ell.\end{aligned}$$

Solution: We have that

$$\dim(\mathfrak{sl}(\ell + 1, \mathbb{C})) = (\ell + 1)^2 - 1 = \ell^2 + 2\ell.$$

The minus 1 comes from the fact that $\mathfrak{sl}(\ell + 1, \mathbb{C})$ is isomorphic to $\mathfrak{gl}(\ell + 1, \mathbb{C})/\mathbb{C}$ (due to the trace map). Next, we note that

$$\mathfrak{so}(2\ell + 1, \mathbb{C}) = \left\{ \begin{pmatrix} 0 & c^t & -b^t \\ b & m & p \\ -c & q & -m^t \end{pmatrix} : p = -p^t \text{ and } q = -q^t \right\}.$$

Therefore,

$$\dim(\mathfrak{so}(2\ell + 1, \mathbb{C})) = \ell^2 + 2 \cdot \frac{\ell(\ell - 1)}{2} + 2 \cdot \ell = 2\ell^2 + \ell.$$

Here, we are really just adding up the number of dimensions for m, b, c , then twice the skew-symmetric matrices. Next,

$$\mathfrak{sp}(2\ell, \mathbb{C}) = \left\{ \begin{pmatrix} m & p \\ q & -m^t \end{pmatrix} : p = p^t \text{ and } q = q^t \right\}.$$

Similarly, we then calculate

$$\dim(\mathfrak{sp}(2\ell, \mathbb{C})) = \ell^2 + 2 \cdot \frac{\ell(\ell - 1)}{2} = 2\ell^2 - \ell.$$

Next, we have that

$$\mathfrak{so}(2\ell, \mathbb{C}) = \left\{ \begin{pmatrix} m & p \\ q & -m^t \end{pmatrix} : p = -p^t \text{ and } q = -q^t \right\}.$$

Therefore,

$$\dim(\mathfrak{so}(2\ell, \mathbb{C})) = \ell^2 + 2 \cdot \frac{\ell(\ell - 1)}{2} = 2\ell^2 - \ell.$$

□

Exercise 12.8.2. Show that the Lie algebras $\mathfrak{sp}(4, \mathbb{C})$ and $\mathfrak{so}(5, \mathbb{C})$ are isomorphic. (For instance, use the root space decomposition to show that they have bases affording the same structure constants.)

Solution: We use the root space decomposition to get a basis, as suggested. For $\mathfrak{so}(5, \mathbb{C})$, we use the numbering $0, \dots, 4$. $H = \text{Span}\{h_1 := e_{1,1} - e_{3,3}, h_2 := e_{2,2} - e_{4,4}\}$. Further, each of the obvious basis elements are eigenvectors for the action of H . That is, we extend H by the following:

$$\begin{aligned}m_{1,2} &= e_{1,2} - e_{4,3}, & m_{2,1} &= e_{2,1} - e_{3,4} \\ p_{1,4} &= e_{1,4} - e_{2,3}, & q_{3,2} &= e_{3,2} - e_{4,1} \\ b_1 &= e_{1,0} - e_{0,3}, & b_2 &= e_{2,0} - e_{0,4} \\ c_1 &= e_{0,1} - e_{3,0}, & c_2 &= e_{0,2} - e_{4,0}\end{aligned}$$

Next we get a suitable basis for $\mathfrak{sp}(4, \mathbb{C})$. $H = \text{Span} \{h_1 := e_{1,1} - e_{3,3}, h_2 := e_{2,2} - e_{4,4}\}$. We extend this by the obvious basis elements, since they work.

$$\begin{aligned} m_{1,2} &= e_{1,2} - e_{4,3}, & m_{2,1} &= e_{2,1} - e_{3,4} \\ p_{1,4} &= e_{1,4} + e_{2,3}, & q_{3,2} &= e_{4,1} + e_{3,2} \\ p_{1,1} &= e_{1,3}, & p_{2,2} &= e_{2,4} \\ q_{1,1} &= e_{3,1}, & q_{2,2} &= e_{4,2} \end{aligned}$$

Note that we have the following eigenvalue tables.

root	ε_i	$-\varepsilon_i$	$\varepsilon_i - \varepsilon_j$	$\varepsilon_i + \varepsilon_j$	$-(\varepsilon_i + \varepsilon_j)$
eigenvector	b_i	c_i	$m_{ij} \ (i \neq j)$	$p_{ij} \ (i < j)$	$q_{ji} \ (i < j)$

We also have

root	$\varepsilon_i - \varepsilon_j$	$\varepsilon_i + \varepsilon_j$	$-(\varepsilon_i + \varepsilon_j)$	$2\varepsilon_i$	$-2\varepsilon_i$
eigenvector	$m_{ij} \ (i \neq j)$	$p_{ij} \ (i < j)$	$q_{ji} \ (i < j)$	p_{ii}	q_{ii}

We almost get everything to match up, but the one issue is with b_i, c_i and p_{ii}, q_{ii} . For this, we replace b_i, c_i with 2 times it. $\square$

Exercise 12.8.3. This exercise gives a way to establish the semisimplicity of the classical Lie algebras using the Killing form.

1. Let L be a classical Lie algebra, and let H be the subalgebra of diagonal matrices, with eigenspace decomposition

$$L = H \oplus \bigoplus_{\alpha \in \Phi} L_{\alpha},$$

so H is self-centralizing. Assume also that the following conditions hold:

- (a) For each $\alpha \in \Phi$, the space L_{α} is 1-dimensional. If $\alpha \in \Phi$, then $-\alpha \in \Phi$.
- (b) For each $\alpha \in \Phi$, the space $[L_{\alpha}, L_{-\alpha}]$ is non-zero.
- (c) The Killing form restricted to H is non-degenerate, and for $h \in H$, if $\kappa(h, h') = 0$ then $h = 0$.

Show that the Killing form of L is then non-degenerate.

In the earlier sections of this chapter, we have found the roots with respect to H explicitly. We can make use of this and find the Killing form restricted to H explicitly.

2. Use the root space decomposition of $\mathfrak{sl}(\ell + 1, \mathbb{C})$ to show that if κ is the Killing form of $\mathfrak{sl}(\ell + 1, \mathbb{C})$, then

$$\kappa(h, h') = 2(\ell + 1)\text{tr}(hh')$$

for all $h, h' \in H$.

Hence, show that condition (c) above holds for the restriction of κ to H and deduce that $\mathfrak{sl}(\ell + 1, \mathbb{C})$ is semisimple.

3. Use similar methods to prove that the orthogonal and symplectic Lie algebras are semisimple.

Solution: 1. Let $x \in L$. We want to show that if $\kappa(x, y) = 0$ for all $y \in L$, then $x = 0$. First, let $y = h' \in H$. Further, write $x = h + \sum_{\alpha} x_{\alpha}$.

$$0 = \kappa(x, h') = \kappa(h, h') + \sum_{\alpha} \kappa(x_{\alpha}, h').$$

Since $\kappa(x_\alpha, h') = 0$, we get that $0 = \kappa(h, h')$, so $h = 0$ by nondegeneracy of κ restricted to H . Next, let $y = x_{-\alpha}$. It follows then that $\kappa(x_\alpha, x_{-\alpha}) = 0$. However, this forces x_α to equal zero then. This is because $[L_\alpha, L_{-\alpha}] \neq 0$, so if $x_\alpha \neq 0$, then $[x_\alpha, x_{-\alpha}] = h'' \neq 0$, for some $h'' \in H$. Then, we have that

$$0 \neq \kappa(h'', h'') = \kappa(h'', [x_\alpha, x_{-\alpha}]) = \alpha(h'')\kappa(x_\alpha, x_{-\alpha}).$$

Therefore, $\kappa(x_\alpha, x_\alpha) \neq 0$ is $x_\alpha \neq 0$. In this manner, we get that $x_\alpha = 0$, for all α , and so we attain that $x = 0$. Thus, κ is nondegenerate on L .

2. Let $h = \text{diag}(a_1, \dots, a_{\ell+1})$, $h' = \text{diag}(a'_1, \dots, a'_{\ell+1})$. We have that

$$\kappa(h, h') = \text{tr}(\text{ad } h \circ \text{ad } h') = \sum_{\alpha} \alpha(h)\alpha(h') = 2 \sum_{i < j} (a_i - a_j)(a'_i - a'_j).$$

We now split this sum into the following:

$$2 \sum_{i < j} (a_i a'_i + a_j a'_j) - 2 \sum_{i < j} (a_i a'_j + a_j a'_i).$$

Then, in the first sum, we note that every $a_i a'_i$ appears ℓ times as a whole, so we get that

$$2 \sum_{i < j} (a_i a'_i + a_j a'_j) = 2\ell \sum_{i=1}^{\ell+1} a_i a'_i = 2\ell \text{tr}(hh').$$

Next, for the second summand.

$$2 \sum_{i < j} (a_i a'_j + a_j a'_i) = 2 \sum_{i=1}^{\ell+1} a_i \sum_{j \neq i} a'_j = 2 \sum_{i=1}^{\ell+1} a_i (-a'_i) = -2 \sum_{i=1}^{\ell+1} a_i a'_i = -2\text{tr}(hh').$$

Therefore, the total sum is as desired:

$$\kappa(h, h') = 2(\ell + 1)\text{tr}(hh').$$

Therefore, κ restricted to H is nondegenerate! It is not zero unless one of the h were zero to begin with.

3. Let's do $\mathfrak{so}(2\ell + 1, \mathbb{C})$ next. We have $h = (0, a_1, \dots, a_\ell, -a_1, \dots, -a_\ell)$ and $h' = (0, a'_1, \dots, a'_\ell, -a'_1, \dots, -a'_\ell)$. We do the same analysis:

$$\begin{aligned} \kappa(h, h') &= \text{tr}(\text{ad } h \circ \text{ad } h') \\ &= \sum_{\alpha} \alpha(h)\alpha(h') \\ &= 2 \sum_{i=1}^{\ell} a_i a'_i + 2 \sum_{i < j} (a_i - a_j)(a'_i - a'_j) + 2 \sum_{i < j} (a_i + a_j)(a'_i + a'_j) \\ &= 2 \sum_{i=1}^{\ell} a_i a'_i + 4 \sum_{i < j} (a_i a'_i + a_j a'_j) \\ &= 2 \sum_{i=1}^{\ell} a_i a'_i + 4(\ell - 1) \sum_{i=1}^{\ell} a_i a'_i \\ &= (4\ell - 2) \sum_{i=1}^{\ell} a_i a'_i \\ &= (2\ell - 1)\text{tr}(hh'). \end{aligned}$$

As for $\mathfrak{so}(2\ell, \mathbb{C})$, the term $2 \sum_{i=1}^{\ell} a_i a'_i$ is not present. In this case, we have

$$\kappa(h, h') = 2(\ell - 1) \operatorname{tr}(hh').$$

Finally, we do $\mathfrak{sp}(2\ell, \mathbb{C})$. Let $h = (a_1, \dots, a_\ell, -a_1, \dots, -a_\ell)$, $h' = (a'_1, \dots, a'_\ell, -a'_1, \dots, -a'_\ell)$. Then, we have as follows:

$$\begin{aligned} \kappa(h, h') &= \operatorname{tr}(\operatorname{ad} h \circ \operatorname{ad} h') \\ &= \sum_{\alpha} \alpha(h) \alpha(h') \\ &= 4 \sum_{i=1}^{\ell} a_i a'_i + 2 \sum_{i < j} (a_i - a_j)(a'_i - a'_j) + 2 \sum_{i < j} (a_i + a_j)(a'_i + a'_j) \\ &= 4 \sum_{i=1}^{\ell} a_i a'_i + 4 \sum_{i < j} (a_i a'_i + a_j a'_j) \\ &= 4 \sum_{i=1}^{\ell} a_i a'_i + 4(\ell - 1) \sum_{i=1}^{\ell} a_i a'_i \\ &= 4\ell \sum_{i=1}^{\ell} a_i a'_i \\ &= 2\ell \operatorname{tr}(hh'). \end{aligned}$$

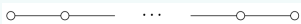
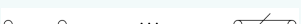
Thus, the same result holds! □

13 The Classification of Root Systems

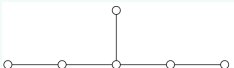
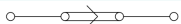
§13.1 Classification of Dynkin Diagrams

Theorem 13.1.1

Every connected Dynkin diagram is one of the following:

- A_n ($n \geq 1$): 
- B_n ($n \geq 2$): 
- C_n ($n \geq 3$): 
- D_n ($n \geq 4$): 

where each of the diagrams above has n vertices, or one of the five exceptional diagrams:

- E_6 : 
- E_7 : 
- E_8 : 
- F_4 : 
- G_2 : 

Remark 13.1.2 — Note that there are no repetitions in this list. For example, we have not included C_2 in the list, as it is the same diagram as B_2 , and so the associated root systems are isomorphic.

Let Δ be a connected Dynkin diagram. As a first approximation, we shall determine the possible underlying graphs for Δ , ignoring for the moment any arrows that may appear. To find these graphs, we do not need to know that they come from root systems. Instead it is convenient to work with more general sets of vectors.

Definition 13.1.3. Let E be a real inner-product space with inner product $(-, -)$. A subset A of E consisting of linearly independent vectors $v_1, v_2, \dots, v_n$ is said to be **admissible** if it satisfies the following conditions:

1. $(v_i, v_i) = 1$ for all i and $(v_i, v_j) \leq 0$ if $i \neq j$.
2. If $i \neq j$, then $4(v_i, v_j)^2 \in \{0, 1, 2, 3\}$.

To the admissible set A , we associate the graph Γ_A with vertices labeled by the vectors $v_1, \dots, v_n$, and with $d_{ij} := 4(v_i, v_j)^2 \in \{0, 1, 2, 3\}$ edges between v_i and v_j for $i \neq j$.

Example 13.1.4

Suppose that B is a base of a root system. Set

$$A := \{\alpha / \sqrt{(\alpha, \alpha)} : \alpha \in B\}.$$

Then A is easily seen to be an admissible set. Moreover, the graph Γ_A is the Coxeter graph of B (as defined two chapters ago).

We now find all the connected graphs that correspond to admissible sets. Let A be an admissible set in the real inner-product space E with connected graph $\Gamma = \Gamma_A$. The first easy observation we make is that any subset of A is also admissible. We shall use this several times below.

Lemma 13.1.5

The number of pairs of vertices joined by at least one edge is at most $|A| - 1$.

Proof. Suppose $A = \{v_1, \dots, v_n\}$. Set $v = \sum_{i=1}^n v_i$. As A is linearly independent, $v \neq 0$. Hence

$$(v, v) = n + 2 \sum_{i < j} (v_i, v_j) > 0$$

and so

$$n > \sum_{i < j} -2(v_i, v_j) = \sum_{i < j} \sqrt{d_{ij}} \geq N,$$

where N is the number of pairs $\{v_i, v_j\}$ such that $d_{ij} \geq 1$; this is the number that interests us. $\square$

Corollary 13.1.6

The graph Γ does not contain any cycles.

Proof. Suppose that Γ does have a cycle. Let A' be the subset of A consisting of the vectors involved in this cycle. Then A' is an admissible set with the same number (or more) edges as vertices, in contradiction to the previous lemma. $\square$

Lemma 13.1.7

No vertex of Γ is incident to four or more edges.

Proof. Take a vertex v of Γ , and let $v_1, v_2, \dots, v_k$ be all the vertices in Γ joined to v . Since Γ does not contain any cycles, we must have $(v_i, v_j) = 0$ for $i \neq j$. Consider the subspace U with basis $v_1, v_2, \dots, v_k, v$. The Gram–Schmidt process allows us to extend

$v_1, \dots, v_k$ to an orthonormal basis of U , say by adjoining v_0 ; necessarily $(v, v_0) \neq 0$. We may express v in terms of this orthonormal basis as

$$v = \sum_{i=0}^k (v, v_i) v_i.$$

By assumption, v is a unit vector, so expanding (v, v) gives

$$1 = (v, v) = \sum_{i=0}^k (v, v_i)^2.$$

Since $(v, v_0)^2 > 0$, this shows that

$$\sum_{i=1}^k (v, v_i)^2 < 1.$$

Now, as A is admissible and $(v, v_i) \neq 0$, we know that $(v, v_i)^2 \geq \frac{1}{4}$ for $1 \leq i \leq k$. Hence $k \leq 3$. $\square$

An immediate corollary of this lemma is the following.

Corollary 13.1.8

If Γ is connected and has a triple edge, then $\Gamma = \text{---}\text{---}\text{---}$.

Lemma 13.1.9 (Shrinking Lemma)

Suppose Γ has a subgraph which is a line, that is, of the form



where there are no multiple edges between the vertices shown. Define

$$A' = (A \setminus \{v_1, v_2, \dots, v_k\}) \cup \{v\} \quad \text{where } v = \sum_{i=1}^k v_i.$$

Then A' is admissible and the graph $\Gamma_{A'}$ is obtained from Γ_A by shrinking the line to a single vertex.

Proof. Clearly A' is linearly independent, so we need only verify the conditions on the inner products. By assumption, we have $2(v_i, v_{i+1}) = -1$ for $1 \leq i \leq k-1$ and $(v_i, v_j) = 0$ for $i \neq j$ otherwise. This allows us to calculate (v, v) . We find that

$$(v, v) = k + 2 \sum_{i=1}^{k-1} (v_i, v_{i+1}) = k - (k-1) = 1.$$

Suppose that $w \in A$ and $w \neq v_i$ for $1 \leq i \leq k$. Then w is joined to at most one of $v_1, \dots, v_k$ (otherwise there would be a cycle). Therefore either $(w, v) = 0$ or $(w, v) = (w, v_i)$ for some $1 \leq i \leq k$ and then $4(w, v)^2 \in \{0, 1, 2, 3\}$, so A' satisfies the defining conditions for an admissible set. These remarks also determine the graph $\Gamma_{A'}$. $\square$

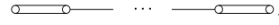
Definition 13.1.10. Say that a vertex of Γ is a **branch vertex** if it is incident to three or more edges; by Lemma 13.1.7 such a vertex is incident to exactly three edges.

Lemma 13.1.11

The graph Γ has

- (i) no more than one double edge,
- (ii) no more than one branch vertex, and
- (iii) not both a double edge and a branch vertex.

Proof. Suppose Γ has two (or more) double edges. Since Γ is connected, it has a subgraph consisting of two double edges connected by a line of the form



By the Shrinking Lemma, we obtain an admissible set with graph



which contradicts Lemma 13.1.7. The proofs of the remaining two parts are very similar, so we leave them to the reader. $\square$

For the final steps of the proof of Theorem ??, we shall need the following calculation of an inner product.

Lemma 13.1.12

Suppose that Γ has a line as a subgraph:



Let $v = \sum_{i=1}^p i v_i$. Then $(v, v) = \frac{p(p+1)}{2}$.

Proof. The shape of the subgraph tells us that $2(v_i, v_{i+1}) = -1$ for $1 \leq i \leq p-1$ and that $(v_i, v_j) = 0$ for $i \neq j$ otherwise, so

$$(v, v) = \sum_{i=1}^p i^2 + 2 \sum_{i=1}^{p-1} (v_i, v_{i+1}) i(i+1) = \sum_{i=1}^p i^2 - \sum_{i=1}^{p-1} i(i+1) = p^2 - \sum_{i=1}^{p-1} i,$$

which is equal to $p(p+1)/2$. $\square$

Proposition 13.1.13

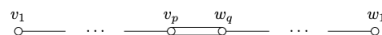
If Γ has a double edge, then Γ is one of



or



Proof. By Lemma 13.1.11, any such Γ has the form



where, without loss of generality, $p \geq q$. Let $v = \sum_{i=1}^p i v_i$ and $w = \sum_{i=1}^q i w_i$. By the calculation above, we have

$$(v, v) = \frac{p(p+1)}{2}, \quad (w, w) = \frac{q(q+1)}{2}.$$

We see from the graph that $4(v_p, w_q)^2 = 2$ and $(v_i, w_j) = 0$ in all other cases. Hence

$$(v, w)^2 = (pv_p, qw_q)^2 = \frac{p^2 q^2}{2}.$$

As v and w are linearly independent, the Cauchy-Schwarz inequality implies that $(v, w)^2 < (v, v)(w, w)$. Substituting, we get $2pq < (p+1)(q+1)$, and hence

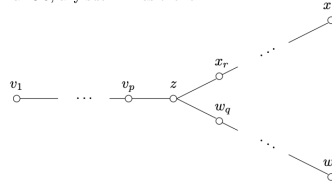
$$(p-1)(q-1) = pq - p - q + 1 < 2$$

so either $q = 1$ or $p = q = 2$. □

Proposition 13.1.14

If Γ has a branch point, then either Γ is D_n for some $n \geq 4$ or Γ is E_6 , E_7 , or E_8 .

Proof. By Lemma 13.1.11, any such Γ has the form



where, without loss of generality, $p \geq q \geq r$. We must show that either $q = r = 1$ or $q = 2$, $r = 1$, and $p \leq 4$.

As in the proof of the last proposition, we let $v = \sum_{i=1}^p i v_i$, $w = \sum_{i=1}^q i w_i$, and $x = \sum_{i=1}^r i x_i$. Then v, w, x are pairwise orthogonal. Let $\hat{v} = v/\|v\|$, $\hat{w} = w/\|w\|$, and $\hat{x} = x/\|x\|$. The space U spanned by v, w, x, z has as an orthonormal basis

$$\{\hat{v}, \hat{w}, \hat{x}, z_0\}$$

for some choice of z_0 which will satisfy $(z, z_0) \neq 0$. We may write

$$z = (z, \hat{v})\hat{v} + (z, \hat{w})\hat{w} + (z, \hat{x})\hat{x} + (z, z_0)z_0.$$

As z is a unit vector and $(z, z_0) \neq 0$, we get

$$(z, \hat{v})^2 + (z, \hat{w})^2 + (z, \hat{x})^2 < 1.$$

We know the lengths of v, w, x from Lemma ???. Furthermore, $(z, v)^2 = (z, pv_p)^2 = p^2/4$, and similarly $(z, w)^2 = q^2/4$ and $(z, x)^2 = r^2/4$. Substituting these into the previous inequality gives

$$\frac{2p^2}{4p(p+1)} + \frac{2q^2}{4q(q+1)} + \frac{2r^2}{4r(r+1)} < 1.$$

By elementary steps, this is equivalent to

$$\frac{1}{p+1} + \frac{1}{q+1} + \frac{1}{r+1} > 1.$$

Since $\frac{1}{p+1} \leq \frac{1}{q+1} \leq \frac{1}{r+1} \leq \frac{1}{2}$, we have $1 < \frac{3}{r+1}$ and hence $r < 2$, so we must have $r = 1$. Repeating this argument gives that $q < 3$, so $q = 1$ or $q = 2$. If $q = 2$, then we see that $p < 5$. On the other hand, if $q = 1$, then there is no restriction on p . $\square$

We have now found all connected graphs which come from admissible sets. We return to our connected Dynkin diagram Δ . We saw in Example 13.3 that the Coxeter graph of Δ , say $\bar{\Delta}$, must appear somewhere in our collection. If Δ has no multiple edges, then, by Proposition 13.1.14, $\Delta = \bar{\Delta}$ is one of the graphs listed in Theorem 13.1.1.

If Δ has a double edge, then Proposition 13.1.13 tells us that there are two possibilities for $\bar{\Delta}$. In the case of B_2 and F_4 , we get essentially the same graph whichever way we put the arrow; otherwise there are two different choices, giving B_n and C_n for $n \geq 3$. Finally, if Δ has a triple edge, then Corollary 13.1.8 tells us that $\Delta = G_2$. This completes the proof of Theorem 13.1.1.

§13.2 Constructions

We now want to show that all the Dynkin diagrams listed in Theorem 13.1 actually occur as the Dynkin diagram of some root system. Our analysis of the classical Lie algebras $\mathfrak{sl}_{\ell+1}$, $\mathfrak{so}_{2\ell+1}$, $\mathfrak{sp}_{2\ell}$, and $\mathfrak{so}_{2\ell}$ in Chapter 12 gives us constructions of root systems of types A , B , C , and D , respectively. We discuss the Weyl groups of these root systems in Appendix D. For the exceptional Dynkin diagrams G_2 , F_4 , E_6 , E_7 , and E_8 , we have to do more work. For completeness, we give constructions of all the corresponding root systems, but as those of type E are rather large and difficult to work with, we do not go into any details for this type. In each case, we shall take for the underlying space E a subspace of a Euclidean space $\mathbb{R}^m$. Let ϵ_i be the vector with 1 in position i and 0 elsewhere.

When describing bases, we shall follow the pattern established in Chapter 12 by taking as many simple roots as possible from the set $\{\alpha_1, \dots, \alpha_{m-1}\}$, where

$$\alpha_i := \epsilon_i - \epsilon_{i+1}.$$

For these elements, we have

$$\langle \alpha_i, \alpha_j \rangle = \begin{cases} 2 & i = j, \\ -1 & |i - j| = 1, \\ 0 & \text{otherwise,} \end{cases}$$

so the corresponding part of the Dynkin diagram is a line,

$$\cdots \alpha_{i-1} \quad \bigcirc - \bigcirc \quad \alpha_i \quad \bigcirc - \bigcirc \quad \alpha_{i+1} \cdots$$

and the corresponding part of the Cartan matrix is

$$\begin{pmatrix} \vdots & & \vdots & & \\ \cdots & 2 & -1 & 0 & \cdots \\ \cdots & -1 & 2 & -1 & \cdots \\ \cdots & 0 & -1 & 2 & \cdots \\ \vdots & & \vdots & & \end{pmatrix}.$$

Both of these will be familiar from root systems of type A .

§13.2.i Type G_2

We have already given one construction of a root system of type G_2 in Example 11.2.6 (iii). We give another here, which is more typical of the other constructions that follow. Let

$$\begin{aligned} E &= \left\{ v = \sum_{i=1}^3 c_i \epsilon_i \in \mathbb{R}^3 : \sum c_i = 0 \right\}, \\ I &= \left\{ m_1 \epsilon_1 + m_2 \epsilon_2 + m_3 \epsilon_3 \in \mathbb{R}^3 : m_1, m_2, m_3 \in \mathbb{Z} \right\}, \\ R &= \{ \alpha \in I \cap E : (\alpha, \alpha) = 2 \text{ or } (\alpha, \alpha) = 6 \}. \end{aligned}$$

This is motivated by noting that the ratio of the length of a long root to the length of a short root in a root system of type G_2 is $\sqrt{3}$. By direct calculation, one finds that

$$R = \{ \pm(\epsilon_i - \epsilon_j), i \neq j \} \cup \{ \pm(2\epsilon_i - \epsilon_j - \epsilon_k), \{i, j, k\} = \{1, 2, 3\} \}.$$

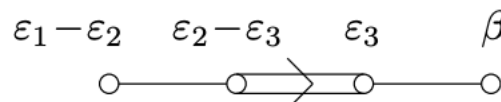
This gives 12 roots in total, as expected from the diagram in Example 11.2.6 (iii). To find a base, we need to find $\alpha, \beta \in R$ of different lengths, making an angle of $5\pi/6$. One suitable choice is $\alpha = \epsilon_1 - \epsilon_2$ and $\beta = \epsilon_2 + \epsilon_3 - 2\epsilon_1$. The Weyl group for G_2 is generated by the simple reflections s_α and s_β . By Exercise 11.5.3, it is the dihedral group of order 12.

§13.2.ii Type F_4

Since the Dynkin diagram F_4 contains the Dynkin diagram B_3 , we might hope to construct the corresponding root system by extending the root system of B_3 . Therefore we look for $\beta \in \mathbb{R}^4$ so that

$$B = (\epsilon_1 - \epsilon_2, \epsilon_2 - \epsilon_3, \epsilon_3, \beta)$$

is a base with Cartan numbers given by the labeled Dynkin diagram:



It is easy to see that the only possible choices for β are

$$\beta = -\frac{1}{2}(\epsilon_1 + \epsilon_2 + \epsilon_3) \pm \frac{1}{2}\epsilon_4.$$

Therefore it seems hopeful to set

$$R = \{ \pm\epsilon_i : 1 \leq i \leq 4 \} \cup \{ \pm\epsilon_i \pm \epsilon_j : 1 \leq i \neq j \leq 4 \} \cup \left\{ \frac{1}{2}(\pm\epsilon_1 \pm \epsilon_2 \pm \epsilon_3 \pm \epsilon_4) \right\}.$$

One can check directly that axioms (R1) up to (R4) hold; see Exercise ???. It remains to check that

$$\begin{aligned} \beta_1 &= \epsilon_1 - \epsilon_2, \\ \beta_2 &= \epsilon_2 - \epsilon_3, \\ \beta_3 &= \epsilon_3, \\ \beta_4 &= \frac{1}{2}(-\epsilon_1 - \epsilon_2 - \epsilon_3 + \epsilon_4), \end{aligned}$$

defines a base for R .

Note that R has 48 elements, so we need to find 24 positive roots. Each ϵ_i is a positive root, and if $1 \leq i < j \leq 3$ then so are $\epsilon_i - \epsilon_j$ and $\epsilon_i + \epsilon_j$. Furthermore, for $1 \leq i \leq 3$, also $\epsilon_4 \pm \epsilon_i$ are positive roots.

This already gives us 16 roots. In total, there are 16 roots of the form $\frac{1}{2}(\sum \pm \epsilon_i)$. As one would expect, half of these turn out to be positive roots. Obviously each must have a summand equal to β_4 . There are 3 positive roots of the form $\beta_4 + \epsilon_j$, and also 3 of the form $\beta_4 + \epsilon_j + \epsilon_k$. Then there is β_4 itself, and finally

$$\beta_4 + \epsilon_1 + \epsilon_2 + \epsilon_3 = \frac{1}{2} \sum \epsilon_i.$$

The Weyl group is known to have order $2^7 3^2$, but its structure is too complicated to be discussed here.

§13.2.iii Type E

To construct the root systems of types E , it will be convenient to first construct a root system of type E_8 and then to find root systems of types E_6 and E_7 inside it.

Let $E = \mathbb{R}^8$ and let

$$R = \{\pm \epsilon_i \pm \epsilon_j : i < j\} \cup \left\{ \frac{1}{2} \sum_{i=1}^8 \pm \epsilon_i \right\},$$

where in the second set an even number of $+$ signs are chosen.

The first set in the union contributes 112 roots and the second 128, giving 240 roots in all. Assuming that R is a root system, we claim that a base for R is given by $B = \{\beta_1, \beta_2, \dots, \beta_8\}$, where

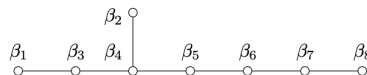
$$\begin{aligned} \beta_1 &= \frac{1}{2} \left(-\epsilon_1 - \epsilon_8 + \sum_{i=2}^7 \epsilon_i \right), \\ \beta_2 &= -\epsilon_1 - \epsilon_2, \\ \beta_i &= \epsilon_{i-2} - \epsilon_{i-1} \quad \text{for } 3 \leq i \leq 8. \end{aligned}$$

To see that B is a base, one first verifies that the roots $\pm(\epsilon_i - \epsilon_j)$ for $i < j$ can be written as linear combinations of the elements of B with positive coefficients. The remaining positive roots are those of the form

$$\frac{1}{2} \left(-\epsilon_8 + \sum_{i=1}^7 \pm \epsilon_i \right),$$

where there are an odd number of $+$ signs chosen in the sum. To check this, subtract off β_1 , and then verify that the result is a positive integral linear combination of the remaining roots. (This is where the condition on the signs comes in.)

The labeled Dynkin diagram is



Omitting the root β_8 gives a base for a root system of type E_7 , and omitting the roots β_7 and β_8 gives a base for a root system of type E_6 . We leave it to the keen reader to explicitly construct these root systems. More details may be found in Bourbaki, Lie Groups and Lie Algebras.

14 Simple Lie Algebras

Lemma 14.0.1

Suppose that Φ is a root system and that $\Phi = \Phi_1 \cup \Phi_2$ where $(\alpha, \beta) = 0$ for all $\alpha \in \Phi_1, \beta \in \Phi_2$.

- (i) If $\alpha \in \Phi_1$ and $\beta \in \Phi_2$, then $\alpha + \beta \notin \Phi$.
- (ii) If $\alpha, \alpha' \in \Phi_1$ and $\alpha + \alpha' \in \Phi$, then $\alpha + \alpha' \in \Phi_1$.

Proof. For (i), note that $(\alpha, \alpha + \beta) = (\alpha, \alpha) \neq 0$, so $\alpha + \beta \notin \Phi_2$. Similarly, $(\beta, \alpha + \beta) = (\beta, \beta) \neq 0$, so $\alpha + \beta \notin \Phi_1$.

To prove (ii), we suppose for a contradiction that $\alpha + \alpha' \in \Phi_2$. Remembering that $-\alpha' \in \Phi_1$, we have $\alpha = -\alpha' + (\alpha + \alpha')$, so α can be expressed as the sum of a root in Φ_1 and a root in Φ_2 . This contradicts the previous part. $\square$

Proposition 14.0.2

Let L be a complex semisimple Lie algebra with Cartan subalgebra H and root system Φ . If L is simple, then Φ is irreducible.

Proof. By the root space decomposition, we may write L as

$$L = H \oplus \bigoplus_{\alpha \in \Phi} L_{\alpha}.$$

Suppose that Φ is reducible, with $\Phi = \Phi_1 \cup \Phi_2$, where Φ_1 and Φ_2 are non-empty and $(\alpha, \beta) = 0$ for all $\alpha \in \Phi_1$ and $\beta \in \Phi_2$. We shall show that the root spaces L_{α} for $\alpha \in \Phi_1$ generate a proper ideal of L , and so L is not simple.

For each $\alpha \in \Phi_1$ we have defined a Lie subalgebra $\mathfrak{sl}(\alpha) \cong \mathfrak{sl}(2, \mathbb{C})$ of L with standard basis $\{e_{\alpha}, f_{\alpha}, h_{\alpha}\}$. Let

$$I := \text{Span}\{e_{\alpha}, f_{\alpha}, h_{\alpha} : \alpha \in \Phi_1\}.$$

The root space decomposition shows that I is a non-zero proper subspace of L .

We claim that I is an ideal of L ; it is a subspace by definition, so we only have to show that $[x, a] \in I$ for all $x \in L$ and $a \in I$. For this it suffices to take $a = e_{\alpha}$ and $a = f_{\alpha}$ for $\alpha \in \Phi_1$ since these elements generate I . Moreover, we may assume that x lies in one of the summands of the root space decomposition of L .

If $x \in H$, then $[x, e_{\alpha}] = \alpha(x)e_{\alpha} \in I$ and similarly $[x, f_{\alpha}] = -\alpha(x)f_{\alpha} \in I$. Suppose that $x \in L_{\beta}$. Then, for any $\alpha \in \Phi_1$, $[x, e_{\alpha}] \in L_{\alpha+\beta}$ by Lemma 10.1(i). If $\beta \in \Phi_2$, then by Lemma 14.1(a) above, we know that $\alpha + \beta$ is not a root, so $L_{\alpha+\beta} = 0$, and hence $[x, e_{\alpha}] \in I$. Otherwise $\beta \in \Phi_1$, and then by Lemma 14.1(b) we know that $\alpha + \beta \in \Phi_1$, so $L_{\alpha+\beta} \subseteq I$, by the definition of I . Similarly, one shows that $[x, f_{\alpha}] \in I$. (Alternatively, one may argue that as f_{α} is a scalar multiple of $e_{-\alpha}$, it is enough to look at the elements e_{α} .) $\square$

§14.1 Serre’s Theorem

Serre’s Theorem is a way to describe a complex semisimple Lie algebra by generators and relations that depend only on data from its Cartan matrix.

§14.1.i Generators

Let L be a complex semisimple Lie algebra with Cartan subalgebra H and root system Φ . Suppose that Φ has as a base $\{\alpha_1, \dots, \alpha_\ell\}$. For each i between 1 and ℓ let e_i, f_i, h_i be a standard basis of $\mathfrak{sl}(\alpha_i)$. We ask whether the e_i, f_i, h_i for $1 \leq i \leq \ell$ might already generate L ; that is, can every element of L be obtained by repeatedly taking linear combinations and Lie brackets of these elements?

Example 14.1.1

Let $L = \mathfrak{sl}(\ell + 1, \mathbb{C})$. We shall show that the elements $e_{i,i+1}$ and $e_{i+1,i}$ for $1 \leq i \leq \ell$ already generate L as a Lie algebra. By taking the commutators $[e_{i,i+1}, e_{i+1,i}]$ we get a basis for the Cartan subalgebra H of diagonal matrices. For $i + 1 < j$, we have $[e_{i,i+1}, e_{i+1,j}] = e_{i,j}$, and hence by induction we get all $e_{i,j}$ with $i < j$. Similarly, we may obtain all $e_{i,j}$ with $i > j$.

It is useful to look at these in terms of roots. Recall that the root system of L with respect to H has as a base $\alpha_1, \dots, \alpha_\ell$, where $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$. For $i < j$, we have $\text{Span}\{e_{i,j}\} = L_\beta$, where $\beta = \alpha_i + \gamma$ and $\gamma = \alpha_{i+1} + \dots + \alpha_{j-1}$. This can be expressed neatly using reflections since

$$s_{\alpha_i}(\gamma) = \gamma - \langle \gamma, \alpha_i \rangle \alpha_i = \gamma + \alpha_i = \beta.$$

In fact, this method gives a general way to obtain any non-zero root space. To show this, we only need to remind ourselves of some earlier results.

Lemma 14.1.2

Let L be a complex semisimple Lie algebra, and let $\{\alpha_1, \dots, \alpha_\ell\}$ be a base of the root system. Suppose $\{e_i, f_i, h_i\}$ is a standard basis of $\mathfrak{sl}(\alpha_i)$. Then L can be generated, as a Lie algebra, by $\{e_1, \dots, e_\ell, f_1, \dots, f_\ell\}$.

Proof. We first show that every element of H can be obtained. Since $h_i = [e_i, f_i]$, it is sufficient to prove that H is spanned by $h_1, \dots, h_\ell$. Recall that we have identified H with H^* , via the Killing form κ , so that $\alpha_i \in H^*$ corresponds to the element $t_{\alpha_i} \in H$. As H^* is spanned by the roots $\alpha_1, \dots, \alpha_\ell$, H has as a basis $\{t_{\alpha_i} : 1 \leq i \leq \ell\}$. By Lemma 10.6, h_i is a non-zero scalar multiple of t_{α_i} . Hence $\{h_1, \dots, h_\ell\}$ is a basis for H .

Now let $\beta \in \Phi$. We want to show that L_β is contained in the Lie subalgebra generated by the e_i and the f_i . Call this subalgebra $\tilde{L}$. By Proposition 11.14, we know that $\beta = w(\alpha_j)$, where w is a product of reflections s_{α_i} for some base elements α_i . Hence, by induction on the number of reflections, it is enough to prove the following: If $\beta = s_{\alpha_i}(\gamma)$ for some $\gamma \in \Phi$ with $L_\gamma \subseteq \tilde{L}$, then $L_\beta \subseteq \tilde{L}$.

By hypothesis, $\beta = \gamma - \langle \gamma, \alpha_i \rangle \alpha_i$. In Proposition 10.10, we looked at the $\mathfrak{sl}(\alpha_i)$ -submodule of L defined by

$$\bigoplus_k L_{\gamma+k\alpha_i},$$

where the sum is over all $k \in \mathbb{Z}$ such that $\gamma + k\alpha_i \in \Phi$, and the module structure is given by the adjoint action of $\mathfrak{sl}(\alpha_i)$. We proved that this is an irreducible $\mathfrak{sl}(\alpha_i)$ -module. If $0 \neq e_\gamma \in L_\gamma$, then by applying powers of ade or adf we may obtain $e_{\gamma+k\alpha_i}$ whenever $\gamma + k\alpha_i \in \Phi$. Hence, if we take $k = \langle \gamma, \alpha_i \rangle$, then we will obtain e_β . Hence L_β is contained in $\tilde{L}$. $\square$

§14.1.ii Relations

Next, we search for relations satisfied by the e_i, f_i, h_i . These should only involve information which can be obtained from the Cartan matrix. We write $c_{ij} = \langle \alpha_i, \alpha_j \rangle$. Note that since the angle between any two base elements is obtuse (see Exercise 11.3), $c_{ij} \leq 0$ for all $i \neq j$.

Lemma 14.1.3

The elements e_i, f_i, h_i for $1 \leq i \leq \ell$ satisfy the following relations.

- (S3) $[h_i, h_j] = 0$ for all i, j ;
- (S3) $[h_i, e_j] = c_{ji}e_j$ and $[h_i, f_j] = -c_{ji}f_j$ for all i, j ;
- (S3) $[e_i, f_i] = h_i$ for each i and $[e_i, f_j] = 0$ if $i \neq j$;
- (S3) $(\text{ade}_i)^{1-c_{ji}}(e_j) = 0$ and $(\text{adf}_i)^{1-c_{ji}}(f_j) = 0$ if $i \neq j$.

Proof. We know H is a Cartan subalgebra and hence it is abelian, so (S1) holds. Condition (S2) follows from

$$[h_i, e_j] = \alpha_j(h_i)e_j = \langle \alpha_j, \alpha_i \rangle e_j = c_{ji}e_j,$$

while the first part of (S3) follows from the isomorphism of $\mathfrak{sl}(\alpha_i)$ with $\mathfrak{sl}(2, \mathbb{C})$. If $i \neq j$, we have $[e_i, f_j] \in L_{\alpha_i - \alpha_j}$; see Lemma 10.1(i). But since $\alpha_1, \dots, \alpha_\ell$ form a base for Φ , $\alpha_i - \alpha_j \notin \Phi$. Therefore $L_{\alpha_i - \alpha_j} = 0$. This proves the second part of (S3).

To prove (S4), we fix α_i, α_j in the base and consider

$$M = \bigoplus_k L_{\alpha_j+k\alpha_i},$$

where the sum is taken over all $k \in \mathbb{Z}$ such that $\alpha_j + k\alpha_i \in \Phi$. As before, this is an $\mathfrak{sl}(\alpha_i)$ -module. Since $\alpha_j - \alpha_i \notin \Phi$, the sum only involves $k \geq 0$ and $k = 0$ does occur. Thus the smallest eigenvalue of $\text{ad } h_i$ on M is $\langle \alpha_j, \alpha_i \rangle = c_{ji}$. By the classification of irreducible $\mathfrak{sl}(2, \mathbb{C})$ -modules, the largest eigenvalue of $\text{ad } h_i$ must be $-c_{ji}$.

An $\text{ad } h_i$ eigenvector with eigenvalue $-c_{ji}$ is given by $x = (\text{ade}_i)^{-c_{ji}}(e_j)$, so applying ade_i to x gives zero. This proves the first part of (S4). In fact, we have even proved that $1 - c_{ji}$ is the minimal integer $r \geq 0$ such that $(\text{ade}_i)^r(e_j) = 0$.

The other part of (S4) is proved by the same method. (Alternatively, one might note that the set of $-\alpha_j$ also is a base for the root system with standard basis $f_i, e_i, -h_i$.) $\square$

Serre's Theorem says that these relations completely determine the Lie algebra.

Theorem 14.1.4 (Serre’s Theorem)

Let C be the Cartan matrix of a root system. Let L be the complex Lie algebra which is generated by elements e_i, f_i, h_i for $1 \leq i \leq \ell$, subject to the relations (S1) to (S4). Then L is finite-dimensional and semisimple with Cartan subalgebra H spanned by $\{h_1, \dots, h_\ell\}$, and its root system has Cartan matrix C .

We immediately give our main application. Suppose that L is a complex semisimple Lie algebra with Cartan matrix C . By Lemma 14.5 this Lie algebra satisfies the Serre relations, so we can deduce that it must be isomorphic to the Lie algebra in Serre’s Theorem with Cartan matrix C . Hence, up to isomorphism, there is just one Lie algebra for each root system.

Serre’s Theorem also solves the problem of constructing Lie algebras with the exceptional root systems G_2, F_4, E_6, E_7 , and E_8 . Just apply it with the Cartan matrix for the type required! Moreover, it shows that, up to isomorphism, there is just one exceptional Lie algebra for each type.

Remark 14.1.5 — Combining Lemma 14.1.3 and Theorem 14.1.4, we have the following: Given the Cartan matrix of a root system, one can write down generators and relations for a semisimple Lie algebra with this root system. Conversely, any semisimple Lie algebra with this root system is isomorphic to this explicit one.

Example 14.1.6

The root system of type A_2 corresponds to e, f, h with $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$, which is just $\mathfrak{sl}(2, \mathbb{C})$.

Remark 14.1.7 — We note that going the converse of this example and getting that it is finite dimensional is quite difficult, and not necessarily true. The constraints are really doing something.

Before going on to the proof, we introduce an example illustrating all of this.

Example 14.1.8

Consider $\mathfrak{sl}(3, \mathbb{C})$. Let H be the subalgebra of $\mathfrak{sl}(3, \mathbb{C})$ of diagonal matrices of trace 0. Let $h \in H$ have diagonal entries a_1, a_2, a_3 . Then,

$$[h, e_{ij}] = (a_i - a_j)e_{ij}.$$

Thus, the e_{ij} are eigenvectors for the action of $\text{ad } h$, for all $h \in H$. Then, if $x \in \mathfrak{sl}(3, \mathbb{C})$ has $[h, x] = 0$, for all $h \in H$, we see that $x \in H$. For this, we can just see that $x = x_0 + \sum_{i \neq j} x_{ij}e_{ij}$, with $x_0 \in H$. Then,

$$[h, x] = \sum_{i \neq j} x_{ij}(a_i - a_j)e_{ij},$$

Thus, the x_{ij} (for $i \neq j$) better be 0. Thus, H is abelian, and all $h \in H$ are diagonalizable. Now denote $L_{ij} := \mathbb{C}e_{ij}$ to be the 1-dimensional vector spaces with $[h, L_{ij}] \subset L_{ij}$, for all $h \in H$. Then,

$$\mathfrak{sl}(3, \mathbb{C}) = H \bigoplus_{i \neq j} L_{ij}.$$

Now, H is 2-dimensional, and $\mathfrak{sl}(3, \mathbb{C})$ is 8-dimensional, so there are 6 of the L_{ij} s. We will take e_{12}, e_{23} as our base for the root system. Also, note that $[h, e_{ij}] = (a_i - a_j)e_{ij}$ and $[h, e_{ji}] = (a_j - a_i)e_{ji}$. Recall that a root of H is a map $\alpha \in H^*$ which satisfies $[h, e_\alpha] = \alpha(h)e_\alpha$, for all $h \in H$, where e_α is a basis for the root space L_α . Therefore, the root corresponding to e_{ij} is the map

$$\begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} \mapsto a_i - a_j.$$

Thus,

roots	α	β	$\alpha + \beta$	$-\alpha$	$-\beta$	$-\alpha - \beta$
eigenvectors	e_{12}	e_{23}	e_{13}	e_{21}	e_{32}	e_{31}

Thus, indeed, e_{12}, e_{23} was a valid basis, and this gives us a good starting point to work out the root system, by setting α to the root corresponding to e_{12} , and β to the root corresponding to e_{23} . Then, we just need to find the $\langle \beta, \alpha \rangle$. We note that by Exercise 10.6.6, we get that $\beta(h\alpha) = \langle \beta, \alpha \rangle$. We note that if L is a semisimple Lie algebra, with root spaces $L_\alpha, L_{-\alpha}$, then $[x_\alpha, x_{-\alpha}] \neq 0$, where $x_\alpha \in L_\alpha$ and $x_{-\alpha} \in L_{-\alpha}$. Then, $h_\alpha \in H$ is the unique multiply of $[x_\alpha, x_{-\alpha}]$ such that $[h_\alpha, x_\alpha] = \alpha(h_\alpha)x_\alpha = 2x_\alpha$. Then, we get that $\alpha(h_\alpha) = 2$, which is what we expect from the formula $\beta(h\alpha) = \langle \beta, \alpha \rangle$. We need to find x_α and $x_{-\alpha}$, but these just come from the eigenvectors above. For instance,

$$[x_\alpha, x_{-\alpha}] = [e_{12}, e_{21}] = e_{11} - e_{22}.$$

Then, $\alpha(e_{11} - e_{22}) = 1 - (-1) = 2$, so we see that $h_\alpha = e_{11} - e_{22}$, and $\beta(h_\alpha) = -1$. With α and β reverse, we get the same answer. Thus, $\langle \beta, \alpha \rangle = -1$ and $\langle \alpha, \beta \rangle = -1$.

Remark 14.1.9 — Now recall Proposition 12.1.2. What we just did then is perform

the analysis that the proposition guides us through.

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